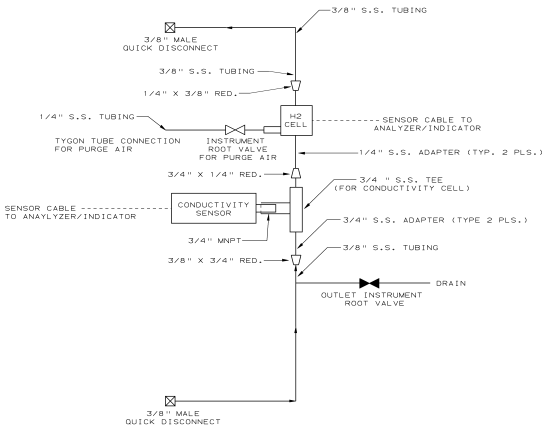
DETAIL S06

PROJECT CLASS 424
CVCS SAMPLING ASSEMBLY



CONINUATION OF NOTES:

16. IN-LINE HYDROGEN ANALYZERS AND CONDUCTIVITY ELEMENTS HAVE REPLACED THE "SAMPLE BOMB" AND WILL PROVIDE CONTINUOUS SAMPLE CAPABILITY. REFERENCE MOC 98-VAM045.

THIS DOCUMENT SUPERSEDES MULTI-SHEET DRAWING 1X4DB140
WORK THIS DRAWING WITH: 1X4DB140-1

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORPORATED PER REA 01-VAA037	06/04/01	JGM	JRP	PDG	JLM
2	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				
1						
0						

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
NUCLEAR SAMPLING SYSTEM-LIQUID
SYSTEM NO. 1212

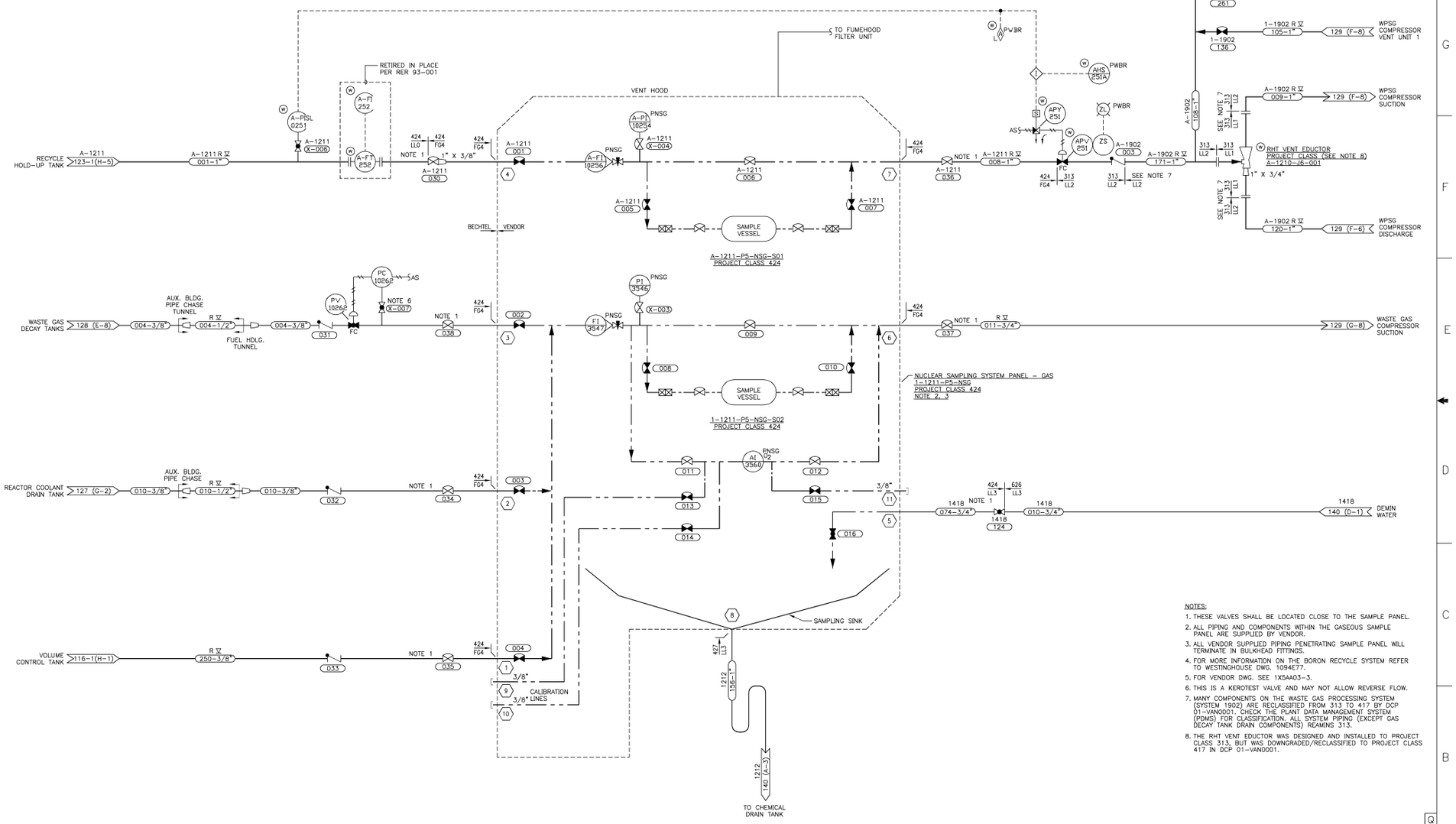
SCALE: NONE

DRAWING NO. 1X4DB140-2 3

REV. 35

JOB NO.10604

SIZE E 34X44



NOTES:

1. THESE VALVES SHALL BE LOCATED CLOSE TO THE SAMPLE PANEL.
2. ALL PIPING AND COMPONENTS WITHIN THE GASEOUS SAMPLE PANEL ARE SUPPLIED BY VENDOR.
3. ALL VENDOR SUPPLIED PIPING PENETRATING SAMPLE PANEL WILL TERMINATE IN BULKHEAD FITTINGS.
4. FOR MORE INFORMATION ON THE BORON RECYCLE SYSTEM REFER TO WESTINGHOUSE DWG. 1094E77.
5. FOR VENDOR DWG. SEE 1X5AA03-3.
6. THIS IS A KEROTEST VALVE AND MAY NOT ALLOW REVERSE FLOW.
7. MANY COMPONENTS ON THE WASTE GAS PROCESSING SYSTEM (SYSTEM 1902) ARE RECLASSIFIED FROM 313 TO 417 BY DCP 01-VAN0001. CHECK THE PLANT DATA MANAGEMENT SYSTEM (PDMS) FOR CLASSIFICATION. ALL SYSTEM PIPING (EXCEPT GAS DECAY TANK DRAIN COMPONENTS) REAMINS 313.
8. THE RHT VENT EDUCTOR WAS DESIGNED AND INSTALLED TO PROJECT CLASS 313, BUT WAS DOWNGRADED/RECLASSIFIED TO PROJECT CLASS 417 IN DCP 01-VAN0001.

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NO.		DATE	DR	CHK	APPV	DTL
20.0	REDRAWN PER ABN-V70009, VER. 1.0	5-10-06	CYN	LCF	SHS	X

CAD 1X4DB141
ACADZ002 CYN-01

SOUTHERN COMPANY

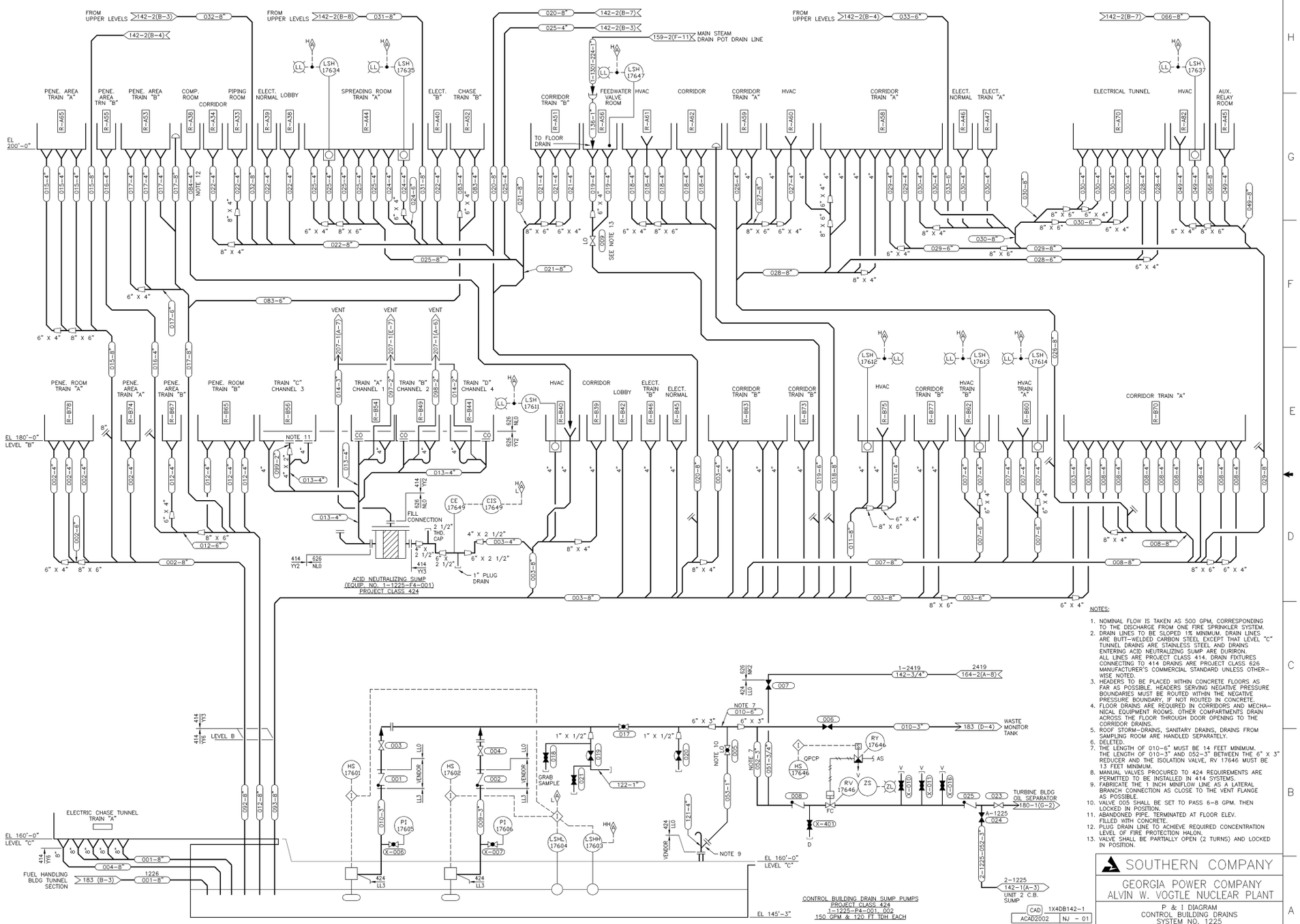
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
NUCLEAR SAMPLING SYSTEM - GASEOUS
SYSTEM NO. 1211

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB141	20.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. NOMINAL FLOW IS TAKEN AS 500 GPM, CORRESPONDING TO THE DISCHARGE FROM ONE FIRE SPRINKLER SYSTEM.
2. DRAIN LINES TO BE SLOPED 1% MINIMUM. DRAIN LINES ARE BUTT-WELDED CARBON STEEL EXCEPT THAT LEVEL "C" TUNNEL DRAINS ARE STAINLESS STEEL AND DRAINS ENTERING ACID NEUTRALIZING SUMP ARE DURIRON. ALL LINES ARE PROJECT CLASS 414. DRAIN FIXTURES CONNECTING TO 414 DRAINS ARE PROJECT CLASS 626 MANUFACTURER'S COMMERCIAL STANDARD UNLESS OTHERWISE NOTED.
3. HEADERS TO BE PLACED WITHIN CONCRETE FLOORS AS FAR AS POSSIBLE. HEADERS SERVING NEGATIVE PRESSURE BOUNDARIES MUST BE ROUTED WITHIN THE NEGATIVE PRESSURE BOUNDARY, IF NOT ROUTED IN CONCRETE.
4. FLOOR DRAINS ARE REQUIRED IN CORRIDORS AND MECHANICAL EQUIPMENT ROOMS. OTHER COMPARTMENTS DRAIN ACROSS THE FLOOR THROUGH DOOR OPENING TO THE CORRIDOR DRAINS.
5. ROOF STORM-DRAINS, SANITARY DRAINS, DRAINS FROM SAMPLING ROOM ARE HANDLED SEPARATELY.
6. DELETED.
7. THE LENGTH OF 010-6" MUST BE 14 FEET MINIMUM. THE LENGTH OF 010-3" AND 052-3" BETWEEN THE 6" X 3" REDUCER AND THE ISOLATION VALVE, RV 17646 MUST BE 13 FEET MINIMUM.
8. MANUAL VALVES PROCURED TO 424 REQUIREMENTS ARE PERMITTED TO BE INSTALLED IN 414 SYSTEMS.
9. FABRICATE THE 1 INCH MINIFLOW LINE AS A LATERAL BRANCH CONNECTION AS CLOSE TO THE VENT FLANGE AS POSSIBLE.
10. VALVE 005 SHALL BE SET TO PASS 6-8 GPM. THEN LOCKED IN POSITION.
11. ABANDONED PIPE. TERMINATED AT FLOOR ELEV. FILLED WITH CONCRETE.
12. PLUG DRAIN LINE TO ACHIEVE REQUIRED CONCENTRATION LEVEL OF FIRE PROTECTION HALON.
13. VALVE SHALL BE PARTIALLY OPEN (2 TURNS) AND LOCKED IN POSITION.

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NO.		DATE	DR	CHK	APPV	DTL
27.0	REDRAWN PER ABN- 52224, VER. 1.0	1-9-06	NJ	LCF	WPE	X

CAD 1X4DB142-1
ACAD2002 NJ - 01

SOUTHERN COMPANY

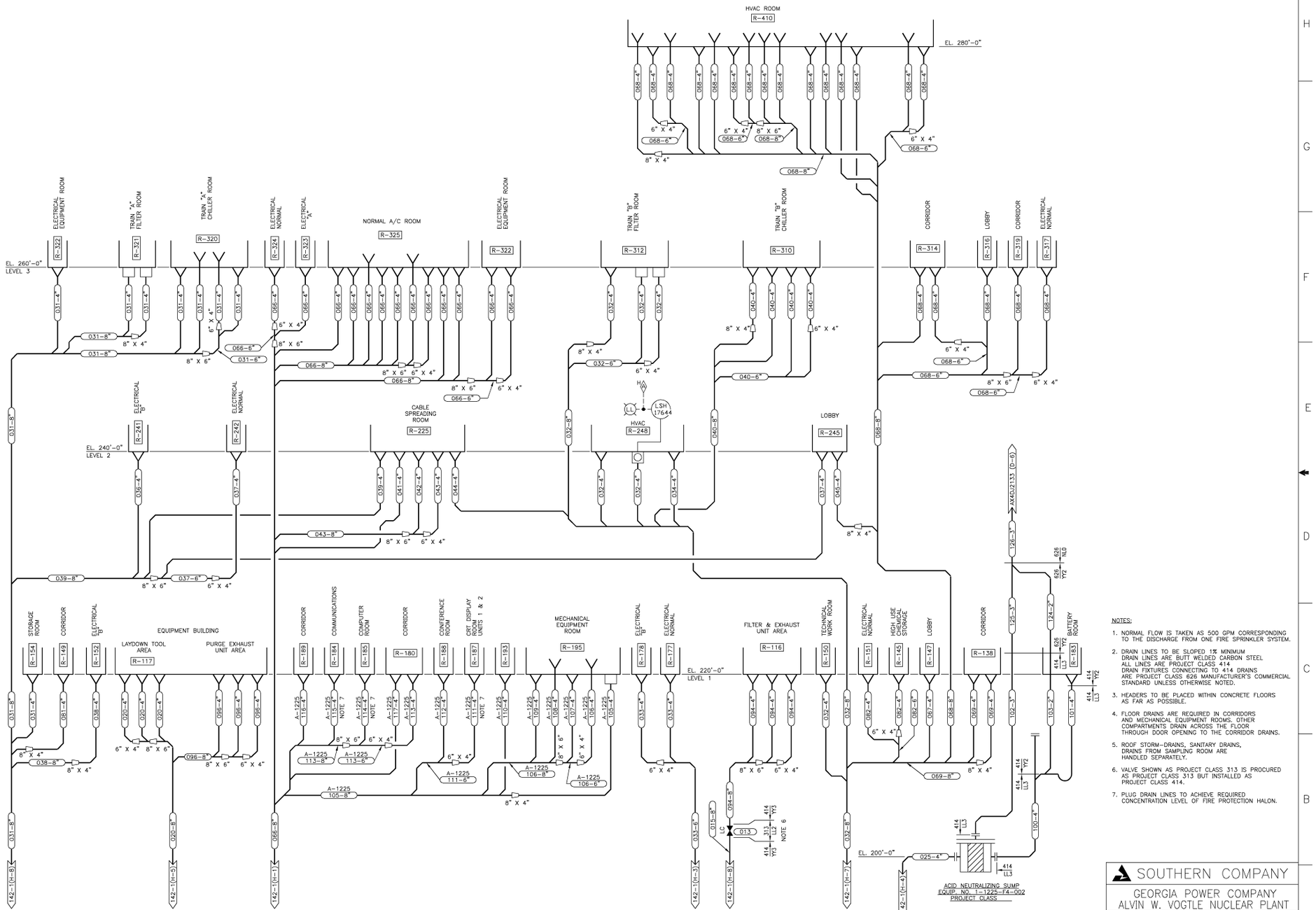
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTROL BUILDING DRAINS
SYSTEM NO. 1225

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB142-1	27.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



CAD 1X4DB142-2
ACAD2002 NJ - 01

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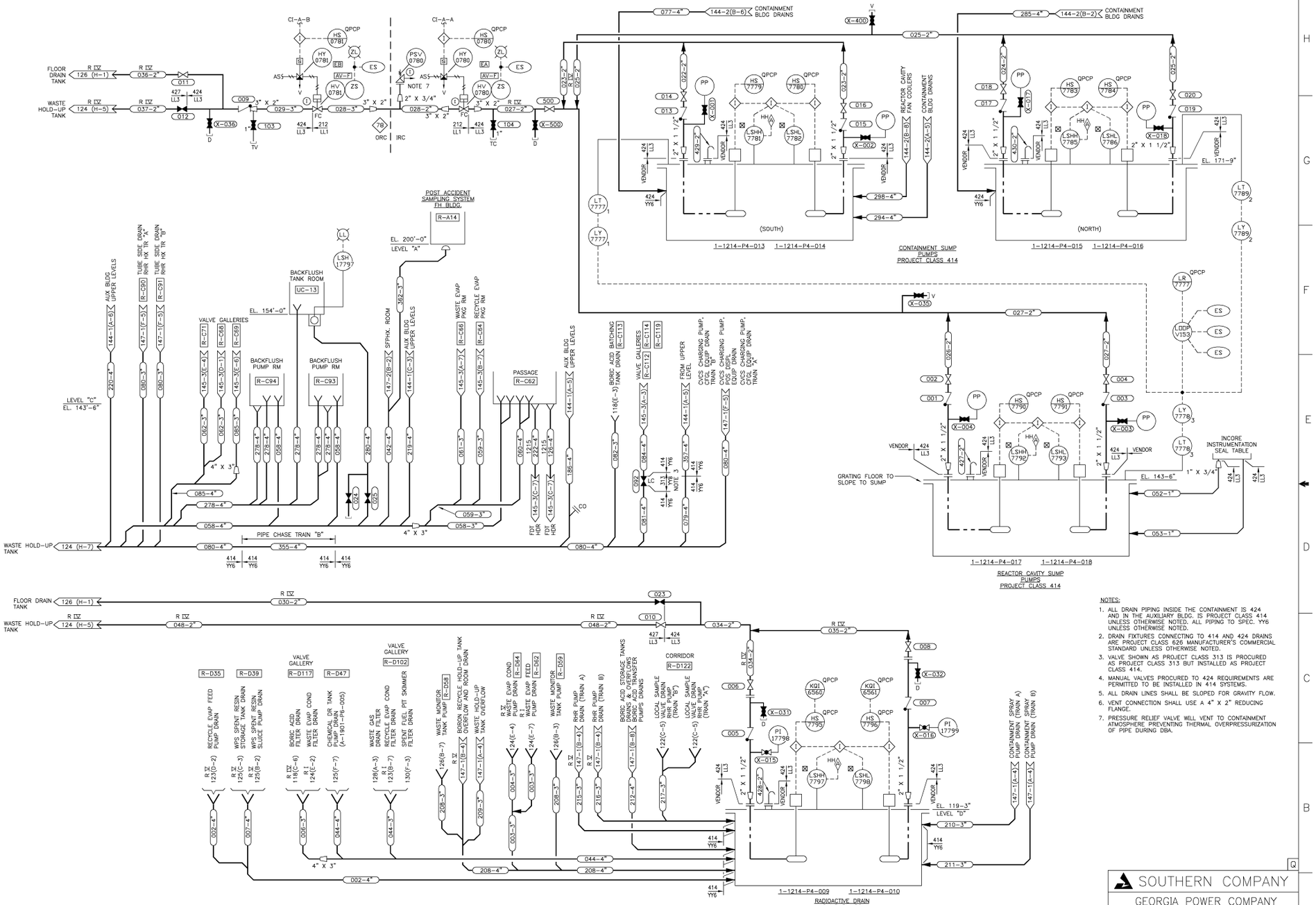
NO.		DATE	DR	CHK	APPV	DTL
8.0	REDRAWN PER ABN- 52270, VER. 1.0	1-23-06	NJ	LCF	WPE	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTROL BUILDING DRAINS
SYSTEM NO. 1225

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB142-2
VER. 8.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. ALL DRAIN PIPING INSIDE THE CONTAINMENT IS 424 AND IN THE AUXILIARY BLDG. IS PROJECT CLASS 414 UNLESS OTHERWISE NOTED. ALL PIPING TO SPEC. YY6 UNLESS OTHERWISE NOTED.
2. DRAIN FIXTURES CONNECTING TO 414 AND 424 DRAINS ARE PROJECT CLASS 626 MANUFACTURER'S COMMERCIAL STANDARD UNLESS OTHERWISE NOTED.
3. VALVE SHOWN AS PROJECT CLASS 313 IS PROCURED AS PROJECT CLASS 313 BUT INSTALLED AS PROJECT CLASS 414.
4. MANUAL VALVES PROCURED TO 424 REQUIREMENTS ARE PERMITTED TO BE INSTALLED IN 414 SYSTEMS.
5. ALL DRAIN LINES SHALL BE SLOPED FOR GRAVITY FLOW.
6. VENT CONNECTION SHALL USE A 4" X 2" REDUCING FLANGE.
7. PRESSURE RELIEF VALVE WILL VENT TO CONTAINMENT ATMOSPHERE PREVENTING THERMAL OVERPRESSURIZATION OF PIPE DURING DBA.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONTAINMENT & AUXILIARY BLDG.
SYSTEM NO. 1214

CAD 1X4DB143
ACADZ002 CYN- 01

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NO.		DATE	DR	CHK	APPV	DTL
25.0	REDRAWN PER ABN-V70013, VER. 1.0.	4-3-06	CYN	LCF	LKB	X

SIZE E 34X44

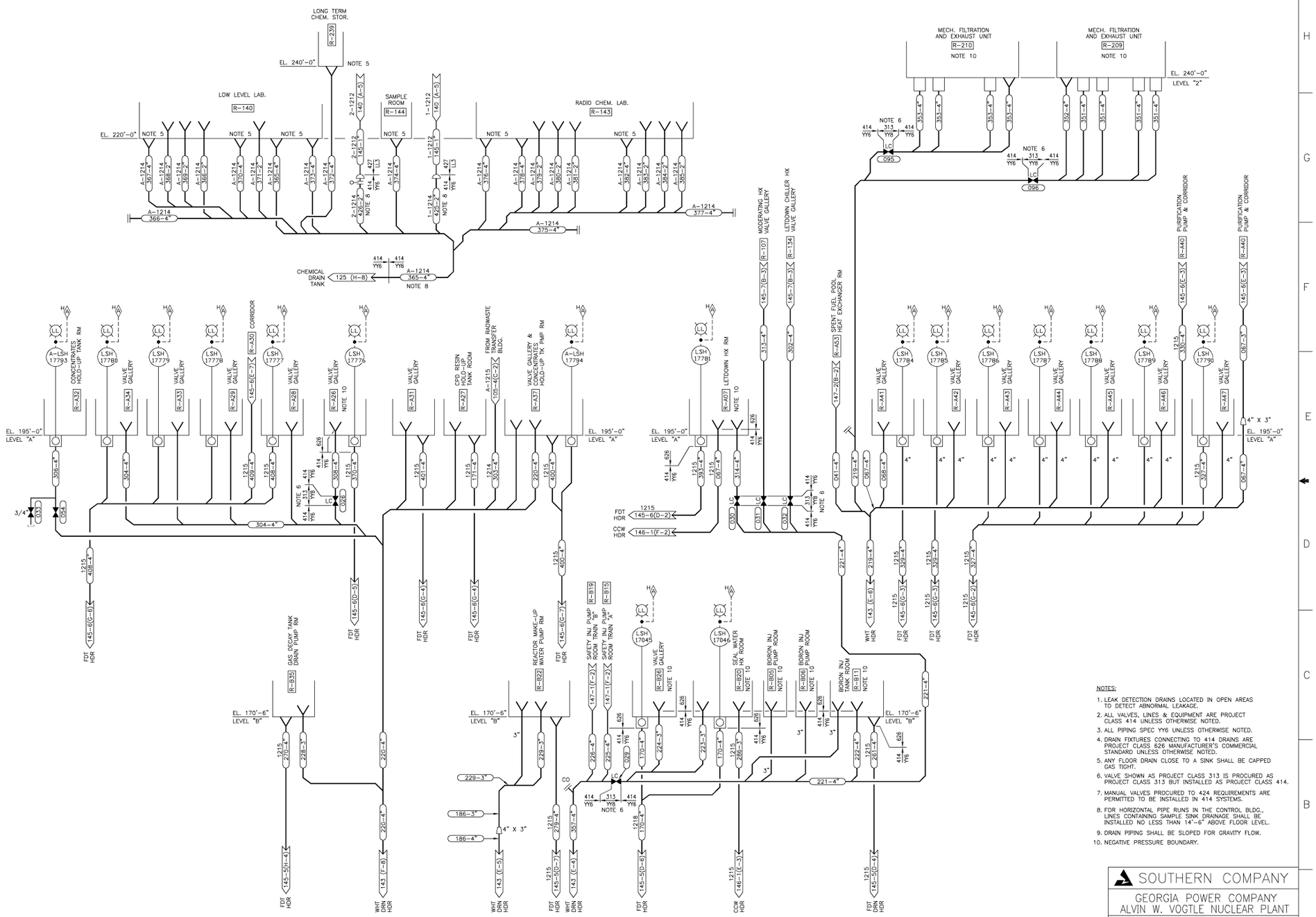
SCALE: NONE

DRAWING NO. 1X4DB143

VER. 25.0

JOB NO.10604

DRAWING CATEGORY: CRITICAL



NOTES:

1. LEAK DETECTION DRAINS LOCATED IN OPEN AREAS TO DETECT ABNORMAL LEAKAGE.
2. ALL VALVES, LINES & EQUIPMENT ARE PROJECT CLASS 414 UNLESS OTHERWISE NOTED.
3. ALL PIPING SPEC YY6 UNLESS OTHERWISE NOTED.
4. DRAIN FIXTURES CONNECTING TO 414 DRAINS ARE PROJECT CLASS 626 MANUFACTURER'S COMMERCIAL STANDARD UNLESS OTHERWISE NOTED.
5. ANY FLOOR DRAIN CLOSE TO A SINK SHALL BE CAPPED GAS TIGHT.
6. VALVE SHOWN AS PROJECT CLASS 313 IS PROCURED AS PROJECT CLASS 313 BUT INSTALLED AS PROJECT CLASS 414.
7. MANUAL VALVES PROCURED TO 424 REQUIREMENTS ARE PERMITTED TO BE INSTALLED IN 414 SYSTEMS.
8. FOR HORIZONTAL PIPE RUNS IN THE CONTROL BLDG., LINES CONTAINING SAMPLE SINK DRAINAGE SHALL BE INSTALLED NO LESS THAN 14'-6" ABOVE FLOOR LEVEL.
9. DRAIN PIPING SHALL BE SLOPED FOR GRAVITY FLOW.
10. NEGATIVE PRESSURE BOUNDARY.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM – CONTAINMENT & AUX BLDG DRAINS – RADIOACTIVE SYSTEM NO. 1214

CAD 1X4DB144-1
ACAD2002 CYN-01

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NO.		DATE	DR	CHK	APPV	DTL
12.0	REDRAWN PER ABN-52274, VER. 1.0	2-24-06	CYN	LCF	LKB	X

SIZE E 34X44

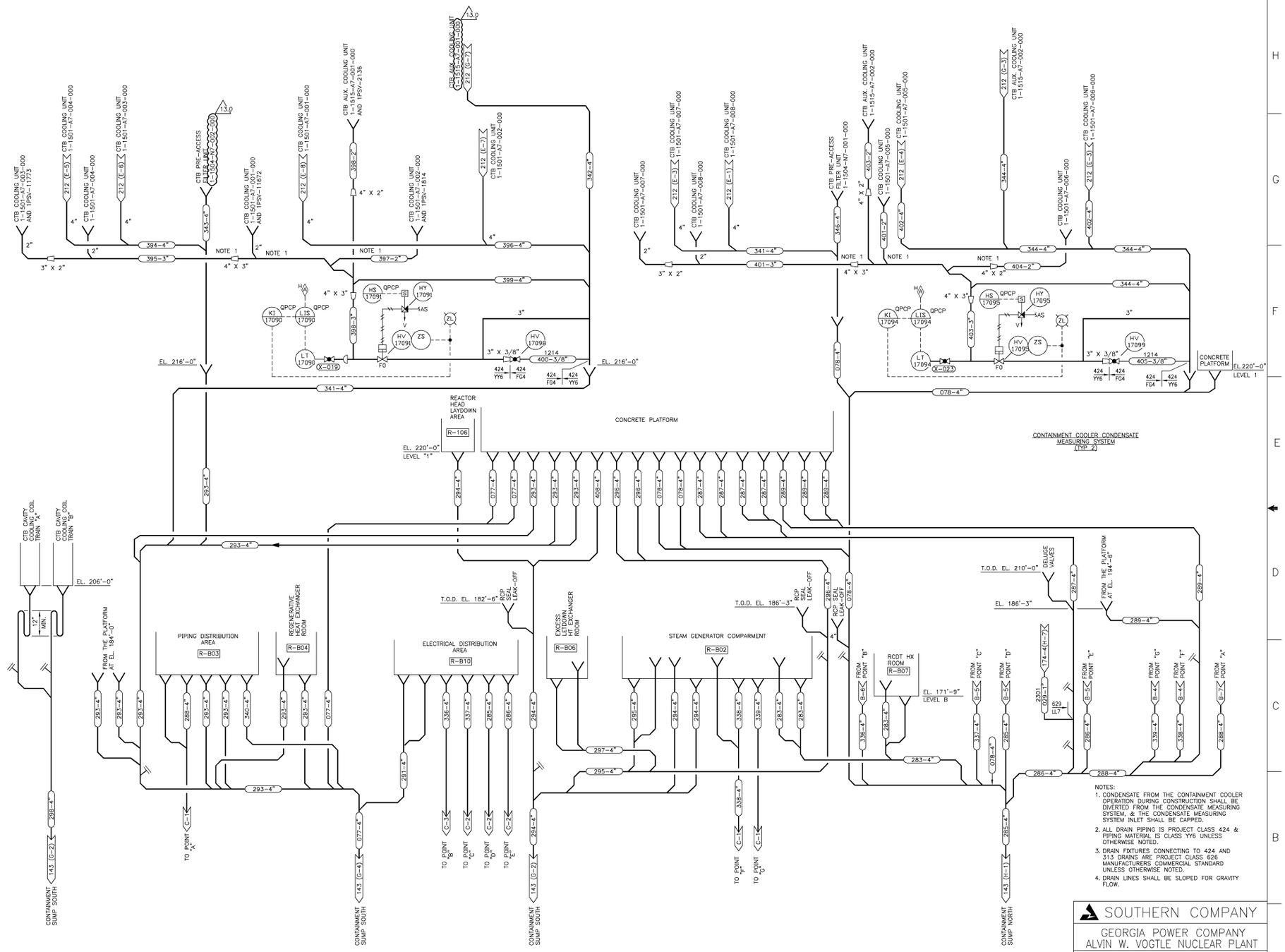
SCALE: NONE

DRAWING NO. 1X4DB144-1

VER. 12.0

JOB NO.10604

DRAWING CATEGORY: CRITICAL



NOTES:
1. CONDENSATE FROM THE CONTAINMENT COOLER OPERATION DURING CONSTRUCTION SHALL BE DIVERTED FROM THE CONDENSATE MEASURING SYSTEM, & THE CONDENSATE MEASURING SYSTEM INLET SHALL BE CAPPED.
2. ALL DRAIN PIPING IS PROJECT CLASS 424 & PIPING MATERIAL IS CLASS YY6 UNLESS OTHERWISE NOTED.
3. DRAIN FITTURES CONNECTING TO 424 AND 313 DRAINS ARE PROJECT CLASS 626 MANUFACTURERS COMMERCIAL STANDARD UNLESS OTHERWISE NOTED.
4. DRAIN LINES SHALL BE SLOPED FOR GRAVITY FLOW.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
13.0	REVISED PER ABN-V01497, VER. 1.0.	1-28-09	TJS	LRP	PML	X

SIZE E 34X44

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

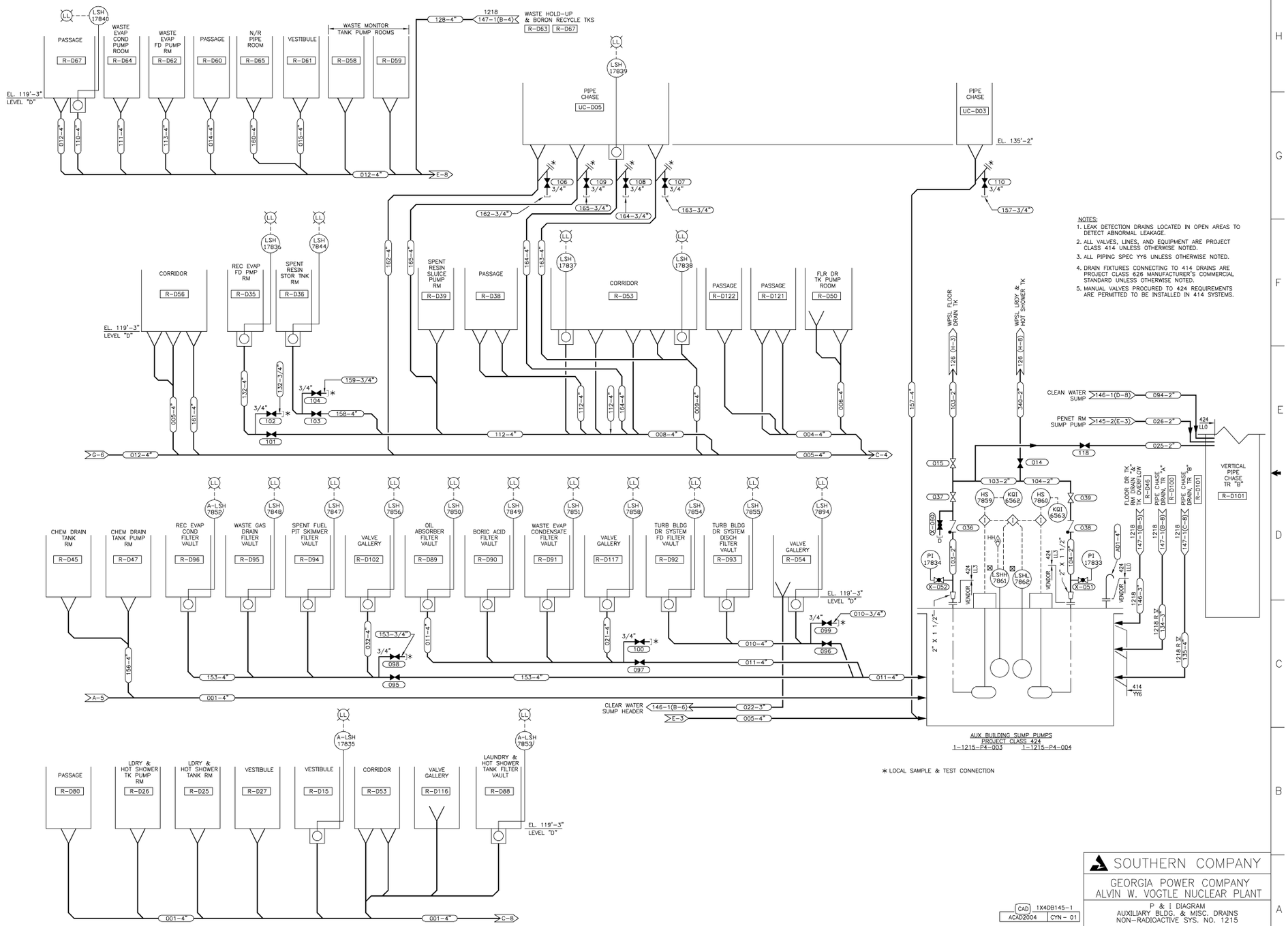
P & I DIAGRAM
CONTAINMENT & AUX. BLDG. DRAINS-
RADIOACTIVE SYSTEM - SYSTEM NO. 1214

CAD 1X4DB144-2
Acad2002 TJS - 02

SCALE: NONE
JOB NO.10604

DRAWING NO. 1X4DB144-2
VER. 13.0

DRAWING CATEGORY: CRITICAL



CAD 1X4DB145-1
ACAD2004 CYN - 01

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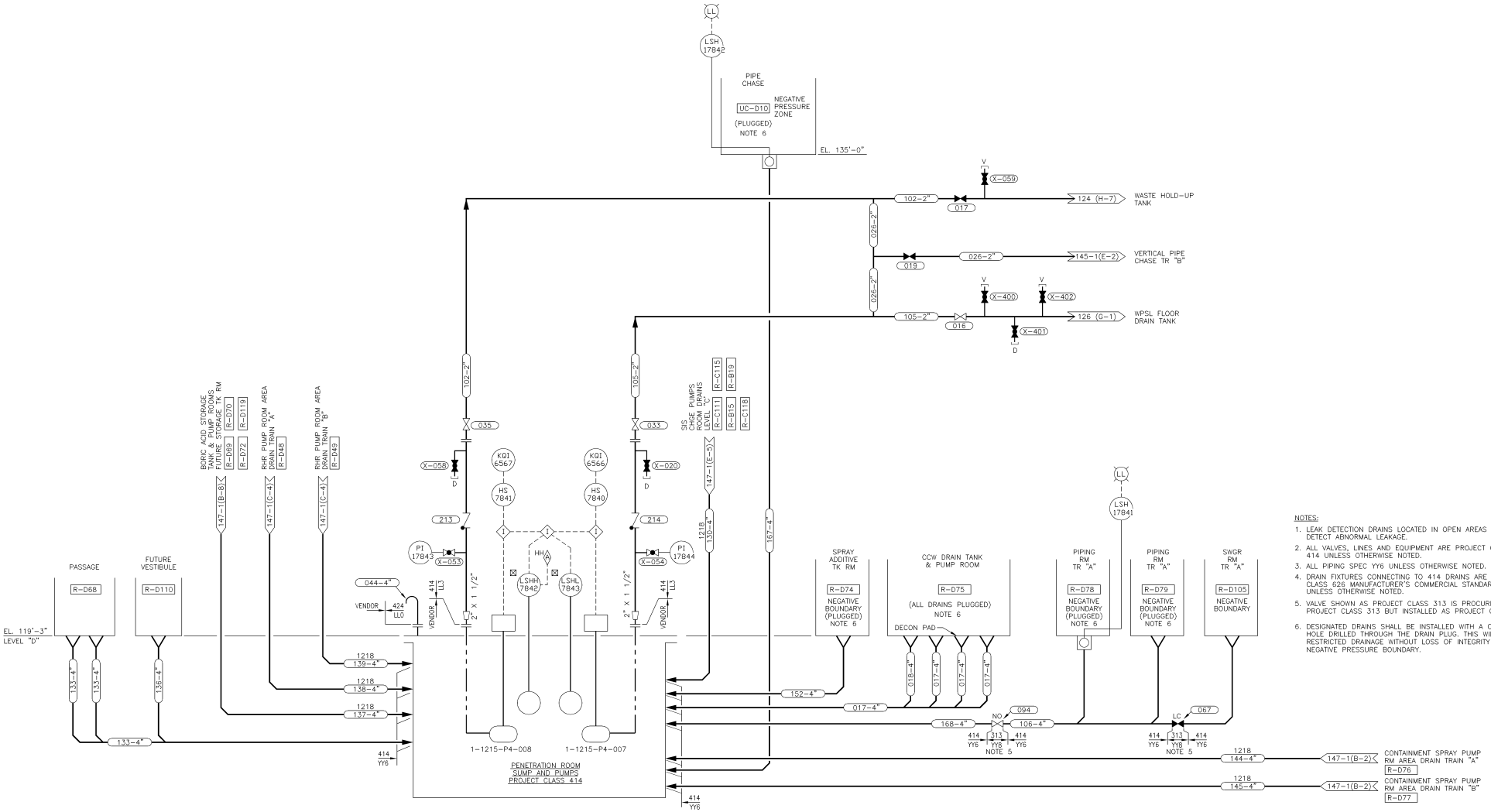
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
15.0	REDRAWN PER ABN-V70140, VER. 1.0	10-30-06	CYN	LCF	WPE	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY BLDG. & MISC. DRAINS
NON-RADIOACTIVE SYS. NO. 1215

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB145-1
VER. 15.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. LEAK DETECTION DRAINS LOCATED IN OPEN AREAS TO DETECT ABNORMAL LEAKAGE.
2. ALL VALVES, LINES AND EQUIPMENT ARE PROJECT CLASS 414 UNLESS OTHERWISE NOTED.
3. ALL PIPING SPEC YY6 UNLESS OTHERWISE NOTED.
4. DRAIN FIXTURES CONNECTING TO 414 DRAINS ARE PROJECT CLASS 626 MANUFACTURER'S COMMERCIAL STANDARD UNLESS OTHERWISE NOTED.
5. VALVE SHOWN AS PROJECT CLASS 313 IS PROCURED AS PROJECT CLASS 313 BUT INSTALLED AS PROJECT CLASS 414.
6. DESIGNATED DRAINS SHALL BE INSTALLED WITH A ONE-INCH HOLE DRILLED THROUGH THE DRAIN PLUG. THIS WILL ALLOW RESTRICTED DRAINAGE WITHOUT LOSS OF INTEGRITY OF THE NEGATIVE PRESSURE BOUNDARY.

CAD 1X4DB145-2
ACAD2004 CYN - 01

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NO.		DATE	DR	CHK	APPV	DTL
19.0	REDRAWN PER ABN-V70077, VER. 1.0	06-28-16	CYN	LCF	WPE	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY BLDG. & MISC. DRAINS
NON-RADIOACTIVE - SYS NO. 1215

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB145-2
VER. 19.0
SIZE E 34X44
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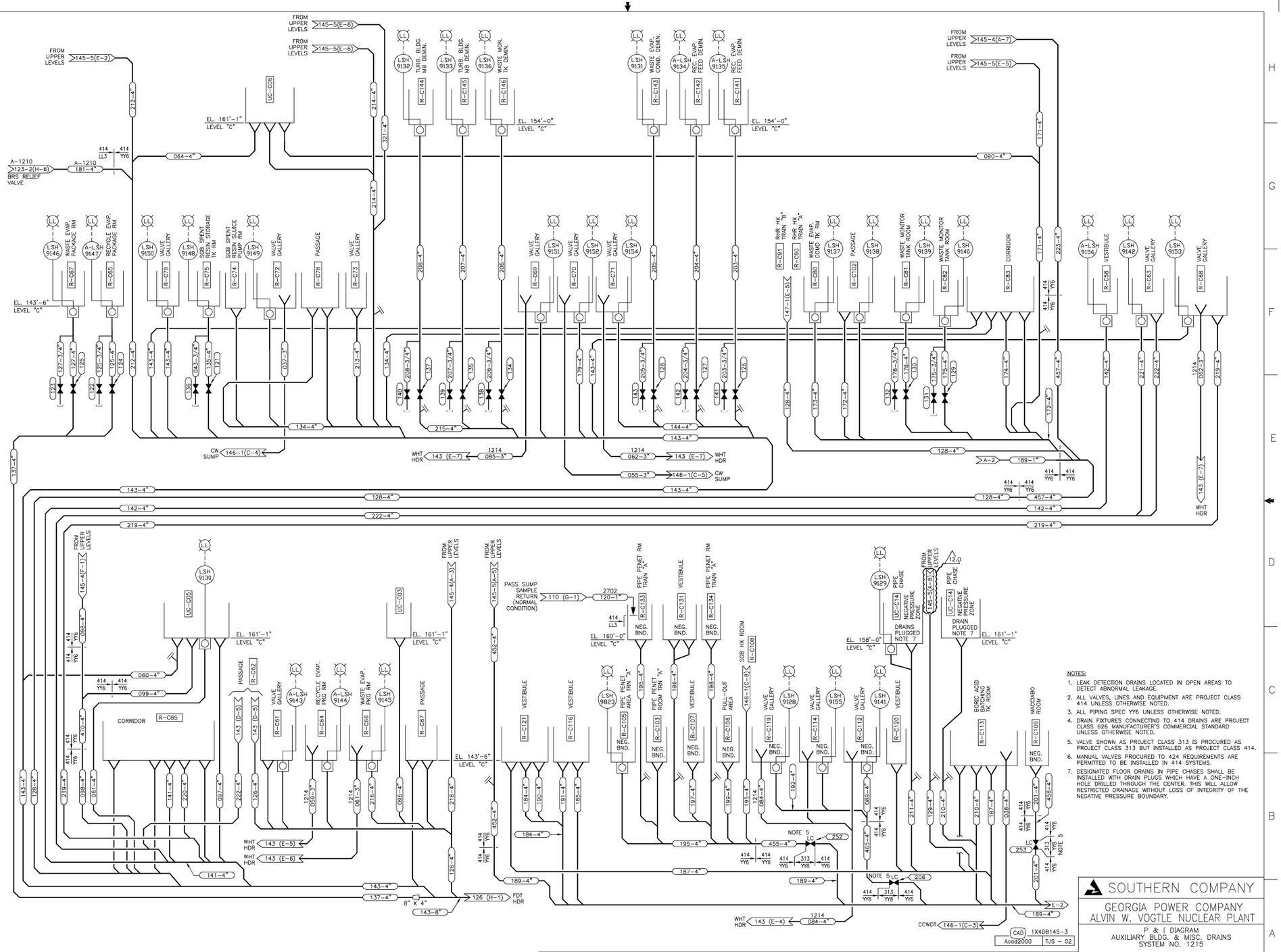

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NO.	VERSIONS	DATE	DR	CHK	APPV
12.0	REVISED PER ABN-V01110, VER. 1.0.	3-27-08	TJS	LRP	PML

CAD 1X4DB145-3 Acad2000 TJS - 02

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

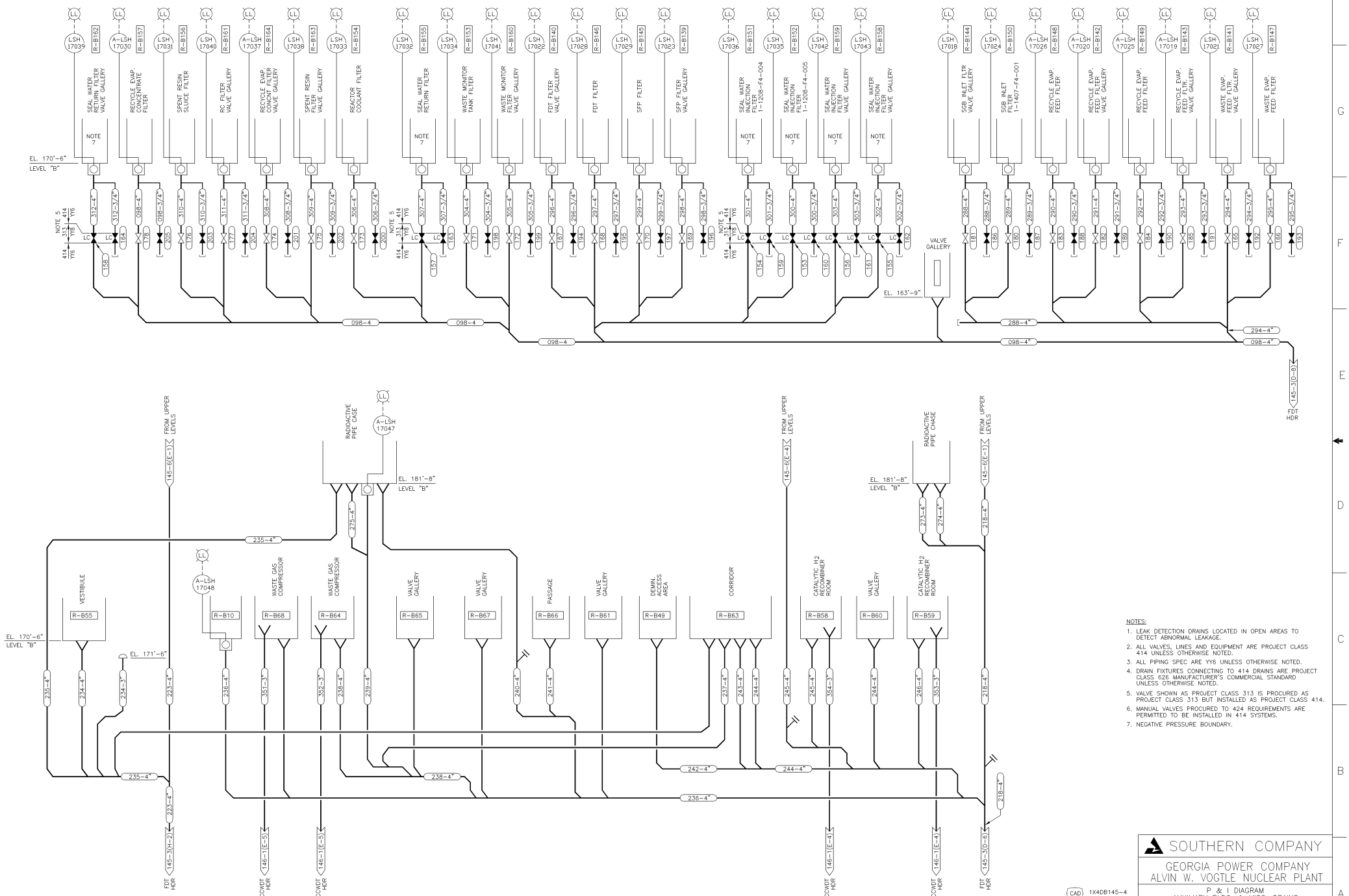
P & I DIAGRAM
AUXILIARY BLDG. & MISC. DRAINS
SYSTEM NO. 1215

SCALE: NONE | DRAWING NO. 1X4DB145-3 | VER. 12.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

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2. ALL VALVES, LINES AND EQUIPMENT ARE PROJECT CLASS 414 UNLESS OTHERWISE NOTED.
3. ALL PIPING SPEC ARE YY6 UNLESS OTHERWISE NOTED.
4. DRAIN FIXTURES CONNECTING TO 414 DRAINS ARE PROJECT CLASS 626 MANUFACTURER'S COMMERCIAL STANDARD UNLESS OTHERWISE NOTED.
5. VALVE SHOWN AS PROJECT CLASS 313 IS PROCURED AS PROJECT CLASS 313 BUT INSTALLED AS PROJECT CLASS 414.
6. MANUAL VALVES PROCURED TO 424 REQUIREMENTS ARE PERMITTED TO BE INSTALLED IN 414 SYSTEMS.
7. NEGATIVE PRESSURE BOUNDARY.

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NO.		DATE	DR	CHK	APPV	DTL
8.0	REDRAWN PER ABN- V70077, VER. 1.0	06-28-16	CYN	LCF	WPE	X

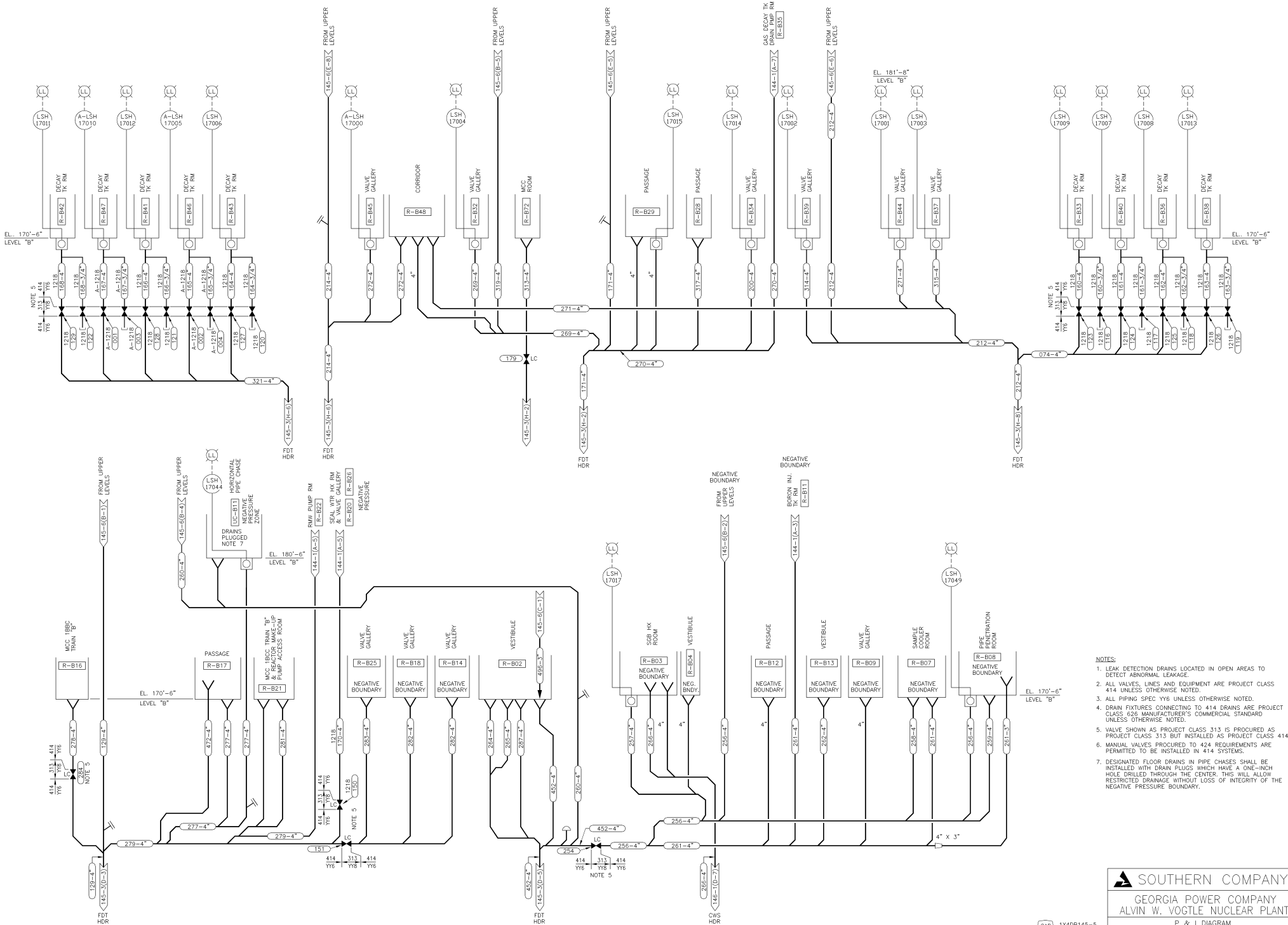
CAD 1X4DB145-4
ACAD2004 CYN - 01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY BLDG. & MISC. DRAINS
SYSTEM NO. 1215

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB145-4
VER. 8.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. LEAK DETECTION DRAINS LOCATED IN OPEN AREAS TO DETECT ABNORMAL LEAKAGE.
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3. ALL PIPING SPEC YY6 UNLESS OTHERWISE NOTED.
4. DRAIN FIXTURES CONNECTING TO 414 DRAINS ARE PROJECT CLASS 626 MANUFACTURER'S COMMERCIAL STANDARD UNLESS OTHERWISE NOTED.
5. VALVE SHOWN AS PROJECT CLASS 313 IS PROCURED AS PROJECT CLASS 313 BUT INSTALLED AS PROJECT CLASS 414.
6. MANUAL VALVES PROCURED TO 424 REQUIREMENTS ARE PERMITTED TO BE INSTALLED IN 414 SYSTEMS.
7. DESIGNATED FLOOR DRAINS IN PIPE CHASES SHALL BE INSTALLED WITH DRAIN PLUGS WHICH HAVE A ONE-INCH HOLE DRILLED THROUGH THE CENTER. THIS WILL ALLOW RESTRICTED DRAINAGE WITHOUT LOSS OF INTEGRITY OF THE NEGATIVE PRESSURE BOUNDARY.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY BUILDING & MISC. DRAINS
SYSTEM NO. 1215

CAD 1X4DB145-5
ACAD2004 CYN- 01

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NO.		DATE	DR	CHK	APPV	DTL
14.0	REDRAWN PER ABN- V70077, VER. 1.0	06-28-06	CYN	LCF	WPE	X

SCALE: NONE
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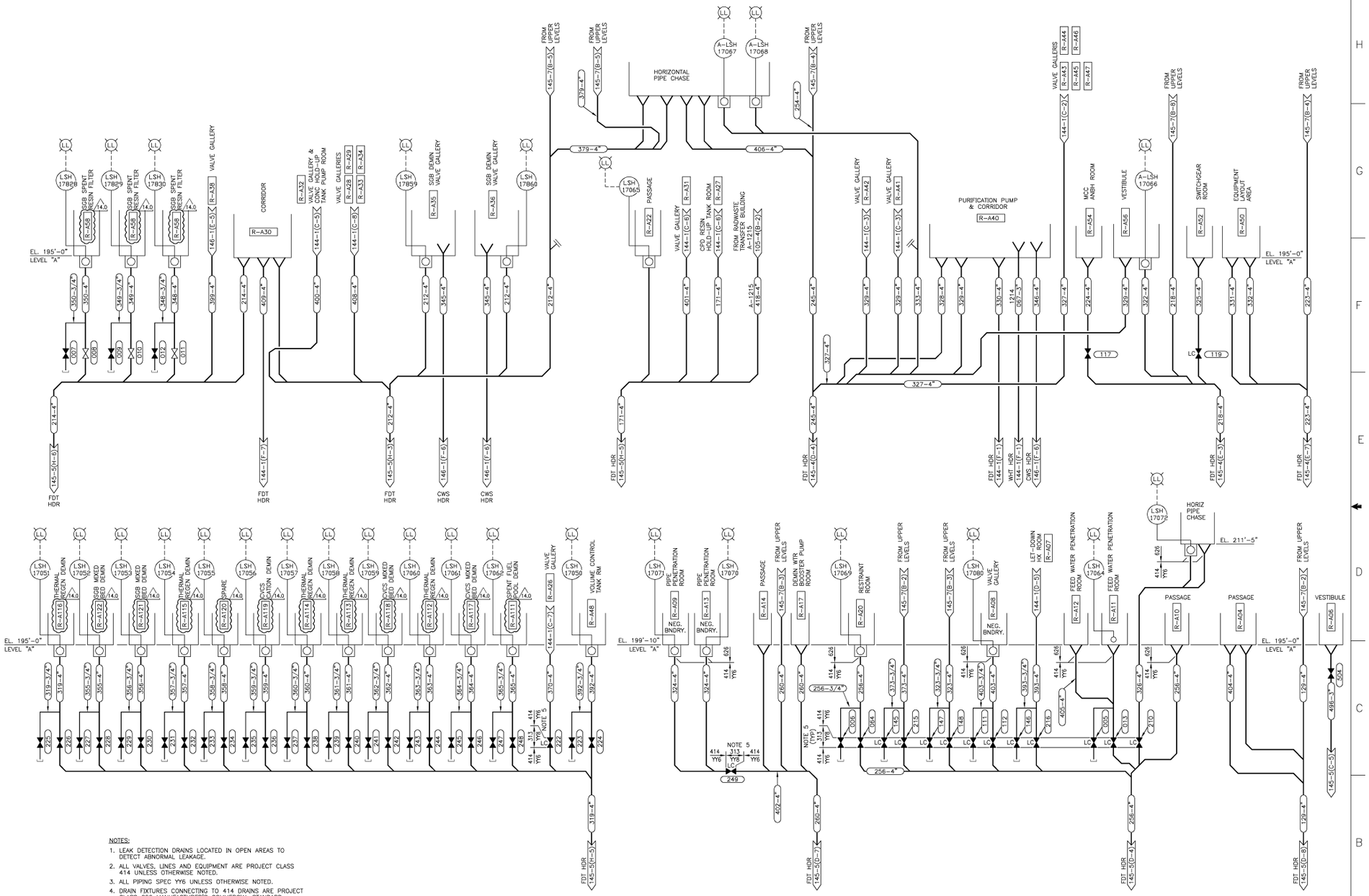
DRAWING CATEGORY: CRITICAL



NOTES:

1. LEAK DETECTION DRAINS LOCATED IN OPEN AREAS TO DETECT ABNORMAL LEAKAGE.
2. ALL VALVES, LINES AND EQUIPMENT ARE PROJECT CLASS 414 UNLESS OTHERWISE NOTED.
3. ALL PIPING SPEC YY6 UNLESS OTHERWISE NOTED.
4. DRAIN FIXTURES CONNECTING TO 414 DRAINS ARE PROJECT CLASS 626 MANUFACTURER'S COMMERCIAL STANDARD UNLESS OTHERWISE NOTED.
5. VALVE SHOWN AS PROJECT CLASS 313 IS PROCURED AS PROJECT CLASS 313 BUT INSTALLED AS PROJECT CLASS 414.
6. MANUAL VALVES PROCURED TO 424 REQUIREMENTS ARE PERMITTED TO BE INSTALLED IN 414 SYSTEMS.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY BUILDING & MISC. DRAINS
SYSTEM NO. 1215

CAD 1X4DB145-6
ACAD2004 BgT - 02

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NO.	VERSION	DATE	DR	CHK	APPV	DTL
14.0	REVISED PER ABN-V00554, VER.1.0	1-10-06	BgT	MWD	WLN	

SIZE E 34X44

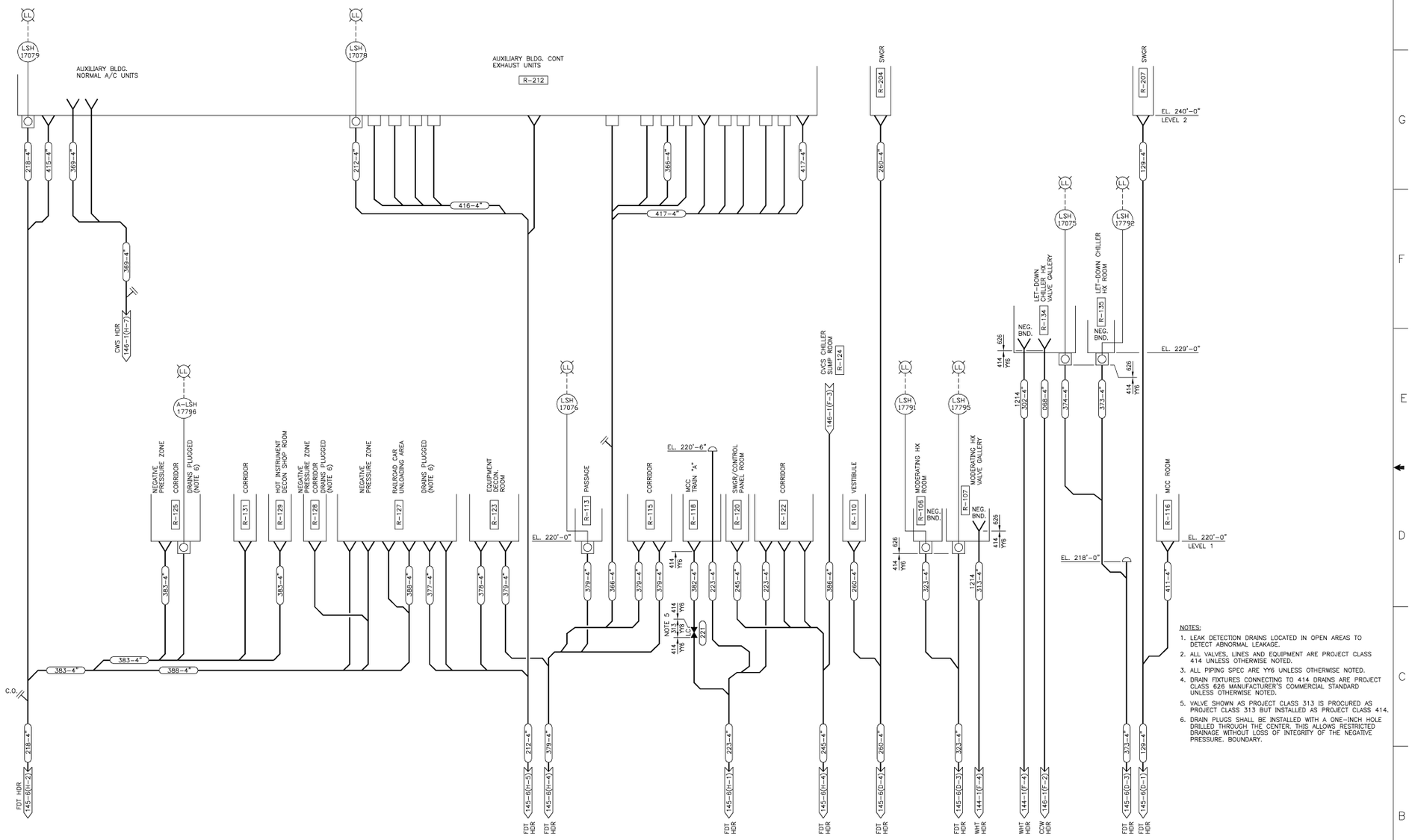
SCALE: NONE

JOB NO.10604

DRAWING NO. 1X4DB145-6

VER. 14.0

DRAWING CATEGORY: CRITICAL



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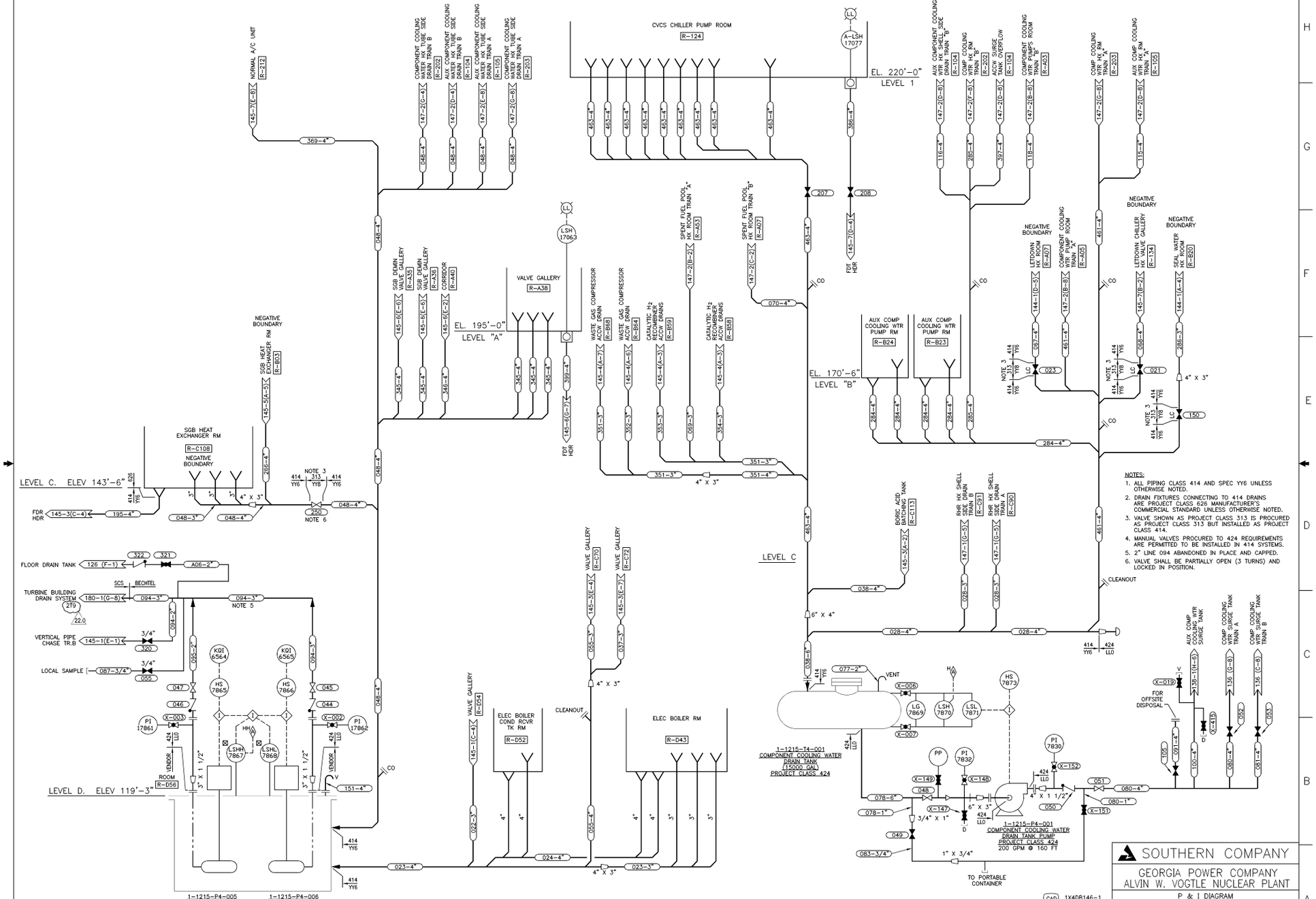
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
11.0	REDRAWN PER ABN-V70103, VER. 1.0	8-1-06	CYN	LCF	WPE	X

CAD 1X4DB145-7
ACADZ002 CYN - 01

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
AUXILIARY BUILDING & MISC. DRAINS
SYSTEM NO. 1215

SCALE: NONE
DRAWING NO. 1X4DB145-7
VER. 11.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY BLDG & MISC DRAINS
NON-RADIOACTIVE SYS NO. 1215

CAD 1X4DB146-1
ACAD2002 CYN- 02

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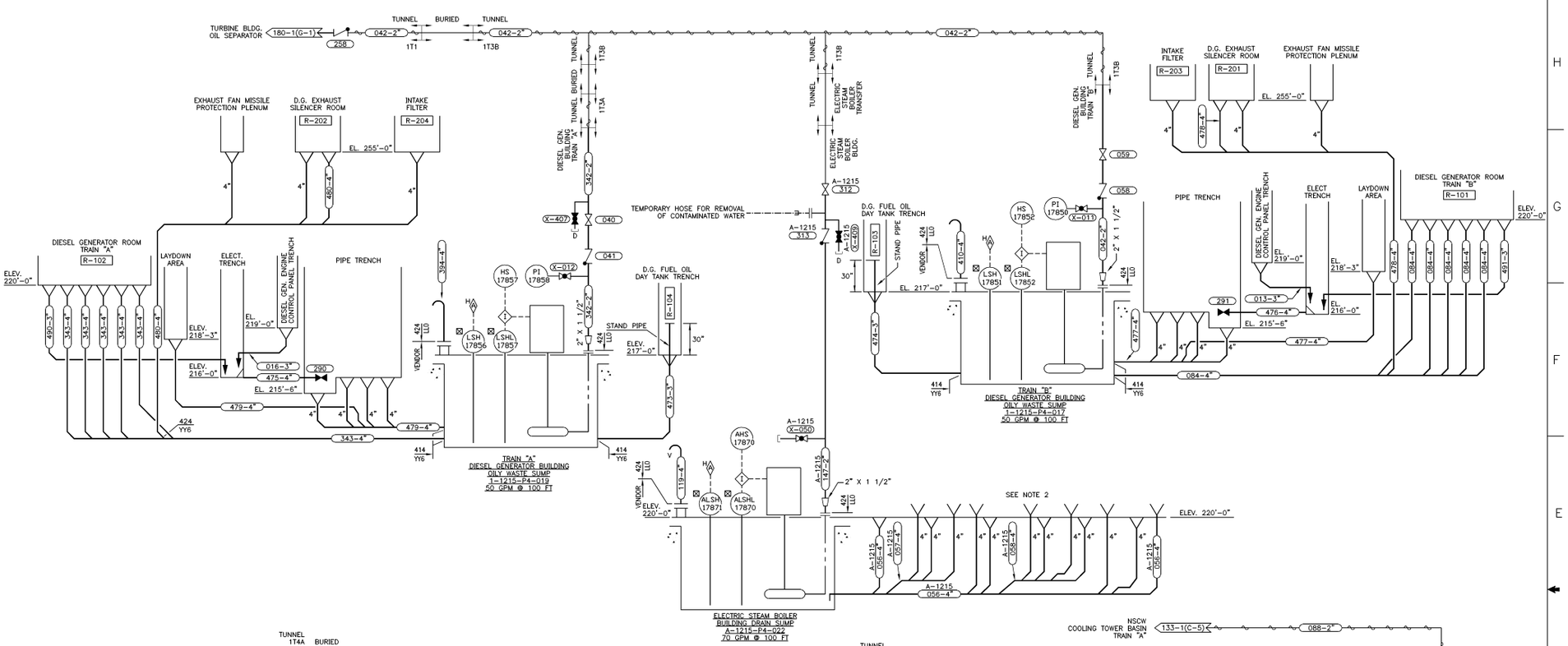
NO.		DATE	DR	CHK	APPV	DTL
22.0	REDRAWN PER ABN-V01516, VER. 1.0	02/21/08	CYN	LCF	PML	X

SIZE E 34X44

SCALE: NONE
JOB NO.10604

DRAWING NO. 1X4DB146-1
VER. 22.0

DRAWING CATEGORY: CRITICAL



NOTES:

1. MANUAL VALVES PROCURED TO 424 REQUIREMENTS ARE PERMITTED TO BE INSTALLED IN 414 SYSTEMS.
2. ELECTRIC STEAM BOILER AND ASSOCIATED COMPONENTS HAVE BEEN REMOVED.
3. HEAT TRACING ON THE FOLLOWING LINES, VALVES AND INSTRUMENTATION IS RETIRED IN PLACE:
   1-1215-086-2" (X-014) (X-405) PI-17848
   1-1215-088-2" (X-013) (X-406) PI-17849

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY BUILDING & MISC. DRAINS
SYSTEM NO. 1215

CAD 1X4DB146-2
ACAD2004 CYN- 02

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB146-2
VER. 25.0

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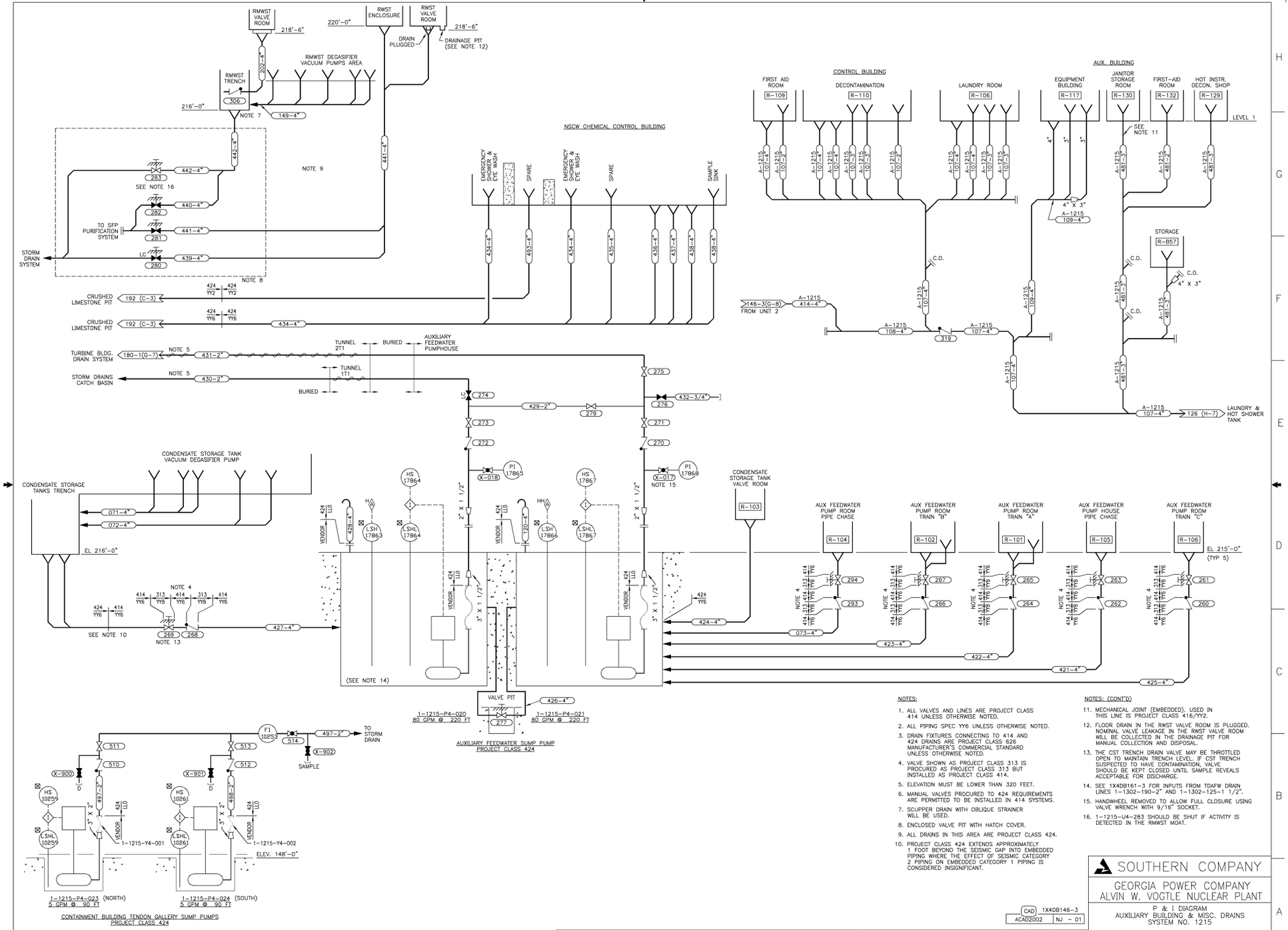
NO.	VERSION	DATE	DR	CHK	APPV	DTL
25.0	REVISED PER ABN-V01818, VER. 1.0.	10/05/09	CYN	SRG	PML	X

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. ALL VALVES AND LINES ARE PROJECT CLASS 414 UNLESS OTHERWISE NOTED.
2. ALL PIPING SPEC YY6 UNLESS OTHERWISE NOTED.
3. DRAIN FIXTURES CONNECTING TO 414 AND 424 DRAINS ARE PROJECT CLASS 626 MANUFACTURER'S COMMERCIAL STANDARD UNLESS OTHERWISE NOTED.
4. VALVE SHOWN AS PROJECT CLASS 313 IS PROCURED AS PROJECT CLASS 313 BUT INSTALLED AS PROJECT CLASS 414.
5. ELEVATION MUST BE LOWER THAN 320 FEET.
6. MANUAL VALVES PROCURED TO 424 REQUIREMENTS ARE PERMITTED TO BE INSTALLED IN 414 SYSTEMS.
7. SCUPPER DRAIN WITH OBLIQUE STRAINER WILL BE USED.
8. ENCLOSED VALVE PIT WITH HATCH COVER.
9. ALL DRAINS IN THIS AREA ARE PROJECT CLASS 424.
10. PROJECT CLASS 424 EXTENDS APPROXIMATELY 1 FOOT BEYOND THE SEISMIC GAP INTO EMBEDDED PIPING WHERE THE EFFECT OF SEISMIC CATEGORY 2 PIPING ON EMBEDDED CATEGORY 1 PIPING IS CONSIDERED INSIGNIFICANT.

NOTES: (CONT'D)

11. MECHANICAL JOINT (EMBEDDED). USED IN THIS LINE IS PROJECT CLASS 416/YY2.
12. FLOOR DRAIN IN THE RWST VALVE ROOM IS PLUGGED. NOMINAL VALVE LEAKAGE IN THE RWST VALVE ROOM WILL BE COLLECTED IN THE DRAINAGE PIT FOR MANUAL COLLECTION AND DISPOSAL.
13. THE CST TRENCH DRAIN VALVE MAY BE THROTTLED OPEN TO MAINTAIN TRENCH LEVEL. IF CST TRENCH SUSPECTED TO HAVE CONTAMINATION, VALVE SHOULD BE KEPT CLOSED UNTIL SAMPLE REVEALS ACCEPTABLE FOR DISCHARGE.
14. SEE 1X4DB161-3 FOR INPUTS FROM TDAFW DRAIN LINES 1-1302-190-2" AND 1-1302-125-1 1/2".
15. HANDWHEEL REMOVED TO ALLOW FULL CLOSURE USING VALVE WRENCH WITH 9/16" SOCKET.
16. 1-1215-U4-283 SHOULD BE SHUT IF ACTIVITY IS DETECTED IN THE RMWST MOAT.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY BUILDING & MISC. DRAINS
SYSTEM NO. 1215

CAD 1X4DB146-3
ACAD2002 NJ - 01

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB146-3	30.0

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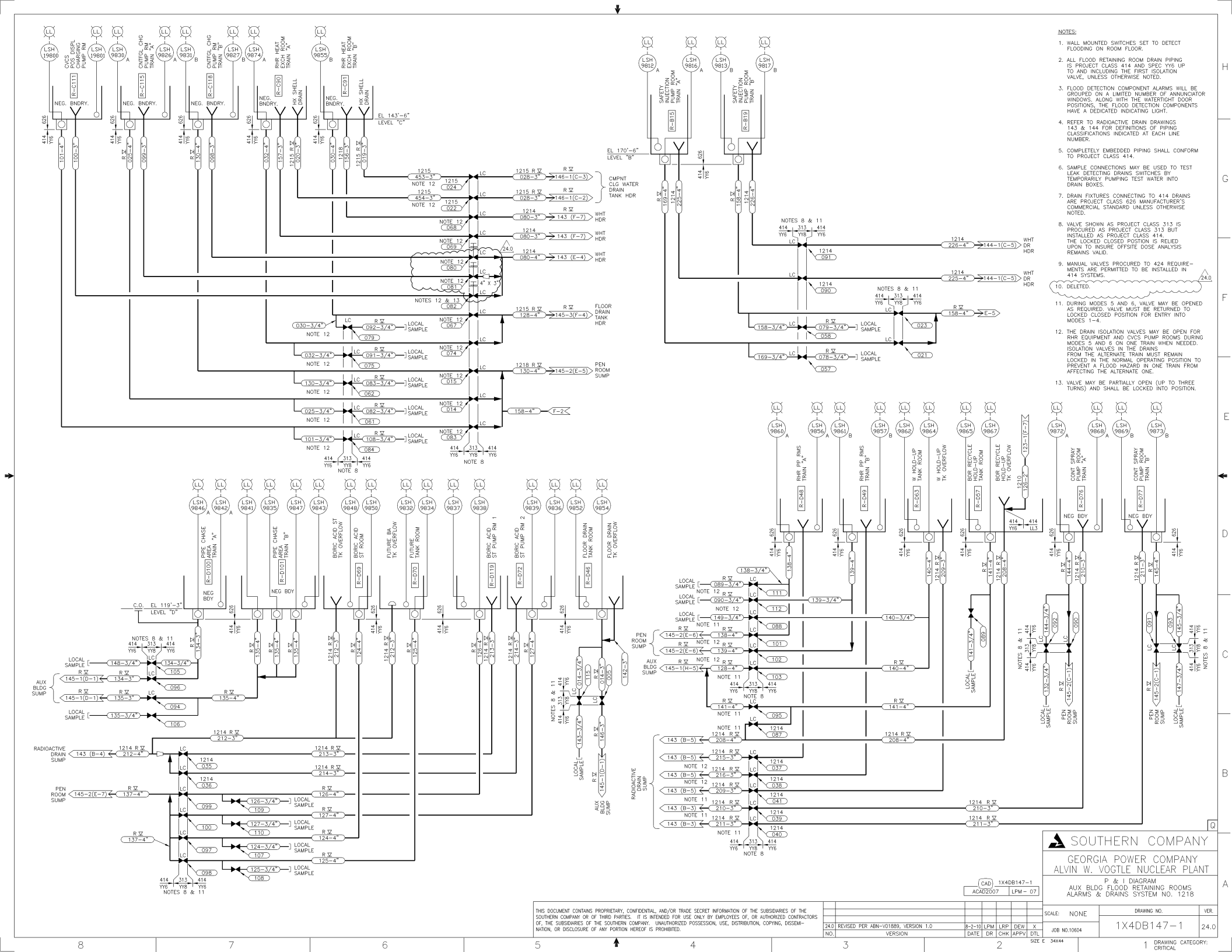
NO.		DATE	DR	CHK	APPV	DTL
30.0	REDRAWN PER ABN- 52270, VER. 1.0	1-23-06	NJ	LCF	WPE	X

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUX BLDG FLOOD RETAINING ROOMS
ALARMS & DRAINS SYSTEM NO. 1218

CAD 1X4DB147-1
ACAD2007 LPM- 07

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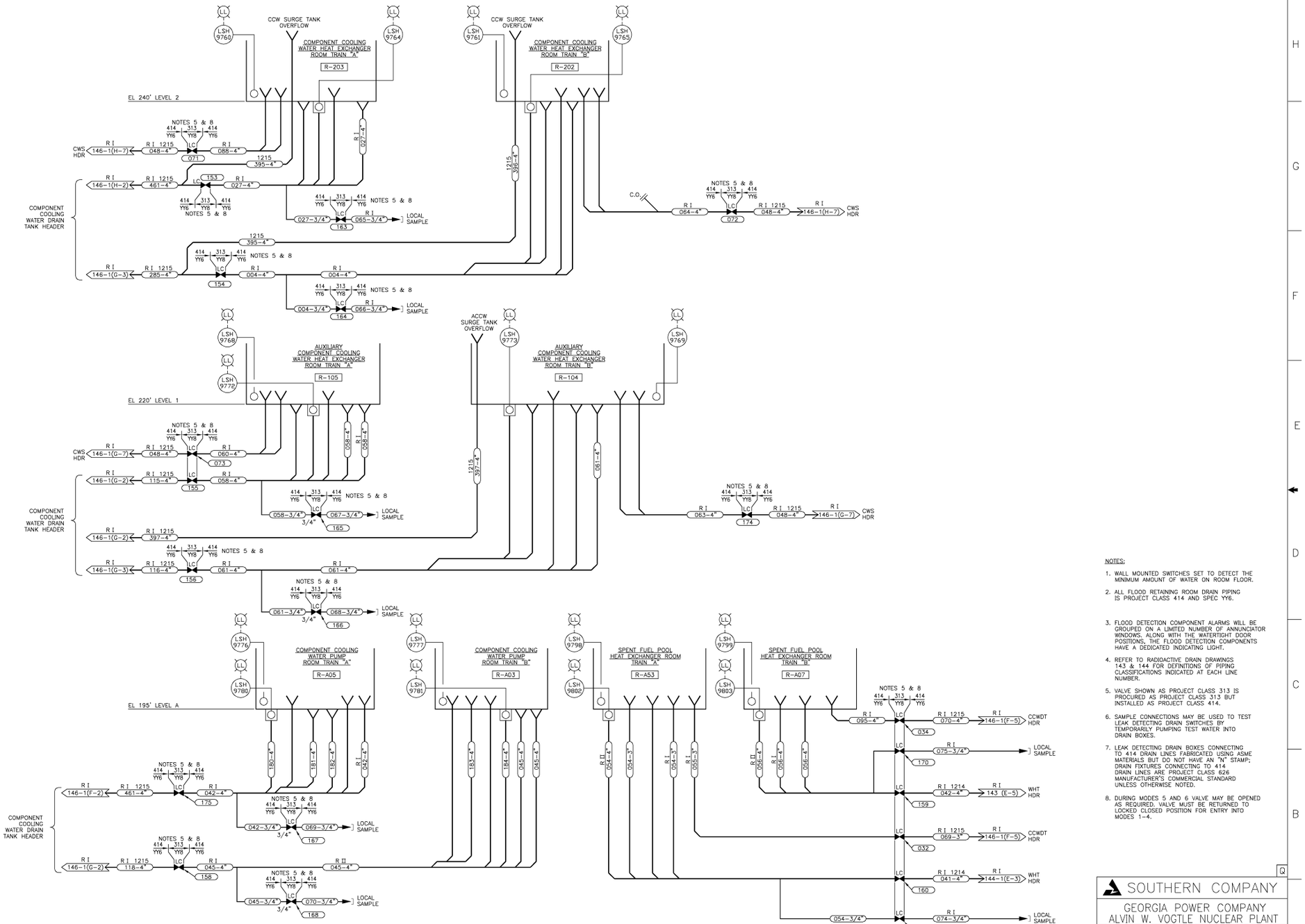
NO.	VERSION	DATE	DR	CHK	APPV	DTL
24.0	REVISED PER ABN-V01889, VERSION 1.0	8-2-10	LPM	LRP	DEW	X

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB147-1
VER. 24.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. WALL MOUNTED SWITCHES SET TO DETECT THE MINIMUM AMOUNT OF WATER ON ROOM FLOOR.
2. ALL FLOOD RETAINING ROOM DRAIN PIPING IS PROJECT CLASS 414 AND SPEC YY6.
3. FLOOD DETECTION COMPONENT ALARMS WILL BE GROUPED ON A LIMITED NUMBER OF ANNUNCIATOR WINDOWS, ALONG WITH THE WATERTIGHT DOOR POSITIONS, THE FLOOD DETECTION COMPONENTS HAVE A DEDICATED INDICATING LIGHT.
4. REFER TO RADIOACTIVE DRAIN DRAWINGS 143 & 144 FOR DEFINITIONS OF PIPING CLASSIFICATIONS INDICATED AT EACH LINE NUMBER.
5. VALVE SHOWN AS PROJECT CLASS 313 IS PROCURED AS PROJECT CLASS 313 BUT INSTALLED AS PROJECT CLASS 414.
6. SAMPLE CONNECTIONS MAY BE USED TO TEST LEAK DETECTING DRAIN SWITCHES BY TEMPORARILY PUMPING TEST WATER INTO DRAIN BOXES.
7. LEAK DETECTING DRAIN BOXES CONNECTING TO 414 DRAIN LINES FABRICATED USING ASME MATERIALS BUT DO NOT HAVE AN "N" STAMP; DRAIN FIXTURES CONNECTING TO 414 DRAIN LINES ARE PROJECT CLASS 626 MANUFACTURER'S COMMERCIAL STANDARD UNLESS OTHERWISE NOTED.
8. DURING MODES 5 AND 6 VALVE MAY BE OPENED AS REQUIRED, VALVE MUST BE RETURNED TO LOCKED CLOSED POSITION FOR ENTRY INTO MODES 1-4.

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NO.		DATE	DR	CHK	APPV	DTL
9.0	REDRAWN PER ABN- 52270, VER. 1.0	1-23-06	NJ	LCF	WPE	X

CAD 1X4DB147-2
ACAD2002 NJ - 01

SOUTHERN COMPANY

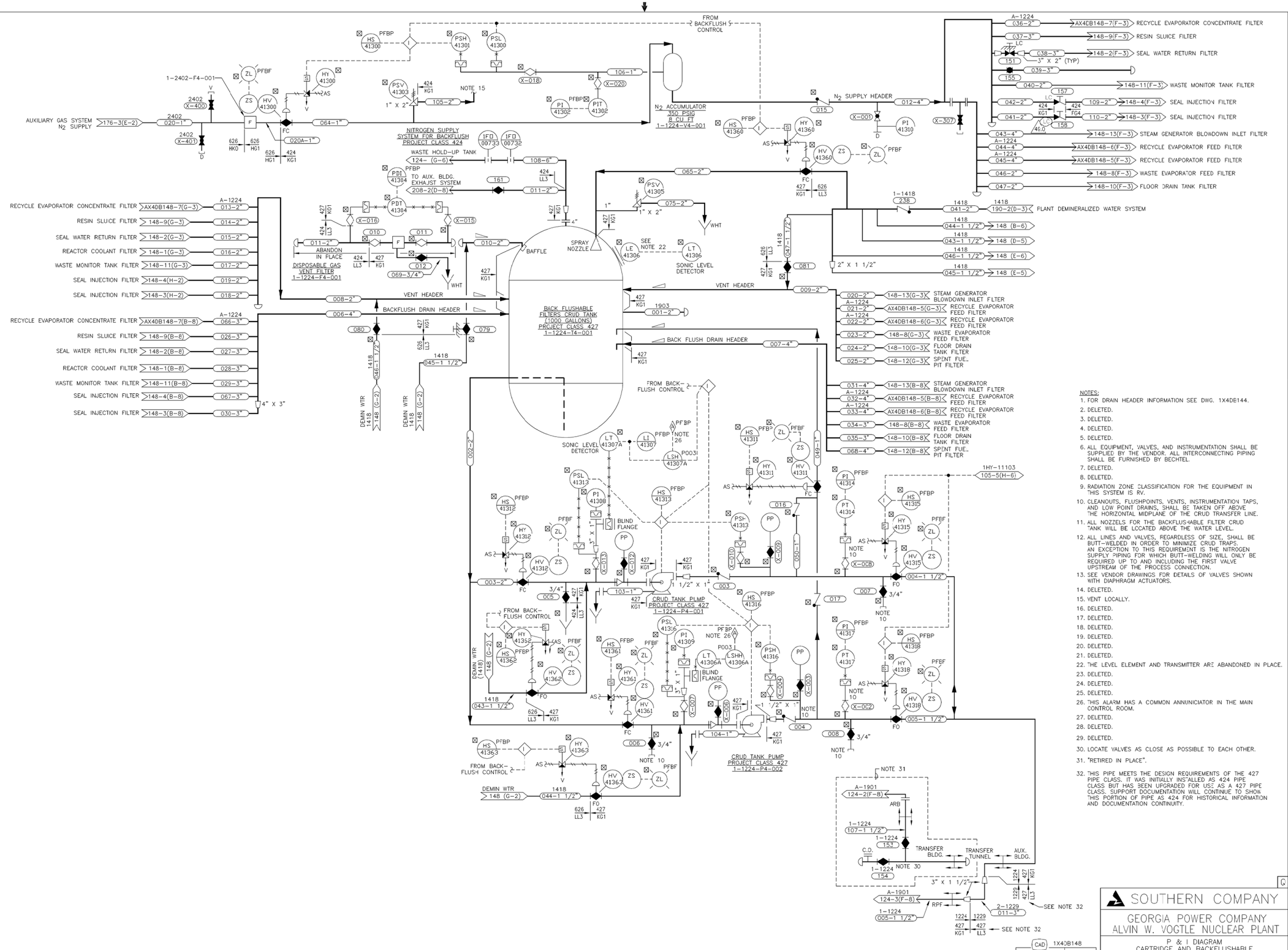
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUX BLDG FLOOD RETAINING ROOMS
ALARMS & DRAINS
SYSTEM NO. 1218

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB147-2	9.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DE144.
2. DELETED.
3. DELETED.
4. DELETED.
5. DELETED.
6. ALL EQUIPMENT, VALVES, AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR. ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
7. DELETED.
8. DELETED.
9. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS RV.
10. CLEANOUTS, FLUSHPOINTS, VENTS, INSTRUMENTATION TAPS, AND LOW POINT DRAINS, SHALL BE TAKEN OFF ABOVE THE HORIZONTAL MIDPLANE OF THE CRUD TRANSFER LINE.
11. ALL NOZZELS FOR THE BACKFLUSHABLE FILTER CRUD TANK WILL BE LOCATED ABOVE THE WATER LEVEL.
12. ALL LINES AND VALVES, REGARDLESS OF SIZE, SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS. AN EXCEPTION TO THIS REQUIREMENT IS THE NITROGEN SUPPLY PIPING FOR WHICH BUTT-WELDING WILL ONLY BE REQUIRED UP TO AND INCLUDING THE FIRST VALVE UPSTREAM OF THE PROCESS CONNECTION.
13. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
14. DELETED.
15. VENT LOCALLY.
16. DELETED.
17. DELETED.
18. DELETED.
19. DELETED.
20. DELETED.
21. DELETED.
22. THE LEVEL ELEMENT AND TRANSMITTER ARE ABANDONED IN PLACE.
23. DELETED.
24. DELETED.
25. DELETED.
26. THIS ALARM HAS A COMMON ANNUNICIATOR IN THE MAIN CONTROL ROOM.
27. DELETED.
28. DELETED.
29. DELETED.
30. LOCATE VALVES AS CLOSE AS POSSIBLE TO EACH OTHER.
31. "RETIRED IN PLACE".
32. THIS PIPE MEETS THE DESIGN REQUIREMENTS OF THE 427 PIPE CLASS. IT WAS INITIALLY INSTALLED AS 424 PIPE CLASS BUT HAS BEEN UPGRADED FOR USE AS A 427 PIPE CLASS. SUPPORT DOCUMENTATION WILL CONTINUE TO SHOW THIS PORTION OF PIPE AS 424 FOR HISTORICAL INFORMATION AND DOCUMENTATION CONTINUITY.

CAD 1X4DB148
ACAD2004 DRW- 02

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
49.0	REVISED PER ABN-V03654, VER. 1.0	06/23/14	PLB	SO	JCG	X

SOUTHERN COMPANY

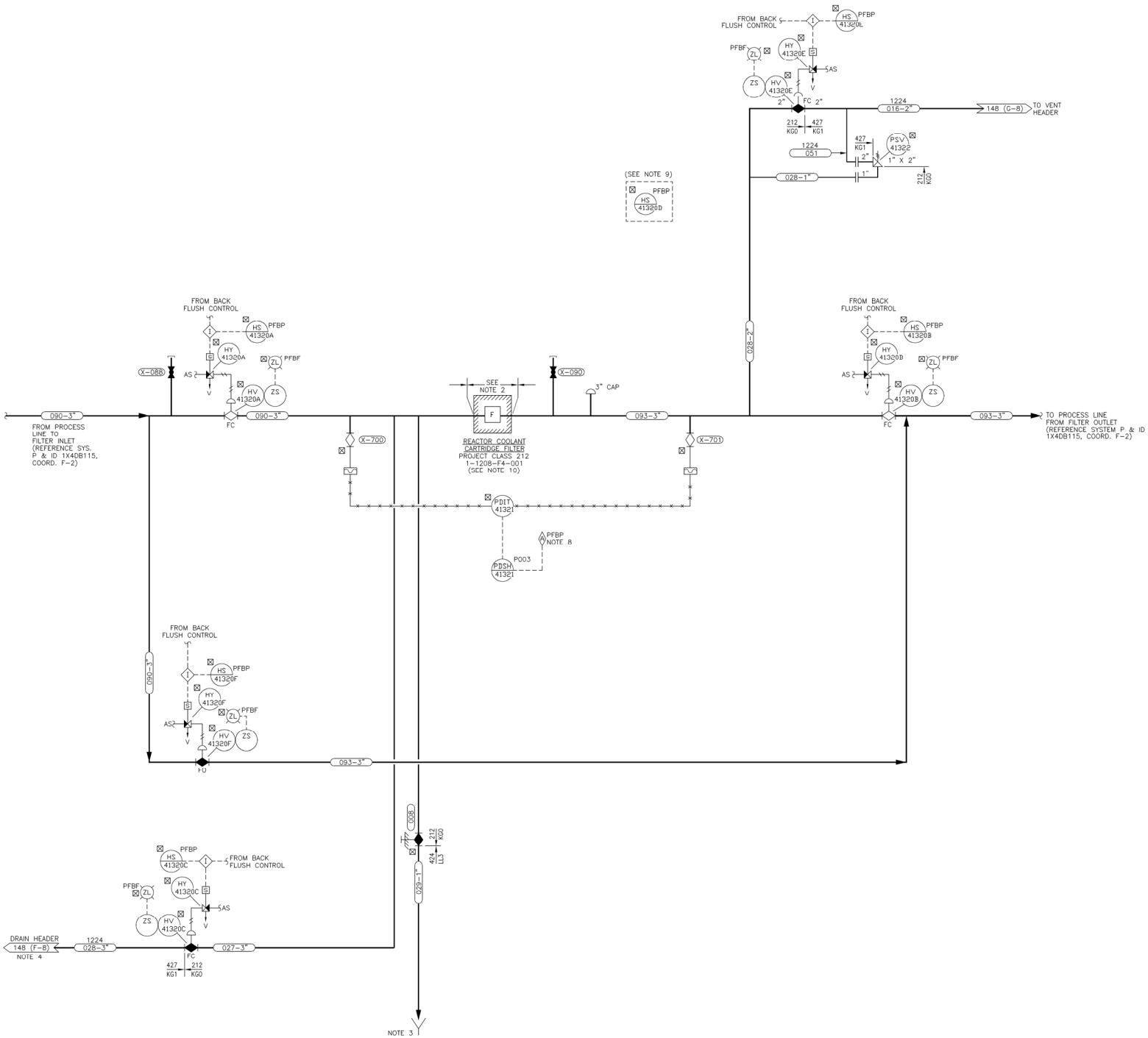
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P. & I. DIAGRAM
CARTRIDGE AND BACKFLUSHABLE FILTERS SYSTEM SYSTEM NO. 1224

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB148	49.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT, VALVES AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR. ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM 1X4DB115 FOR DRAINAGE SYSTEM.
4. DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS R Σ.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE, SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNICATOR IN THE MAIN CONTROL ROOM.
9. 1HS-41320D ON PANEL 1-1224-P5-FBP IS RETIRED IN PLACE.
10. THIS FILTER HAS BEEN MODIFIED TO UTILIZE PALL TRINITY DISPOSABLE CARTRIDGES.

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NO.		DATE	DR	CHK	APPV	DTL
1.0	REDRAWN PER ABN-V70077, VER. 1.0	06-28-06	CYN	LCF	LKB	X

CAD 1X4DB148-1
ACAD2004 CYN-01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
REACTOR COOLANT FILTER
SYSTEM NO. 1208

SCALE: NONE

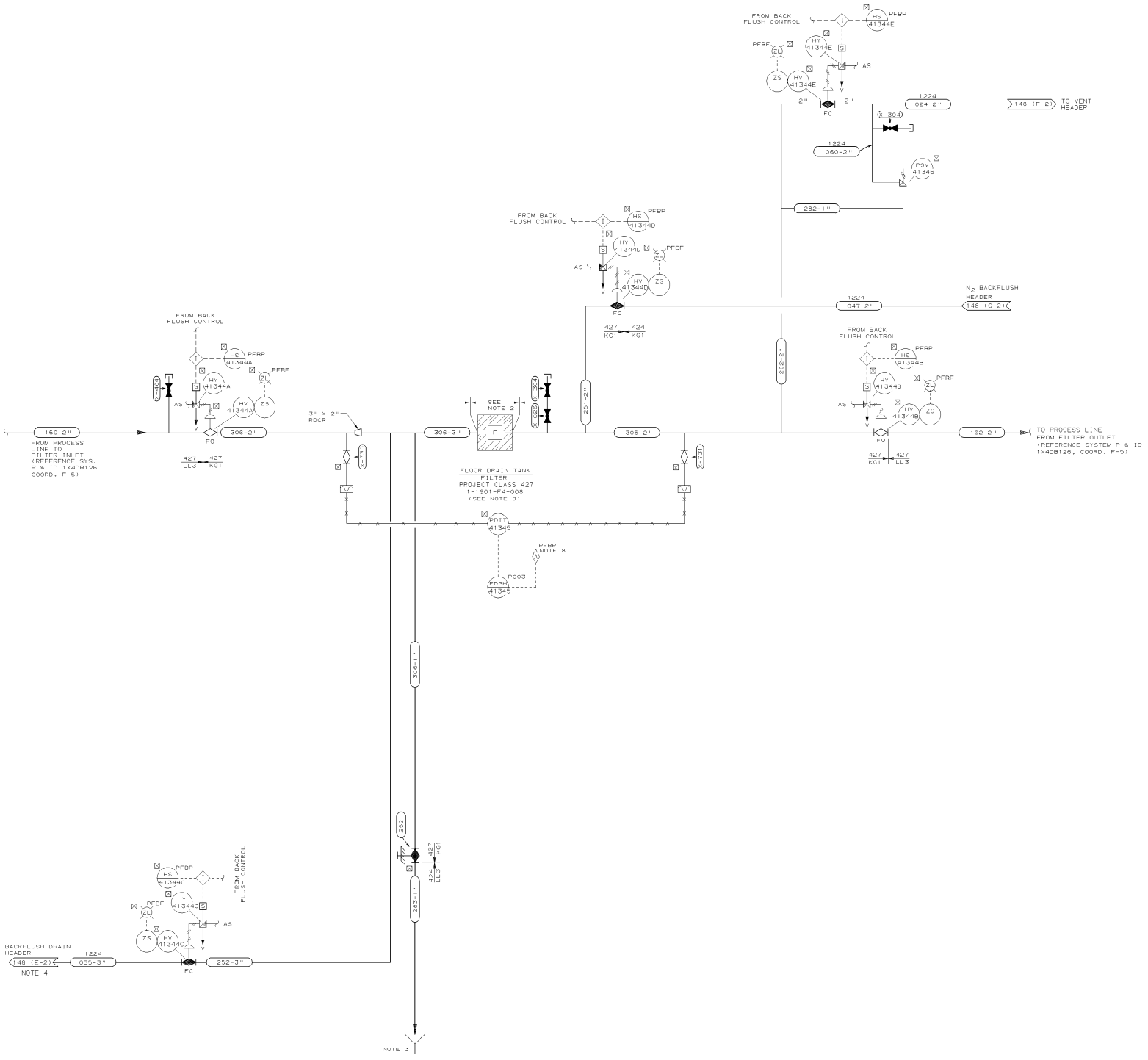
DRAWING NO. 1X4DB148-1

VER. 1.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT,VALVES,AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR.ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM 1X4DB126 FOR DRAINAGE SYSTEM.
4. BACKFLUSH DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS RV.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE,SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNICATOR IN THE MAIN CONTROL ROOM.
9. FILTER ELEMENTS HAVE BEEN REMOVED FROM THIS UNIT.

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NO.	REVISIONS	DATE	DR	CHK	APPV
	INCORP. PER ABN-33045	4-2-98	LVL	JGM	MWD

WFP DTL

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
FLOOR DRAIN TANK FILTER SYSTEM
SYSTEM NO. 1901

SCALE: NONE

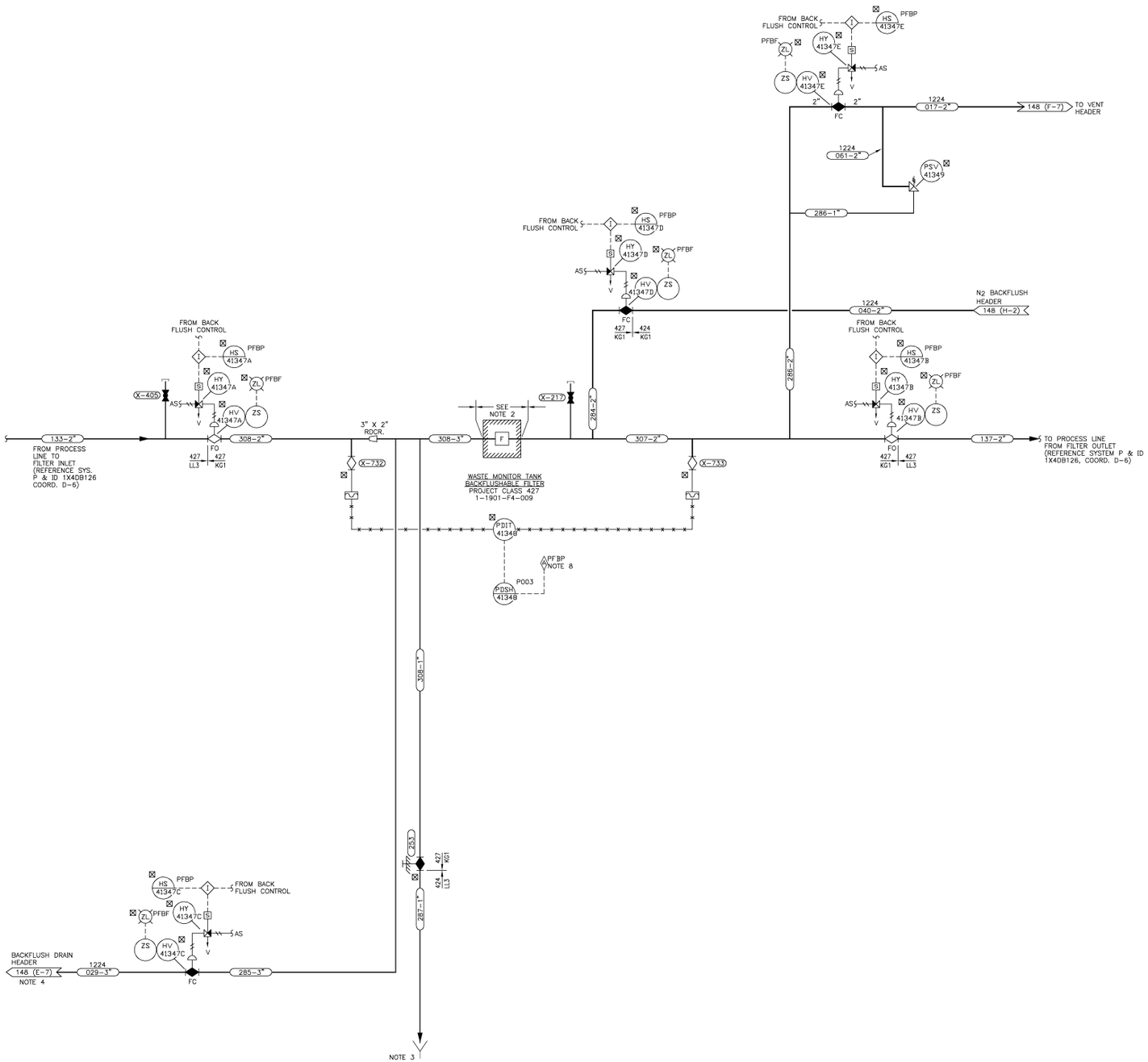
JOB NO.10604

DRAWING NO. 1X4DB148-10

REV. 0

SIZE E 36x44

CAD NAME 14B14810



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT, VALVES AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR. ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM 1X4DB126 FOR DRAINAGE SYSTEM.
4. BACKFLUSH DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS R3Z.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE, SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNCIATOR IN THE MAIN CONTROL ROOM.

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NO.	VERSION	DATE	DR	CHK	APPV	DTL
1.0	REDRAWN PER ABN-V70079, VER. 1.0	7-11-06	CYN	LCF	WPE	X

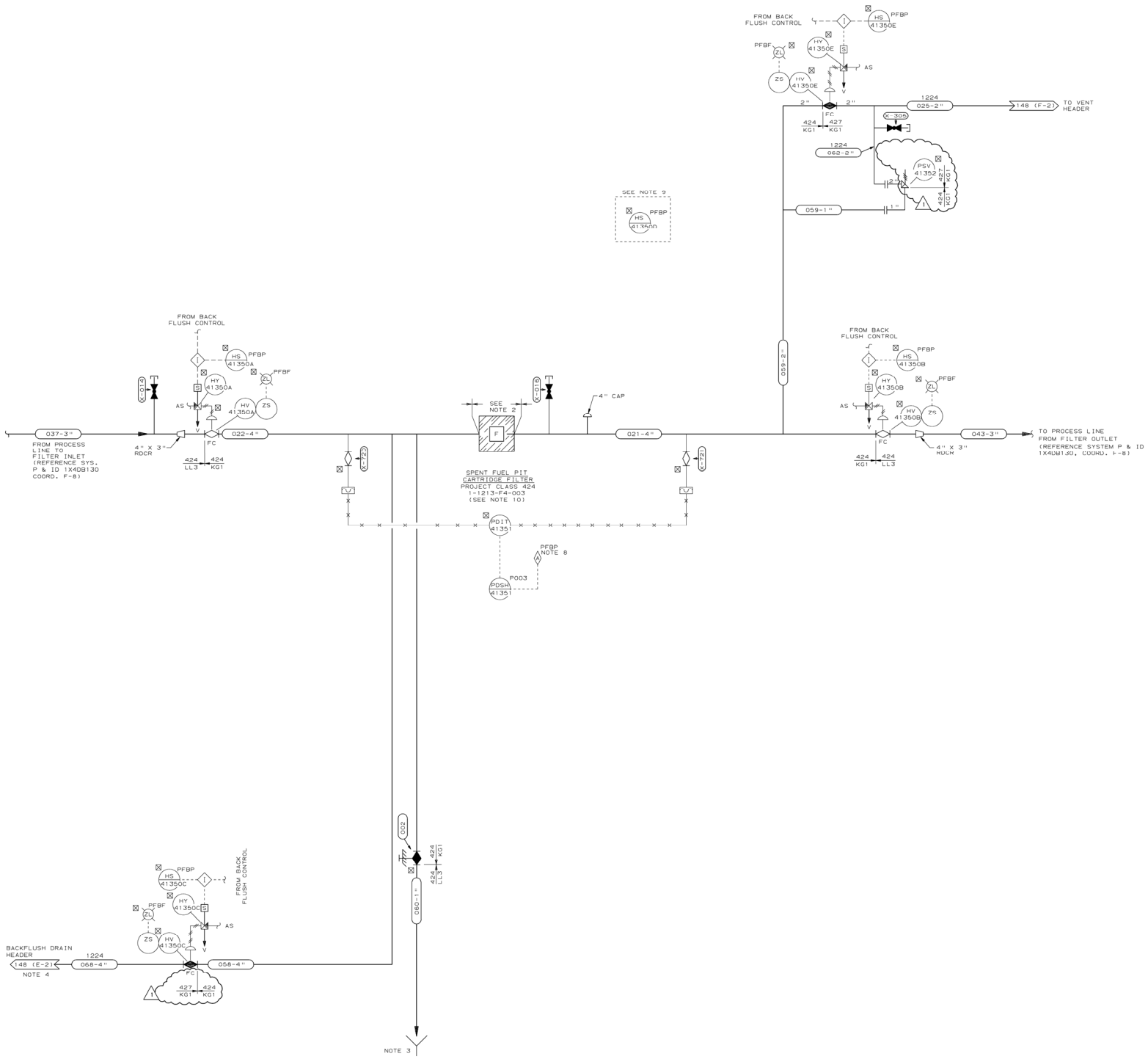
CAD 1X4DB148-11
ACAD2004 CYN-01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
WASTE MONITOR TANK BACKFLUSHABLE FILTER SYSTEM SYSTEM NO. 1901

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB148-11
VER. 1.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT,VALVES,AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR,ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM 1X4DB130 FOR DRAINAGE SYSTEM.
4. DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS RV.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE,SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNICATOR IN THE MAIN CONTROL ROOM.
9. 1HS-41350D ON PANEL 1-1224-P5-FBP IS RETIRED IN PLACE.
10. THIS FILTER HAS BEEN MODIFIED TO UTILIZE PALL TRINITY DISPOSABLE CARTRIDGES.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
1	INCORP. PER ABN-36988 & ABN-37014	06/01/02	PM	GLB	ELC	

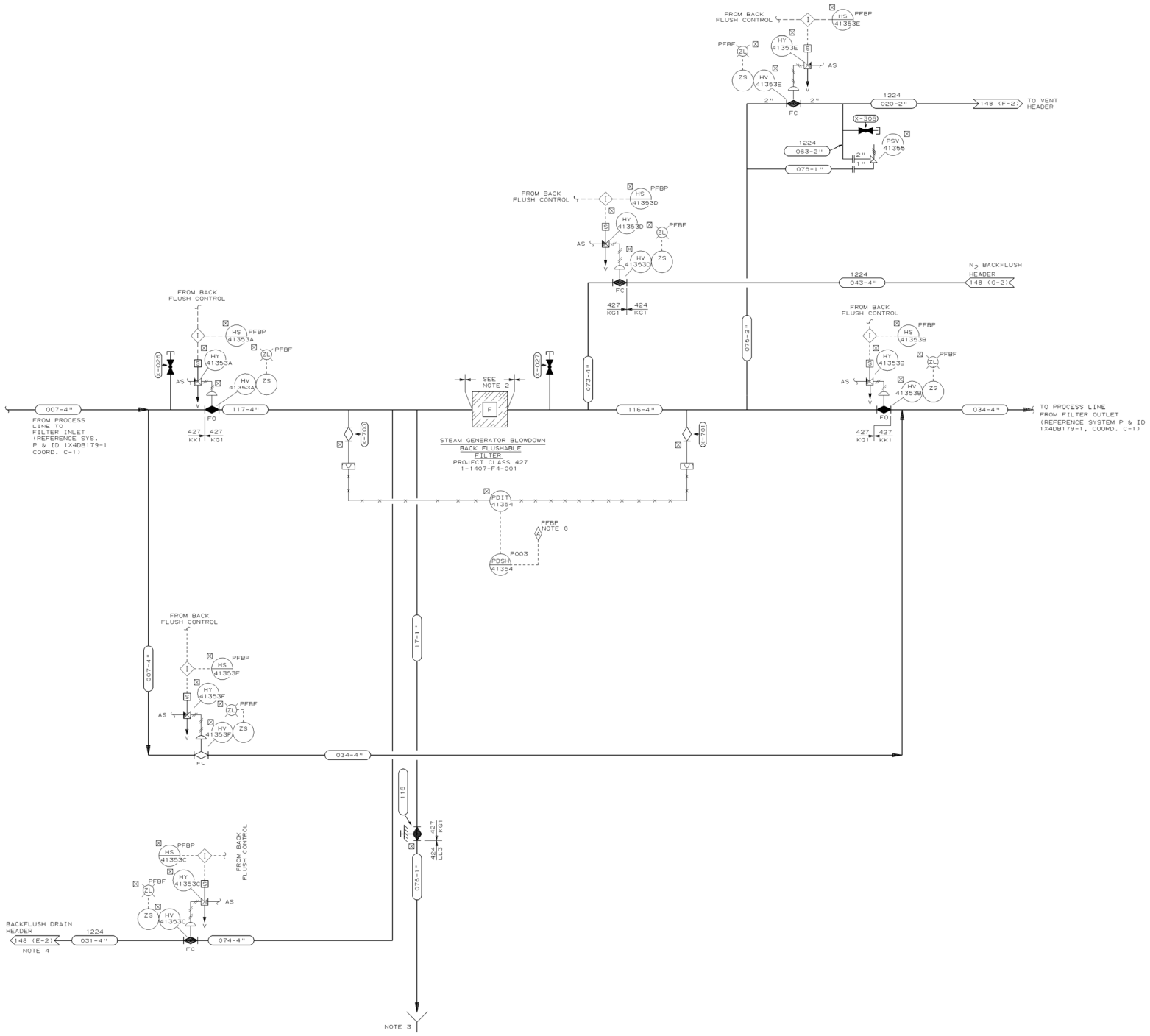
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SPENT FUEL PIT
FILTER SYSTEM SYSTEM NO. 1213

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	1X4DB148-12	1

SIZE E 34X44



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT,VALVES,AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR.ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM 1X4DB179-1 FOR DRAINAGE SYSTEM.
4. BACKFLUSH DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS RV.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE,SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNICATOR IN THE MAIN CONTROL ROOM.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4						
3						
2	INCORP. PER ABN-33048	4/4/98	LVL	JGM	MWD	WFP
1						

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

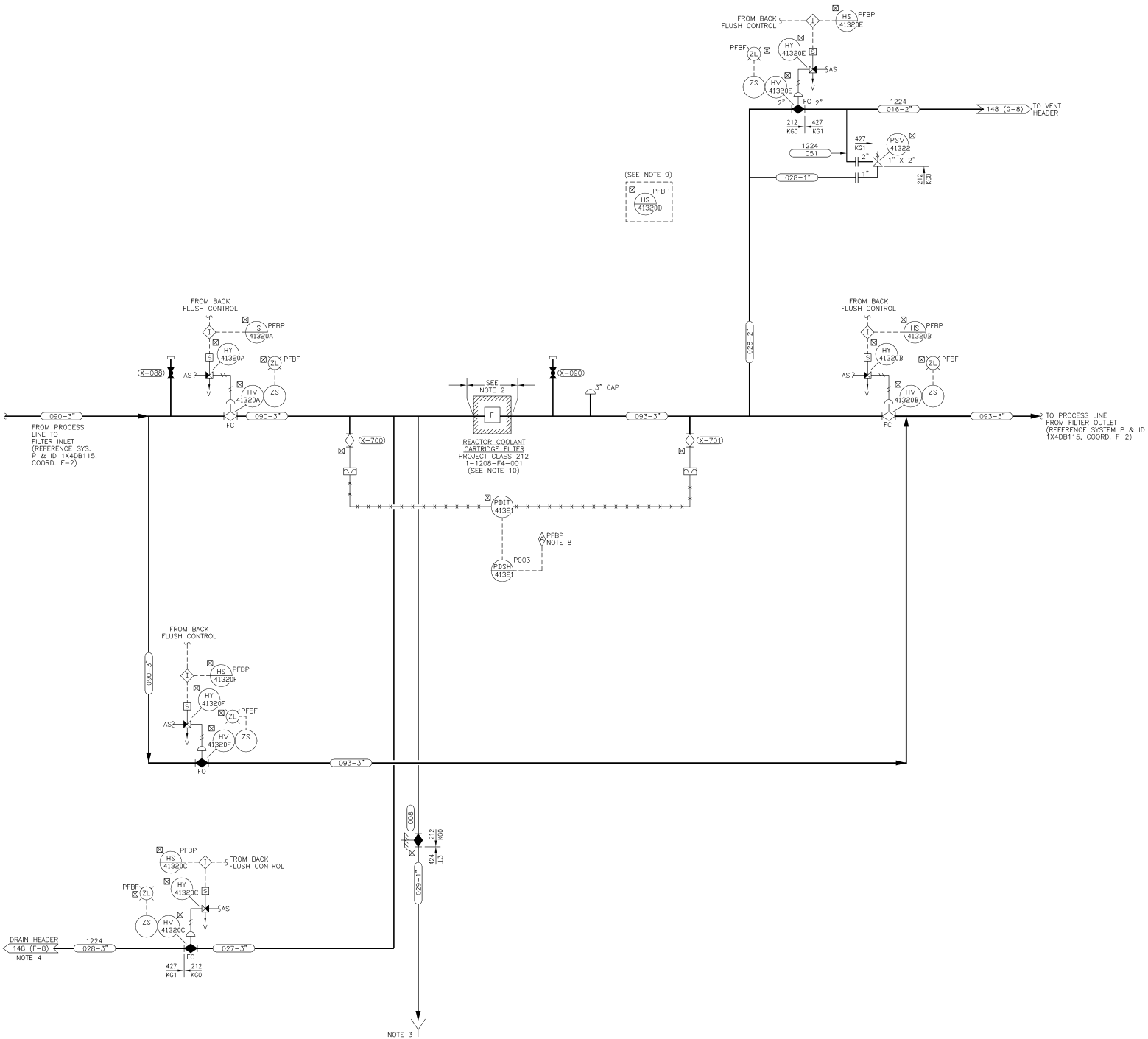
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
STEAM GENERATOR BLOWDOWN
BACKFLUSHABLE FILTER
SYSTEM NO. 1407

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	1X4DB148-13	0

SIZE E 34x44

CAD NAME 14B14813



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT, VALVES AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR. ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM 1X4DB115 FOR DRAINAGE SYSTEM.
4. DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS R 3Z.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE, SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNICATOR IN THE MAIN CONTROL ROOM.
9. 1HS-41320D ON PANEL 1-1224-P5-FBP IS RETIRED IN PLACE.
10. THIS FILTER HAS BEEN MODIFIED TO UTILIZE PALL TRINITY DISPOSABLE CARTRIDGES.

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NO.		DATE	DR	CHK	APPV	DTL
1.0	REDRAWN PER ABN-V70077, VER. 1.0	06-28-06	CYN	LCF	LKB	X

CAD 1X4DB148-1
ACAD2004 | CYN-01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
REACTOR COOLANT FILTER
SYSTEM NO. 1208

SCALE: NONE

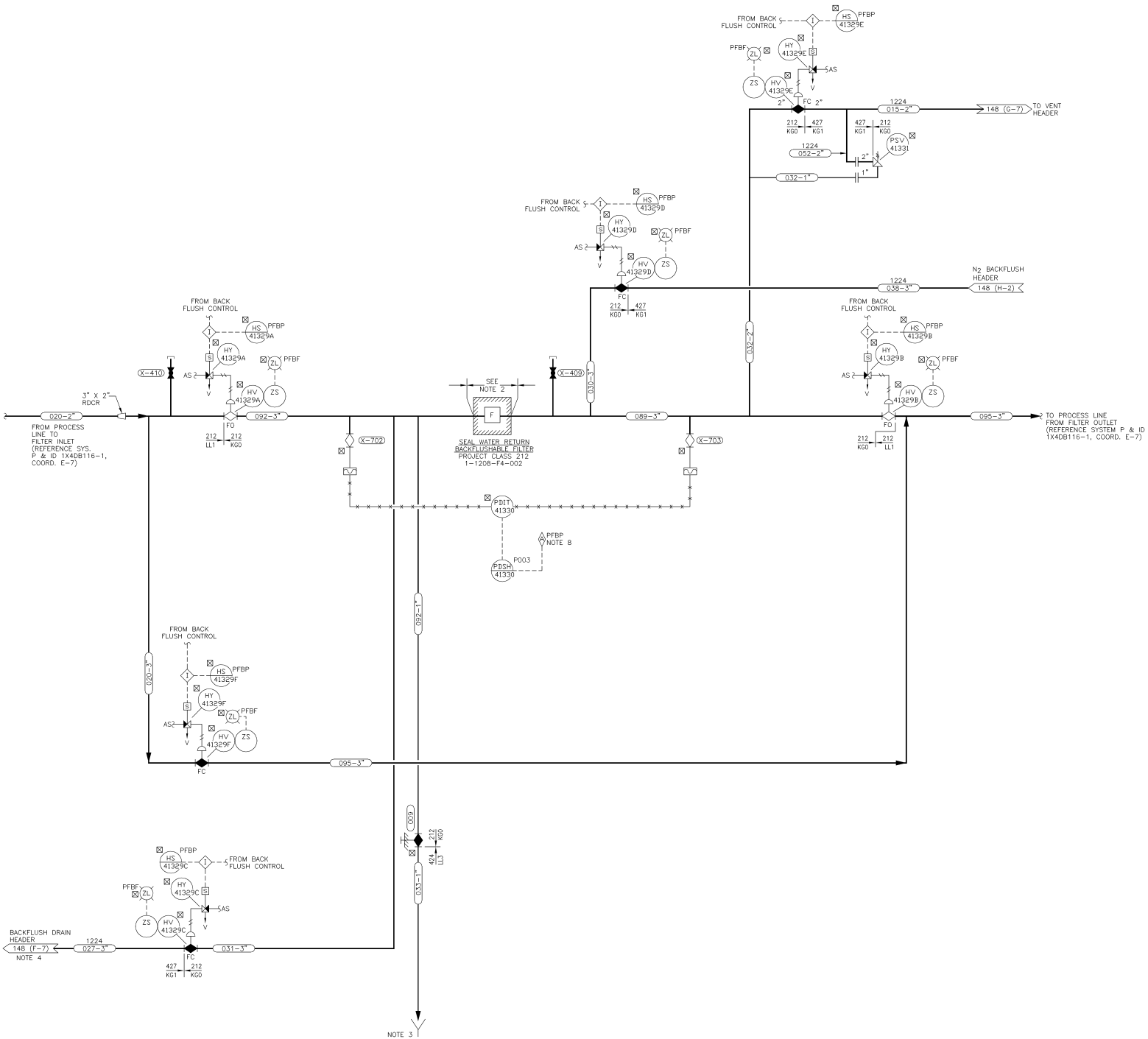
DRAWING NO. 1X4DB148-1

VER. 1.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT, VALVES AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR. ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM 1X4DB116-1 FOR DRAINAGE SYSTEM.
4. BACKFLUSH DRAIN LINES WILL BE SLOPED TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS R Ⅳ.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE, SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNICATOR IN THE MAIN CONTROL ROOM.

CAD 1X4DB148-2
ACAD2004 | CYN - 01

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NO.		DATE	DR	CHK	APPV	DTL
1.0	REDRAWN PER ABN-V70077, VER. 1.0	06-28-06	CYN	LCF	LKB	X

SOUTHERN COMPANY

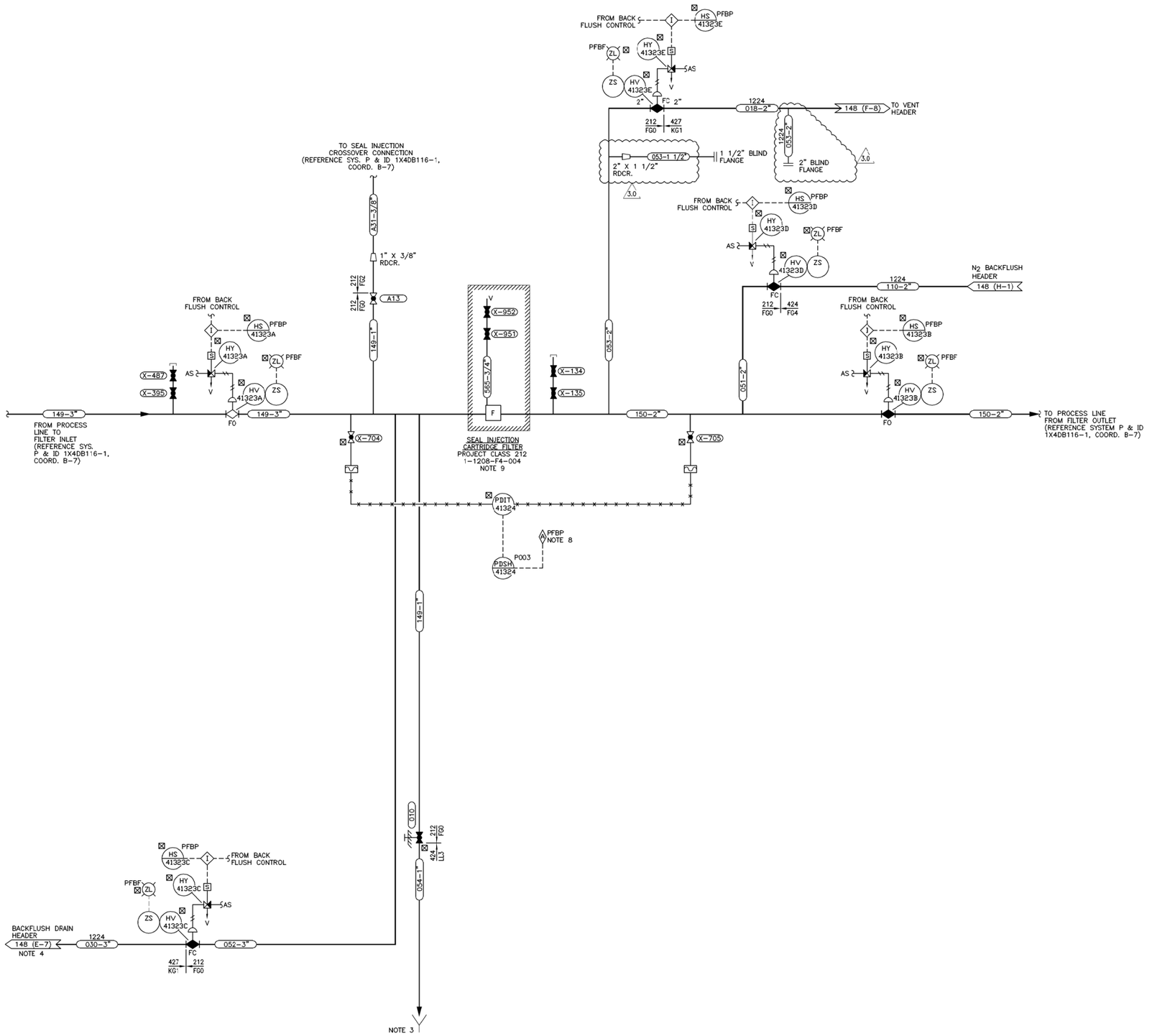
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SEAL WATER RETURN
BACKFLUSHABLE FILTER
SYSTEM NO. 1208

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB148-2	1.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. DELETED.
3. SEE SYSTEM P & I DIAGRAM 1X4DB116-1 FOR DRAINAGE SYSTEM.
4. BACKFLUSH DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NC POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS R Ⅳ.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE, SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNICATOR IN THE MAIN CONTROL ROOM.
9. FILTER IS SHOWN OUT OF SERVICE. FOR OVERPRESSURE PROTECTION, THE SEAL INJECTION CARTRIDGE FILTER VESSELS SHALL NOT BE ISOLATED SUCH THAT A VOLUME OF WATER WOULD BECOME TRAPPED.

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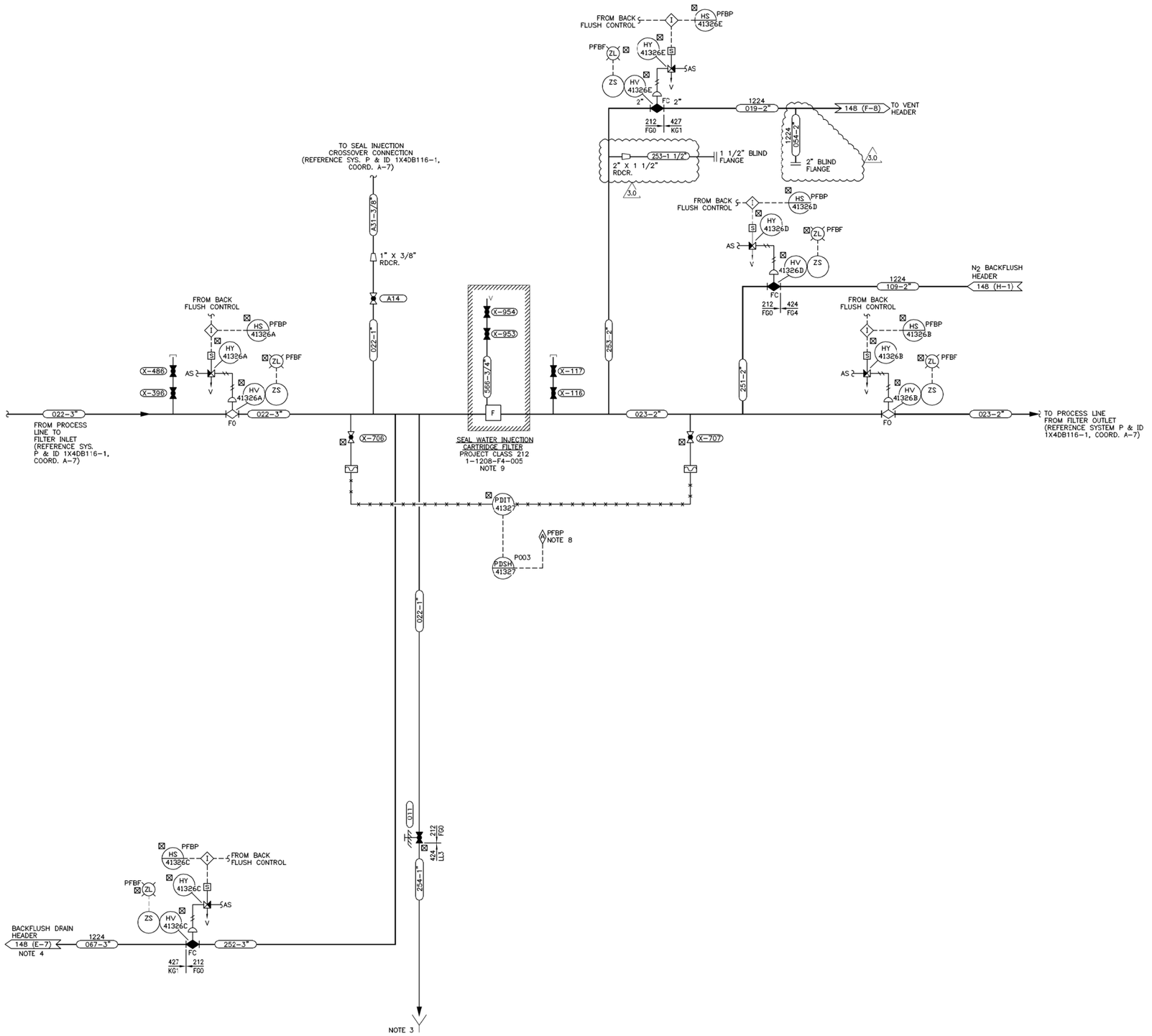
NO.	VERSION	DATE	DR	CHK	APPV	DTL
3.0	REVISED PER ABN-V03750, VER. 1.0	09-24-14	DCP	MWD	WLN	X

CAD 1X4DB148-3
ACAD2004 DCP- 03

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SEAL WATER INJECTION CARTRIDGE FILTER
SYSTEM NO. 1208

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB148-3
VER. 3.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. DELETED.
3. SEE SYSTEM P & I DIAGRAM 1X4DB116-1 FOR DRAINAGE SYSTEM.
4. BACKFLUSH DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS R Ⅳ.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE, SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNICATOR IN THE MAIN CONTROL ROOM.
9. FILTER IS SHOWN IN SERVICE. FOR OVERPRESSURE PROTECTION, THE SEAL INJECTION CARTRIDGE FILTER VESSELS SHALL NOT BE ISOLATED SUCH THAT A VOLUME OF WATER WOULD BECOME TRAPPED. TO PLACE FILTER IN STANDBY, 1HV41326B MAY BE CLOSED.

CAD 1X4DB148-4
ACAD2004 DCP- 03

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NO.	VERSION	DATE	DR	CHK	APPV	DTL
3.0	REVISED PER ABN-V03760, VER. 1.0	09-25-14	DCP	MWD	WLN	X

SIZE E 34X44

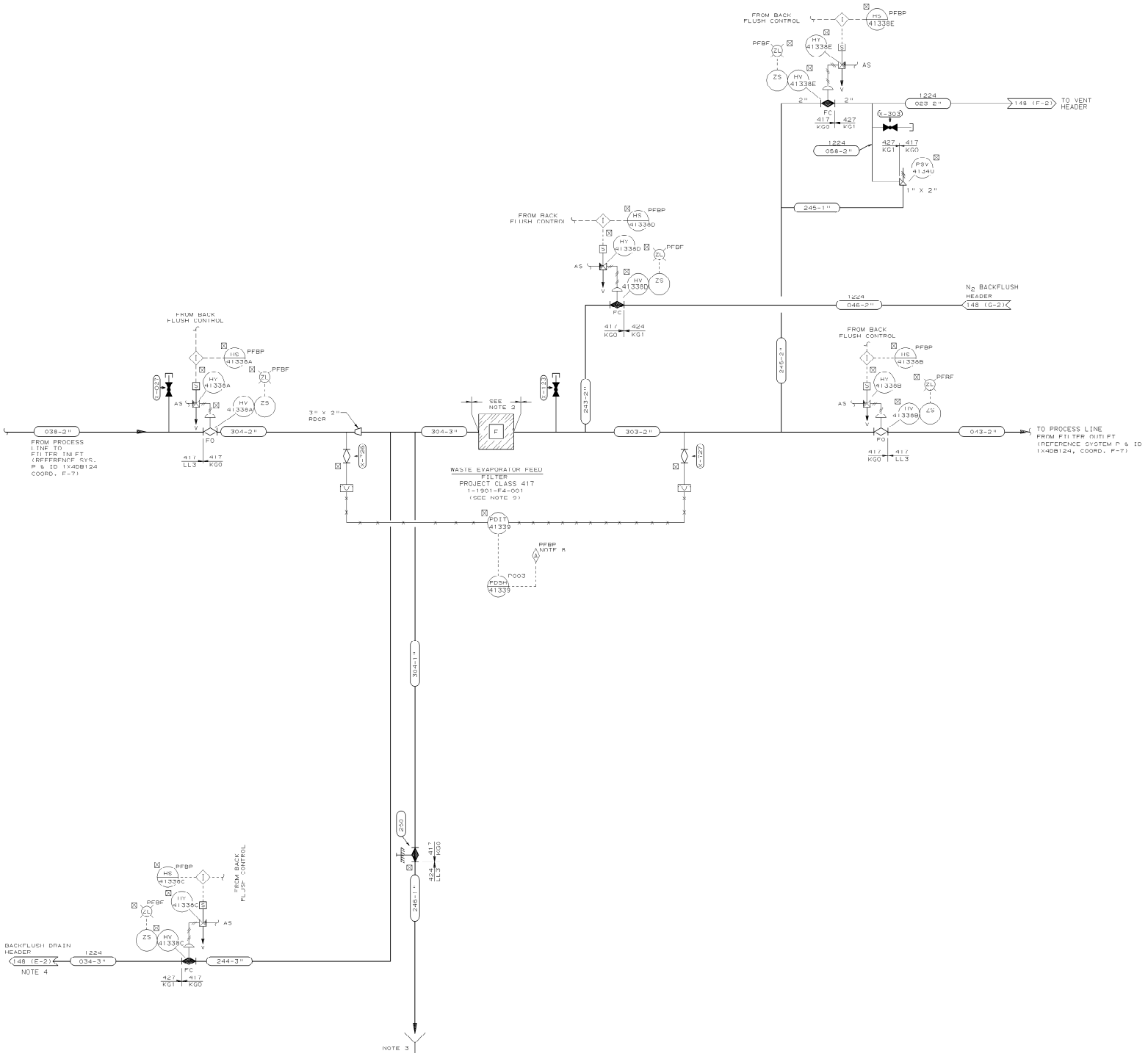
SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SEAL WATER INJECTION
CARTRIDGE FILTER
SYSTEM NO. 1208

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB148-4	3.0

1 DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT,VALVES,AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR.ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM 1X4DB124 FOR DRAINAGE SYSTEM.
4. BACKFLUSH DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS RV.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE,SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNICATOR IN THE MAIN CONTROL ROOM.
9. FILTER ELEMENTS HAVE BEEN REMOVED FROM THIS UNIT.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
2	INCORP. PER ABN-33043	4-2-98	LVL	JGM	GLR	WFP

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

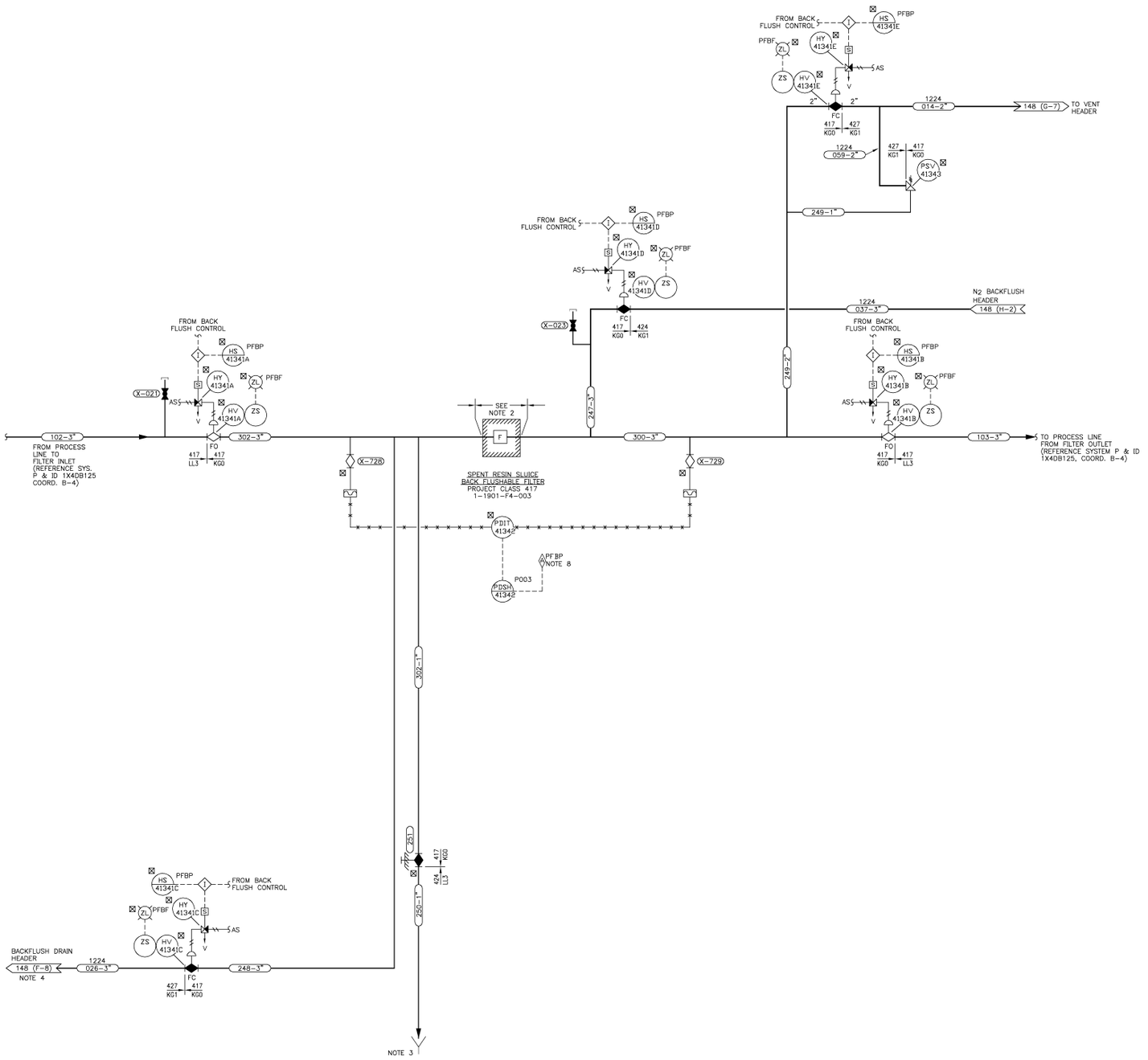
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
WASTE EVAPORATOR FEED FILTER
SYSTEM NO. 1901

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	1X4DB148-8	0

SIZE E 34x44

CAD NAME 1408148B



NOTES:

1. FOR DRAIN HEADER INFORMATION SEE DWG. 1X4DB143 AND 1X4DB144-1.
2. ALL EQUIPMENT, VALVES AND INSTRUMENTATION SHALL BE SUPPLIED BY THE VENDOR. ALL INTERCONNECTING PIPING SHALL BE FURNISHED BY BECHTEL.
3. SEE SYSTEM P & I DIAGRAM 1X4DB125 FOR DRAINAGE SYSTEM.
4. BACKFLUSH DRAIN LINES WILL SLOPE TO THE BACKFLUSHABLE FILTER CRUD TANK WITH NO POCKETS.
5. RADIATION ZONE CLASSIFICATION FOR THE EQUIPMENT IN THIS SYSTEM IS R3Z.
6. ALL LINES AND VALVES, REGARDLESS OF SIZE, SHALL BE BUTT-WELDED IN ORDER TO MINIMIZE CRUD TRAPS.
7. SEE VENDOR DRAWINGS FOR DETAILS OF VALVES SHOWN WITH DIAPHRAGM ACTUATORS.
8. THIS ALARM HAS A COMMON ANNUNCIATOR IN THE MAIN CONTROL ROOM.

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NO.	VERSION	DATE	DR	CHK	APPV	DTL
1.0	REDRAWN PER ABN-V70079, VER. 1.0	7-11-06	CYN	LCF	WPE	X

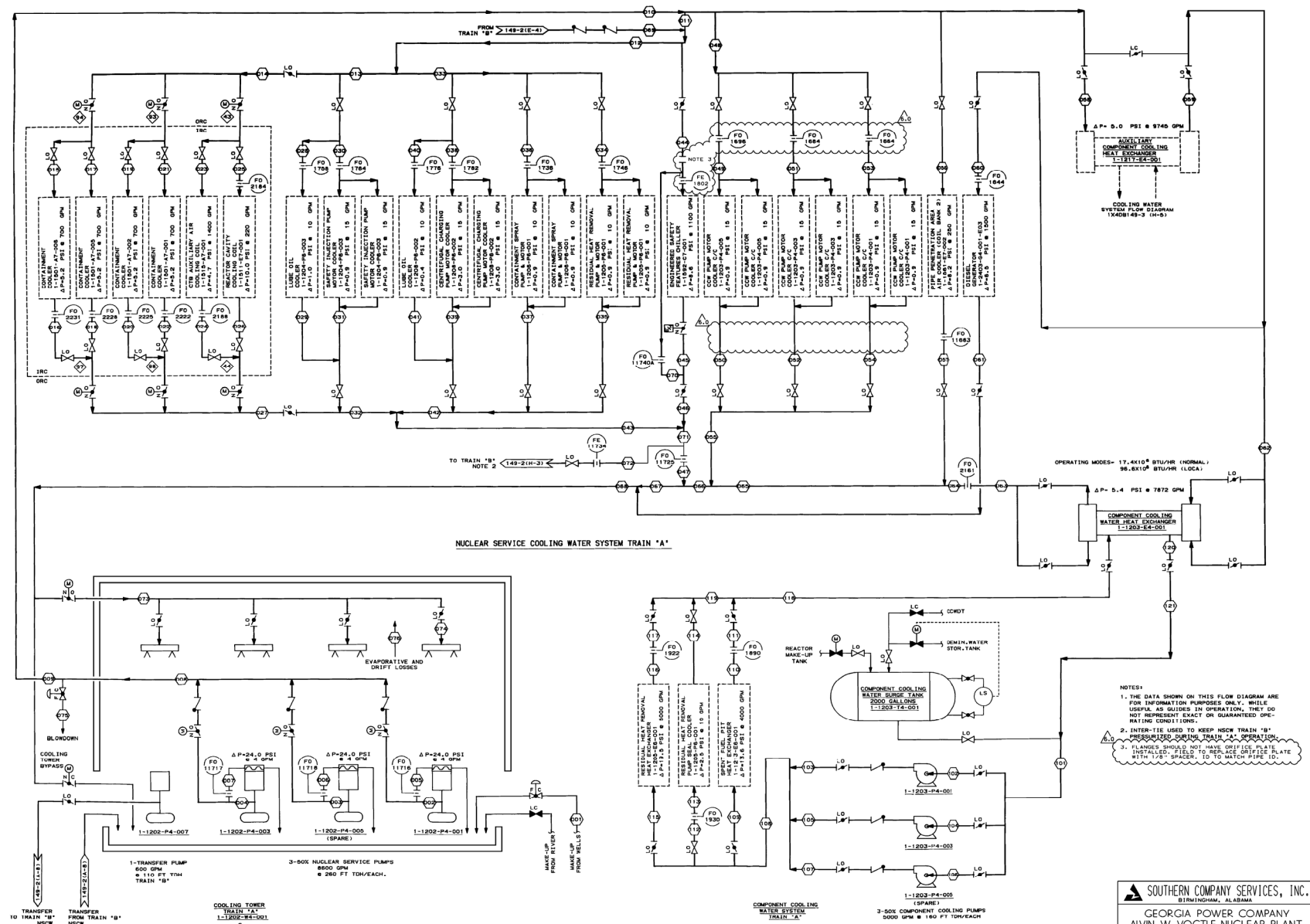
CAD 1X4DB148-9
ACAD2004 CYN-01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SPENT RESIN SLUICE BACKFLUSHABLE FILTER SYSTEM SYSTEM NO. 1901

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB148-9
VER. 1.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



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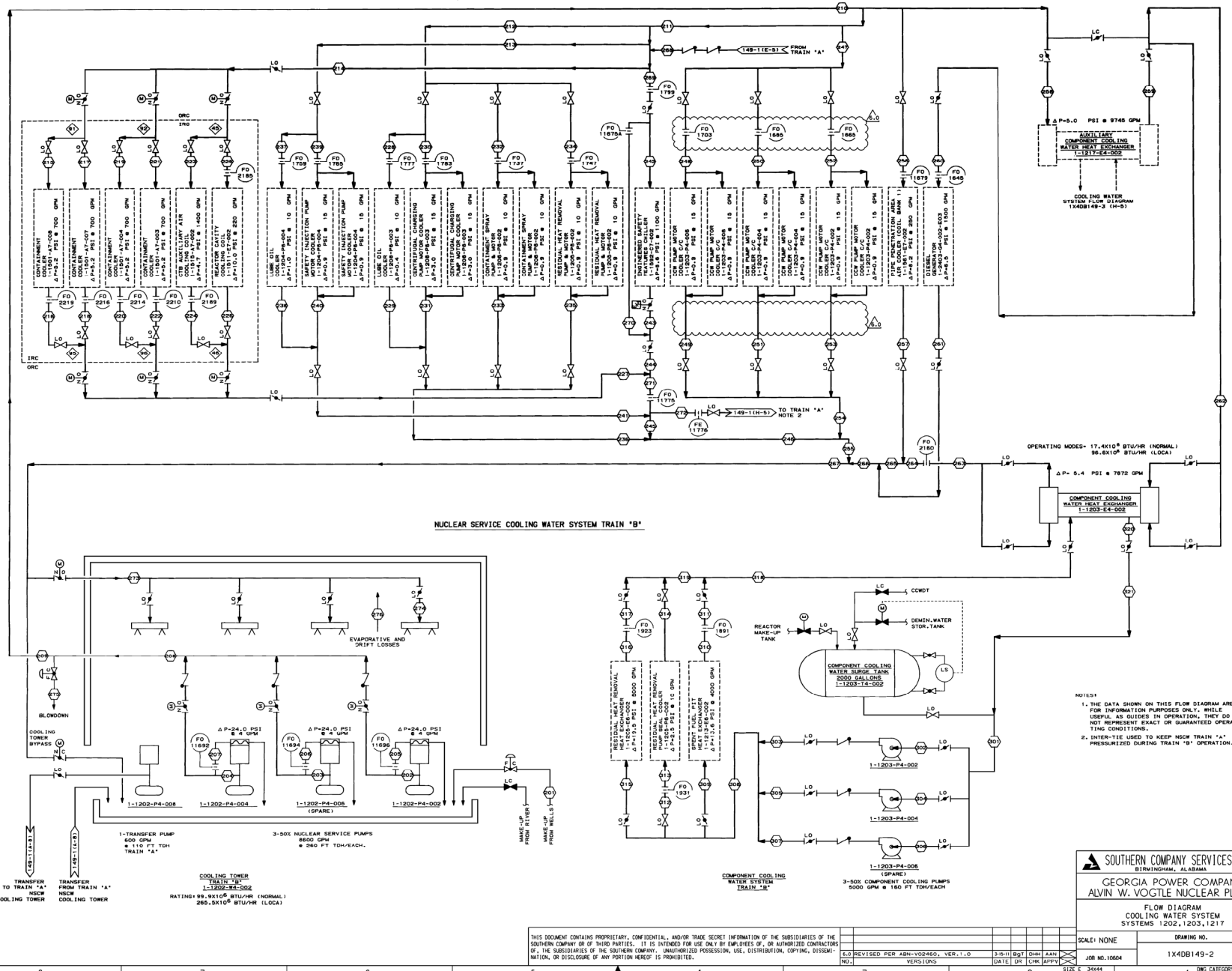
NO.	VERSIONS	DATE	DR	CHK	APPV
6.0	REVISED PER ABN-V02460, VER.1.0	3-15-11	BGT	DHH	AAN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
COOLING WATER SYSTEM
SYSTEMS 1202,1203,1217

SCALE: NONE
SIZE E 34x44
DRAWING NO. 1X4DB149-1
VER. 6.0
JOB NO. 10604
DWG CATEGORY: CRITICAL



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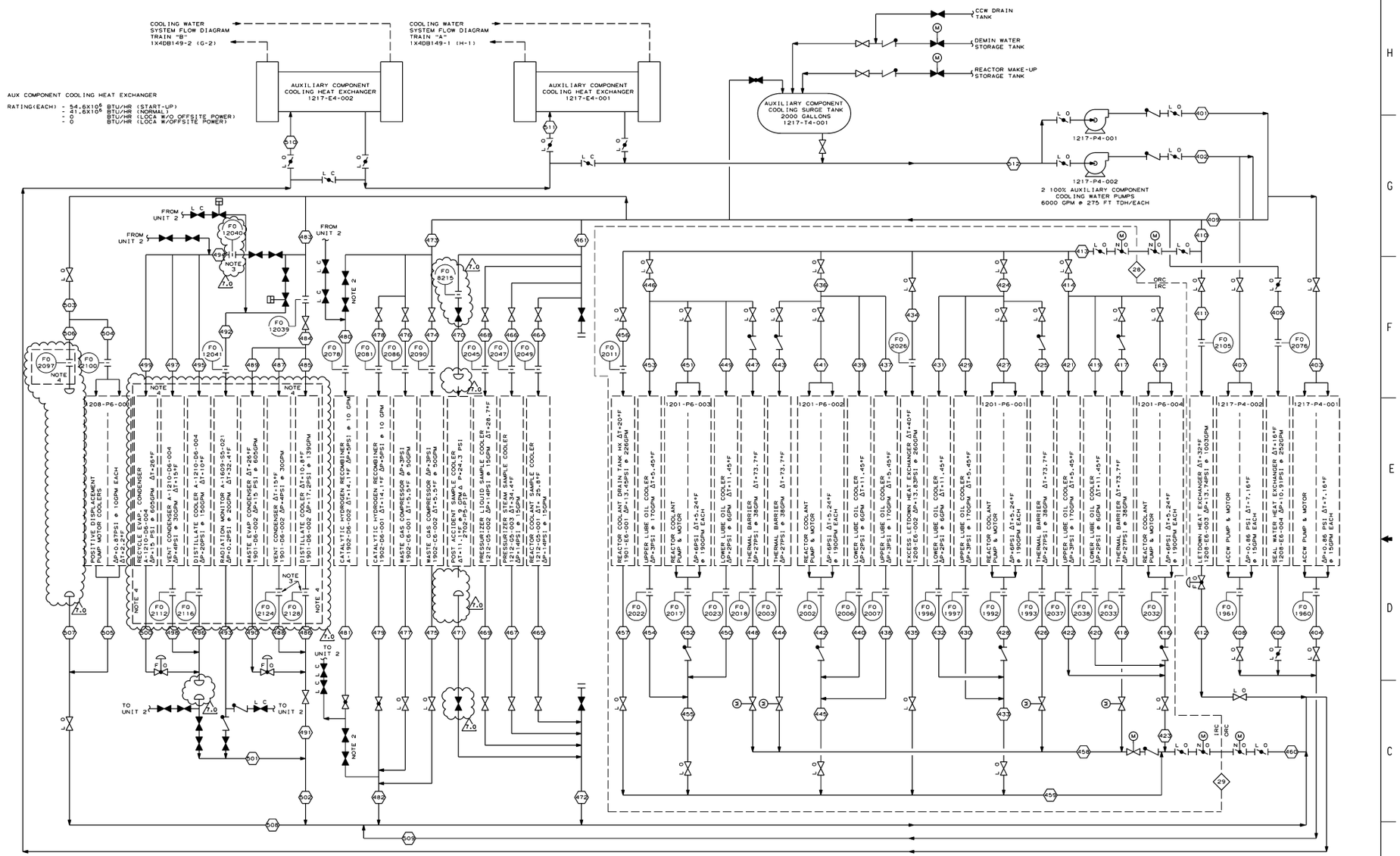
NO.	VERSIONS	DATE	DR	CHK	APPV
6.0	REVISED PER ABN-V02460, VER.1.0	3-15-11	BGT	DHH	AAN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
COOLING WATER SYSTEM
SYSTEMS 1202,1203,1217

SCALE: NONE
DRAWING NO. 1X4DB149-2
VER. 6.0
JOB NO. 10604
SIZE E 34x44
DWG CATEGORY: CRITICAL



AUXILIARY COMPONENT COOLING WATER SYSTEM

NOTE

1. THE DATA SHOWN ON THIS FLOW DIAGRAM ARE FOR INFORMATION PURPOSES ONLY, AND WHILE USEFUL AS GUIDES IN OPERATION, DO NOT REPRESENT EXACT OR GUARANTEED OPERATING CONDITIONS.
2. THESE VALVES WILL BE LOCKED CLOSED WHEN COOLING WATER IS SUPPLIED FROM UNIT 2.
3. BLANK PLATE
4. "RETIRED IN PLACE"

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
COOLING WATER SYSTEM
SYSTEMS 1202, 1203, 1217

SCALE: NONE | DRAWING NO. 1X4DB149-3 | VER. 7.0
JOB NO.10604

NO.	VERSIONS	DATE	DR	CHK	APPV
7.0	REVISED PER ABN-V02460, VER.1.0	3-15-11	BgT	DHH	AAN

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SIZE E 34X44 | DWG CATEGORY: CRITICAL

NUCLEAR SERVICE COOLING WATER SYSTEM TRAIN "A" (NOTE 2)

MODES OF OPERATION	PARAMETER		001	002	003	004	005	006	007	008	009	010	011	012	013	014	015	016	017	018	019	020	021	022	023	024	025	026	027	028	029	030	031	032	033	034	035	036	037	038	039	040	041	042	043	044	045	046	047	048	049	050	051	052	053	054	055	056
100 % POWER	FLOW	GPM	287	8700	0	8700	5	0	5	17390	17330	17330	6558	4863	4761	4710	746	746	746	746	746	746	746	1491	1491	234	234	4710	13	13	38	38	4760	102	26	26	26	26	37	38	13	13	102	4863	1695	1100	1695	6558	114	38	38	38	38	38	38	114	268
	TEMP	°F	70	90	90	90	90	90	90	90	90	90	90	90	90	90	90	97	90	97	90	97	90	97	90	96	90	94	97	90	90	90	90	97	90	90	90	90	90	90	97	90	102	94	97	90	90	90	95	90	90	97	90	97	90	90	95	90
PEAK MSLBA CONDITION (NOTE 4)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	3318	3200	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	14	40	40	3200	107	27	27	27	27	40	40	13	13	107	3307	1727	1100	1727	5034	120	40	40	40	40	40	40	120	283
	TEMP	°F	70	92	92	92	92	92	92	92	92	92	92	92	92	92	92	255	92	255	92	255	92	255	92	92	92	92	255	92	95	92	96	252	92	92	92	119	92	97	92	98	92	104	104	247	92	101	98	196	92	92	95	92	95	92	94	92
PEAK LOCA CONDITION (NOTE 5)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	3318	3200	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	14	40	40	3200	107	27	27	27	27	40	40	13	13	107	3307	1727	1100	1727	5034	120	40	40	40	40	40	40	120	283
	TEMP	°F	70	91	91	91	91	91	91	91	91	91	91	91	91	91	91	266	91	266	91	266	91	266	91	91	91	91	266	91	94	91	95	263	91	91	91	118	91	96	91	97	91	102	103	258	91	100	97	202	91	91	94	91	94	91	93	91
RECIRC-MODE POST-LOCA (NOTE 6)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	3318	3200	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	14	40	40	3200	107	27	27	27	27	40	40	13	13	107	3307	1727	1100	1727	5034	120	40	40	40	40	40	40	120	283
	TEMP	°F	70	95	95	95	95	95	95	95	95	95	95	95	95	95	95	242	95	242	95	242	95	242	95	95	95	95	242	95	99	95	99	239	95	95	95	123	95	101	95	102	95	107	107	235	95	105	101	188	95	95	99	95	99	95	95	98	95

NUCLEAR SERVICE COOLING WATER SYSTEM TRAIN "A" (CONTINUED) (NOTE 2)

MODES OF OPERATION	PARAMETER		057	058	059	060	061	062	063	064	065	066	067	068	069	070	071	072	073	074	075	076
100 % POWER	FLOW	GPM	268	10390	10390	1718	1718	8672	8672	8672	8940	9054	15612	17330	0	595	6558	0	17330	4333	65	201
	TEMP	°F	90	90	98	98	98	98	102	102	102	102	99	99	95	90	95	95	99	99	90	90
PEAK MSLBA CONDITION (NOTE 4)	FLOW	GPM	283	10952	10952	1811	1811	9141	9141	9141	9424	9544	14578	16389	0	627	5034	0	16389	4097	0	472
	TEMP	°F	101	92	93	93	112	93	146	146	145	144	133	131	92	92	196	196	131	131	92	104
PEAK LOCA CONDITION (NOTE 5)	FLOW	GPM	283	10952	10952	1811	1811	9141	9141	9141	9424	9544	14578	16389	0	627	5034	0	16389	4097	0	477
	TEMP	°F	100	91	91	91	110	91	95	95	95	95	132	129	91	91	202	202	129	129	91	104
RECIRC-MODE POST-LOCA (NOTE 6)	FLOW	GPM	283	10952	10952	1811	1811	9141	9141	9141	9424	9544	14578	16389	0	627	5034	0	16389	4097	0	432
	TEMP	°F	105	95	95	95	114	95	129	129	128	128	148	144	95	95	188	188	144	144	95	110

COMPONENT COOLING WATER SYSTEM TRAIN "A"

MODES	PARAMETER		101	102	103	104	105	106	107	108	109	110	111	112	113	114	115	116	117	118	119	120	121
100% POWER	FLOW	GPM	8979	4490	4490	4490	4490	0	0	8979	3982	3988	3988	10	10	10	4987	4987	4987	8987	4997	8979	8979
	TEMP	°F	100	100	100	100	100	100	100	100	100	108	108	100	100	100	100	100	100	104	100	100	100
2 TRAIN START OF RECIRC POST-LOCA	FLOW	GPM	8978	4489	4489	4489	4489	0	0	8978	3981	3985	3985	10	10	10	4987	5062	5062	9057	5070	8978	8978
	TEMP	°F	102	102	102	102	102	102	102	102	102	106	106	102	102	107	102	156	156	134	156	102	102
1 TRAIN COOLDOWN: OFFSITE POWER AVAIL.	FLOW	GPM	9020	4510	4510	4510	4510	0	0	9020	4000	4010	4010	10	10	10	5010	5122	5122	9132	5132	9020	9020
	TEMP	°F	120	120	120	120	120	120	120	120	120	129	129	120	120	125	120	189	189	162	189	120	120
SINGLE TRAIN COOLDN W/LOP	FLOW	GPM	9002	4501	4501	4501	4501	0	0	9002	3992	4000	4000	10	10	10	5000	5077	5077	9167	5175	9002	9002
	TEMP	°F	111	111	111	111	111	111	111	111	111	120	120	111	111	116	111	202	202	165	201	111	111

NUCLEAR SERVICE COOLING WATER SYSTEM TRAIN "B" (NOTE 2)

MODES OF OPERATION	PARAMETER		201	202	203	204	205	206	207	208	209	210	211	212	213	214	215	216	217	218	219	220	221	222	223	224	225	226	227	228	229	230	231	232	233	234	235	236	237	238	239	240	241	242	243	244	245	246	247	248	249	250	251	252	253	254	255	256
100 % POWER	FLOW	GPM	287	8700	0	8700	5	0	5	17390	17330	17330	6558	102	51	4710	746	746	746	746	746	746	746	1491	1491	234	234	4710	13	13	37	37	26	26	26	26	102	13	13	38	38	51	1100	1100	1695	6456	6558	114	38	38	38	38	38	38	114	6672	268
	TEMP	°F	70	90	90	90	90	90	90	90	90	90	90	90	90	90	90	97	90	97	90	97	90	97	90	96	90	94	97	90	102	90	97	90	90	90	90	94	90	90	90	90	90	90	90	90	95	95	90	90	97	90	97	90	90	95	95	90
PEAK MSLBA CONDITION (NOTE 4)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	107	54	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	13	40	40	27	27	27	27	107	14	14	40	40	54	1100	1100	1727	4927	5034	120	40	40	40	40	40	40	120	5154	283
	TEMP	°F	70	92	92	92	92	92	92	92	92	92	92	92	92	92	92	255	92	255	92	255	92	255	92	92	92	92	255	92	104	92	98	92	97	92	119	104	92	95	92	96	96	92	101	98	198	196	92	92	95	92	95	92	92	94	193	92
PEAK LOCA CONDITION (NOTE 5)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	107	54	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	13	40	40	27	27	27	27	107	14	14	40	40	54	1100	1100	1727	4927	5034	120	40	40	40	40	40	40	120	5154	283
	TEMP	°F	70	91	91	91	91	91	91	91	91	91	91	91	91	91	91	266	91	266	91	266	91	266	91	91	91	91	266	91	102	91	97	91	96	91	118	103	91	94	91	95	94	91	100	97	205	202	91	91	94	91	94	91	91	93	200	91
RECIRC-MODE POST-LOCA (NOTE 6)	FLOW	GPM	0	8200	0	8200	6	0	6	16389	16389	16389	5034	107	54	3146	787	787	787	787	787	787	787	0	0	0	0	3146	14	13	40	40	27	27	27	27	107	14	14	40	40	54	1100	1100	1727	4927	5034	120	40	40	40	40	40	40	120	5154	283
	TEMP	°F	70	95	95	95	95	95	95	95	95	95	95	95	95	95	95	242	95	242	95	242	95	242	95	95	95	95	242	95	107	95	102	95	101	95	123	107	95	99	95	99	99	95	105	101	190	188	95	95	99	95	99	95	95	98	186	95

NUCLEAR SERVICE COOLING WATER SYSTEM TRAIN "B" (CONTINUED) (NOTE 2)

MODES OF OPERATION	PARAMETER		257	258	259	260	261	262	263	264	265	266	267	268	269	270	271	272	273	274	275	276
100 % POWER	FLOW	GPM	268	10390	10390	1718	1718	8672	8672	8672	8940	10658	17330	0	1695	595	6405	0	17330	4333	65	200
	TEMP	°F	90	90	98	98	98	102	102	102	101	99	95	90	90	95	95	99	99	99	90	90
PEAK MSLBA CONDITION (NOTE 4)	FLOW	GPM	283	10952	10952	1811	1811	9141	9141	9141	9424	11235	16389	0	1727	627	4873	0	16389	4097	0	472
	TEMP	°F	101	92	93	93	112	93	146	146	145	139	131	92	92	92	199	199	131	131	92	104
PEAK LOCA CONDITION (NOTE 5)	FLOW	GPM	283	10952	10952	1811	1811	9141	9141	9141	9424	11235	16389	0	1727	627	4873	0	16389	4097	0	477
	TEMP	°F	100	91	91	91	110	91	95	95	95	97	129	91	91	91	206	206	129	129	91	104
RECIRC-MODE POST-LOCA (NOTE 6)	FLOW	GPM	283	10952	10952	1811	1811	9141	9141	9141	9424	11235	16389	0	1727	627	4873	0	16389	4097	0	432
	TEMP	°F	105	95	95	95	114	95	129	129	128	125	144	95	95	95	191	191	144	144	95	110

COMPONENT COOLING WATER SYSTEM TRAIN "B"

MODES	PARAMETER		301	302	303	304	305	306	307	308	309	310	311	312	313	314	315	316	317	318	319	320	321
100% POWER	FLOW	GPM	8979	4490	4490	4490	4490	0	0	8979	3982	3988	3988	10	10	10	4987	4987	4987	8987	4997	8979	8979
	TEMP	°F	100	100	100	100	100	100	100	100	100	108	108	100	100	100	100	100	100	104	100	100	100
2 TRAIN START OF RECIRC POST-LOCA	FLOW	GPM	8978	4489	4489	4489	4489	0	0	8978	3981	3985	3985	10	10	10	4987	5062	5062	9057	5070	8978	8979
	TEMP	°F	102	102	102	102	102	102	102	102	102	106	106	102	102	107	102	156	156	134	156	102	102
1 TRAIN COOLDOWN: OFFSITE POWER AVAIL.	FLOW	GPM	9020	4510	4510	4510	4510	0	0	9020	4000	4010	4010	10	10	10	5010	5122	5122	9132	5132	9020	9020
	TEMP	°F	120	120	120	120	120	120	120	120	120	129	129	120	120	125	120	189	189	162	189	120	120
SINGLE TRAIN COOLDN W/LOP	FLOW	GPM	9002	4501	4501	4501	4501	0	0	9002	3992	4000	4000	10	10	10	5000	5077	5077	9167	5175	9002	9002
	TEMP	°F	111	111	111	111	111	111	111	111	111	120	120	111	111	116	111	202	202	165	201	111	111

AUXILIARY COMPONENT COOLING WATER SYSTEM (NOTE 3)

MODES OF OPERATION	PARAMETER		401	402	403	404	405	406	407	408	409	410	411	412	413	414	415	416	417	418	419	420	421	422	423	424	425	426	427	428	429	430	431	432	433	434	435	436	437	438	439	440	441	442	443	444	445	446	447	448	449	450	451	452	453	454	455	456
100 % POWER	FLOW	GPM	0	5465	31	31	258	259	31	31	5403	3205	262	266	2943	611	390	390	41	42	6	6	174	174	571	611	41	42	390	390	174	174	6	6	571	267	267	611	174	174	6	6	390	390	41	42	571	611	41	42	6	6	390	390	174	174	571	232
	TEMP	°F	105	105	105	105	105	120	105	112	105	105	105	155	105	105	105	110	105	173	105	116	105	110	110	105	105	173	105	110	105	110	105	116	110	105	105	105	105	110	105	116	105	110	105	173	110	105	105	173	105	116	105	110	105	110
START -UP	FLOW	GPM	0	6200	31	31	260	261	31	31	6138	3960	1003	1011	2957	614	392	392	41	42	6	6	175	175	574	614	41	42	392	392	175	175	6	6	574	268	271	614	175	175	6	6	392	392	41	42	614	614	41	42	6	6	392	392	175	175	574	233
	TEMP	°F	105	105	105	105	105	120	105	112	105	105	105	137	105	105	105	110	105	173	105	116	105	110	110	105	105	173	105	110	105	110	105	116	110	105	105	144	105	105	110	105	116	105	110	105	173	110	105	105	173	105	116	105	110	105	110	105
SHUT DOWN, LOP	FLOW	GPM	0	6500	30	30	252	253	30	30	6440	4064	1194	1194	2870	596	380	380	40	41	6	6	170	170	556	596	40	41	380	380	170	170	6	6	556	260	260	596	170	170	6	6	380	380	40	41	596	596	40	41	6	6	380	380	170	170	556	226
	TEMP	°F	105	105	105	105	105	121	105	112	105	105	105	105	105	105	105	105	105	175	105	105	105	105	105	105	105	175	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	175	105	105	105	105	105	175	105	105	105	105	105	105
HOT STANDBY WITH OFFSITE POWER	FLOW	GPM	0	5465	31	31	258	259	31	31	5403	3205	262	266	2943	611	390	390	41	42	6	6	174	174	570	611	41	42	390	390	174	174	6	6	571	267	267	611	174	174	6	6	390	390	41	42	570	611	41	42	6	6	390	390	174	174	570	232
	TEMP	°F	105	105	105	105	105	120	105	112	105	105	105	155	105	105	105	105	105	173	105	116	105	110	105	105	105	173	105	110	105	110	105	116	110	105	105	105	105	105	105	116	105	110	105	173	105	105	105	173	105	116	105	110	105	105

CONTINUATION OF AUXILIARY COMPONENT COOLING WATER SYSTEM (NOTE 3)

MODES OF OPERATION	PARAMETER		457	458	459	460	461			464	465	466	467	468	469	470	471	472	473	474	475	476	477	478	479	480	481	482	483	484	485	486	487	488	489	490	491	492	493	494	495	496	497	498	499	500	501	502	503	504	505	506	507	508	509	510	511	512
100 % POWER	FLOW	GPM	233	125	2212	2949	84			15	15	15	15	15	15	10	10	84	122	51	51	51	61	10	10	10	10	122	1630	805	154	154	31	31	620	624	809	20	20	805	154	154	31	31	620	624	829	1639	103	20	20	83	83	103	321	5484	5465	5465
	TEMP	°F	124	173	111	114	105			105	131	105	139	105	134	105	105	125	105	105	110	105	107	105	119	105	105	108	105	105	115	105	120	105	130	127	105	136	105	105	115	105	120	105	130	127	127	105	105	107	105	115	113	118	120	105	105
START -UP	FLOW	GPM	233	125	2225	2965	84			15	15	15	15	15	15	10	10	84	124	52	52	52	52	10	10	10	10	124	1603	791	155	155	31	31	605	609	795	21	21	791	155	155	31	31	605	609	817	1612	105	21	21	84	84	105	323	6227	6200	6200
	TEMP	°F	111	173	114	117	105			105	131	105	139	105	134	105	105	125	105	105	110	105	105	105	119	105	105	108	105	105	115	105	120	105	131	127	105	134	105	105	115	105	120	105	131	128	127	105	105	107	105	115	113	118	123	105	105
SHUT DOWN, LOP	FLOW	GPM	226	122	2154	2873	83			15	15	15	15	15	15	10	10	83	120	50	50	50	50	10	10	10	10	120	1820	900	150	150	30	30	720	720	900	20	20	900	150	150	30	30	720	720	920	1820	101	20	20	81	81	101	313	6507	6500	6500
	TEMP	°F	105	175	105	109	105			105	105	105	105	105	105	105	105	107	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	105	119	107	105	105
HOT STANDBY WITH OFFSITE POWER	FLOW	GPM	232	126	2210	2946	84			15	15	15	15	15	15	10	10	84	122	51	51	51	51	10	10	10	10	122	1630	805	154	154	31	31	620	624	809	20	20	805	154	154	31	31	620	624	829	1639	103	20	20	83	83	103	321	5480	5465	5465
	TEMP	°F	105	173	106	110	105			105	131	105	139	105	134	105	105	125	105	105	110	105	105	105	119	105	105	108	105	105	115	105	120	105	130	127	105	136	105	105	115	105	120	105	130	127	127	105	105	107	105	115	113	118	118	105	105

NOTES:

1. THE DATA SHOWN ON THIS FLOW DIAGRAM ARE FOR INFORMATION PURPOSES ONLY, WHILE USEFUL AS GUIDES IN OPERATION, THEY DO NOT REPRESENT EXACT OR GUARANTEED OPERATING CONDITIONS.
2. ALL NSCW DATA SHOWN ARE FOR ONE-TRAIN OPERATION, WITH ALL ACCW HEAT LOADS ON THE OPERATING TRAIN.
3. ACCW DATA ASSUME ACCW HEAT LOADS REJECTED TO TRAIN B NSCW AND THAT UNIT 1 ACCW IS SUPPLIED TO THE COMMON COMPONENTS.
4. TEMPERATURES SHOWN HERE ARE THE MAXIMUM TEMPERATURES DURING THE MSLBA TRANSIENT. THE MAXIMUM TEMPERATURE AT EACH NODE MAY NOT OCCUR AT THE SAME TIME DURING THE TRANSIENT. THEREFORE THE INLET AND OUTLET TEMPERATURE DIFFERENCES DO NOT NECESSARILY REPRESENT THE ACTUAL TEMPERATURE RISES ACROSS THE COMPONENTS AT A SPECIFIC TIME. (REFER TO CALC. * X4C1202S18 FOR DETAILS).
5. TEMPERATURES SHOWN HERE ARE THE MAXIMUM TEMPERATURES DURING THE LOCA TRANSIENT. THE MAXIMUM TEMPERATURE AT EACH NODE MAY NOT OCCUR AT THE SAME TIME DURING THE TRANSIENT. THEREFORE THE INLET AND OUTLET TEMPERATURE DIFFERENCES DO NOT NECESSARILY REPRESENT THE ACTUAL TEMPERATURE RISES ACROSS THE COMPONENTS AT A SPECIFIC TIME. (REFER TO CALC. * X4C1202S17 FOR DETAILS).
6. TEMPERATURES SHOWN HERE ARE THE MAXIMUM TEMPERATURES DURING THE POST-LOCA, RECIRCULATION TRANSIENT. THE MAXIMUM TEMPERATURE AT EACH NODE MAY NOT OCCUR AT THE SAME TIME DURING THE TRANSIENT. THEREFORE THE INLET AND OUTLET TEMPERATURE DIFFERENCES DO NOT NECESSARILY REPRESENT THE ACTUAL TEMPERATURE RISES ACROSS THE COMPONENTS AT A SPECIFIC TIME. (REFER TO CALC. * X4C1202C17 FOR DETAILS).

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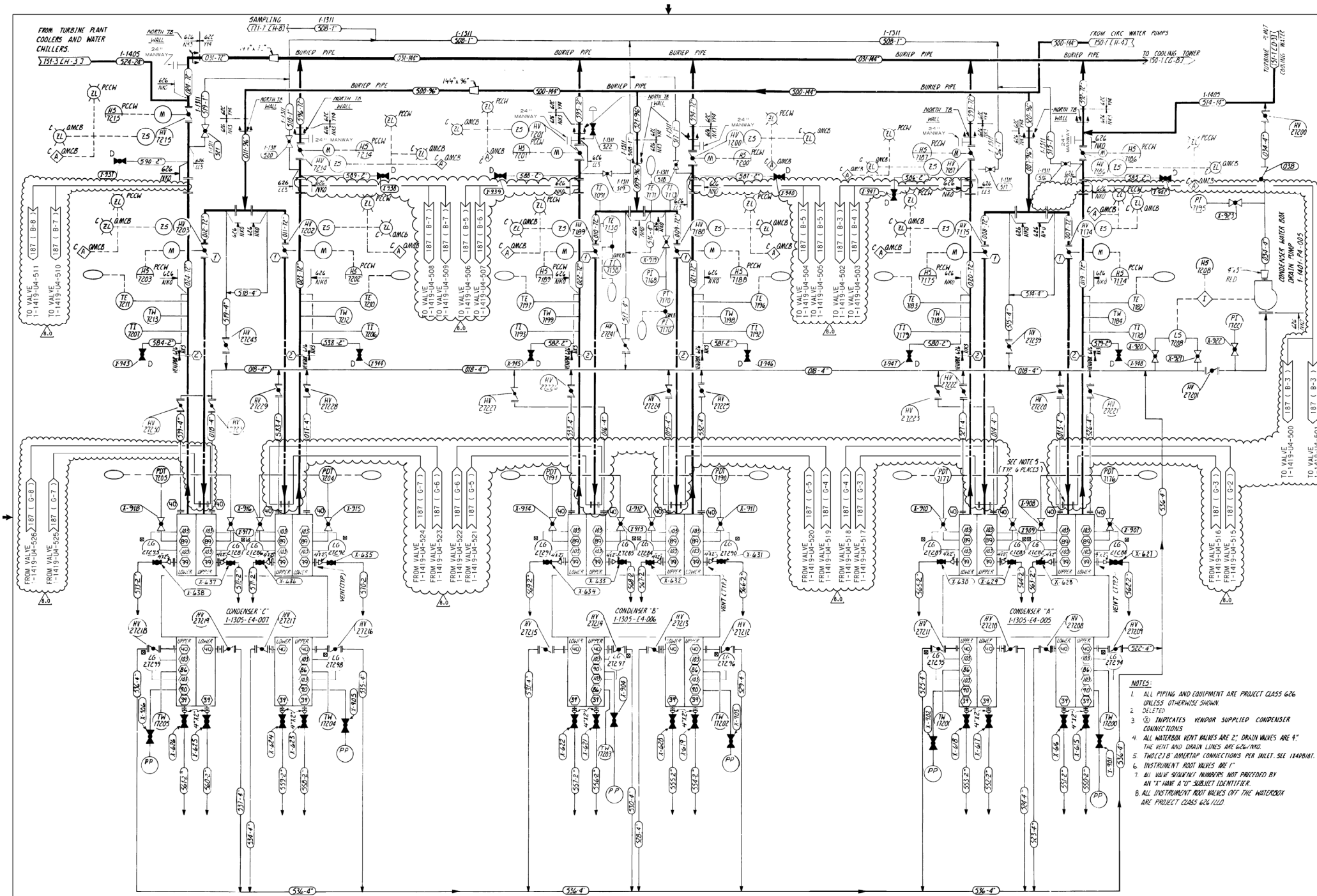
NO.	VERSIONS	DATE	DR	CHK	APPV
6.0	REVISED PER ABN-V02460, VER.1.0	3-15-11	BgT	DHH	AAN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

FLOW DIAGRAM
COOLING WATER SYSTEM
SYSTEMS 1202,1203,1217

SCALE: NONE | DRAWING NO. 1X4DB149-4 | VER. 6.0 | JOB NO.10604 | SIZE E 34X44 | DWG CATEGORY: CRITICAL



NOTES:

1. ALL PIPING AND EQUIPMENT ARE PROJECT CLASS 626 UNLESS OTHERWISE SHOWN.
2. DELETED
3. ⓧ INDICATES VENDOR SUPPLIED CONDENSER CONNECTIONS
4. ALL WATERBOX VENT VALVES ARE 2", DRAIN VALVES ARE 4". THE VENT AND DRAIN LINES ARE 626/NKO.
5. TWO(2)18" ANNUBAR CONNECTIONS PER INLET. SEE 1140B187.
6. INSTRUMENT ROOT VALVES ARE 1".
7. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" HAVE A "U" SUBJECT IDENTIFIER.
8. ALL INSTRUMENT ROOT VALVES OFF THE WATERBOX ARE PROJECT CLASS 626 I.L.D.

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CIRCULATING WATER SYSTEM
SYSTEM NO. 1401

SCALE: NONE | DRAWING NO. 1X4DB150 | VER. 8.0 | JOB NO. 10604

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NO.	VERSIONS	DATE	DR	CHK	APPV
8.0	REVISED PER ABN-V00004, VER. 1.0	3-23-06	DFV	MWD	ASK

SIZE E 34x44

DRAWING CATEGORY: CRITICAL

FROM TURBINE PLANT COOLERS AND WATER CHILLERS.

SAMPLING 171-1 CH-87

TO COOLING TOWER 150-1 EG-87

FROM CIRC WATER PUMPS 150-1 CH-47

TURBINE 2 NAUT 151-1 CD 53 COOLING WATER

CONDENSER "C" 1-1305-E4-007

CONDENSER "B" 1-1305-E4-006

CONDENSER "A" 1-1305-E4-005

CONDENSER WATER BOX DRAIN PUMP 1-1401-P4-005

SEE NOTE 5 (TYP 6 PLACES)

NOTES:

1. ALL PIPING AND EQUIPMENT ARE PROJECT CLASS 626 UNLESS OTHERWISE SHOWN.
2. DELETED
3. (X) INDICATES VENDOR SUPPLIED CONDENSER CONNECTIONS
4. ALL WATERBOX VENT VALVES ARE 2", DRAIN VALVES ARE 4". THE VENT AND DRAIN LINES ARE 626/NK0.
5. TWO (2) 8" AMERTAP CONNECTIONS PER INLET. SEE 1X4PB187.
6. INSTRUMENT ROOT VALVES ARE 1".
7. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" HAVE A "U" SUBJECT IDENTIFIER.
8. ALL INSTRUMENT ROOT VALVES OFF THE WATERBOX ARE PROJECT CLASS 626/LLD.

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NO.	VERSIONS	DATE	DR	CHK	APPV
8.0	REVISED PER ABN-V00004, VER. 1.0	3-23-06	DFV	MWD	ASK

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CIRCULATING WATER SYSTEM
SYSTEM NO. 1401

SCALE: NONE

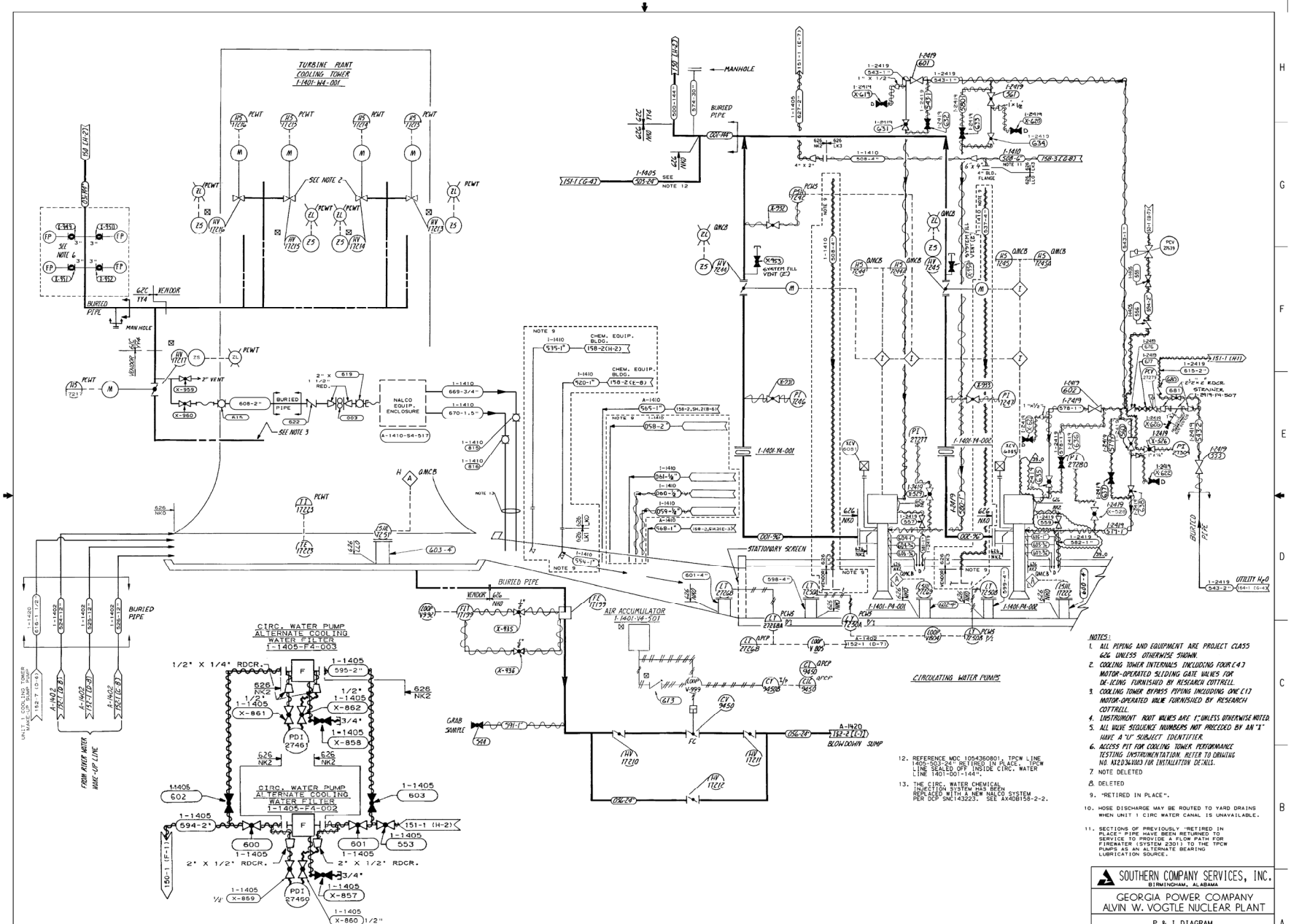
DRAWING NO. 1X4DB150

VER. 8.0

JOB NO. 10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



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NO.	VERSIONS	DATE	DR	CHK	APPV
39.0	REVISED PER ABN 1090653201M003, VER. 2.0	10/27/15	EJ	MWD	WLN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

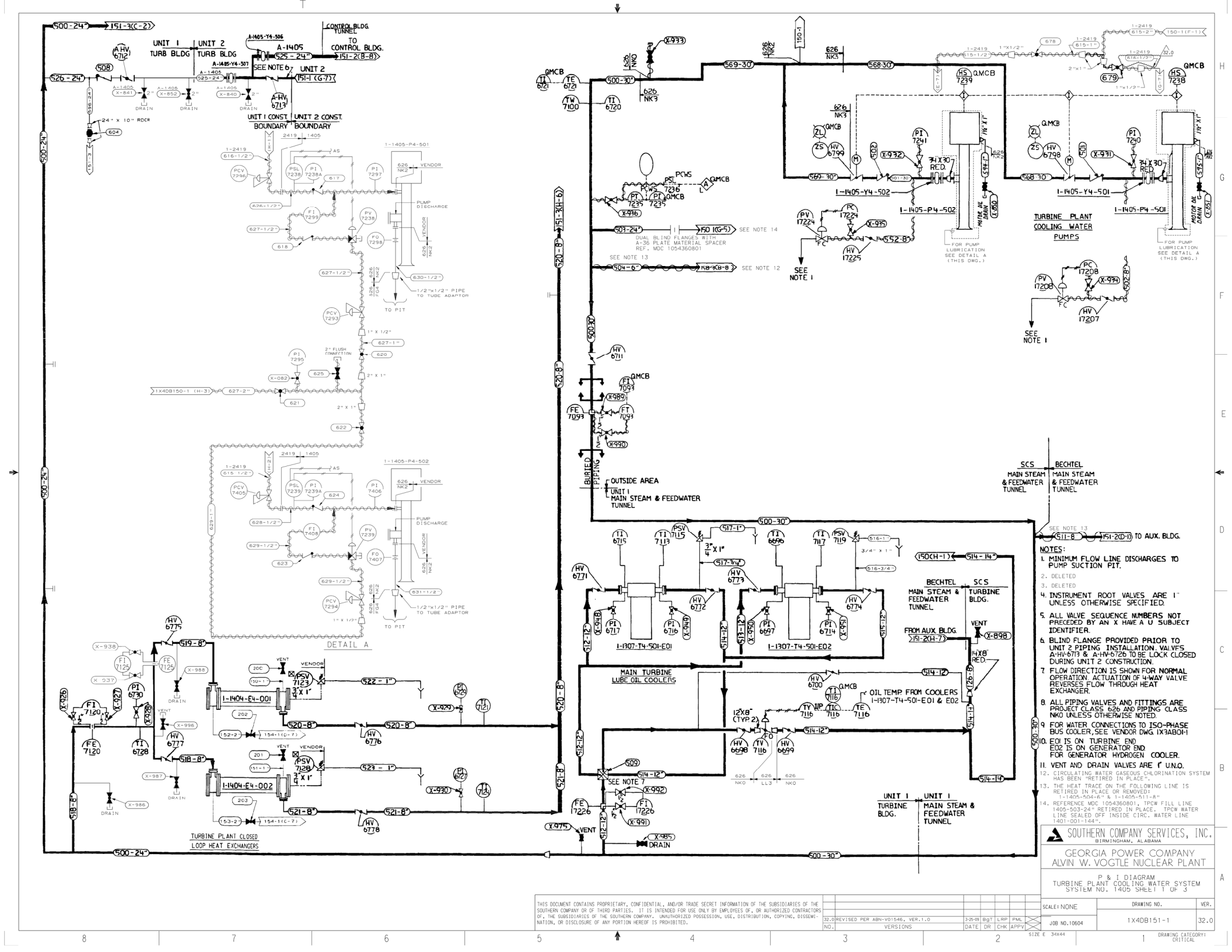
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CIRCULATING WATER SYSTEM
SYSTEM NO. 1401

SCALE: NONE	DRAWING NO.	VER.
	1X4DB150-1	39.0
JOB NO. 10604		

SIZE E 34X44

DRAWING CATEGORY: 1 CRITICAL



NOTES:

1. MINIMUM FLOW LINE DISCHARGES TO PUMP SUCTION PIT.
2. DELETED
3. DELETED
4. INSTRUMENT ROOT VALVES ARE 1" UNLESS OTHERWISE SPECIFIED.
5. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN X HAVE A U SUBJECT IDENTIFIER.
6. BLIND FLANGE PROVIDED PRIOR TO UNIT 2 PIPING INSTALLATION. VALVES A-HV-6713 & A-HV-6726 TO BE LOCK CLOSED DURING UNIT 2 CONSTRUCTION.
7. FLOW DIRECTION IS SHOWN FOR NORMAL OPERATION. ACTUATION OF 4-WAY VALVE REVERSES FLOW THROUGH HEAT EXCHANGER.
8. ALL PIPING VALVES AND FITTINGS ARE PROJECT CLASS 626 AND PIPING CLASS NK0 UNLESS OTHERWISE NOTED.
9. FOR WATER CONNECTIONS TO ISO-PHASE BUS COOLER, SEE VENDOR DWG. 1X3ABO1-1
10. E01 IS ON TURBINE END. E02 IS ON GENERATOR END. FOR GENERATOR HYDROGEN COOLER.
11. VENT AND DRAIN VALVES ARE 1" U.N.O.
12. CIRCULATING WATER GASEOUS CHLORINATION SYSTEM HAS BEEN "RETIRED IN PLACE".
13. THE HEAT TRACE ON THE FOLLOWING LINE IS RETIRED IN PLACE OR REMOVED: 1-1405-504-6" & 1-1405-511-8"
14. REFERENCE MDC 1054360801, TPCW FILL LINE 1405-503-24" RETIRED IN PLACE. TPCW WATER LINE SEALED OFF INSIDE CIRC. WATER LINE 1401-001-144".

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT COOLING WATER SYSTEM
SYSTEM NO. 1405 SHEET 1 OF 3

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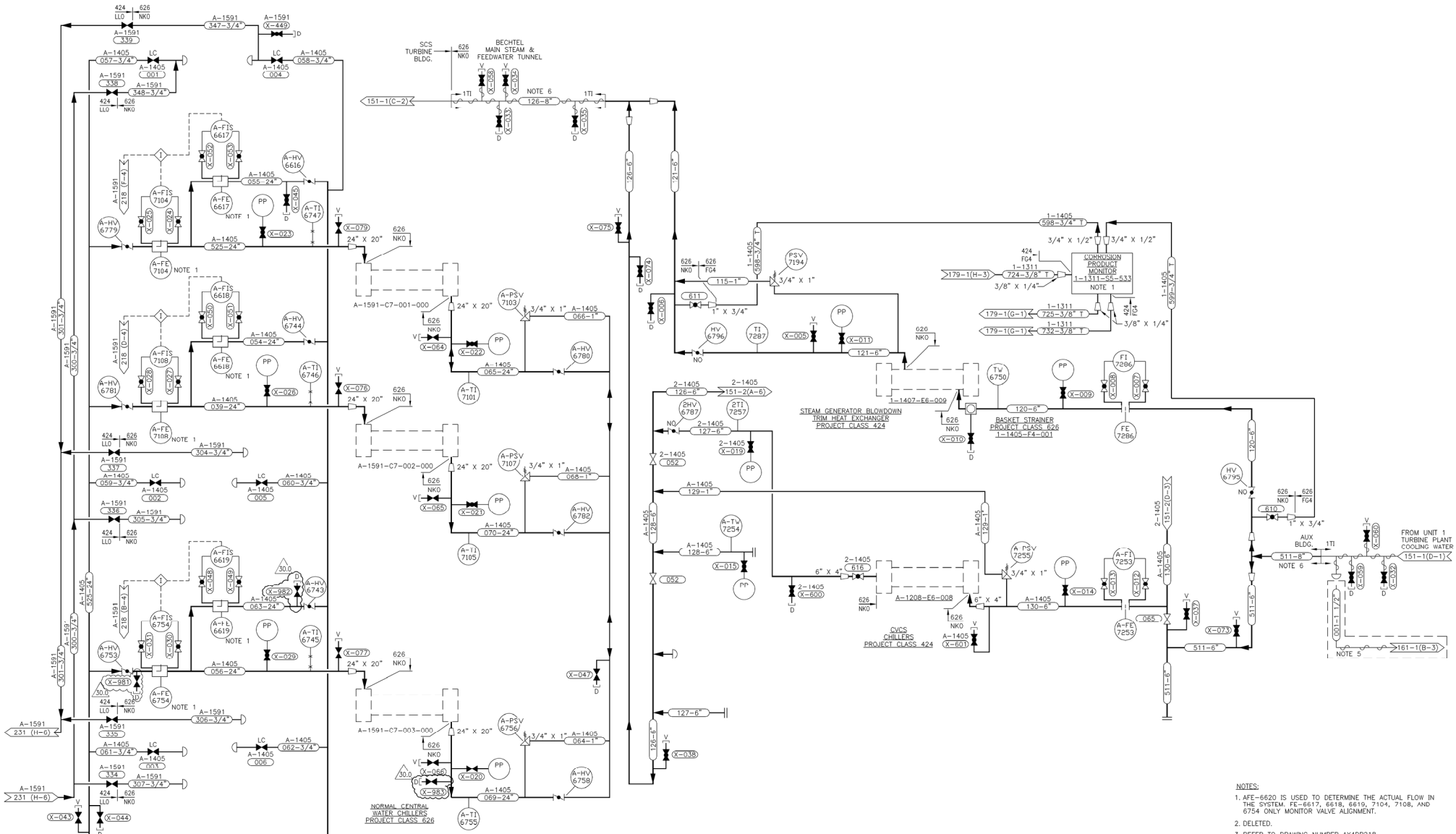
NO.	VERSIONS	DATE	DR	CHK	APPV
32.0	REVISED PER ABN-V01546, VER.1.0	3-25-09	BGT	LRP	PML

SCALE: NONE
JOB NO. 10604
DRAWING NO. 1X4DB151-1
VER. 32.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



OPERATING MODE	CHILLER NO.		VALVE NO. NOTE: ALL VALVES SHALL BE CLOSED EXCEPT AS NOTED								
	POSITION NO. 1	POSITION NO. 2	6779	6781	6753	6780	6782	6758	6616	6744	6743
A	001	002	OPEN			OPEN	OPEN			OPEN	
B	001	003	OPEN			OPEN		OPEN			OPEN
C	002	003		OPEN			OPEN	OPEN			OPEN
D	NS	NS	CHILLER #1 MODE A OR B			CHILLER #2 MODE C OR E			CHILLER #3 MODE F OR G		
E	002	001		OPEN			OPEN	OPEN		OPEN	
F	003	001			OPEN	OPEN		OPEN	OPEN		
G	003	002			OPEN		OPEN	OPEN		OPEN	
SEE P & ID AX4DB218 FOR REQUIRED CORRESPONDING VALVE POSITIONS FOR CHILLED WATER FLOW.											

NOTES:

1. AFE-6620 IS USED TO DETERMINE THE ACTUAL FLOW IN THE SYSTEM. FE-6617, 6618, 6619, 7104, 7108, AND 6754 ONLY MONITOR VALVE ALIGNMENT.
2. DELETED.
3. REFER TO DRAWING NUMBER AX4DB218.
4. FOR CORROSION PROJECT MONITOR SCHEMATIC/ TAGGING SEE DWG. AX5AA01-20004.
5. RETIRED IN PLACE.
6. THE HEAT TRACE ON THE FOLLOWING LINES IS RETIRED IN PLACE OR REMOVED:

   1-1405-126-8" & 1-1405-511-8"

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
30.0	REVISED PER ABN SNC493699M002, VER. 1.0	11/05/13	DCP	MWD	JMR	X

CAD 1X4DB151-2
ACAD2K DCP- 05

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT COOLING WATER SYSTEM
SYSTEM NO. 1405

SCALE: NONE

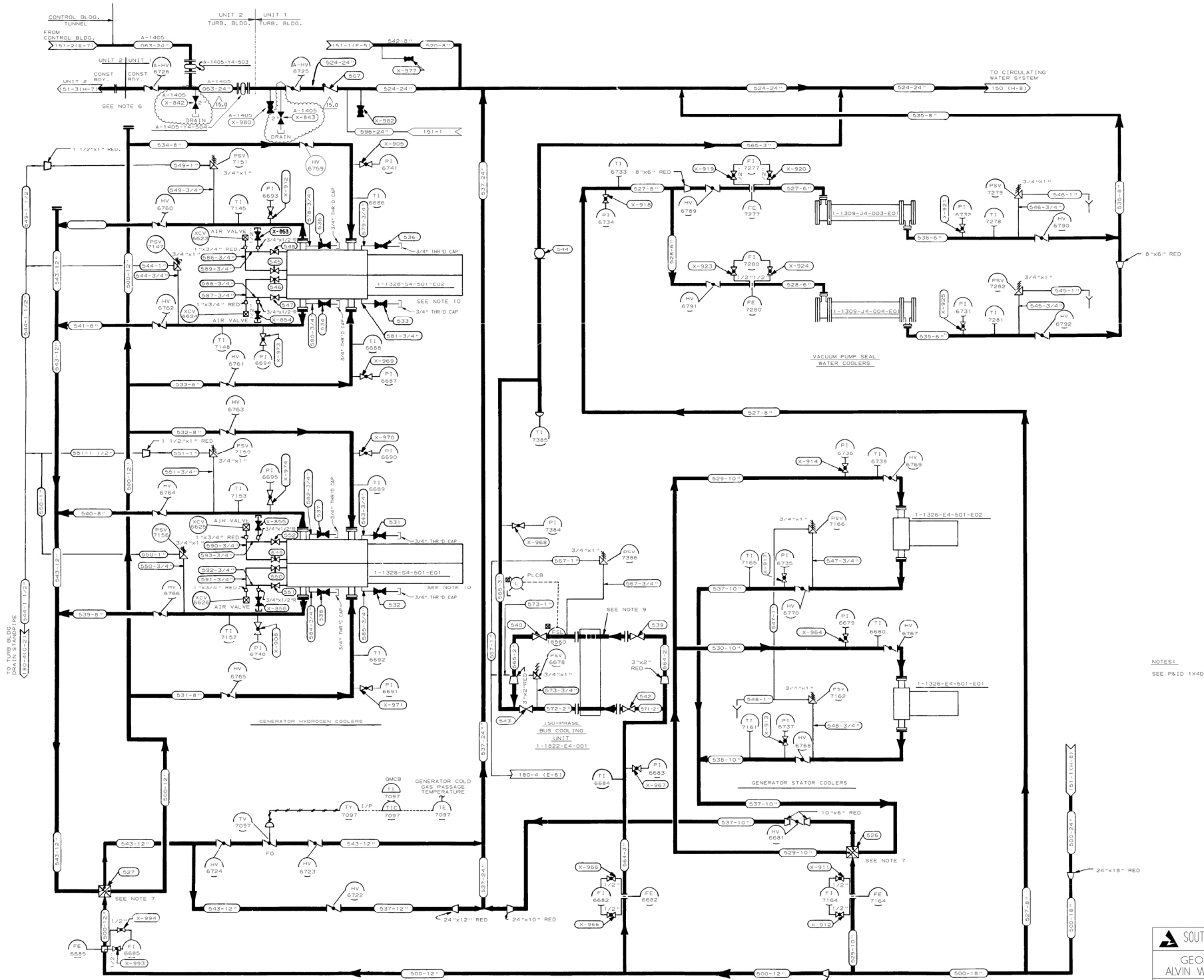
DRAWING NO. 1X4DB151-2

VER. 30.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL


CONTROL BLDG. TUNNEL
FROM CONTROL BLDG. 151-2(E-7)
UNIT 2 TURB. BLDG.
UNIT 1 TURB. BLDG.
A-1405 063-24"
A-1405-Y4-503
UNIT 2 UNIT 1 CONST BOV. CONST BOV.
UNIT 2 151-3(H-7)
SEE NOTE 6
A-1405 X-842
DRAIN
A-1405-Y4-504
A-1405 X-980
A-1405 X-843
DRAIN
A-HV 6726
A-HV 6725
524-24"
507
X-977
X-982
696-24"
151-1
542-8"
520-A"
151-1(F-6)
524-24"
524-24"
TO CIRCULATING WATER SYSTEM
150 (H-8)
535-8"
565-3"
534-8"
1 1/2"x1" RED.
549-1"
PSV 7151
3/4"x1"
X-905
PI 6741
HV 6759
549-3/4"
HV 6760
PI 6693
X-892
TI 7145
TI 6686
548-3/4"
3/4" THR'D CAP
679-3/4"
536
3/4" THR'D CAP
AIR VALVE
X-853
1"x3/4" RED
548
586-3/4"
589-3/4"
588-3/4"
587-3/4"
549
545
1-1328-S4-501-E02
SEE NOTE 10
3/4" THR'D CAP
PSV 7152
3/4"x1"
544-1"
544-3/4"
HV 6762
XCV 6622
AIR VALVE
X-856
1"x3/4" RED
547
3/4"x1/2"RED
541-8"
TI 7148
HV 6761
PI 6694
X-893
680-3/4"
534
3/4" THR'D CAP
533
681-3/4"
TI 6688
X-969
PI 6687
533-8"
HV 6763
532-8"
1 1/2"x1" RED
X-970
551-1 1/2
551-1"
PSV 7155
3/4"x1"
PI 6690
551-3/4"
PI 6695
X-894
TI 7153
TI 6689
682-3/4"
3/4" THR'D CAP
683-3/4"
HV 6764
640-8"
AIR VALVE
X-855
3/4"x1/2"RED
1"x3/4" RED
552
590-3/4"
549
550-1"
PSV 7156
3/4"x1"
550-3/4"
592-3/4"
591-3/4"
550
551
1-1328-S4-501-E01
SEE NOTE 10
3/4" THR'D CAP
531
632
HV 6765
XCV 6620
AIR VALVE
X-856
3/4"x1/2"RED
1"x3/4" RED
TI 7157
PI 6740
X-958
684-3/4"
534
3/4" THR'D CAP
685-3/4"
TI 6692
HV 6765
PI 6691
X-971
539-8"
GENERATOR HYDROGEN COOLERS
TO TURB BLDG DRAIN HEADER
180-4(G-2)
644-1 1/2"
644-1 1/2"
649-1 1/2"
543-12"
500-12"
543-12"
550-1"
543-12"
640-8"
550-1"
543-12"
539-8"
TI 7385
527-8"
565-3"
PLCB
673-1"
SEE NOTE 9
540
539
PSV 6678
3/4"x1"
673-3/4"
672-2"
542
671-2"
543
3"x2" RED
3"x2" RED
3"x2" RED
684-3
685-3
685-2
ISO-PHASE BUS COOLING UNIT 1-1822-E4-001
180-4 (E-6)
TI 6684
PI 6683
X-967
X-968
PI 7384
3/4"x1"
PSV 7386
567-1"
567-3/4"
637-24"
637-24"
637-24"
637-24"
500-12"
500-12"
500-12"
500-18"
500-18"
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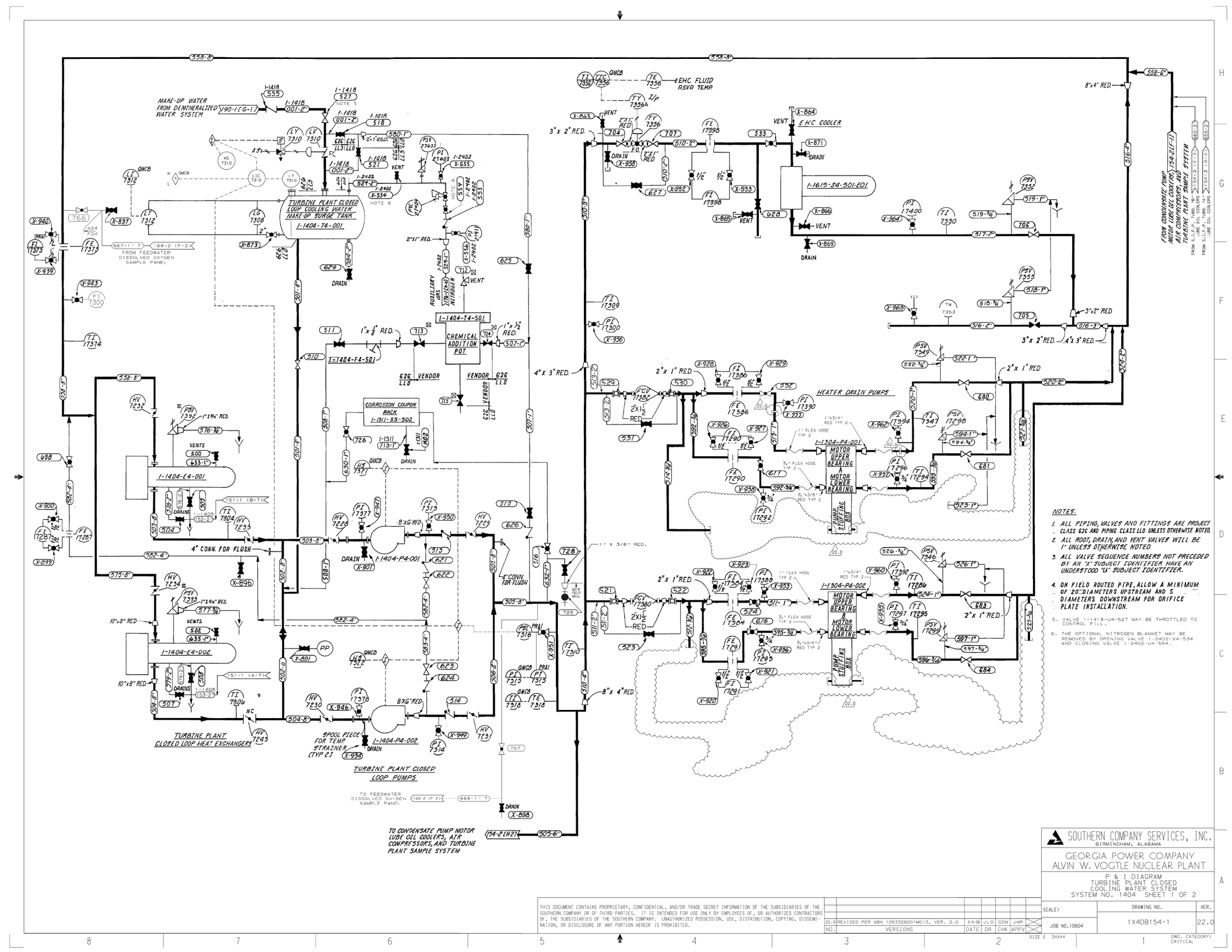
500-18



NOTES:

1. ALL PIPING, VALVES AND FITTINGS ARE PROJECT CLASS 62G AND PIPING CLASS LLD UNLESS OTHERWISE NOTED.
2. ALL ROOT, DRAIN, AND VENT VALVES WILL BE 1" UNLESS OTHERWISE NOTED
3. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" SUBJECT IDENTIFIER HAVE AN UNDERSTOOD "U" SUBJECT IDENTIFIER.
4. ON FIELD ROUTED PIPE, ALLOW A MINIMUM OF 20 DIAMETERS UPSTREAM AND 5 DIAMETERS DOWNSTREAM FOR ORIFICE PLATE INSTALLATION.
5. VALVE 1-1418-U4-527 MAY BE THROTTLED TO CONTROL FILL.
6. THE OPTIONAL NITROGEN BLANKET MAY BE REMOVED BY OPENING VALVE 1-2402-X4-534 AND CLOSING VALVE 1-2402-U4-554.

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NO.	VERSIONS	DATE	DR	CHK	APPV
22.0	REVISED PER ABN 1053328201M013, VER. 2.0	5-6-08	JLO	GDM	JMR

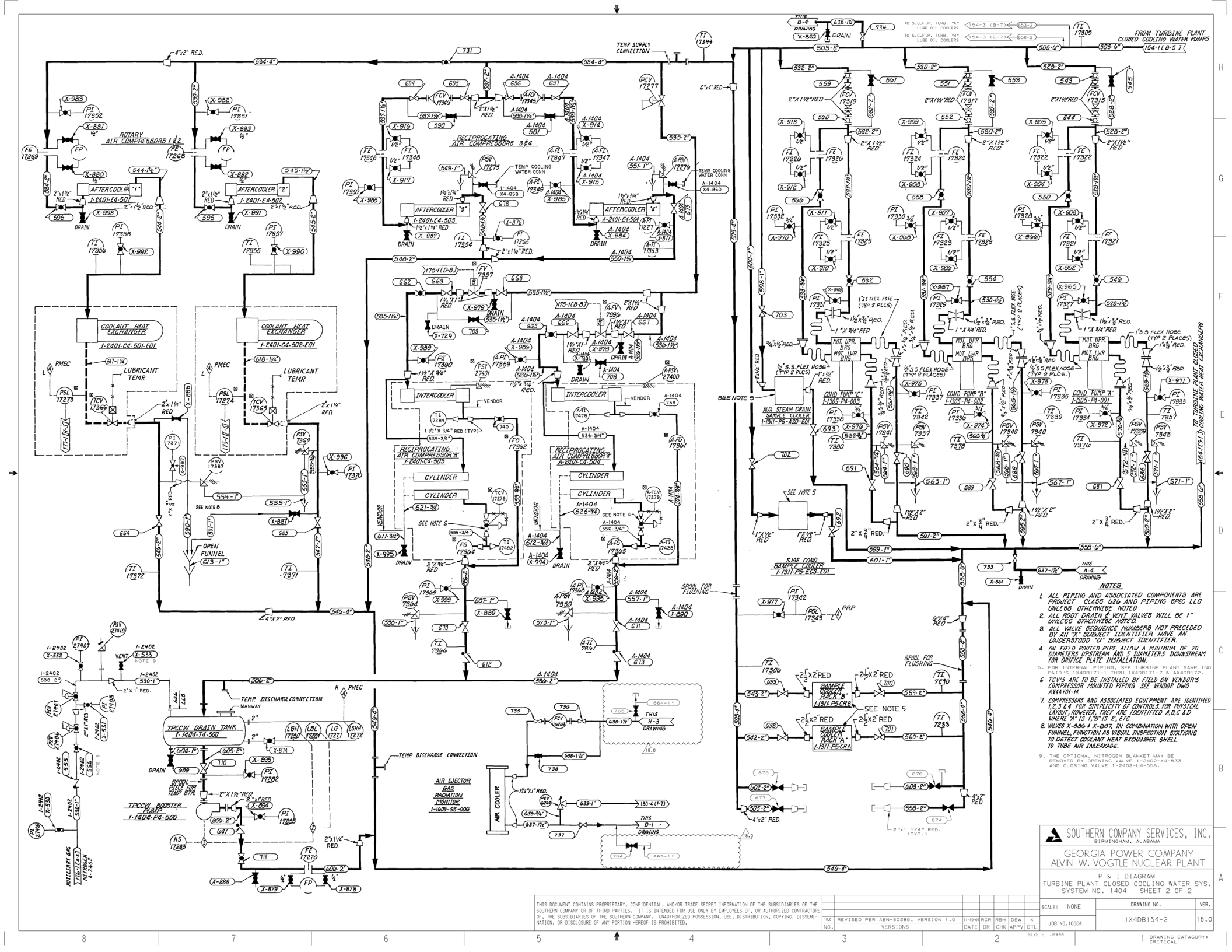
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT CLOSED COOLING WATER SYSTEM
SYSTEM NO. 1404 SHEET 1 OF 2

SCALE:	DRAWING NO.	VER.
JOB NO. 10604	1X4DB154-1	22.0

SIZE E 34x44 | DWG. CATEGORY: CRITICAL



NOTES

1. ALL PIPING AND ASSOCIATED COMPONENTS ARE PROJECT CLASS (2)G AND PIPING SPEC LLD UNLESS OTHERWISE NOTED
2. ALL ROOT DRAIN & VENT VALVES WILL BE 1" UNLESS OTHERWISE NOTED
3. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" SUBJECT IDENTIFIER HAVE AN UNDERSTOOD "U" SUBJECT IDENTIFIER.
4. ON FIELD ROUTED PIPE, ALLOW A MINIMUM OF 20 DIAMETERS UPSTREAM AND 5 DIAMETERS DOWNSTREAM FOR ORIFICE PLATE INSTALLATION.
5. FOR INTERNAL PIPING, SEE TURBINE PLANT SAMPLING P&ID'S 1X4DB171-1 THRU 1X4DB171-7 & AX4DB172.
6. TCV'S ARE TO BE INSTALLED BY FIELD ON VENDOR'S COMPRESSOR MOUNTED PIPING SEE VENDOR DWG A1X4Y01-14.
7. COMPRESSORS AND ASSOCIATED EQUIPMENT ARE IDENTIFIED 1,2,3 & 4 FOR SIMPLICITY OF CONTROLS FOR PHYSICAL LAYOUT, HOWEVER, THEY ARE IDENTIFIED A,B,C & D WHERE "A" IS 1, "B" IS 2, ETC.
8. VALVES X-885 & X-887, IN COMBINATION WITH OPEN FUNNEL, FUNCTION AS VISUAL INSPECTION STATIONS TO DETECT COOLANT HEAT EXCHANGER SHELL TO TUBE AIR INLEAKAGE.
9. THE OPTIONAL NITROGEN BLANKET MAY BE REMOVED BY OPENING VALVE 1-2402-X4-533 AND CLOSING VALVE 1-2402-U4-556.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
18.0	REVISED PER ABN-80385, VERSION 1.0	11-10-04	RCR	RBH	DEW	X

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

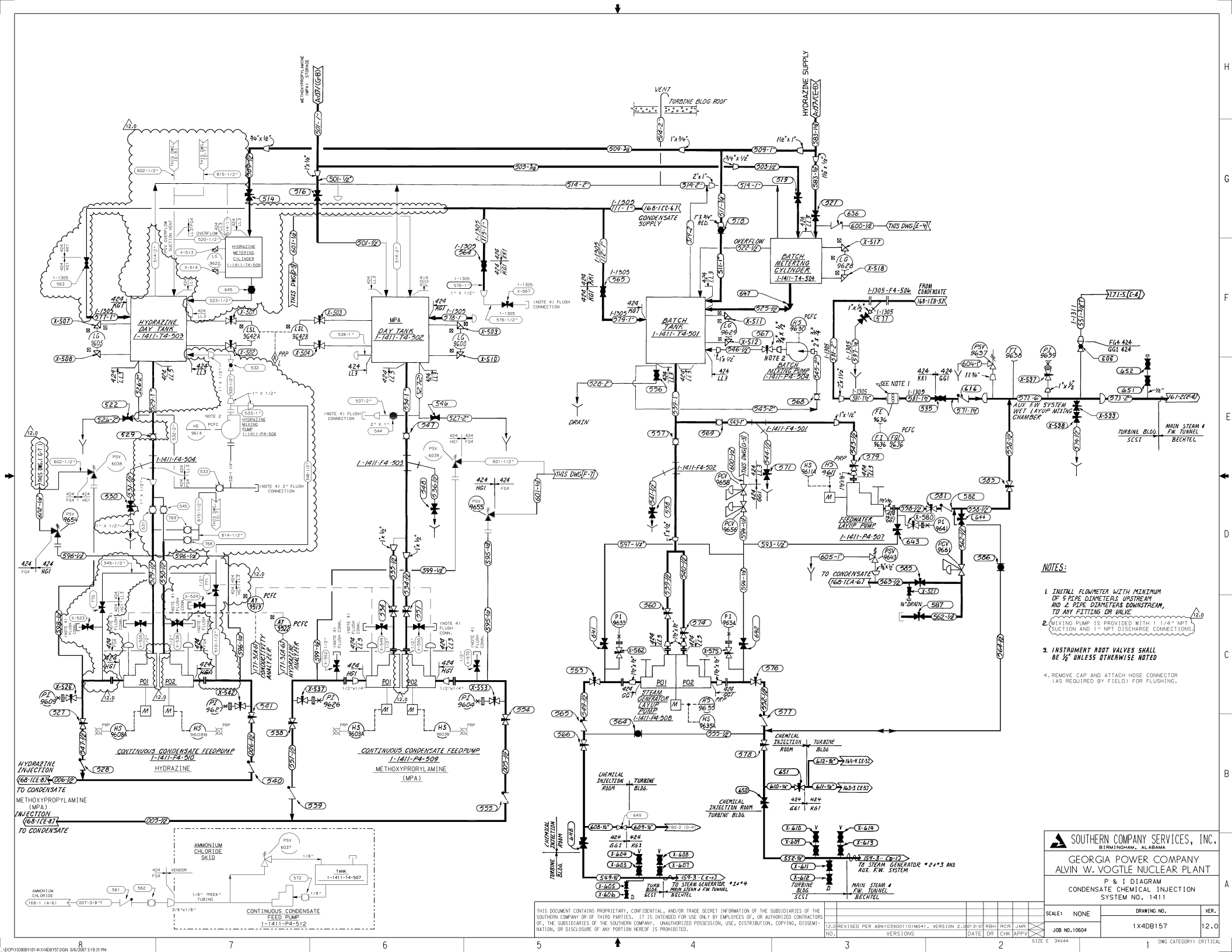
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT CLOSED COOLING WATER SYS.
SYSTEM NO. 1404 SHEET 2 OF 2

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB154-2	18.0

SIZE E 34X44

1 DRAWING CATAGORY: CRITICAL



NOTES:

1. INSTALL FLOWMETER WITH MINIMUM OF 5 PIPE DIAMETERS UPSTREAM AND 2 PIPE DIAMETERS DOWNSTREAM, TO ANY FITTING OR VALVE.
2. MIXING PUMP IS PROVIDED WITH 1 1/4" NPT SUCTION AND 1" NPT DISCHARGE CONNECTIONS.
3. INSTRUMENT ROOT VALVES SHALL BE ½" UNLESS OTHERWISE NOTED.
4. REMOVE CAP AND ATTACH HOSE CONNECTOR (AS REQUIRED BY FIELD) FOR FLUSHING.

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NO.	VERSIONS	DATE	DR	CHK	APPV
12.0	REVISED PER ABN1039001101M041, VERSION 2.	07-31-07	RBH	RCR	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONDENSATE CHEMICAL INJECTION
SYSTEM NO. 1411

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB157
VER. 12.0
SIZE E 34X44
DWG CATEGORY: CRITICAL

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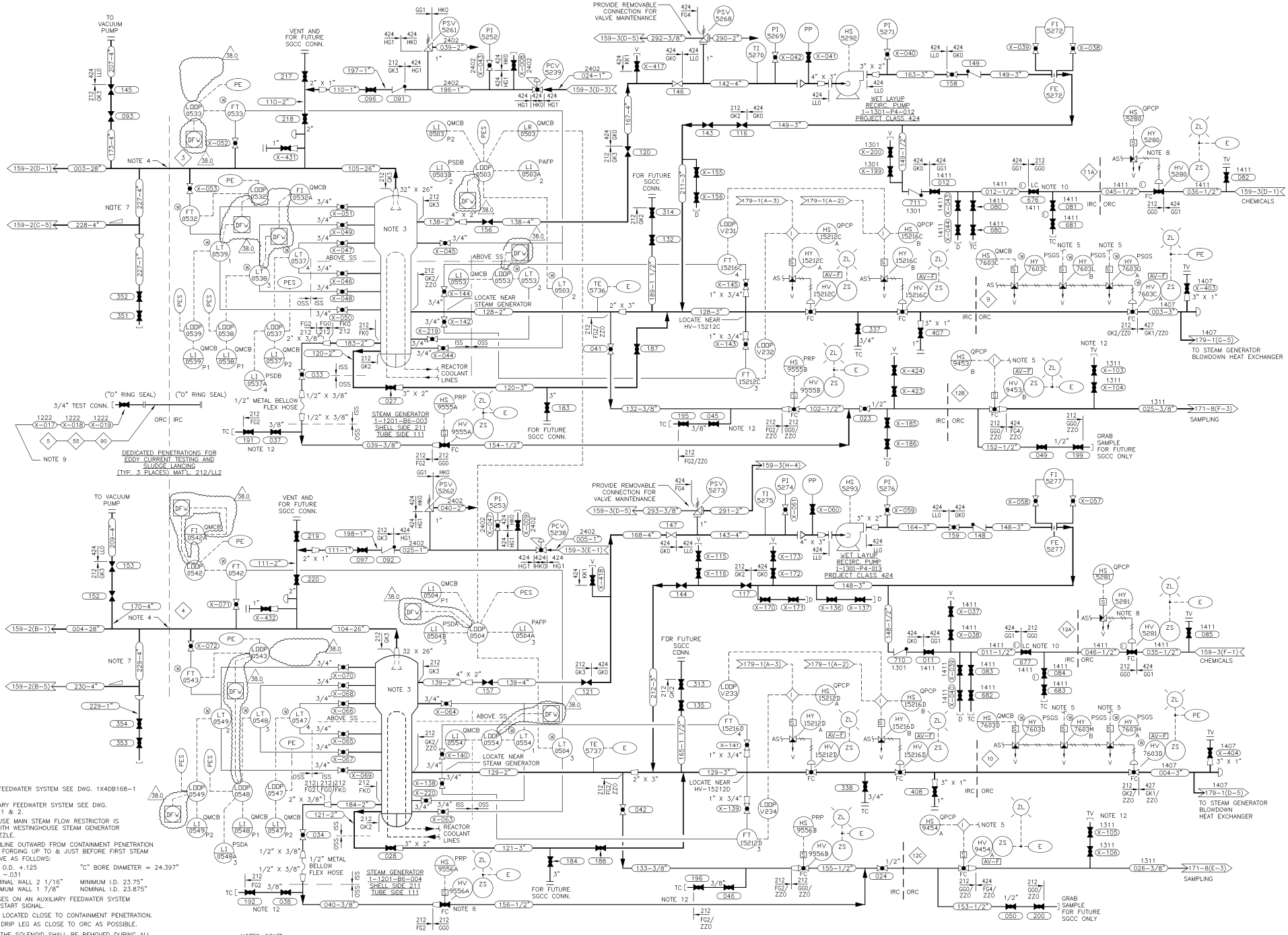

NOTES:

1. FOR MAIN FEEDWATER SYSTEM SEE DWG. 1X4DB168-1 THRU 3.
2. FOR AUXILIARY FEEDWATER SYSTEM SEE DWG. 1X4DB161-1 & 2.
3. WESTINGHOUSE MAIN STEAM FLOW RESTRICTOR IS INTEGRAL WITH WESTINGHOUSE STEAM GENERATOR OUTLET NOZZLE.
4. MAIN STEAMLINE OUTWARD FROM CONTAINMENT PENETRATION TO BE 28" FORGING UP TO & JUST BEFORE FIRST STEAM SAFETY VALVE AS FOLLOWS:
   28" O.D. +.125 −.031 "C" BORE DIAMETER = 24.397"
   NOMINAL WALL 2 1/16" MINIMUM I.D. 23.75"
   MINIMUM WALL 1 7/8" NOMINAL I.D. 23.875"
5. VALVE CLOSES ON AN AUXILIARY FEEDWATER SYSTEM AUTOMATIC START SIGNAL.
6. THIS VALVE LOCATED CLOSE TO CONTAINMENT PENETRATION.
7. LOCATE 4" DRIP LEG AS CLOSE TO ORC AS POSSIBLE.
8. POWER TO THE SOLENOID SHALL BE REMOVED DURING ALL MODES OF OPERATION, EXCEPT DURING WET LAYUP OPERATION AND PERIODIC TESTING OF VALVES, BY MOVING THE SLIDE LINKS OF THE STATES TERMINAL BLOCKS, LOCATED IN THE TERMINAL BOX, TO THE OPEN POSITION.
9. DURING MODE 5 (COLD SHUTDOWN) OR MODE 6 (REFUELING), "OUTAGE FIXTURES" MAY BE INSTALLED ON THESE PENETRATIONS TO FACILITATE SLUDGE LANCING AND EDDY CURRENT TESTING INSIDE CONTAINMENT. SEE DRAWING 1X2D01J019 FOR DETAILS OF "OUTAGE FIXTURES".

NOTES-CON'T:

10. THIS VALVE SHALL REMAIN LOCKED CLOSED DURING ALL MODES OF OPERATION, EXCEPT DURING WET LAYUP.
11. DELETED
12. THESE ARE KEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
MAIN STEAM SYSTEM
SYSTEM NO. 1301

CAD 1X4DB159-1
ACAD2007 JLO – 03

SCALE: NONE
DRAWING NO. 1X4DB159-1
VER. 38.0
JOB NO.10604

NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
38.0	REVISED PER ABN 1039002501J174, VER. 1.0	10/10/12	JLO	ELC	JMR	X

SIZE E 34X44

DRAWING CATEGORY: CRITICAL

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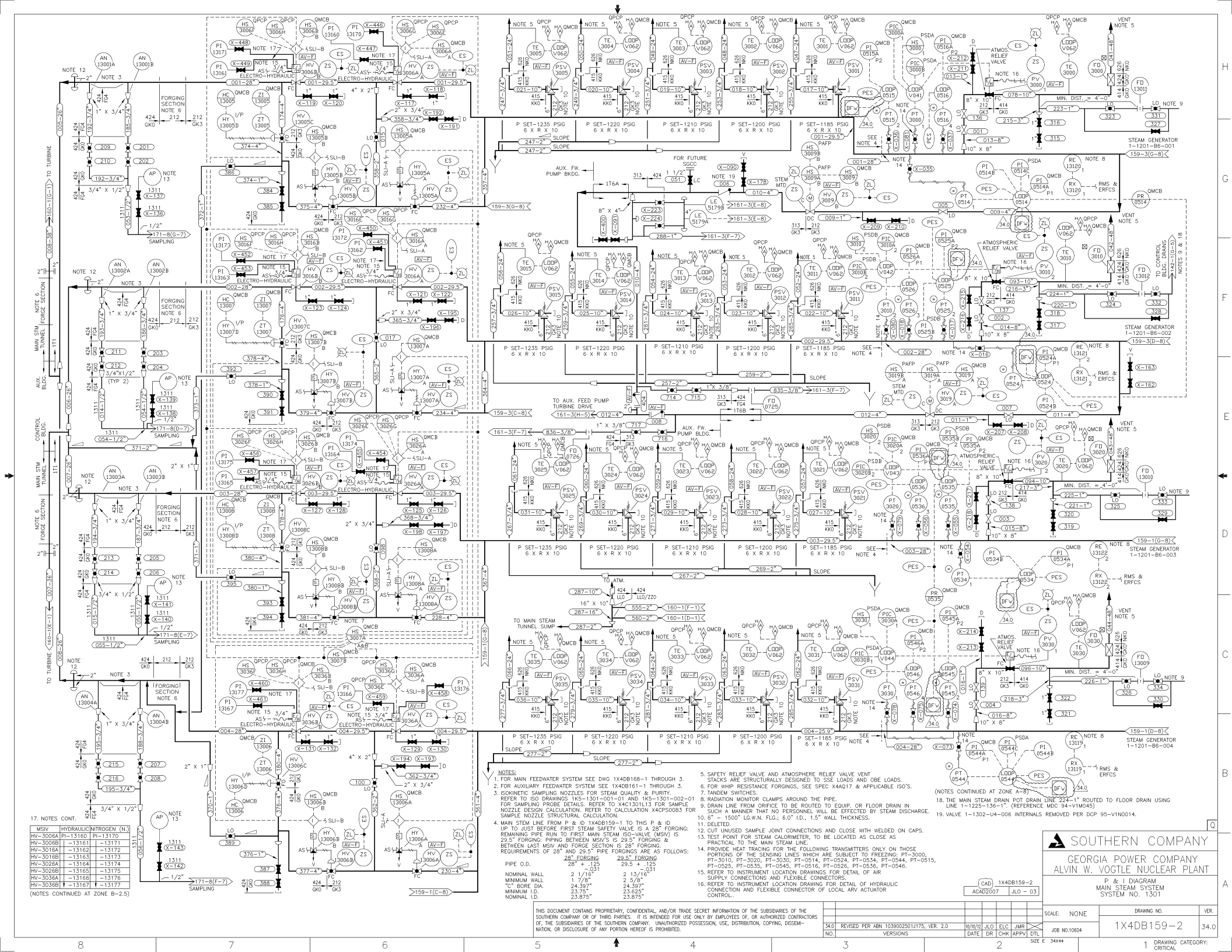

17. NOTES CONT.

MSIV	HYDRAULIC	NITROGEN (N.)
HV-3006A	PI-13160	PI-13170
HV-3006B	-13161	-13171
HV-3016A	-13162	-13172
HV-3016B	-13163	-13173
HV-3026A	-13164	-13174
HV-3026B	-13165	-13175
HV-3036A	-13166	-13176
HV-3036B	-13167	-13177

NOTES:

1. FOR MAIN FEEDWATER SYSTEM SEE DWG 1X4DB168-1 THROUGH 3.
2. FOR AUXILIARY FEEDWATER SYSTEM SEE 1X4DB161-1 THROUGH 3.
3. ISOKINETIC SAMPLING NOZZLES FOR STEAM QUALITY & PURITY. REFER TO ISO DRAWINGS 1KS-1301-001-01 AND 1KS-1301-002-01 FOR SAMPLING PROBE DETAILS. REFER TO X4C1301L13 FOR SAMPLE NOZZLE DESIGN CALCULATION. REFER TO CALCULATION X4CPS0083 FOR SAMPLE NOZZLE STRUCTURAL CALCULATION.
4. MAIN STEM LINE FROM P & ID 1X4DB159-1 TO THIS P & ID UP TO JUST BEFORE FIRST STEAM SAFETY VALVE IS A 28" FORGING: REMAINING PIPE RUN TO FIRST MAIN STEAM ISO-VALVE (MSIV) IS 29.5" FORGING; PIPING BETWEEN MSIV'S IS 29.5" FORGING & BETWEEN LAST MSIV AND FORGE SECTION IS 28" FORGING. REQUIREMENTS OF 28" AND 29.5" PIPE FORGINGS ARE AS FOLLOWS:

	28" FORGING	29.5" FORGING
PIPE O.D.	28" +.125 −.031	29.5" +.125 −.031
NOMINAL WALL	2 1/16"	2 13/16"
MINIMUM WALL	1 7/8"	2 5/8"
"C" BORE DIA.	24.397"	24.397"
MINIMUM I.D.	23.75"	23.625"
NOMINAL I.D.	23.875"	23.875"

5. SAFETY RELIEF VALVE AND ATMOSPHERE RELIEF VALVE VENT STACKS ARE STRUCTURALLY DESIGNED TO SSE LOADS AND OBE LOADS.
6. FOR WHIP RESISTANCE FORGINGS, SEE SPEC X4AQ17 & APPLICABLE ISO'S.
7. TANDEM SWITCHES.
8. RADIATION MONITOR CLAMPS AROUND THE PIPE.
9. DRAIN LINE FROM ORIFICE TO BE ROUTED TO EQUIP. OR FLOOR DRAIN IN SUCH A MANNER THAT NO PERSONNEL WILL BE EFFECTED BY STEAM DISCHARGE.
10. 6" – 1500 LG.W.N. FLG.; 6.0" I.D., 1.5" WALL THICKNESS.
11. DELETED.
12. CUT UNUSED SAMPLE JOINT CONNECTIONS AND CLOSE WITH WELDED ON CAPS.
13. TEST POINT FOR STEAM CALORIMETER, TO BE LOCATED AS CLOSE AS PRACTICAL TO THE MAIN STEAM LINE.
14. PROVIDE HEAT TRACING FOR THE FOLLOWING TRANSMITTERS ONLY ON THOSE PORTIONS OF THE SENSING LINES WHICH ARE SUBJECT TO FREEZING: PT-3000, PT-3010, PT-3020, PT-3030, PT-0514, PT-0524, PT-0534, PT-0544, PT-0515, PT-0525, PT-0535, PT-0545, PT-0516, PT-0526, PT-0536, PT-0546.
15. REFER TO INSTRUMENT LOCATION DRAWINGS FOR DETAIL OF AIR SUPPLY CONNECTIONS AND FLEXIBLE CONNECTORS.
16. REFER TO INSTRUMENT LOCATION DRAWING FOR DETAIL OF HYDRAULIC CONNECTION AND FLEXIBLE CONNECTOR OF LOCAL ARV ACTUATOR CONTROL..
18. THE MAIN STEAM DRAIN POT DRAIN LINE 224–1" ROUTED TO FLOOR DRAIN USING LINE 1–1225–136–1". (REFERENCE MDC 94–V1M045)
19. VALVE 1–1302–U4–006 INTERNALS REMOVED PER DCP 95–V1N0014.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
34.0	REVISED PER ABN 10.39002501J175, VER. 2.0	10/10/12	JLO	ELC	JMR	

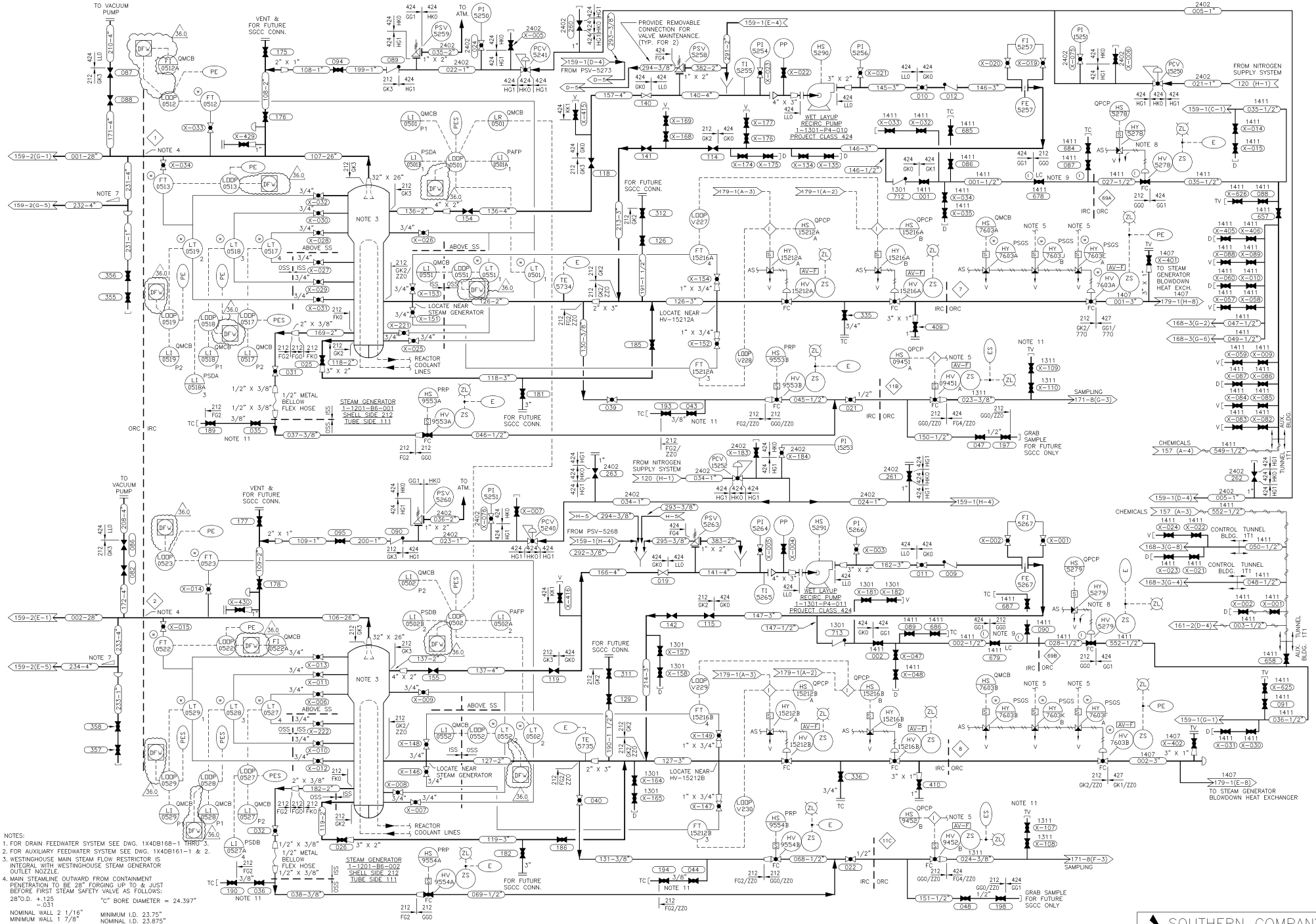
CAD 1X4DB159-2
ACAD2007 JLO - 03

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
MAIN STEAM SYSTEM
SYSTEM NO. 1301

SCALE: NONE
DRAWING NO. 1X4DB159-2
VER. 34.0
JOB NO. J0604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR DRAIN FEEDWATER SYSTEM SEE DWG. 1X4DB168-1 THRU 3.
2. FOR AUXILIARY FEEDWATER SYSTEM SEE DWG. 1X4DB161-1 & 2.
3. WESTINGHOUSE MAIN STEAM FLOW RESTRICTOR IS INTEGRAL WITH WESTINGHOUSE STEAM GENERATOR OUTLET NOZZLE.
4. MAIN STEAMLINE OUTWARD FROM CONTAINMENT PENETRATION TO BE 28" FORGING UP TO & JUST BEFORE FIRST STEAM SAFETY VALVE AS FOLLOWS:
   28"O.D. +.125 −.031 "C" BORE DIAMETER = 24.397"
   NOMINAL WALL 2 1/16" MINIMUM I.D. 23.75"
   MINIMUM WALL 1 7/8" NOMINAL I.D. 23.875"
5. VALVE CLOSES ON AN AUXILIARY FEEDWATER SYSTEM AUTOMATIC START SIGNAL.
6. THIS VALVE LOCATED CLOSE TO CONTAINMENT PENETRATION.
7. LOCATE 4" DRIP LEG AS CLOSE TO ORC AS POSSIBLE.
8. POWER TO THE SOLENOID SHALL BE REMOVED DURING ALL MODES OF OPERATION, EXCEPT DURING WET LAYUP OPERATION AND PERIODIC TESTING OF THE VALVES, BY MOVING THE SLIDE LINKS OF THE STATES TERMINAL BLOCKS, LOCATED IN THE TERMINAL BOX, TO THE OPEN POSITION.
9. THIS VALVE SHALL REMAIN LOCKED CLOSED DURING ALL MODES OF OPERATION, EXCEPT DURING WET LAYUP.
10. DELETED.
11. THESE ARE KEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
36.0	REVISED PER ABN 10390025O1J176, VER. 2.0	10/10/12	JLO	ELC	JMR	X

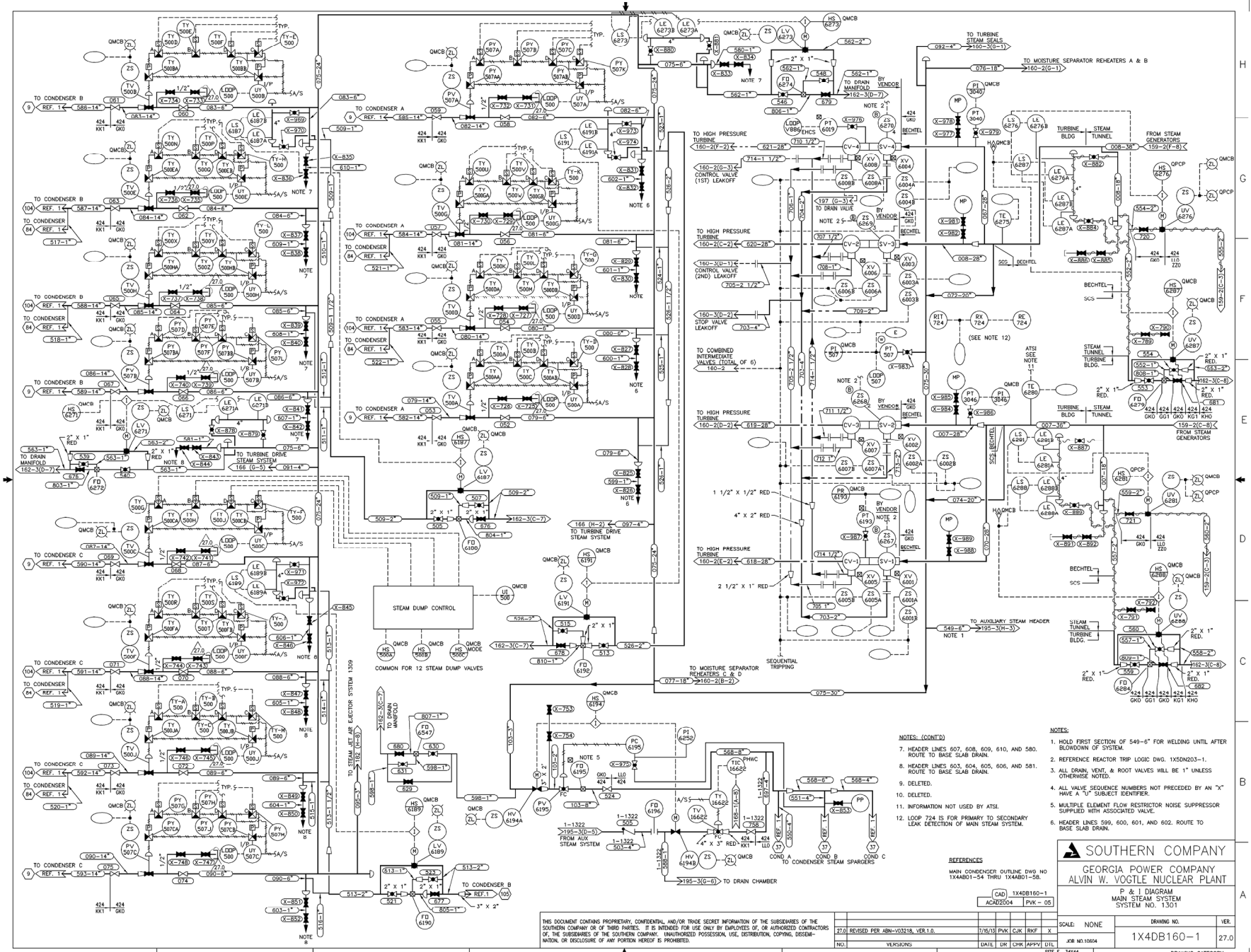
CAD 1X4DB159-3
ACAD2007 JLO – 05

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
MAIN STEAM SYSTEM
SYSTEM NO. 1301

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB159-3
VER. 36.0
SIZE E 34X44
DRAWING CATEGORY: 1 CRITICAL


STEAM DUMP CONTROL
COMMON FOR 12 STEAM DUMP VALVES
TO HIGH PRESSURE TURBINE 160-2(F-2)
TO HIGH PRESSURE TURBINE 160-2(D-2)
TO HIGH PRESSURE TURBINE 160-2(E-2)
TO MOISTURE SEPARATOR REHEATERS A & B
TO MOISTURE SEPARATOR REHEATERS C & D
TO AUXILIARY STEAM HEADER
TO TURBINE DRIVE STEAM SYSTEM
TO TURBINE STEAM SEALS
TO DRAIN MANIFOLD
TO STEAM JET AIR EJECTOR SYSTEM 1309
TO DRAIN CHAMBER
SEQUENTIAL TRIPPING
COND A TO CONDENSER STEAM SPARGERS
COND B
COND C
FROM STEAM GENERATORS
TURBINE BLDG
STEAM TUNNEL
TURBINE BLDG.
STEAM TUNNEL TURBINE BLDG.
BECHTEL
SCS
QMCB
QPCP
PHWC
LLO
ZZO
GKO GG1 GKO KG1 KHO
TO CONDENSER A
TO CONDENSER B
TO CONDENSER C
REF. 1
NOTE 1
NOTE 2
NOTE 5
NOTE 6
NOTE 7
NOTE 8
NOTE 11
NOTE 12
REFERENCES
MAIN CONDENSER OUTLINE DWG NO 1X4AB01-54 THRU 1X4AB01-58.
NOTES:
1. HOLD FIRST SECTION OF 549-6" FOR WELDING UNTIL AFTER BLOWDOWN OF SYSTEM.
2. REFERENCE REACTOR TRIP LOGIC DWG. 1X5DN203-1.
3. ALL DRAIN, VENT, & ROOT VALVES WILL BE 1" UNLESS OTHERWISE NOTED.
4. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" HAVE A "U" SUBJECT IDENTIFIER.
5. MULTIPLE ELEMENT FLOW RESTRICTOR NOISE SUPPRESSOR SUPPLIED WITH ASSOCIATED VALVE.
6. HEADER LINES 599, 600, 601, AND 602. ROUTE TO BASE SLAB DRAIN.
NOTES: (CONT'D)
7. HEADER LINES 607, 608, 609, 610, AND 580. ROUTE TO BASE SLAB DRAIN.
8. HEADER LINES 603, 604, 605, 606, AND 581. ROUTE TO BASE SLAB DRAIN.
9. DELETED.
10. DELETED.
11. INFORMATION NOT USED BY ATSI.
12. LOOP 724 IS FOR PRIMARY TO SECONDARY LEAK DETECTION OF MAIN STEAM SYSTEM.
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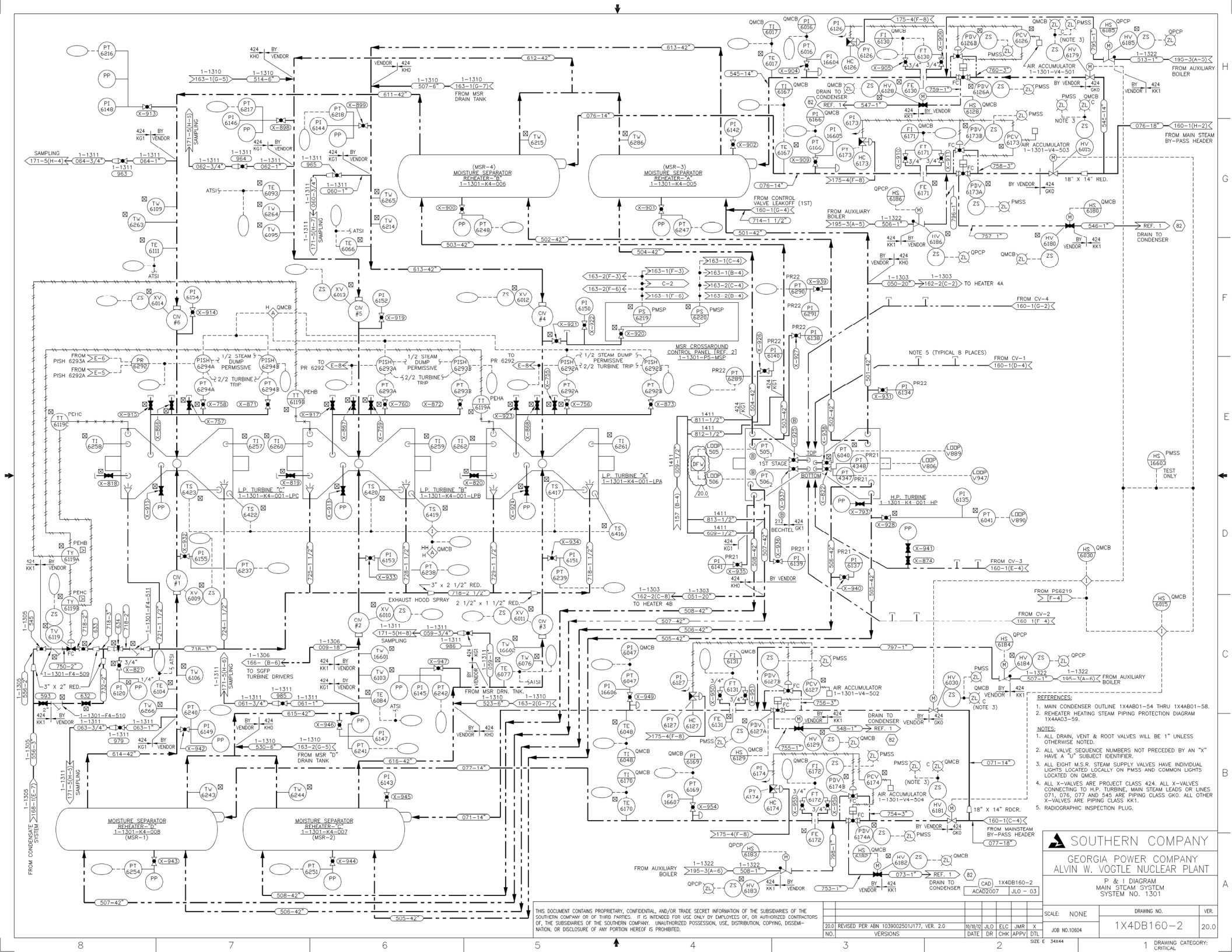
27.0 REVISED PER ABN-V03218, VER.1.0. 7/15/15 PVK CJK RKF X
NO. VERSIONS DATE DR CHK APPV DTL
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
MAIN STEAM SYSTEM
SYSTEM NO. 1301
CAD 1X4DB160-1
ACAD2004 PVK - 05
SCALE: NONE
DRAWING NO. 1X4DB160-1
VER. 27.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL


(MSR-4) MOISTURE SEPARATOR REHEATER "B" 1-1301-K4-006
(MSR-3) MOISTURE SEPARATOR REHEATER "A" 1-1301-K4-005
MOISTURE SEPARATOR REHEATER "D" 1-1301-K4-008 (MSR-1)
MOISTURE SEPARATOR REHEATER "C" 1-1301-K4-007 (MSR-2)
L.P. TURBINE "A" 1-1301-K4-001-LPA
L.P. TURBINE "B" 1-1301-K4-001-LPB
L.P. TURBINE "C" 1-1301-K4-001-LPC
H.P. TURBINE 1-1301-K4-001-HP
MSR CROSSAROUND CONTROL PANEL (REF. 2) 1-1301-P5-MSP
AIR ACCUMULATOR 1-1301-V4-501
AIR ACCUMULATOR 1-1301-V4-502
AIR ACCUMULATOR 1-1301-V4-503
AIR ACCUMULATOR 1-1301-V4-504
FROM MAIN STEAM BY-PASS HEADER
FROM MAINSTEAM BY-PASS HEADER
FROM AUXILIARY BOILER
FROM CV-1
FROM CV-2
FROM CV-3
FROM CV-4
TO HEATER 4A
TO HEATER 4B
DRAIN TO CONDENSER
FROM MSR DRAIN TANK
TO SGFP TURBINE DRIVERS
FROM MSR DRN. TNK.
FROM CONDENSATE SYSTEM
EXHAUST HOOD SPRAY
TEST ONLY
REFERENCES:
1. MAIN CONDENSER OUTLINE 1X4AB01-54 THRU 1X4AB01-58.
2. REHEATER HEATING STEAM PIPING PROTECTION DIAGRAM 1X4AA03-59.
NOTES:
1. ALL DRAIN, VENT & ROOT VALVES WILL BE 1" UNLESS OTHERWISE NOTED.
2. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" HAVE A "U" SUBJECT IDENTIFIER.
3. ALL EIGHT M.S.R. STEAM SUPPLY VALVES HAVE INDIVIDUAL LIGHTS LOCATED LOCALLY ON PMSS AND COMMON LIGHTS LOCATED ON QMCB.
4. ALL X-VALVES ARE PROJECT CLASS 424. ALL X-VALVES CONNECTING TO H.P. TURBINE, MAIN STEAM LEADS OR LINES 071, 076, 077 AND 545 ARE PIPING CLASS GK0. ALL OTHER X-VALVES ARE PIPING CLASS KK1.
5. RADIOGRAPHIC INSPECTION PLUG.
SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
MAIN STEAM SYSTEM
SYSTEM NO. 1301
SCALE: NONE
DRAWING NO. 1X4DB160-2
VER. 20.0
JOB NO.10604
CAD 1X4DB160-2
ACAD2007 JLO - 03
20.0 REVISED PER ABN 1039002501J177, VER. 2.0 10/10/12 JLO ELC JMR X
NO. VERSIONS DATE DR CHK APPV DTL
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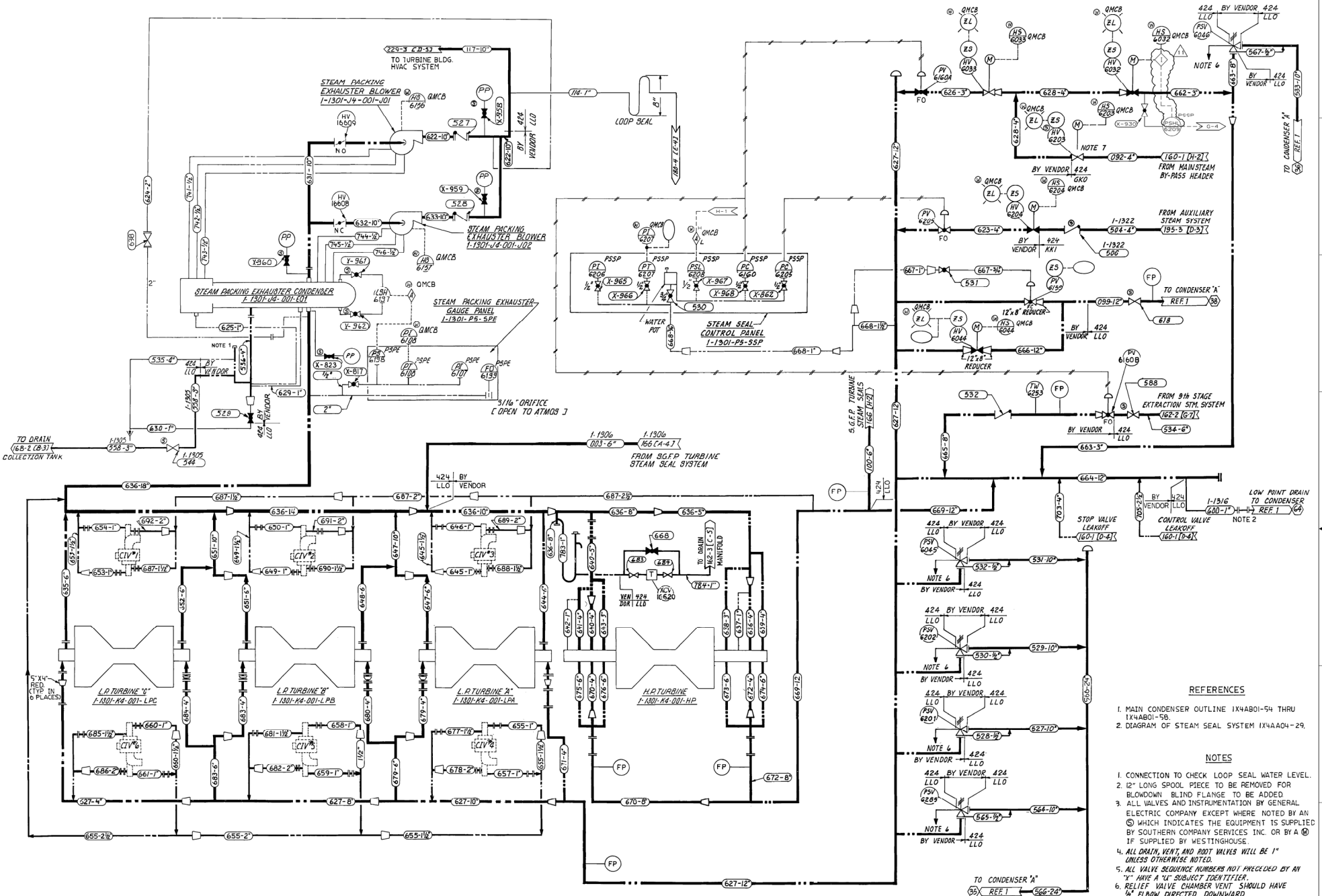
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



REFERENCES

1. MAIN CONDENSER OUTLINE 1X4AB01-54 THRU 1X4AB01-58.
2. DIAGRAM OF STEAM SEAL SYSTEM 1X4AA04-29.

NOTES

1. CONNECTION TO CHECK LOOP SEAL WATER LEVEL.
2. 12" LONG SPOOL PIECE TO BE REMOVED FOR BLOWDOWN. BLIND FLANGE TO BE ADDED.
3. ALL VALVES AND INSTRUMENTATION BY GENERAL ELECTRIC COMPANY EXCEPT WHERE NOTED BY AN Ⓢ WHICH INDICATES THE EQUIPMENT IS SUPPLIED BY SOUTHERN COMPANY SERVICES INC. OR BY A Ⓜ IF SUPPLIED BY WESTINGHOUSE.
4. ALL DRAIN, VENT, AND ROOT VALVES WILL BE 1" UNLESS OTHERWISE NOTED.
5. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" HAVE A "U" SUBJECT IDENTIFIER.
6. RELIEF VALVE CHAMBER VENT SHOULD HAVE ½" ELBOW DIRECTED DOWNWARD.
7. HOLD VALVE HV-6203 FOR WELDING UNTIL AFTER BLOWDOWN OF STEAM SEAL PIPING.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORPORATED PER DCP 98-V1N0046	10-5-00	JGM	RBH	MWD	
2	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

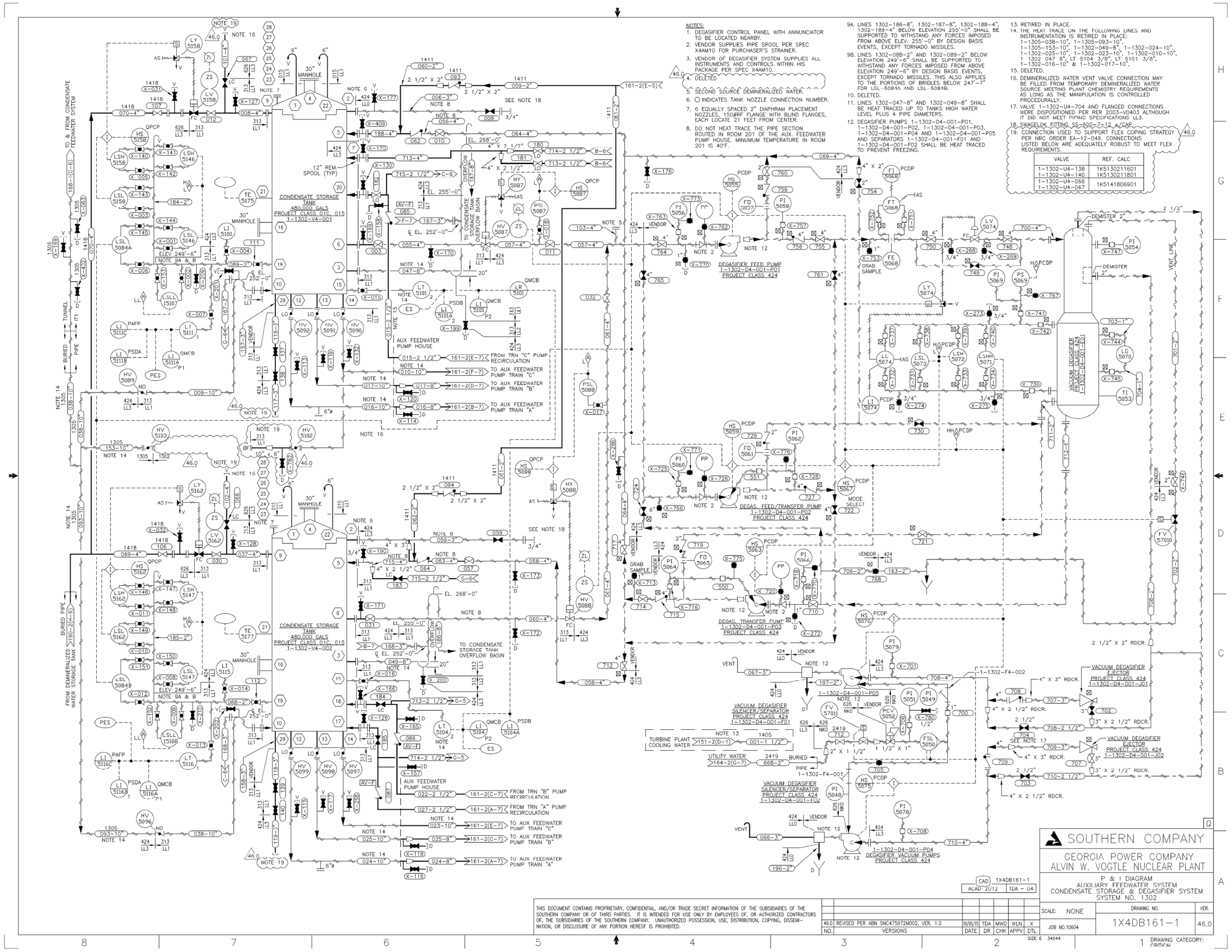
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
MAIN STEAM SYSTEM
TURBINE STEAM SEAL SYSTEM NO. 1301 SHEET 5 OF 5

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DB160-3	11

SIZE E 34X44



46.0	REVISED PER ABN SNC475972M002, VER. 1.0	10/15/15	TDA	MWD	WLN	X	
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL	

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CAD 1X4DB161-1
ACAD 2012 TDA - U4

SOUTHERN COMPANY

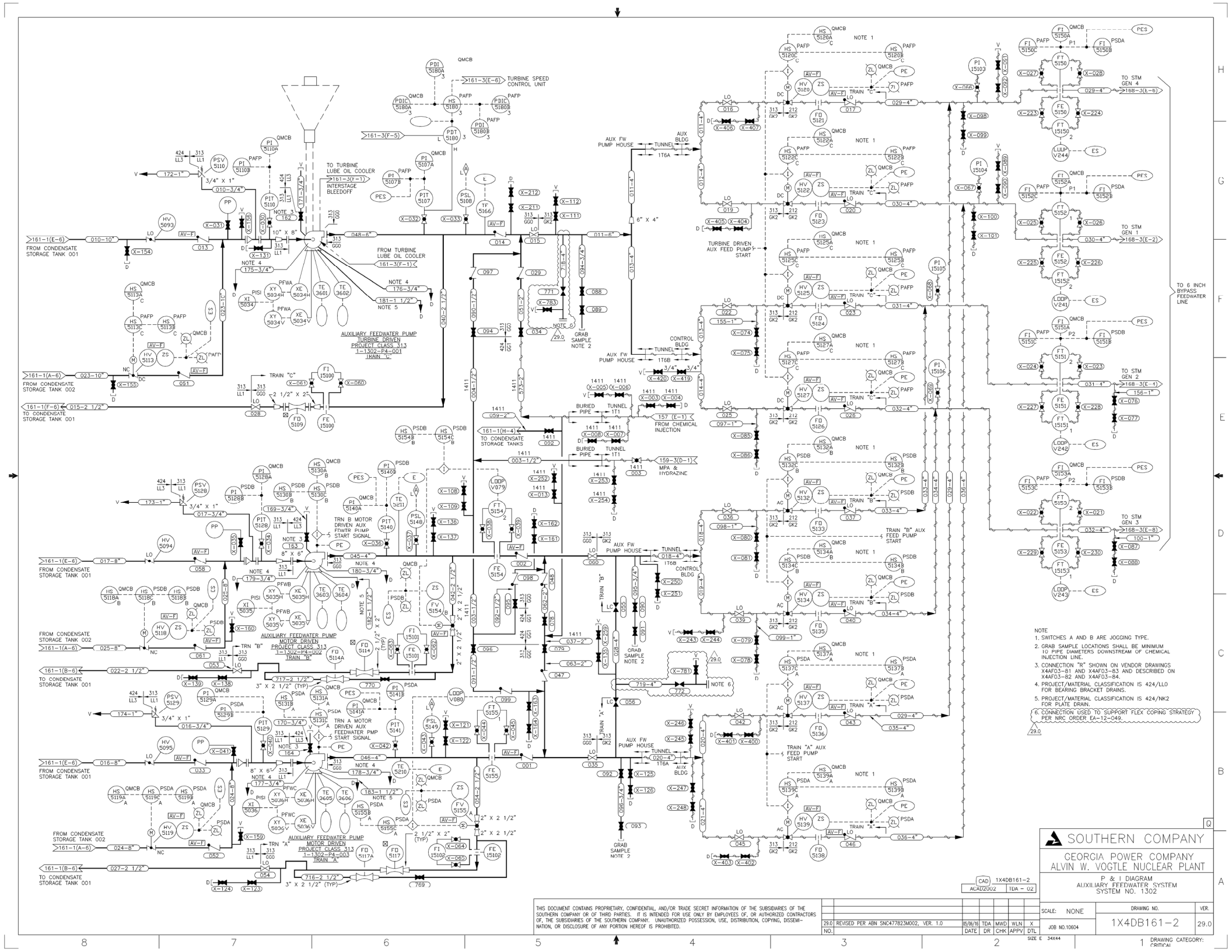
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY FEEDWATER SYSTEM
CONDENSATE STORAGE & DEGASIFIER SYSTEM
SYSTEM NO. 1302

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB161-1	46.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY FEEDWATER SYSTEM
SYSTEM NO. 1302

CAD 1X4DB161-2
ACADZ002 TDA - 02

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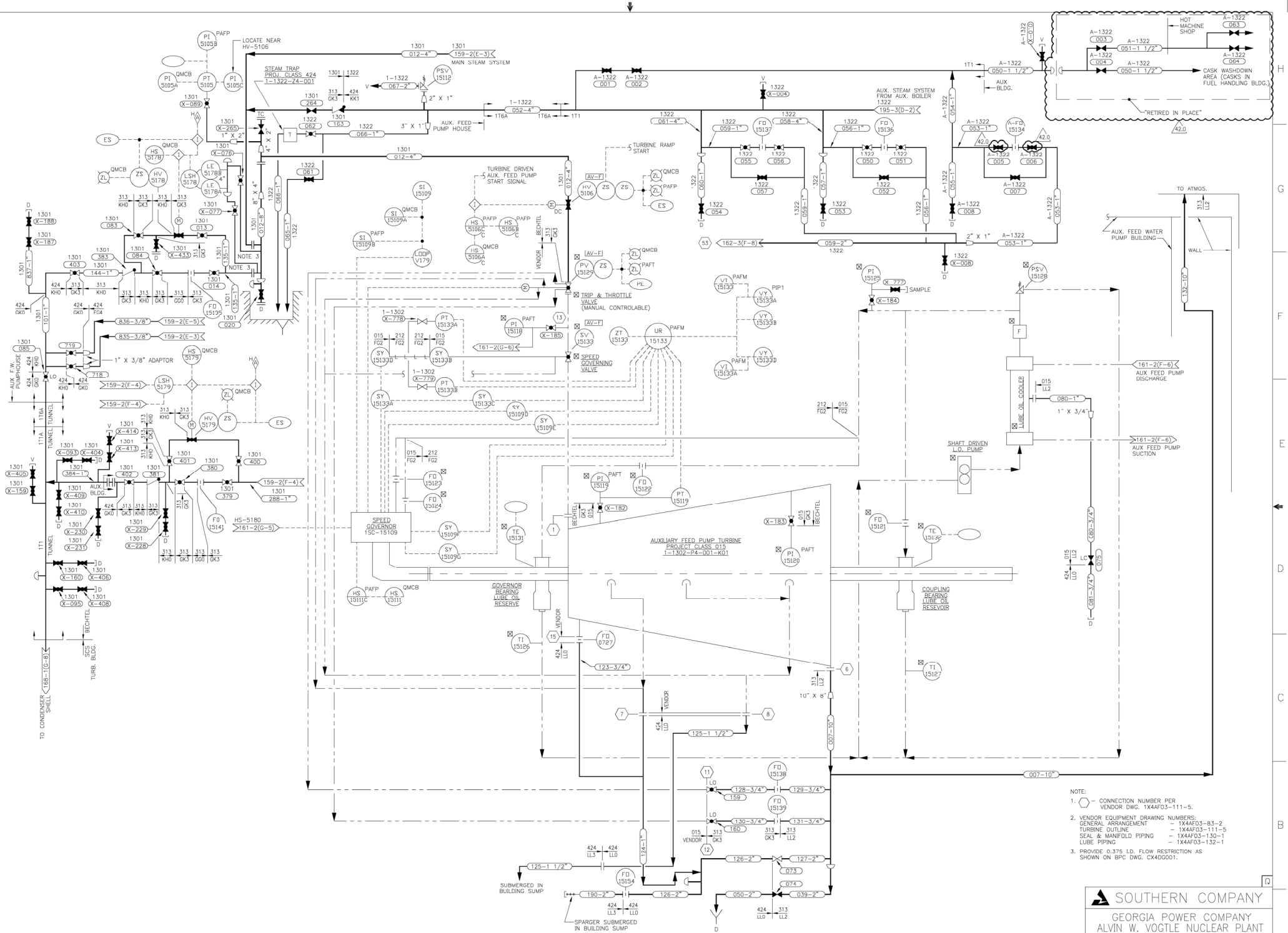
NO.		DATE	DR	CHK	APPV	DTL
29.0	REVISED PER ABN SNC477823M002, VER. 1.0	05/06/16	TDA	MWD	WLN	X

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB161-2
VER. 29.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTE:

1. ⬡ – CONNECTION NUMBER PER VENDOR DWG. 1X4AF03-111-5.
2. VENDOR EQUIPMENT DRAWING NUMBERS:
   - GENERAL ARRANGEMENT – 1X4AF03-83-2
   - TURBINE OUTLINE – 1X4AF03-111-5
   - SEAL & MANIFOLD PIPING – 1X4AF03-130-1
   - LUBE PIPING – 1X4AF03-132-1
3. PROVIDE 0.375 I.D. FLOW RESTRICTION AS SHOWN ON BPC DWG. CX4DG001.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
AUXILIARY FEEDWATER PUMP SYSTEM
(AUX FEEDWATER PUMP TURBINE DRIVER)
SYSTEM 1302

CAD 1X4DB161-3
ACAD2004 KB – 03.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
42.0	REV. PER ABN C100867701M006, VER. 1.0.	07/16/12	KB	RBH	JMR	X

SCALE: NONE

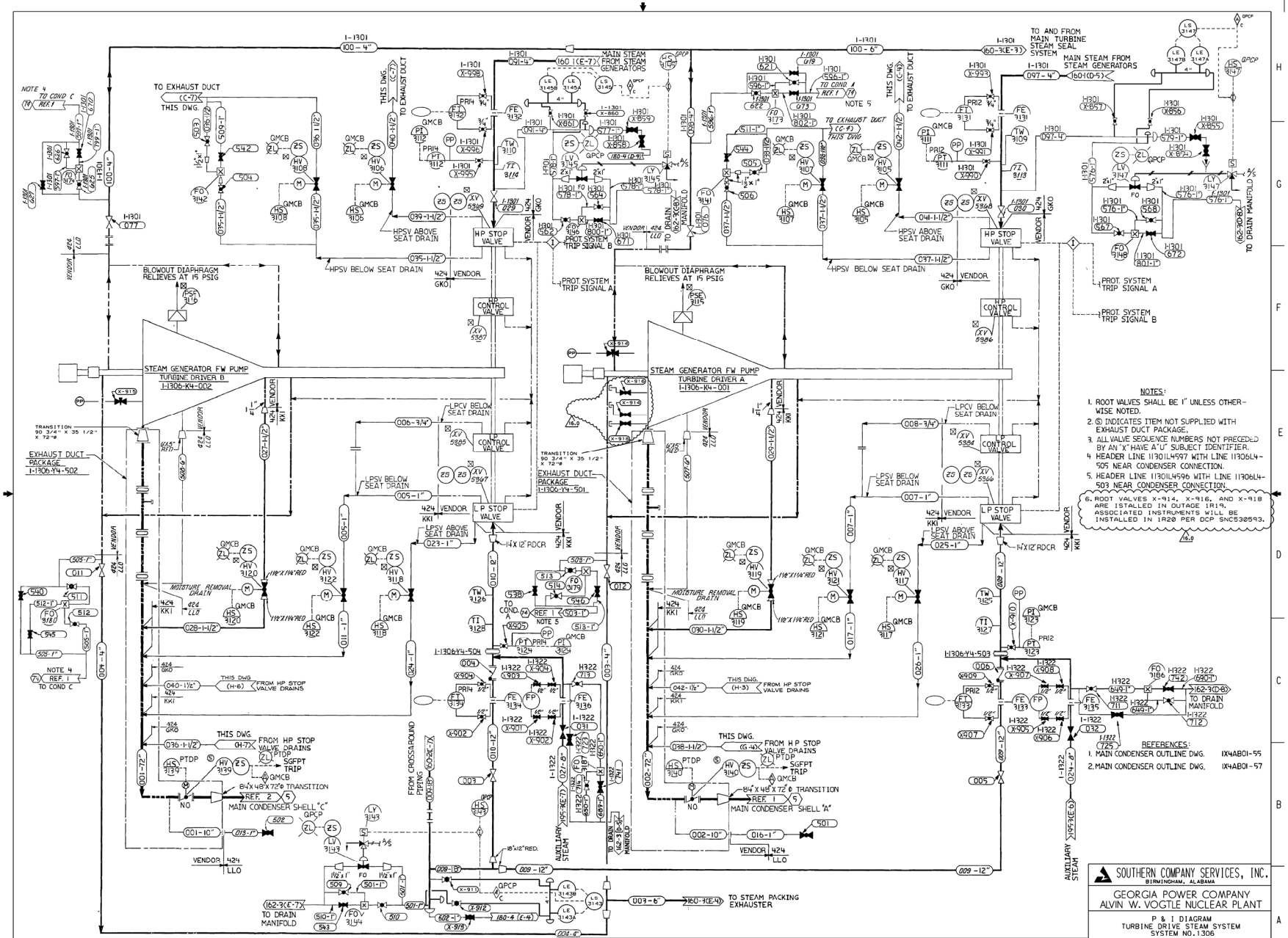
DRAWING NO. 1X4DB161-3

VER. 42.0

JOB NO.10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



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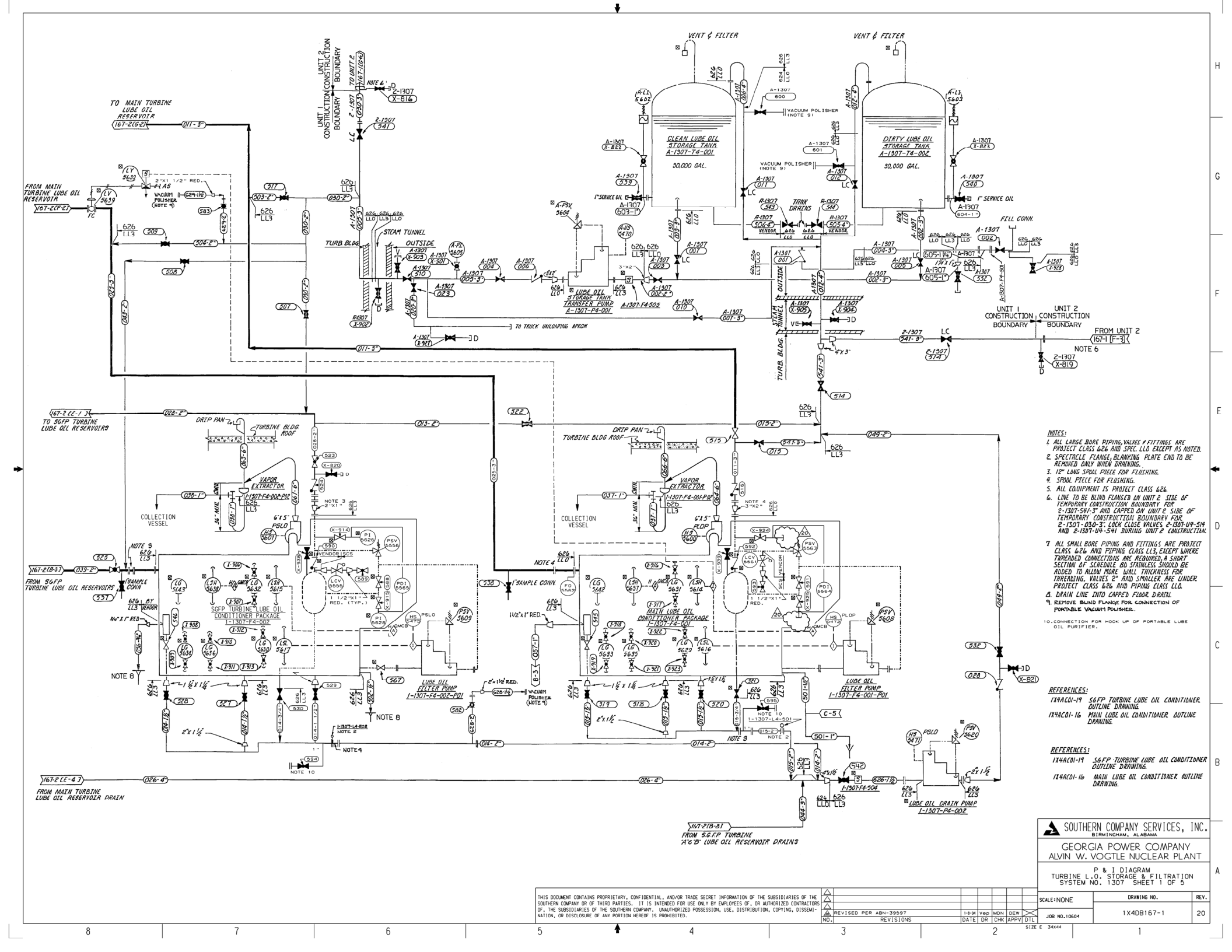
NO.	VERSIONS	DATE	DR	CHK	APPV
16.0	REVISED PER ABN SNC530593J600, VER. 1.0	10/29/19	TDA	ALM	WLN

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE DRIVE STEAM SYSTEM
SYSTEM NO.1306

SCALE: NONE
DRAWING NO. 1X4DB166
VER. 16.0
JOB NO.10604
SIZE E 34x44
DWG CATEGORY: CRITICAL



NOTES:

1. ALL LARGE BORE PIPING, VALVES & FITTINGS ARE PROJECT CLASS 626 AND SPEC LLO EXCEPT AS NOTED.
2. SPECTACLE FLANGE, BLANKING PLATE END TO BE REMOVED ONLY WHEN DRAWING.
3. 12" LONG SPOOL PIECE FOR FLUSHING.
4. SPOOL PIECE FOR FLUSHING.
5. ALL EQUIPMENT IS PROJECT CLASS 626.
6. LINE TO BE BLIND FLANGED ON UNIT 2 SIDE OF TEMPORARY CONSTRUCTION BOUNDARY FOR 2-1307-541-3" AND CAPPED ON UNIT 2 SIDE OF TEMPORARY CONSTRUCTION BOUNDARY FOR 2-1307-030-3". LOCK CLOSE VALVES 2-1307-U4-514 AND 2-1307-U4-541 DURING UNIT 2 CONSTRUCTION.
7. ALL SMALL BORE PIPING AND FITTINGS ARE PROJECT CLASS 626 AND PIPING CLASS L13, EXCEPT WHERE THREADED CONNECTIONS ARE REQUIRED, A SHORT SECTION OF SCHEDULE 80 STAINLESS SHOULD BE ADDED TO ALLOW MORE WALL THICKNESS FOR THREADING. VALVES 2" AND SMALLER ARE UNDER PROJECT CLASS 626 AND PIPING CLASS LLO.
8. DRAIN LINE INTO CAPPED FLOOR DRAIN.
9. REMOVE BLIND FLANGE FOR CONNECTION OF PORTABLE VACUUM POLISHER.
10. CONNECTION FOR HOOK UP OF PORTABLE LUBE OIL PURIFIER.

REFERENCES:

1X4AC01-19 S&FP TURBINE LUBE OIL CONDITIONER OUTLINE DRAWING.
1X4AC01-16 MAIN LUBE OIL CONDITIONER OUTLINE DRAWING.

REFERENCES:

1X4AC01-19 S&FP -TURBINE LUBE OIL CONDITIONER OUTLINE DRAWING.
1X4AC01-16 MAIN LUBE OIL CONDITIONER OUTLINE DRAWING.

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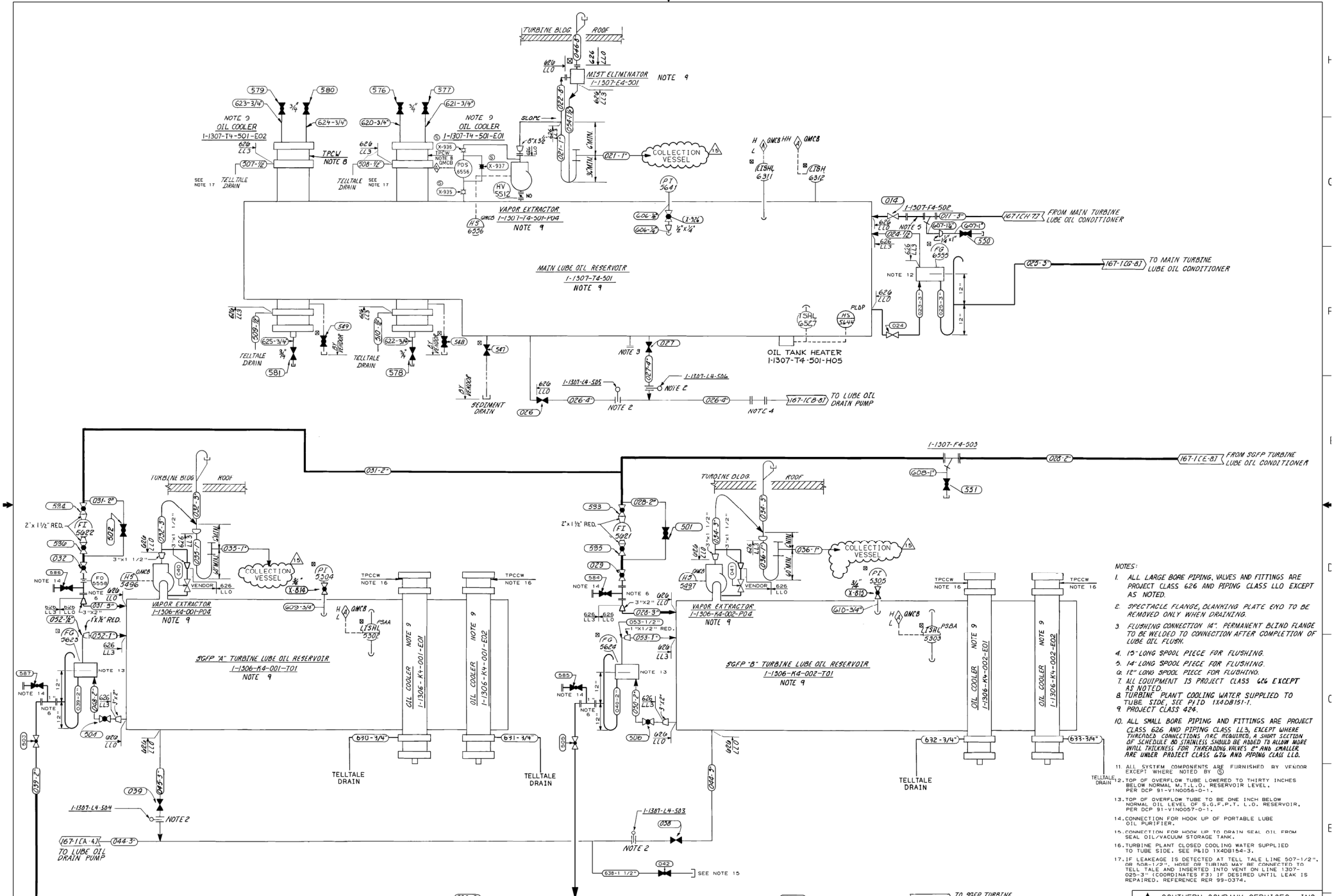
NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	REVISED PER ABN-39597	1-8-04	Vep	MDN	DEW	

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE L.O. STORAGE & FILTRATION
SYSTEM NO. 1307 SHEET 1 OF 5

SCALE: NONE | DRAWING NO. 1X4DB167-1 | REV. 20
JOB NO. 10604
SIZE E 34x44



NOTES:

1. ALL LARGE BORE PIPING, VALVES AND FITTINGS ARE PROJECT CLASS 626 AND PIPING CLASS LLO EXCEPT AS NOTED.
2. SPECTACLE FLANGE, BLANKING PLATE END TO BE REMOVED ONLY WHEN DRAINING.
3. FLUSHING CONNECTION 14". PERMANENT BLIND FLANGE TO BE WELDED TO CONNECTION AFTER COMPLETION OF LUBE OIL FLUSH.
4. 15' LONG SPOOL PIECE FOR FLUSHING.
5. 14' LONG SPOOL PIECE FOR FLUSHING.
6. 12" LONG SPOOL PIECE FOR FLUSHING.
7. ALL EQUIPMENT IS PROJECT CLASS 626 EXCEPT AS NOTED.
8. TURBINE PLANT COOLING WATER SUPPLIED TO TUBE SIDE, SEE P&ID 1X4DB151-1.
9. PROJECT CLASS 626.
10. ALL SMALL BORE PIPING AND FITTINGS ARE PROJECT CLASS 626 AND PIPING CLASS LL3, EXCEPT WHERE THREADED CONNECTIONS ARE REQUIRED, A SHORT SECTION OF SCHEDULE 80 STAINLESS SHOULD BE ADDED TO ALLOW MORE WALL THICKNESS FOR THREADING VALVES 2" AND SMALLER ARE UNDER PROJECT CLASS 626 AND PIPING CLASS LLO.
11. ALL SYSTEM COMPONENTS ARE FURNISHED BY VENDOR EXCEPT WHERE NOTED BY Ⓢ
12. TOP OF OVERFLOW TUBE LOWERED TO THIRTY INCHES BELOW NORMAL M.T.L.O. RESERVOIR LEVEL. PER DCP 91-V1N0056-0-1.
13. TOP OF OVERFLOW TUBE TO BE ONE INCH BELOW NORMAL OIL LEVEL OF S.G.F.P.T. L.O. RESERVOIR, PER DCP 91-V1N0057-0-1.
14. CONNECTION FOR HOOK UP OF PORTABLE LUBE OIL PURIFIER.
15. CONNECTION FOR HOOK UP TO DRAIN SEAL OIL FROM SEAL OIL/VACUUM STORAGE TANK.
16. TURBINE PLANT CLOSED COOLING WATER SUPPLIED TO TUBE SIDE. SEE P&ID 1X4DB154-3.
17. IF LEAKEAGE IS DETECTED AT TELL TALE LINE 507-1/2", OR 508-1/2", HOSE OR TUBING MAY BE CONNECTED TO TELL TALE AND INSERTED INTO VENT ON LINE 1307-025-3" (COORDINATES F3) IF DESIRED UNTIL LEAK IS REPAIRED. REFERENCE RER 99-0374.

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
2	INCORP. PER ABN-39556	12-23-03	KMH	CMM	DEW	

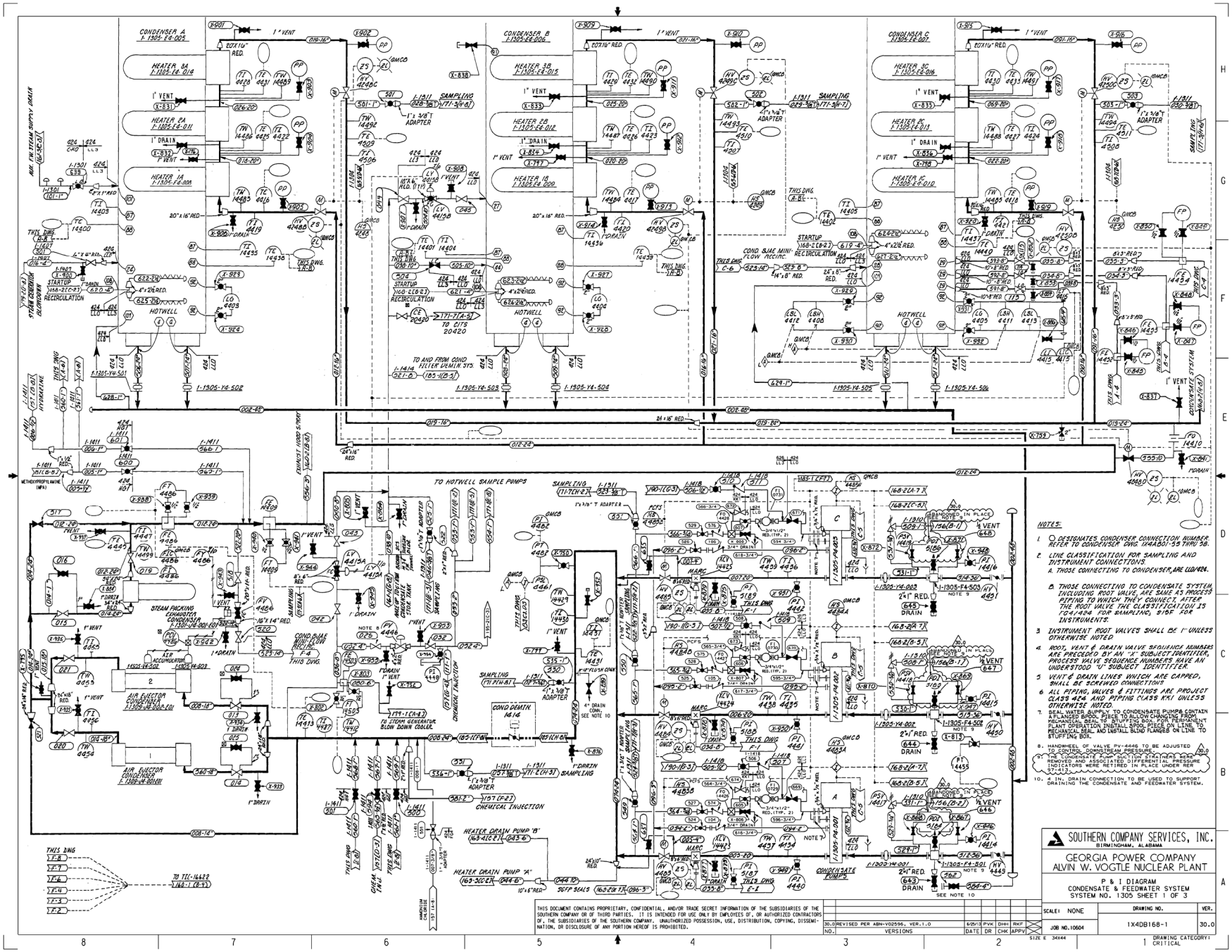
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE L.O. STORAGE & FILTRATION
SYSTEM NO. 1307 SHEET 2 OF 5

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DB167-2	15

SIZE E 34x44


CONDENSER A
1-1305-E4-005
CONDENSER B
1-1305-E4-006
CONDENSER C
1-1305-E4-007
HEATER 3A
1-1305-E4-014
HEATER 2A
1-1305-E4-011
HEATER 1A
1-1305-E4-008
HEATER 3B
1-1305-E4-015
HEATER 2B
1-1305-E4-012
HEATER 1B
1-1305-E4-009
HEATER 3C
1-1305-E4-016
HEATER 2C
1-1305-E4-013
HEATER 1C
1-1305-E4-010
HOTWELL
AIR EJECTOR CONDENSER
1-1305-E4-002 EOI
AIR EJECTOR CONDENSER
1-1305-E4-001 EOH
STEAM PACKING EXHAUSTER CONDENSER
1-1305-E4-003 EOJ
AUX. FW STEAM SUPPLY DRAIN
STEAM GENERATOR BLOWDOWN RECIRCULATION
STARTUP RECIRCULATION
COND. & AE MINI-FLOW RECIRC.
TO HOTWELL SAMPLE PUMPS
CONDENSATE PUMPS
HEATER DRAIN PUMP "A"
HEATER DRAIN PUMP "B"
CHEMICAL INJECTION
TO TTC-16622
SEAL WATER SUPPLY
CONDENSATE SYSTEM
NOTES:
1. ○ DESIGNATES CONDENSER CONNECTION NUMBER. REFER TO CONDENSER DWG 1X4AB01-55 THRU 58.
2. LINE CLASSIFICATION FOR SAMPLING AND INSTRUMENT CONNECTIONS.
A. THOSE CONNECTING TO CONDENSATE, ARE LLD/424.
Δ THOSE CONNECTING TO CONDENSATE SYSTEM, INCLUDING ROOT VALVE, ARE SAME AS PROCESS PIPING TO WHICH THEY CONNECT. AFTER THE ROOT VALVE THE CLASSIFICATION IS FC4/424 FOR SAMPLING, SIDE FOR INSTRUMENTS.
3. INSTRUMENT ROOT VALVES SHALL BE 1" UNLESS OTHERWISE NOTED.
4. ROOT, VENT & DRAIN VALVE SEQUENCE NUMBERS ARE PRECEDED BY AN "X" SUBJECT IDENTIFIER, PROCESS VALVE SEQUENCE NUMBERS HAVE AN UNDERSTOOD "V" SUBJECT IDENTIFIER.
5. VENT & DRAIN LINES WHICH ARE CAPPED, SHALL BE SCREWED CONNECTION.
6. ALL PIPING, VALVES & FITTINGS ARE PROJECT CLASS 424 AND PIPING CLASS KK1 UNLESS OTHERWISE NOTED.
7. SEAL WATER SUPPLY TO CONDENSATE PUMPS CONTAIN A FLANGED SPOOL PIECE TO ALLOW CHANGING FROM MECHANICAL SEAL TO STUFFING BOX. FOR PERMANENT PLANT OPERATION INSTALL SPOOL PIECE ON LINE TO MECHANICAL SEAL AND INSTALL BLIND FLANGES ON LINE TO STUFFING BOX.
8. HANDWHEEL OF VALVE PV-4446 TO BE ADJUSTED TO CONTROL DOWNSTREAM PRESSURE.
9. FOR CONDENSATE PUMP OPERATION, ORIFICES ARE REMOVED AND ASSOCIATED DIFFERENTIAL PRESSURE INDICATORS WERE RETIRED IN PLACE UNDER RER 97-413.
10. 4 IN. DRAIN CONNECTION TO BE USED TO SUPPORT DRAINING THE CONDENSATE AND FEEDWATER SYSTEM.
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
CONDENSATE & FEEDWATER SYSTEM
SYSTEM NO. 1305 SHEET 1 OF 3
SCALE: NONE
DRAWING NO. 1X4DB168-1
VER. 30.0
JOB NO. 10604
SIZE E 34X44
DRAWING CATEGORY: 1 CRITICAL
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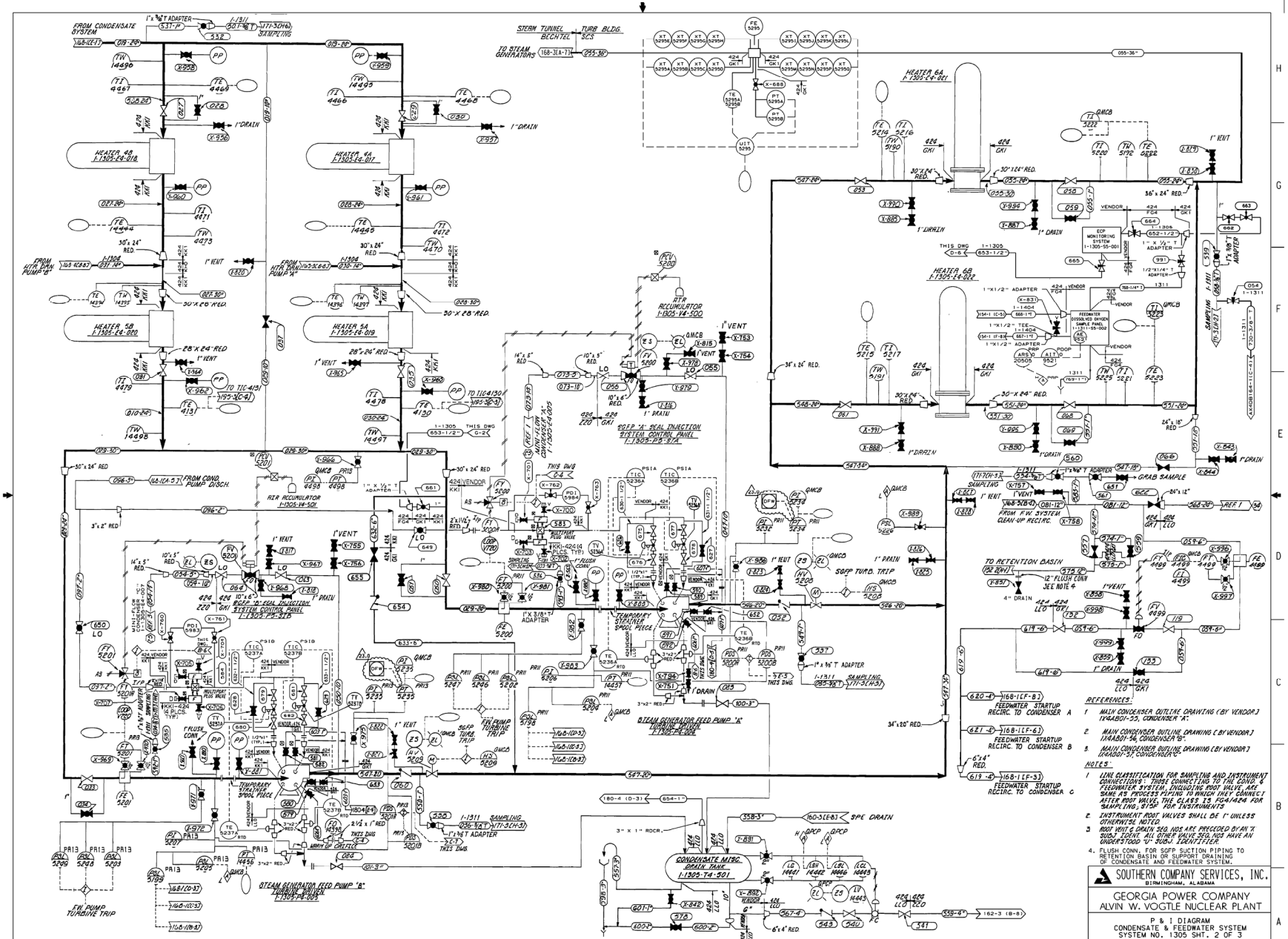
NO.0 REVISED PER ABN-V02596, VER.1.0 6/25/13 PVK DHH RKP
NO. VERSIONS DATE DR CHK APPV



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NO.	VERSIONS	DATE	DR	CHK	APPV
23.0	REVISED PER ABN 10390025011178, VER. 2.0	10/15/12	JLO	ELC	JMR

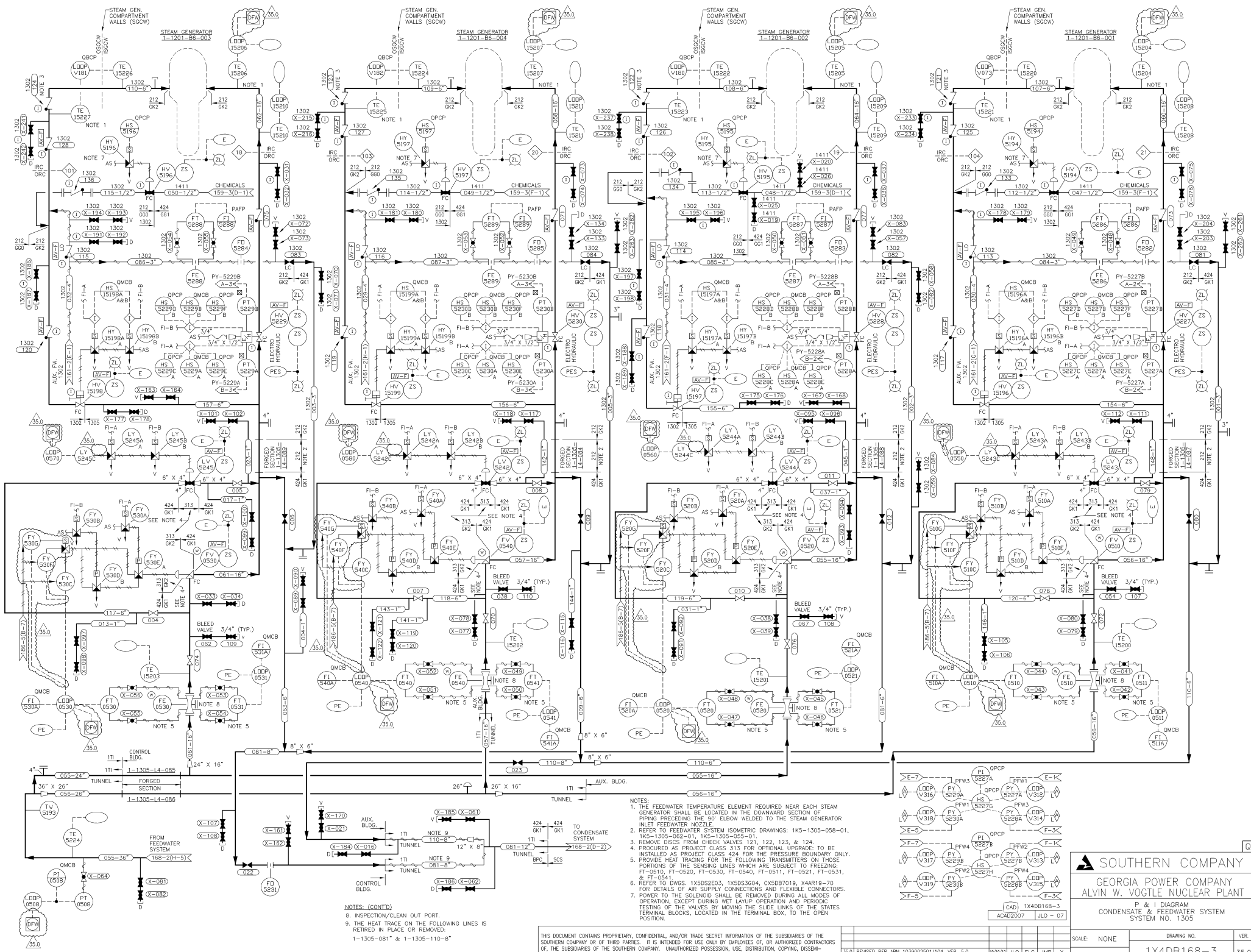
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONDENSATE & FEEDWATER SYSTEM
SYSTEM NO. 1305 SHT. 2 OF 3

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	1X4DB168-2	23.0

Size: E JX44 — DWG CATEGORY: CRITICAL



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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
35.0	REVISED PER ABN 1039002501J104, VER. 5.0	10/10/12	JLO	ELC	JMR	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONDENSATE & FEEDWATER SYSTEM
SYSTEM NO. 1305

CAD: 1X4DB168-3
ACAD2007 | JLO - 07

SCALE: NONE
JOB NO.10604
DRAWING NO.: 1X4DB168-3
VER.: 35.0

SIZE E 34x44

1 DRAWING CATEGORY: CRITICAL

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
46.0	REVISED PER ABN-V03084, VER. 2.0	9-20-12	JJB	LCF	AAN	X

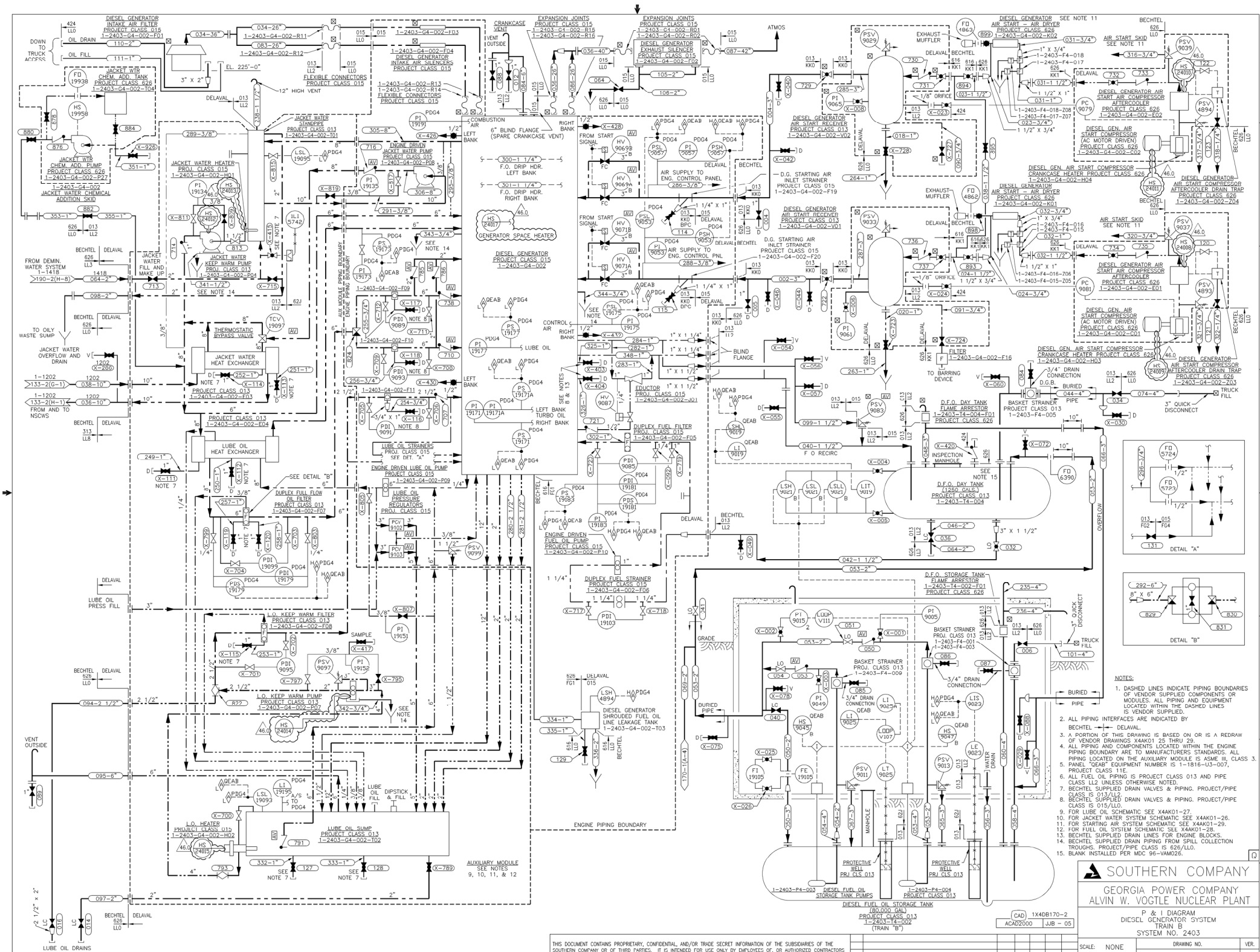
CAD 1X4DB170-1
ACAD2000 JJB – 06

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
DIESEL GENERATOR SYSTEM
TRAIN A
SYSTEM NO. 2403

SCALE: NONE
DRAWING NO. 1X4DB170-1
VER. 46.0
JOB NO. 10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. DASHED LINES INDICATE PIPING BOUNDARIES OF VENDOR SUPPLIED COMPONENTS OR MODULES. ALL PIPING AND EQUIPMENT LOCATED WITHIN THE DASHED LINES IS VENDOR SUPPLIED.
2. ALL PIPING INTERFACES ARE INDICATED BY BECHTEL ⟷ DELAVAL.
3. A PORTION OF THIS DRAWING IS BASED ON OR IS A REDRAW OF VENDOR DRAWINGS X4AK01 25 THRU 29.
4. ALL PIPING AND COMPONENTS LOCATED WITHIN THE ENGINE PIPING BOUNDARY ARE TO MANUFACTURERS STANDARDS. ALL PIPING LOCATED ON THE AUXILIARY MODULE IS ASME III, CLASS 3.
5. PANEL "QEAB" EQUIPMENT NUMBER IS 1-1816-U3-007, PROJECT CLASS 11E.
6. ALL FUEL OIL PIPING IS PROJECT CLASS 013 AND PIPE CLASS LL2 UNLESS OTHERWISE NOTED.
7. BECHTEL SUPPLIED DRAIN VALVES & PIPING. PROJECT/PIPE CLASS IS 013/LL2.
8. BECHTEL SUPPLIED DRAIN VALVES & PIPING. PROJECT/PIPE CLASS IS 015/LLO.
9. FOR LUBE OIL SCHEMATIC SEE X4AK01-27.
10. FOR JACKET WATER SYSTEM SCHEMATIC SEE X4AK01-26.
11. FOR STARTING AIR SYSTEM SCHEMATIC SEE X4AK01-29.
12. FOR FUEL OIL SYSTEM SCHEMATIC SEE X4AK01-28.
13. BECHTEL SUPPLIED DRAIN LINES FOR ENGINE BLOCKS.
14. BECHTEL SUPPLIED DRAIN PIPING FROM SPILL COLLECTION TROUGHS. PROJECT/PIPE CLASS IS 626/LLO.
15. BLANK INSTALLED PER MDC 96-VAM026.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
DIESEL GENERATOR SYSTEM
TRAIN B
SYSTEM NO. 2403

CAD 1X4DB170-2
ACAD2000 | JJB - 05

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB170-2	46.0

NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
46.0	REVISED PER ABN-V03084, VER. 2.0	9-20-12	JJB	LCF	AAN	X

SIZE E 34X44

DRAWING CATEGORY: CRITICAL

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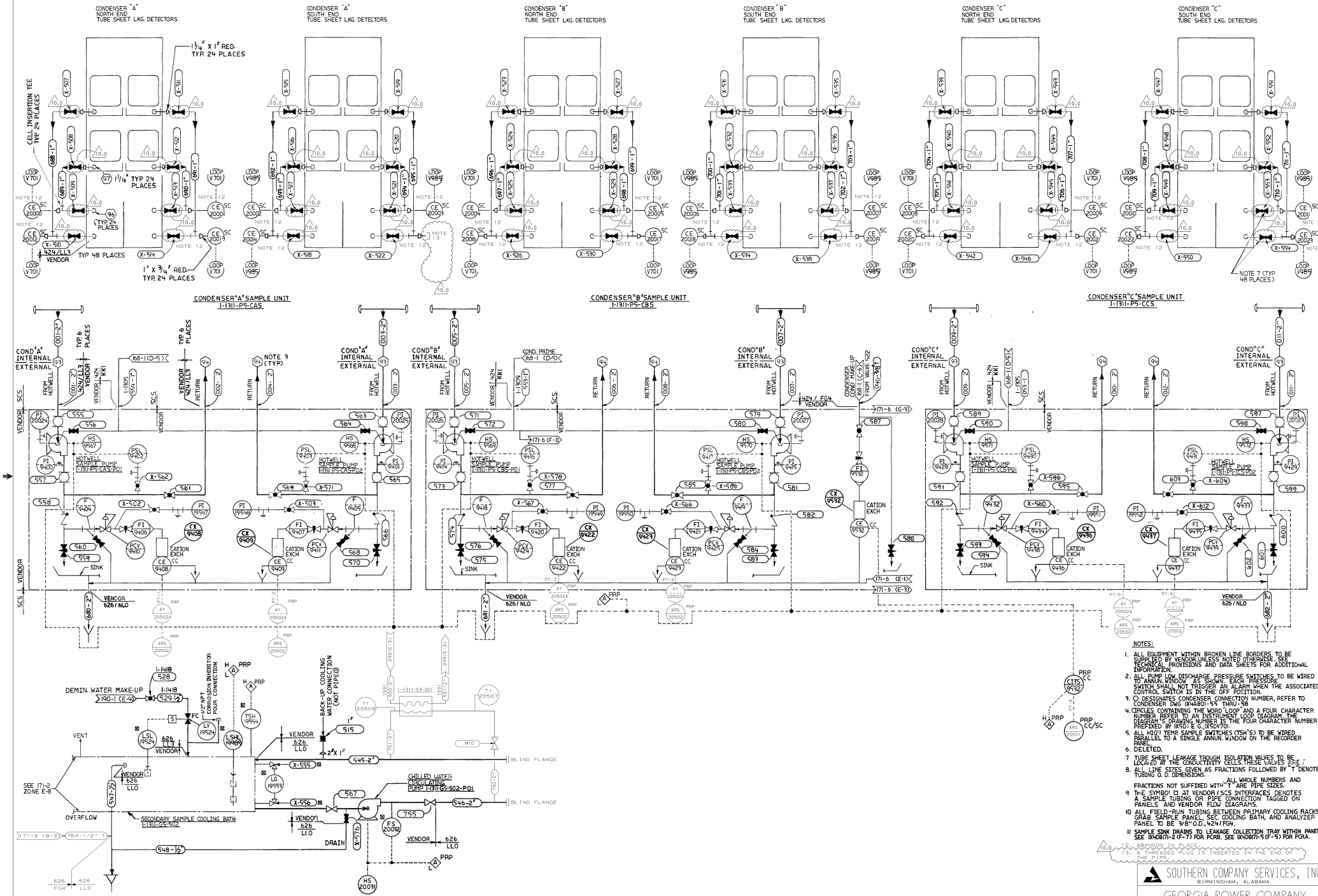

NOTES:

1. ALL EQUIPMENT WITHIN BROKEN LINE BORDERS TO BE SUPPLIED BY VENDOR UNLESS NOTED OTHERWISE. SEE TECHNICAL PROVISIONS AND DATA SHEETS FOR ADDITIONAL INFORMATION.
2. ALL PUMP LOW DISCHARGE PRESSURE SWITCHES TO BE WIRED TO ANNUN. WINDOW AS SHOWN. EACH PRESSURE SWITCH SHALL NOT TRIGGER AN ALARM WHEN THE ASSOCIATED CONTROL SWITCH IS IN THE OFF POSITION.
3. ○ DESIGNATES CONDENSER CONNECTION NUMBER, REFER TO CONDENSER DWG 1X4AB01-S5 THRU-S8
4. CIRCLES CONTAINING THE WORD 'LOOP' AND A FOUR CHARACTER NUMBER REFER TO AN INSTRUMENT LOOP DIAGRAM. THE DIAGRAM'S DRAWING NUMBER IS THE FOUR CHARACTER NUMBER PREFIXED BY 1X5D: E.G.,1X5DV701.
5. ALL HIGH TEMP SAMPLE SWITCHES (TSH'S) TO BE WIRED PARALLEL TO A SINGLE ANNUN. WINDOW ON THE RECORDER PANEL.
6. DELETED.
7. TUBE SHEET LEAKAGE TROUGH ISOLATION VALVES TO BE LOCATED AT THE CONDUCTIVITY CELLS. THESE VALVES ARE ¾".
8. ALL LINE SIZES GIVEN AS FRACTIONS FOLLOWED BY "T" DENOTE TUBING O.D. DIMENSIONS.

   ALL WHOLE NUMBERS AND FRACTIONS NOT SUFFIXED WITH "T" ARE PIPE SIZES.
9. THE SYMBOL □ AT VENDOR / SCS INTERFACES DENOTES A SAMPLE TUBING OR PIPE CONNECTION TAGGED ON PANELS AND VENDOR FLOW DIAGRAMS.
10. ALL FIELD-RUN TUBING BETWEEN PRIMARY COOLING RACKS, GRAB SAMPLE PANEL, SEC. COOLING BATH, AND ANALYZER PANEL TO BE 3/8" O.D., 424/FG4.
11. SAMPLE SINK DRAINS TO LEAKAGE COLLECTION TRAY WITHIN PANEL. SEE 1X4DB171-2 (F-7) FOR PCRB. SEE 1X4DB171-5 (F-5) FOR PCRA.
12. ABANDON IN PLACE.
13. A THREADED PLUG IS INSERTED IN THE END OF THE PIPE.

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NO.	VERSIONS	DATE	DR	CHK	APPV
10.0	REVISED PER ABN-V01084, VER 1.0	03/17/08	ELC	SRG	PML

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

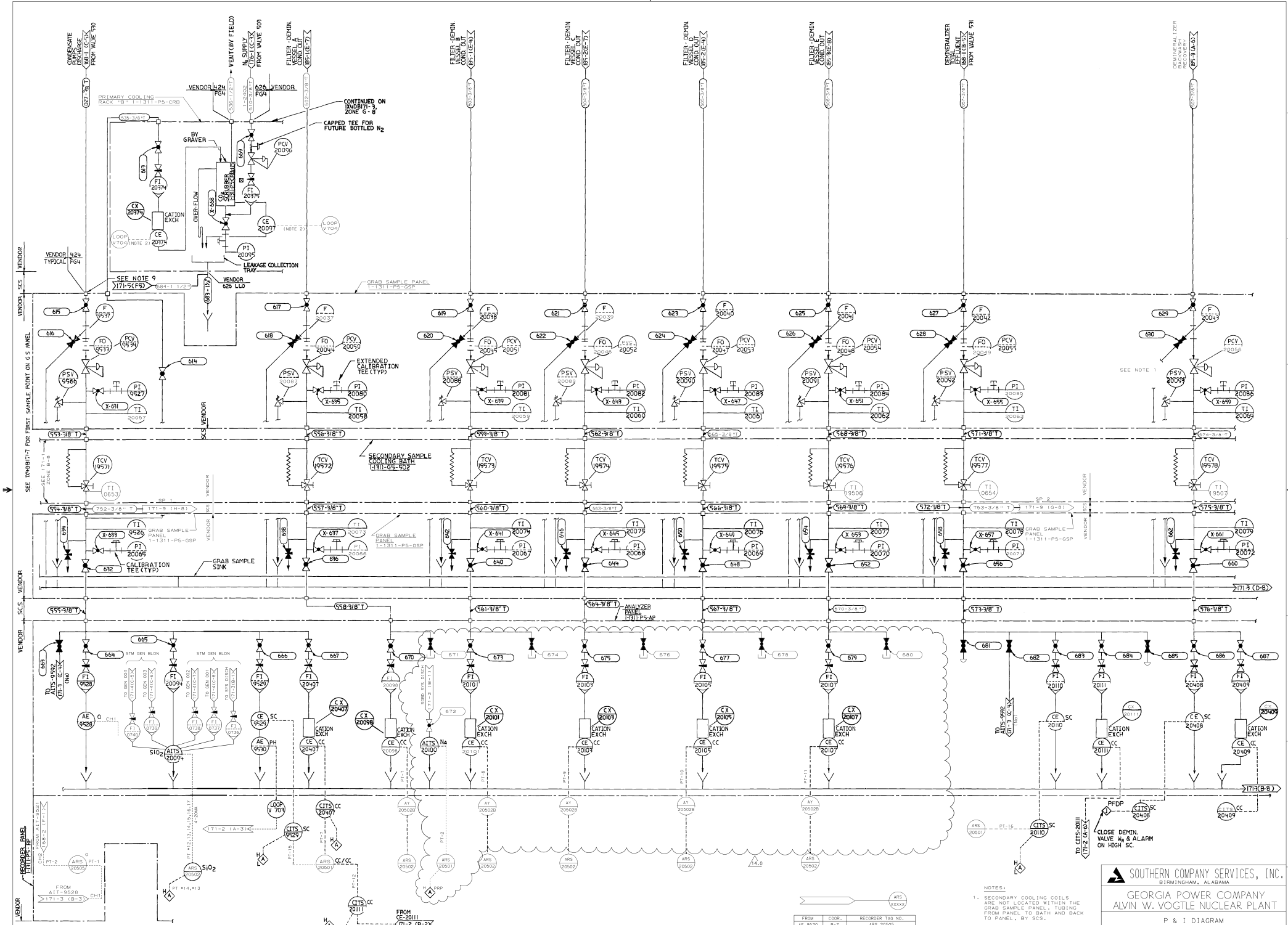
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT SAMPLING
SYSTEM NO. 1311

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	1X4DB171-1	10.0

SIZE E 34X44

DWG CATEGORY: CRITICAL



FROM	COOR.	RECORDER TAG NO.
AE 9530	B-7	ARS 20505

NOTES:
1. SECONDARY COOLING COILS ARE NOT LOCATED WITHIN THE GRAB SAMPLE PANEL. TUBING FROM PANEL TO BATH AND BACK TO PANEL, BY SCS.
2. ABANDON IN PLACE.

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT SAMPLING
SYSTEM NO. 1311

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB171-2	14.0

NO.	VERSIONS	DATE	DR	CHK	APPV
14.0	REVISED PER ABN-V00192, VER 1.0	06/20/06	ELC	RBH	DEW

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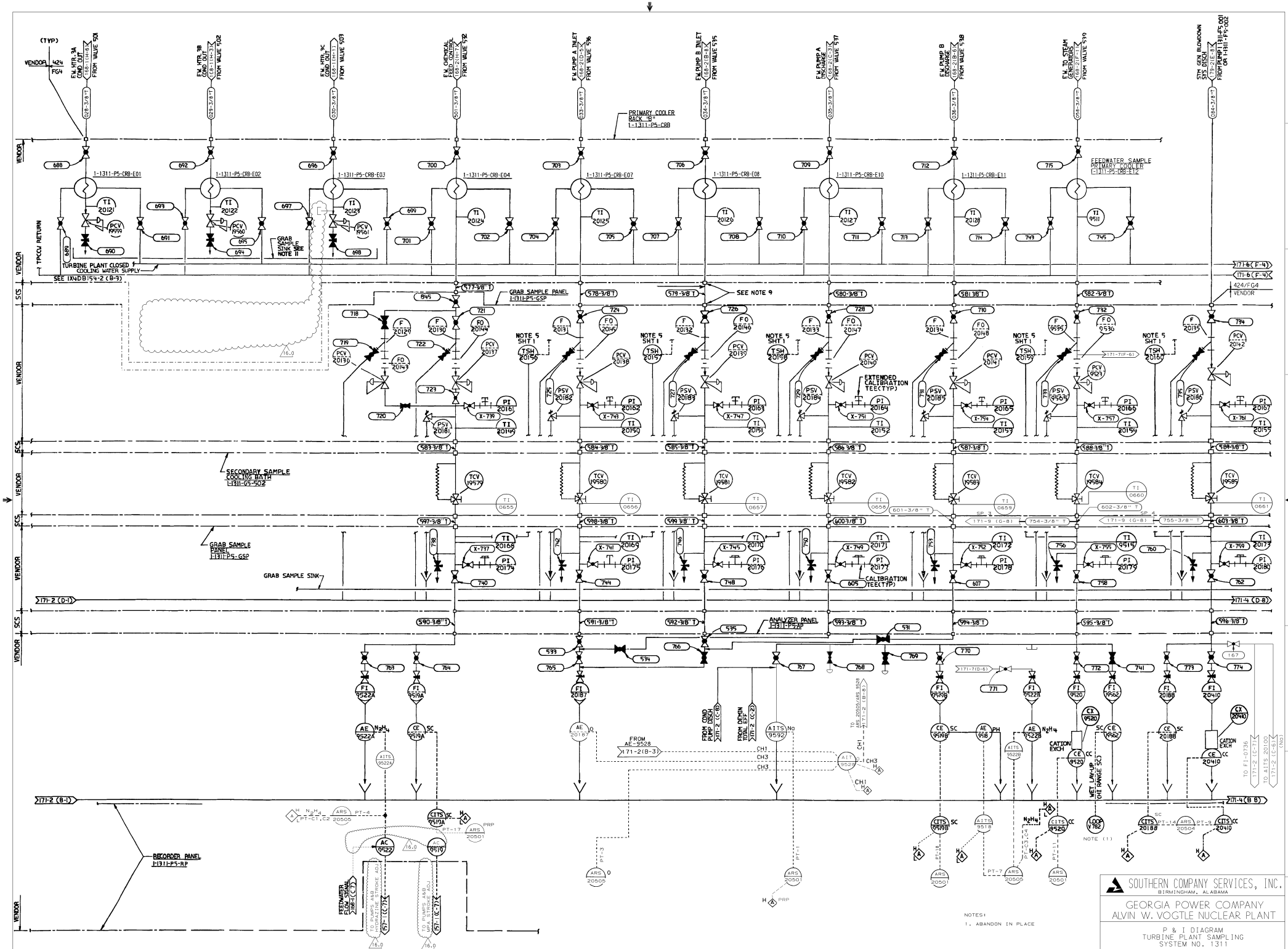
SIZE E 34X44

DWG CATEGORY: CRITICAL



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT SAMPLING
SYSTEM NO. 1311

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB171-3	16.0

NO.	VERSIONS	DATE	DR	CHK	APPV
16.0	REVISED PER ABN1039001101M004, VER 1.0	10/31/08	ELC	RLP	JMR

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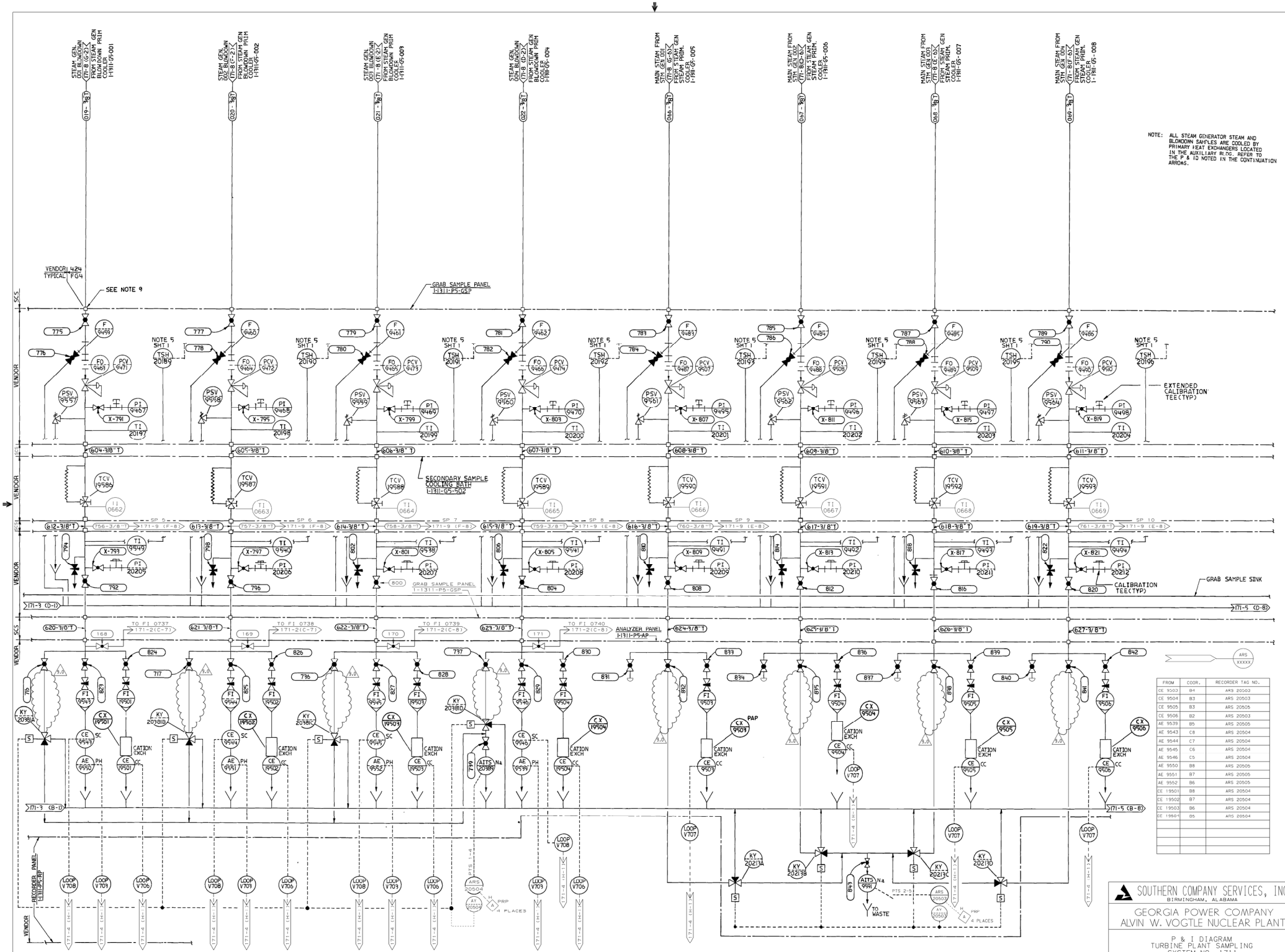
SIZE E 34X44

DRAWING CATEGORY: CRITICAL



FROM	COOR.	RECORDER TAG NO.
CE 9503	B4	ARS 20503
CE 9504	B3	ARS 20503
CE 9505	B3	ARS 20505
CE 9506	B2	ARS 20503
AE 9539	B5	ARS 20505
AE 9543	C8	ARS 20504
AE 9544	C7	ARS 20504
AE 9545	C6	ARS 20504
AE 9546	C5	ARS 20504
AE 9550	B8	ARS 20505
AE 9551	B7	ARS 20505
AE 9552	B6	ARS 20505
CE 19501	B8	ARS 20504
CE 19502	B7	ARS 20504
CE 19503	B6	ARS 20504
CE 19504	B5	ARS 20504

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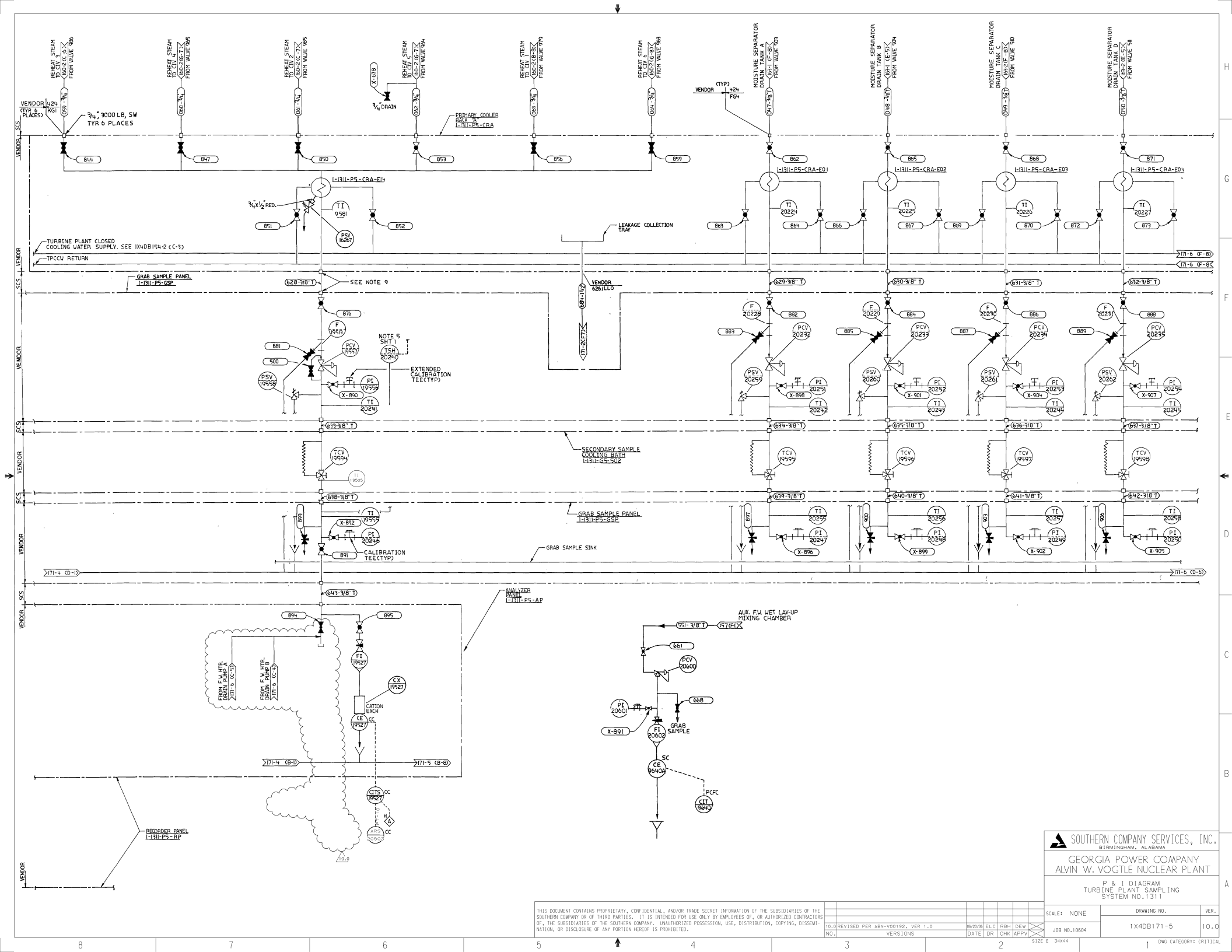
NO.	VERSIONS	DATE	DR	CHK	APPV
9.0	REVISED PER ABN-V00192, VER 1.0	06/20/06	ELC	RBH	DEW

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT SAMPLING
SYSTEM NO. 1311

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB171-4
VER. 9.0
SIZE E 34X44
DWG CATEGORY: CRITICAL



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT SAMPLING
SYSTEM NO.1311

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB171-5	10.0

NO.	VERSIONS	DATE	DR	CHK	APPV
10.0	REVISED PER ABN-V00192, VER 1.0	06/20/08	ELC	RBH	DEW

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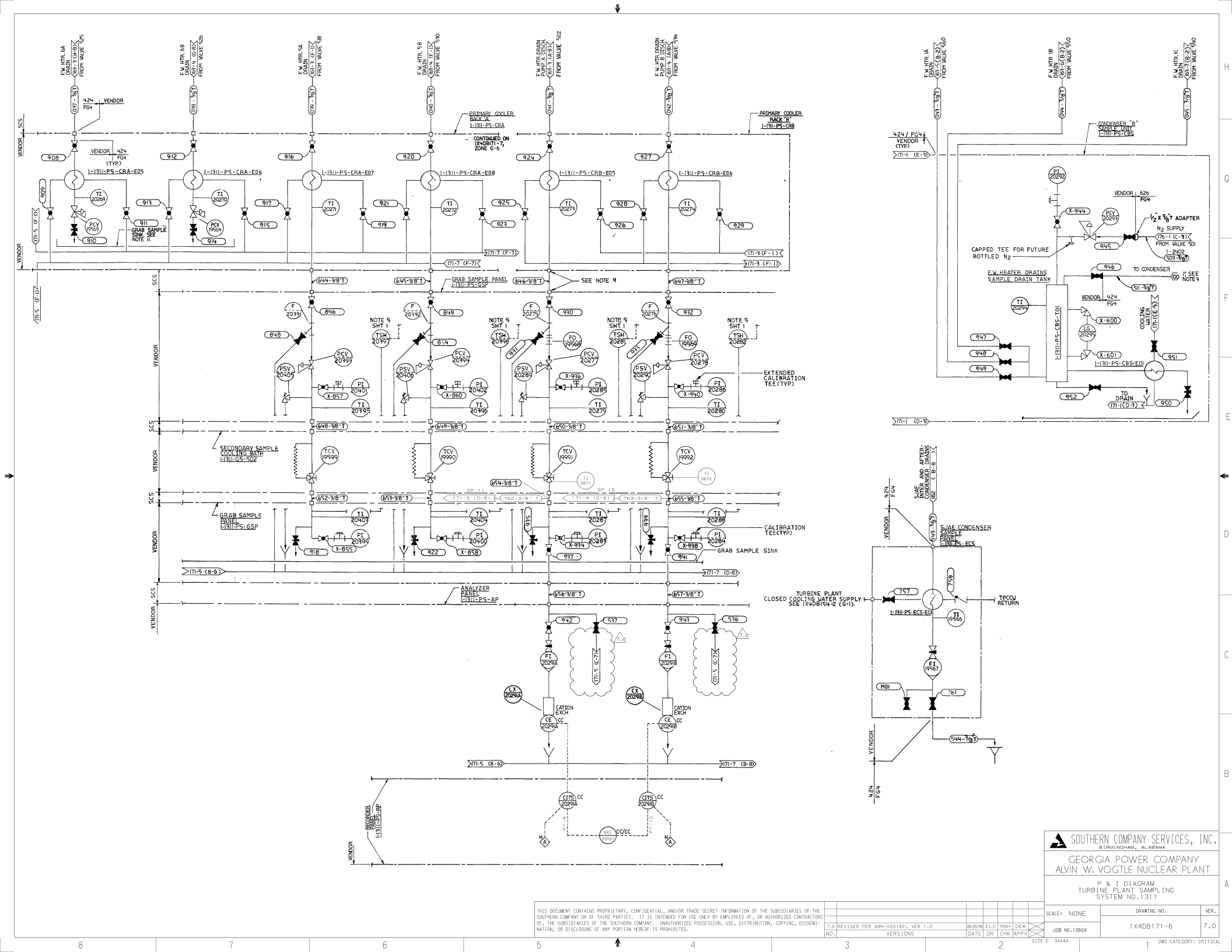
SIZE E 34X44

DWG CATEGORY: CRITICAL



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NO.	VERSIONS	DATE	DR	CHK	APPV
7.0	REVISED PER ABN-V00192, VER 1.0	06/20/08	ELC	RBH	DEW

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT SAMPLING
SYSTEM NO. 1311

SCALE: NONE

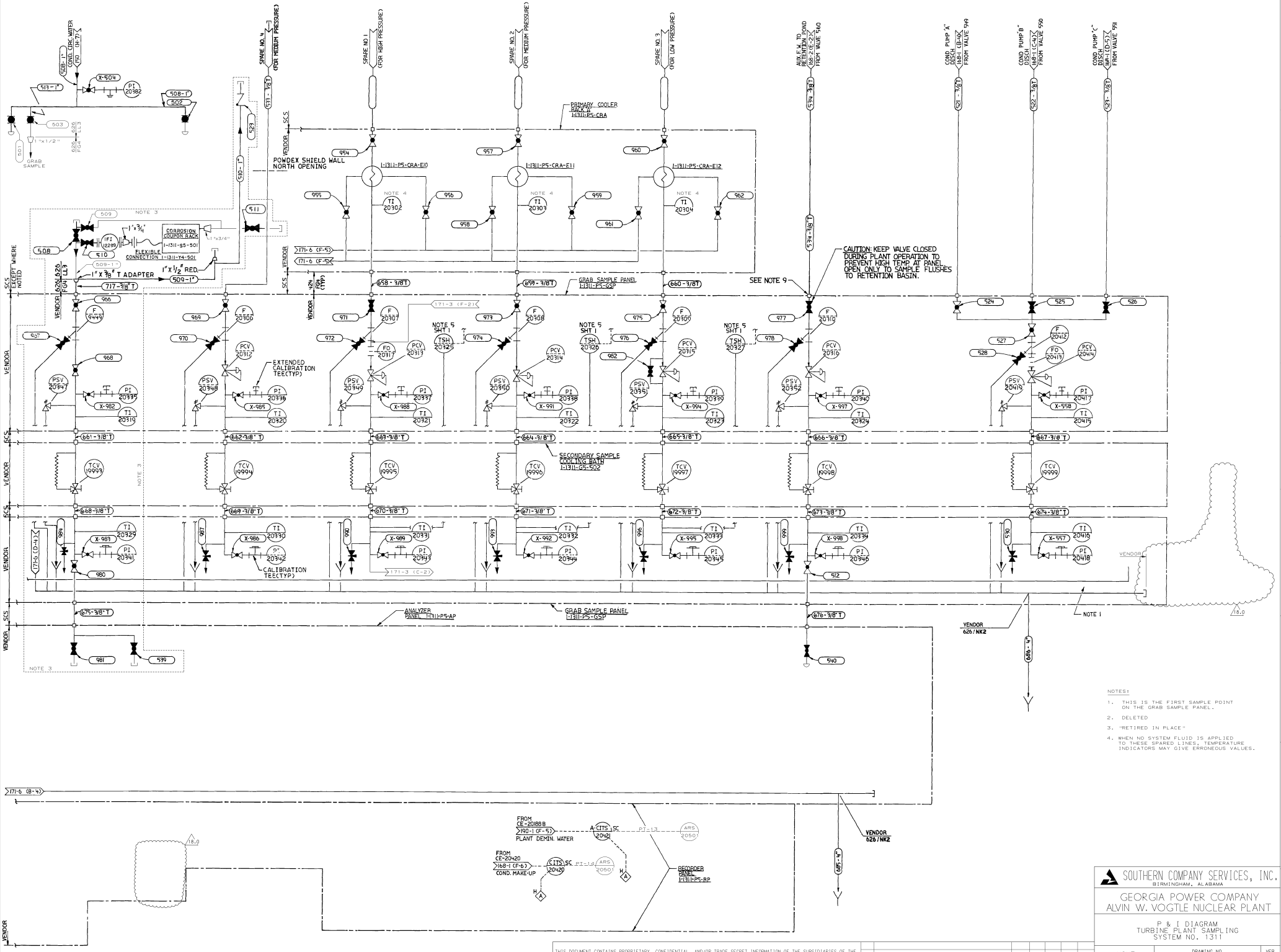
DRAWING NO. 1X4DB171-6

VER. 7.0

JOB NO. 10604

SIZE E 34X44

DWG CATEGORY: CRITICAL



NOTES:

1. THIS IS THE FIRST SAMPLE POINT ON THE GRAB SAMPLE PANEL.
2. DELETED
3. "RETIRED IN PLACE"
4. WHEN NO SYSTEM FLUID IS APPLIED TO THESE SPARED LINES, TEMPERATURE INDICATORS MAY GIVE ERRONEOUS VALUES.

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NO.	VERSIONS	DATE	DR	CHK	APPV
18.0	REVISED PER ABN1039001101.J009, VER 2.0	11/29/06	ELC	RBH	JMR

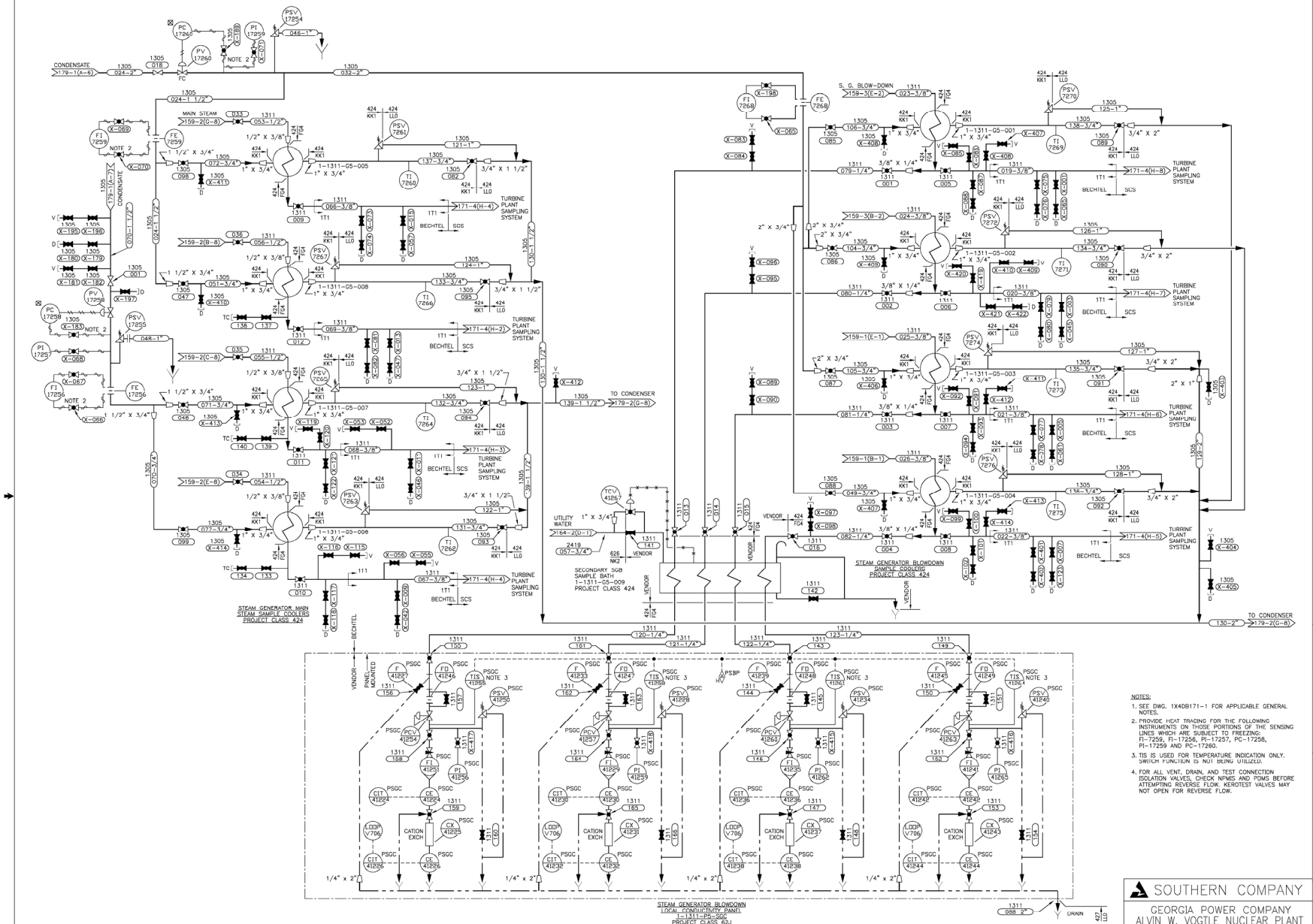
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT SAMPLING
SYSTEM NO. 1311

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB171-7	18.0

SIZE E 34X44 — DWG CATEGORY: CRITICAL



NOTES:

1. SEE DWG. 1X4DB171-1 FOR APPLICABLE GENERAL NOTES.
2. PROVIDE HEAT TRACING FOR THE FOLLOWING INSTRUMENTS ON THOSE PORTIONS OF THE SENSING LINES WHICH ARE SUBJECT TO FREEZING: FI-7259, FI-17256, PI-17257, PC-17258, PI-17259 AND PC-17260.
3. TIS IS USED FOR TEMPERATURE INDICATION ONLY. SWITCH FUNCTION IS NOT BEING UTILIZED.
4. FOR ALL VENT, DRAIN, AND TEST CONNECTION ISOLATION VALVES, CHECK NPMIS AND POMS BEFORE ATTEMPTING REVERSE FLOW. KEROTEST VALVES MAY NOT OPEN FOR REVERSE FLOW.

CAD 1X4DB171-8
ACAD2004 CYN- 01

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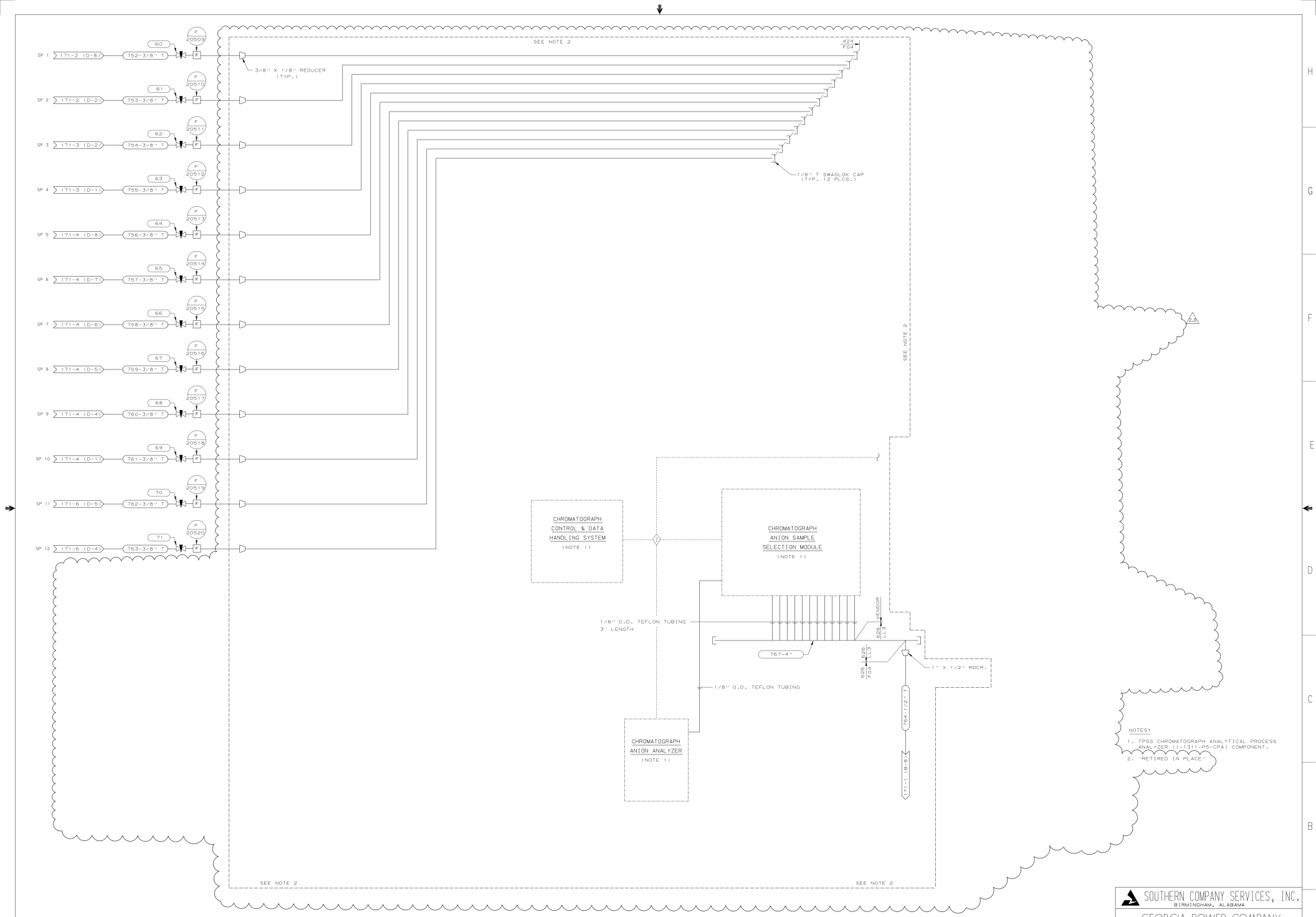
NO.	VERSIONS	DATE	DN	CHK	APPV	UTL
20.0	REDRAWN PER ABN-V70204, VER. 1.0	12-06-08	CYN	LCF	SHS	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT SAMPLING
SYSTEM 1305, 1311

SCALE: NONE
SIZE E 34X44
DRAWING NO. 1X4DB171-8
VER. 20.0
JOB NO.10604
1 DRAWING CATEGORY: CRITICAL



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NO.	VERSIONS	DATE	DR	CHK	APPV
2.0	REVISED PER ABN1039001101M038, VER 2.0	11/29/06	ELC	RBH	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

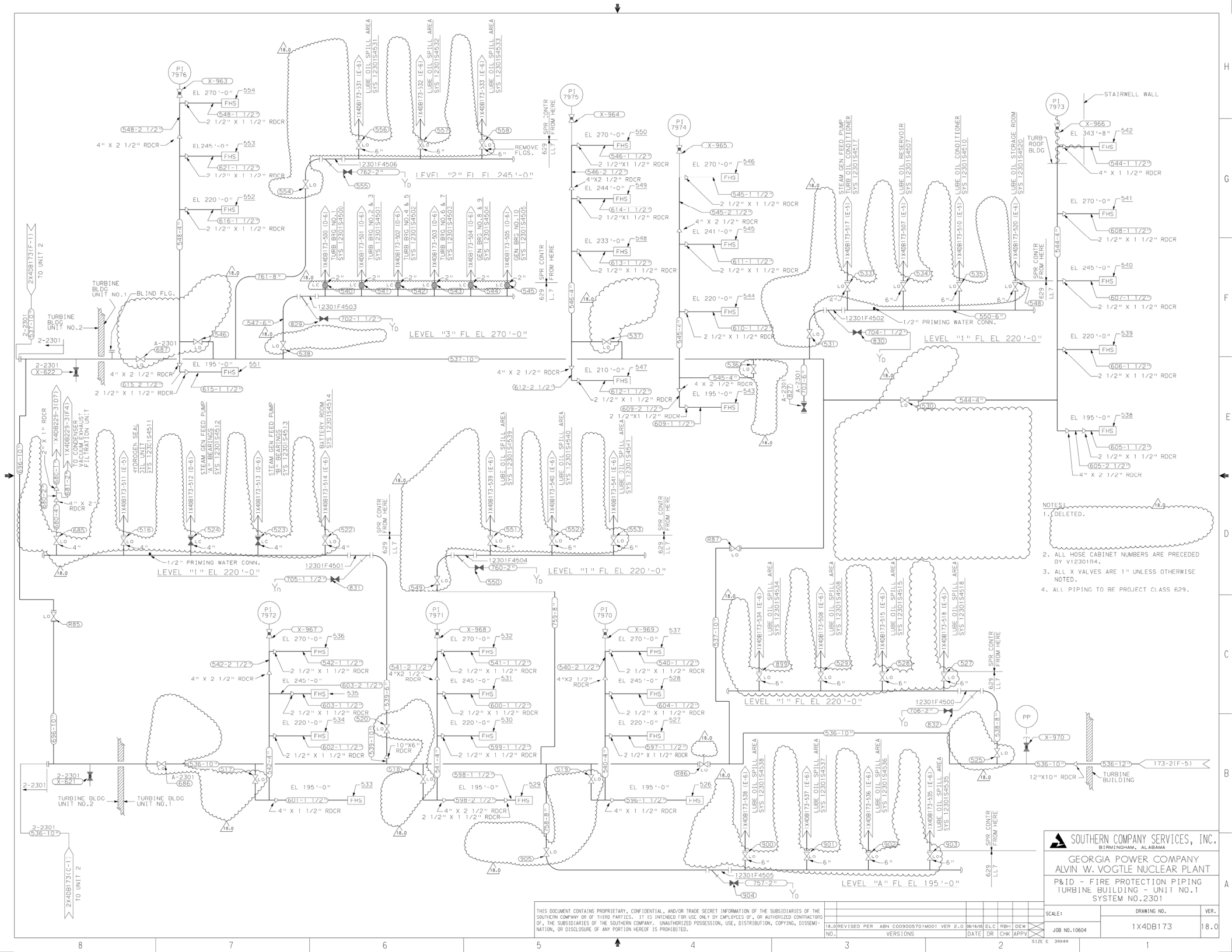
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE PLANT SAMPLING
SYSTEM NO. 1311

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB171-9	2.0

SIZE E 34X44

DWG CATEGORY: CRITICAL



NOTES:

1. DELETED.
2. ALL HOSE CABINET NUMBERS ARE PRECEDED BY V12301R4.
3. ALL X VALVES ARE 1" UNLESS OTHERWISE NOTED.
4. ALL PIPING TO BE PROJECT CLASS 629.

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NO.	VERSIONS	DATE	DR	CHK	APPV
18.0	REVISED PER ABN C00900570 1M001 VER 2.0	08/16/05	ELC	RBH	DEW

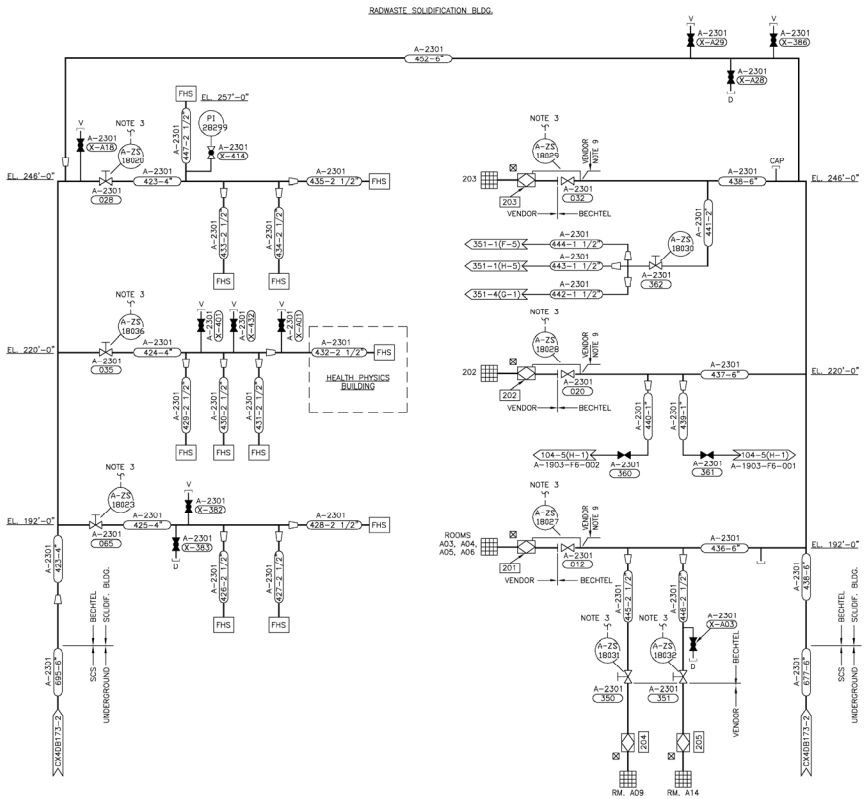
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

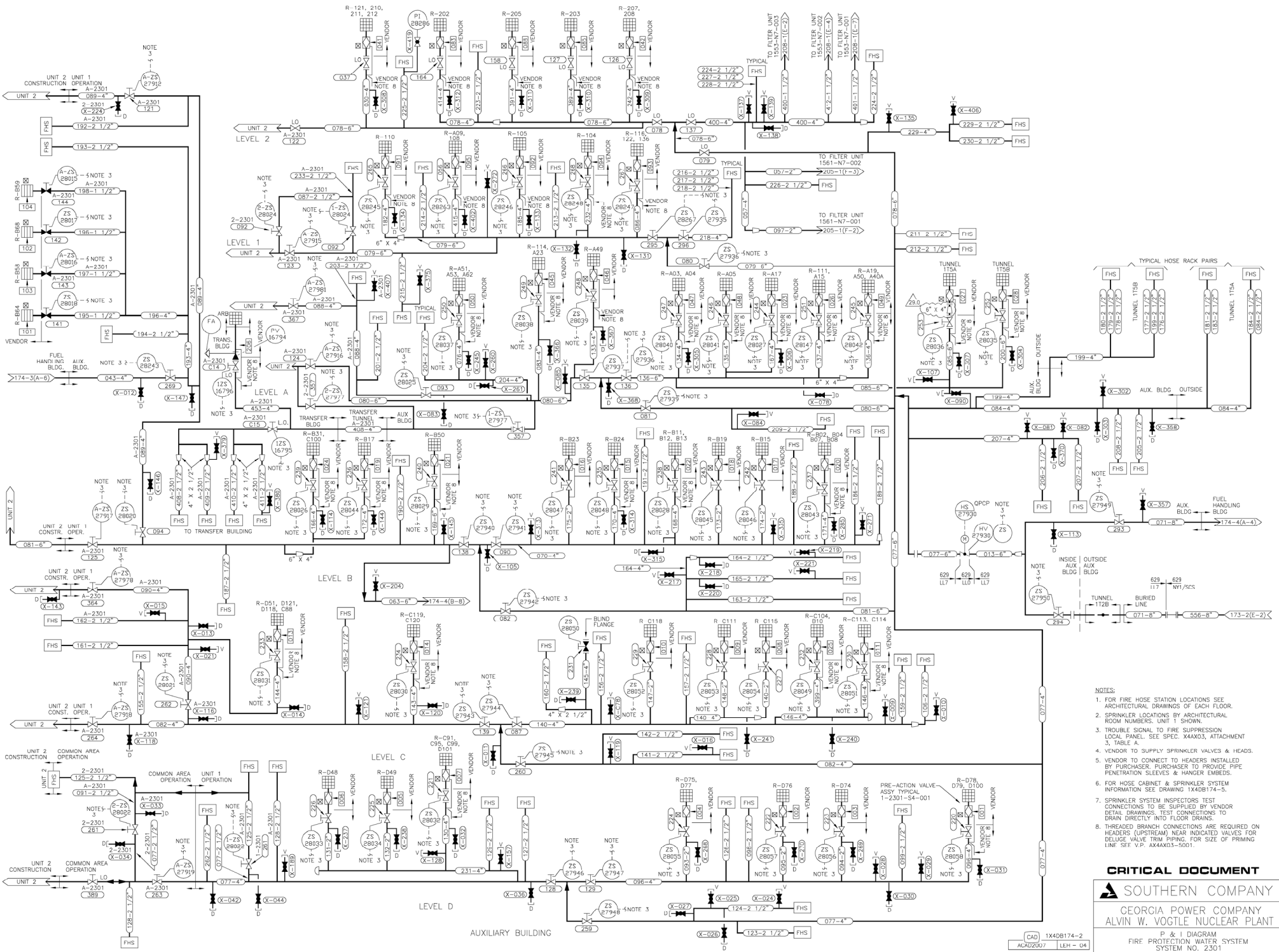
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P&ID - FIRE PROTECTION PIPING
TURBINE BUILDING - UNIT NO.1
SYSTEM NO.2301

SCALE:	DRAWING NO.	VER.
JOB NO.10604	1X4DB173	18.0

SIZE E 34X44


RADWASTE SOLIDIFICATION BLDG.
A-2301 452-6"
A-2301 (X-A29)
A-2301 (X-A28)
A-2301 (X-A16)
A-ZS 18605
A-2301 423-4"
A-2301 435-2 1/2"
A-2301 (X-A14)
A-2301 028
A-2301 433-2 1/2"
A-2301 434-2 1/2"
A-2301 424-4"
A-2301 432-2 1/2"
A-2301 035
A-2301 429-2 1/2"
A-2301 430-2 1/2"
A-2301 431-2 1/2"
HEALTH PHYSICS BUILDING
A-2301 425-4"
A-2301 (X-382)
A-2301 428-2 1/2"
A-2301 065
A-2301 (X-383)
A-2301 426-2 1/2"
A-2301 427-2 1/2"
A-2301 695-6"
SCS UNDERGROUND
BECHTEL SOLIDIF. BLDG.
CX4DB173-2
A-2301 (X-A30)
A-2301 (X-A31)
A-2301 (X-A32)
A-2301 (X-A33)
A-2301 (X-A34)
A-2301 (X-A35)
A-2301 (X-A36)
A-2301 (X-A37)
A-2301 (X-A38)
A-2301 (X-A39)
A-2301 (X-A40)
A-2301 (X-A41)
A-2301 (X-A42)
A-2301 (X-A43)
A-2301 (X-A44)
A-2301 (X-A45)
A-2301 (X-A46)
A-2301 (X-A47)
A-2301 (X-A48)
A-2301 (X-A49)
A-2301



NOTES:

1. FOR FIRE HOSE STATION LOCATIONS SEE ARCHITECTURAL DRAWINGS OF EACH FLOOR.
2. SPRINKLER LOCATIONS BY ARCHITECTURAL ROOM NUMBERS. UNIT 1 SHOWN.
3. TROUBLE SIGNAL TO FIRE SUPPRESSION LOCAL PANEL. SEE SPEC. X4AX03, ATTACHMENT 3, TABLE A.
4. VENDOR TO SUPPLY SPRINKLER VALVES & HEADS.
5. VENDOR TO CONNECT TO HEADERS INSTALLED BY PURCHASER. PURCHASER TO PROVIDE PIPE PENETRATION SLEEVES & HANGER EMBEDS.
6. FOR HOSE CABINET & SPRINKLER SYSTEM INFORMATION SEE DRAWING 1X4DB174-5.
7. SPRINKLER SYSTEM INSPECTORS TEST CONNECTIONS TO BE SUPPLIED BY VENDOR DETAIL DRAWINGS. TEST CONNECTIONS TO DRAIN DIRECTLY INTO FLOOR DRAINS.
8. THREADED BRANCH CONNECTIONS ARE REQUIRED ON HEADERS (UPSTREAM) NEAR INDICATED VALVES FOR DELUGE VALVE TRIM PIPING. FOR SIZE OF PRIMING LINE SEE V.P. AX4AX03-5001.

CRITICAL DOCUMENT

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
FIRE PROTECTION WATER SYSTEM
SYSTEM NO. 2301

CAD 1X4DB174-2
ACAD2007 LEH – U4

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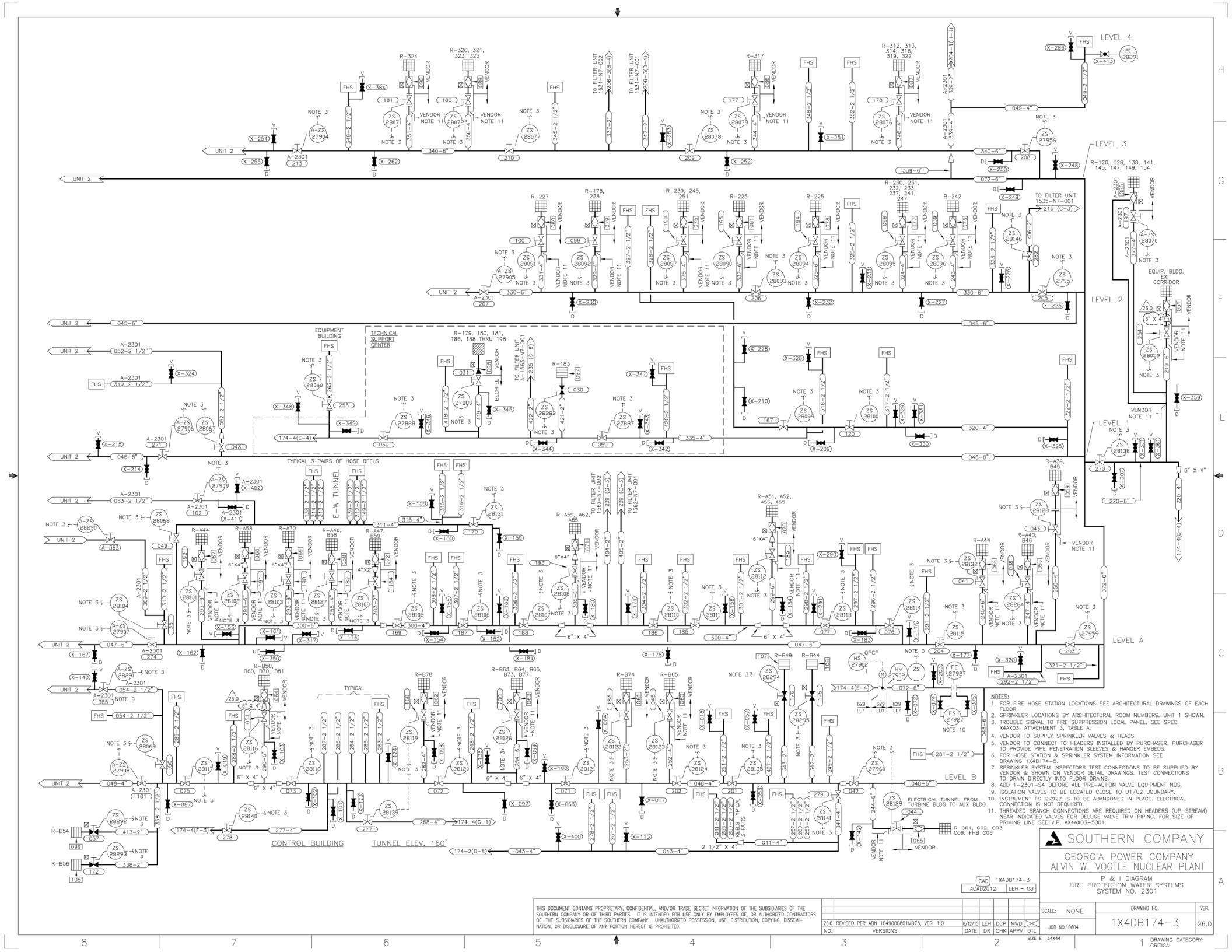
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
29.0	REVISED PER ABN 1049000801M081, VER. 1.0	6/12/15	LEH	DCP	MWD	

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB174-2
VER. 29.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. FOR FIRE HOSE STATION LOCATIONS SEE ARCHITECTURAL DRAWINGS OF EACH FLOOR.
2. SPRINKLER LOCATIONS BY ARCHITECTURAL ROOM NUMBERS. UNIT 1 SHOWN.
3. TROUBLE SIGNAL TO FIRE SUPPRESSION LOCAL PANEL. SEE SPEC. X4AX03, ATTACHMENT 3, TABLE A.
4. VENDOR TO SUPPLY SPRINKLER VALVES & HEADS.
5. VENDOR TO CONNECT TO HEADERS INSTALLED BY PURCHASER. PURCHASER TO PROVIDE PIPE PENETRATION SLEEVES & HANGER EMBEDS.
6. FOR HOSE STATION & SPRINKLER SYSTEM INFORMATION SEE DRAWING 1X48174-5.
7. SPRINKLER SYSTEM INSPECTORS TEST CONNECTIONS TO BE SUPPLIED BY VENDOR & SHOWN ON VENDOR DETAIL DRAWINGS. TEST CONNECTIONS TO DRAIN DIRECTLY INTO FLOOR DRAINS.
8. ADD 1-2301-S4 BEFORE ALL PRE-ACTION VALVE EQUIPMENT NOS.
9. ISOLATION VALVES TO BE LOCATED CLOSE TO U1/U2 BOUNDARY.
10. INSTRUMENT FS-27927 IS TO BE ABANDONED IN PLACE. ELECTRICAL CONNECTION IS NOT REQUIRED.
11. THREADED BRANCH CONNECTIONS ARE REQUIRED ON HEADERS (UP-STREAM) NEAR INDICATED VALVES FOR DELUGE VALVE TRIM PIPING. FOR SIZE OF PRIMING LINE SEE V.P. AX4AX03-5001.

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
26.0	REVISED PER ABN 1049000801M075, VER. 1.0	6/12/15	LEH	DCP	MWD	

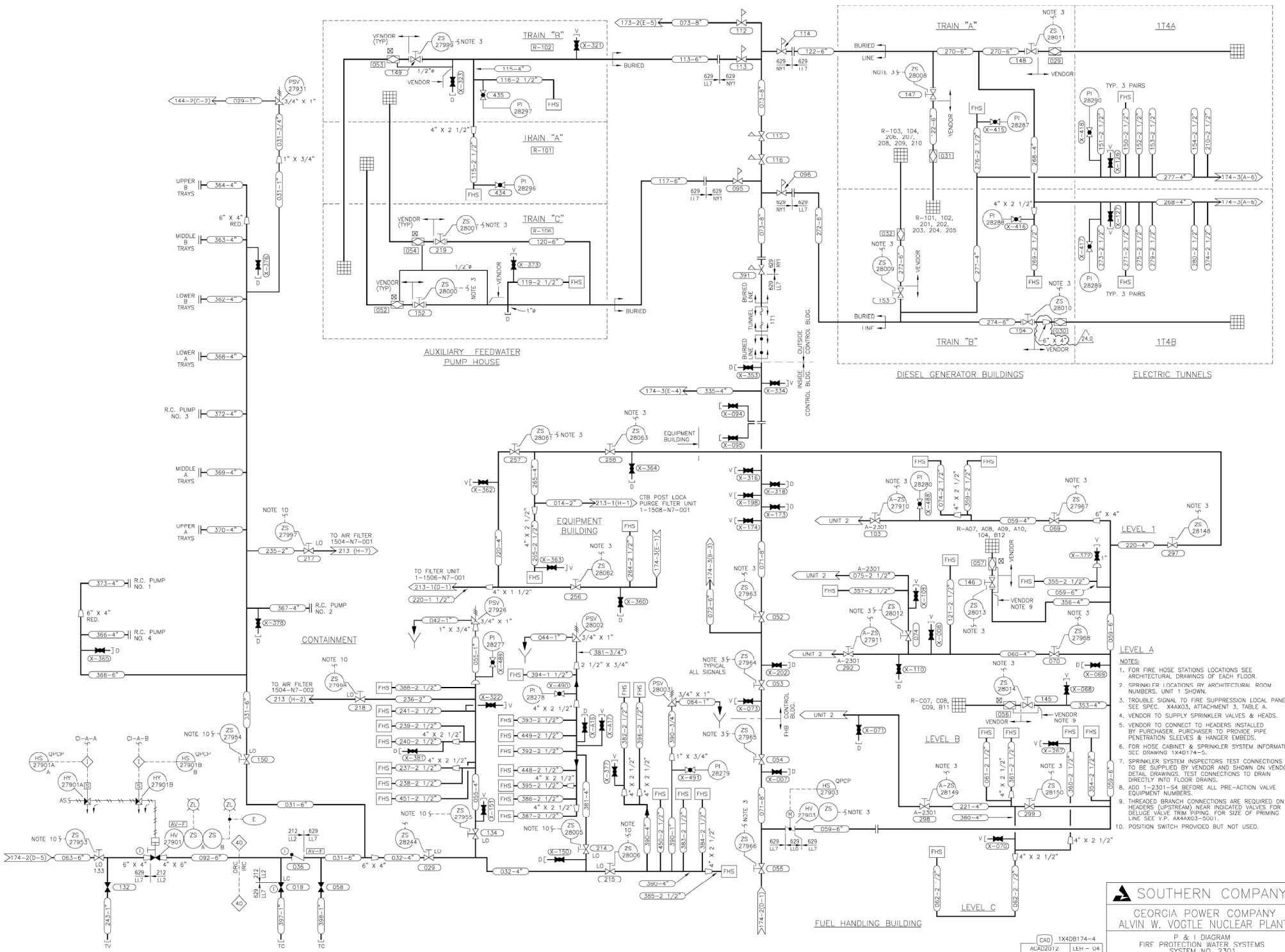
SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
FIRE PROTECTION WATER SYSTEMS
SYSTEM NO. 2301

CAD 1X4DB174-3
ACADZ012 LEH – 08

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB174-3
VER. 26.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
FIRE PROTECTION WATER SYSTEMS
SYSTEM NO. 2301

CAD 1X4DB174-4
ACADZ012 LEH – U4

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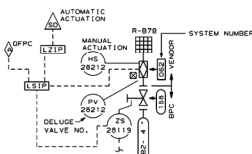
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
24.0	REVISED PER ABN 1049000801M076, VER. 1.0	6/12/15	LEH	DCP	MWD	

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB174-4
VER. 24.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL

SYSTEM NO.	BLDG	LEVEL	ZS NO.	HV NO.	VALVE NO.	LINE NO.	LINE SIZE	FIRE ZONE NUMBERS	DETECTOR TYPE	DELUGE VALVE PI NUMBER	ROOM NUMBERS	LSIP PANEL I.D. NUMBER
001 *	AB	D	28056	28151	220	096	4"	1,3	SD	28151	RD70,D78,D79,D100	1-2301-G3-F01
002	AB	D	28057	28152	222	095	2"	4	SD	28152	RD76	1-2301-G3-F01
003	AB	D	28056	28153	223	094	2"	1	SD	28153	RD74	1-2301-G3-F01
004	AB	D	28055	28154	224	093	2"	5	SD	28154	RD77	1-2301-G3-F02
005	AB	D	28034	28155	225	132	2"	9	SD	28155	RD49	1-2301-G3-F03
006	AB	D	28033	28156	226	131	2"	10	SD	28156	RD48	1-2301-G3-F03
007 *	AB	D/C	28032	28157	221	130	4"	11	SD	28157	RD101,C95,C99	1-2301-G3-F03
008	AB	C	28054	28158	227	140	2"	20	PE	28158	RC115	1-2301-G3-F04
009	AB	C	28053	28159	228	148	2"	21	PE	28159	RC111	1-2301-G3-F04
010	AB	C	28052	28160	229	147	2"	19	PE	28160	RC118	1-2301-G3-F05
011	AB	C	28051	28161	230	146	4"	14	SD	28161	RC113,C114	1-2301-G3-F06
013 *	AB	C/D	28031	28163	233	144	4"	17,12	SD	28163	RC88,D51,D121,D118	1-2301-G3-F07
014	AB	C	28030	28164	234	143	4"	14	SD	28164	RC119,C120	1-2301-G3-F08
015	AB	B	28048	28165	235	170	4"	30	SD	28165	RB824	1-2301-G3-F09
016	AB	B	28047	28166	241	175	2"	33	SD	28166	RB23	1-2301-G3-F09
017	AB	B	28046	28167	242	174	2"	32	SD	28167	RB15	1-2301-G3-F10
018	AB	B	28045	28168	243	173	2"	31	SD	28168	RB19	1-2301-G3-F10
019	AB	B	28044	28169	236	172	4"	34	SD	28169	RB17	1-2301-G3-F12
020 *	AB	B	28043	28170	237	171	4"	26	SD	28170	RB02,B04,B07,B08	1-2301-G3-F11
021 *	AB	B	28029	28171	240	169	6"	11	SD	28171	RB50	1-2301-G3-F12
022	AB	B	28028	28172	238	168	4"	171	SD	28172	RB11,B12,B13	1-2301-G3-F10
023	AB	A	28027	28173	244	167	4"	39	SD	28173	RA17	1-2301-G3-F15
024	AB	B/C	28026	28174	239	166	4"	25	SD	28174	RB31,C100	1-2301-G3-F12
025	AB	B/C	28049	28175	232	399	4"	23	SD	28175	RB10,C104	1-2301-G3-F06
026	AB	A/1	28147	28176	251	137	4"	33	SD	28176	RA15,111	1-2301-G3-F14
027 *	TUNNEL	—	28036	28177	253	085	6"	145	SD	28177	T5A	1-2301-G3-F41
028 *	TUNNEL	—	28035	28178	252	200	6"	146	SD	28178	T5B	1-2301-G3-F42
029 *	TUNNEL	—	28011	28179	148	270	6"	143	SD	28179	T14A	1-2301-G3-F43
030 *	TUNNEL	—	28010	28180	154	274	6"	144	SD	28180	T14B	1-2301-G3-F43
031 *	DGB	1	28008	28181	147	122	6"	162,164	IR	28181		1-2301-G3-F53
032 *	DGB	1	28009	28182	153	272	6"	161,163 (21.0)	IR	28182		1-2301-G3-F53
041 *	AB	1/2		28254	037	030	4"	17	SD	28254	R121,211,210,212	1-2301-G3-F23
042	AB	2		28255	126	242	4"	11,149	SD	28255	R207,208	1-2301-G3-F23
045	AB	A/1	28038	28195	249	083	4"	25	SD	28195	RA23,114	1-2301-G3-F17
046	AB	A	28039	28196	248	133	4"	11	SD	28196	RA49	1-2301-G3-F18
047 *	AB	A	28040	28197	247	134	4"	37	SD	28197	RA03	1-2301-G3-F16
048 *	AB	A	28041	28198	246	135	4"	36	SD	28198	RA05,A04	1-2301-G3-F15
049 *	AB	A	28042	28199	245	136	4"	39,38	SD	28199	RA19,A50,A40A	1-2301-G3-F14
050 *	AB	A/B	28037	28200	250	076	4"	17	SD	28200	RA51,B62,A53	1-2301-G3-F19
051 *	EB	1	28059	28201	254	219	6"	141	SD	28201	R118	1-2301-G3-F55
052	AFPH	1	28000	28202	152	117	6"	156	SD	28202		1-2301-G3-F54
053	AFPH	1	27999	28203	149	113	6"	155	SD	28203		1-2301-G3-F54
054	AFPH	1	28001	28204	219	120	6"	157	SD	28204		1-2301-G3-F54
NOTE 5 A055 *	CB	1	28070	28205	197	377	4"	118,109	SD	28205	R120,141,146,128,149, 154,145,138	1-2301-G3-F34
056 *	FHB	B/C	28014	28206	145	353	4"	15	SD	28206	RC07,C08,C09,B11	1-2301-G3-F45
057 *	FHB	1/A/B	28013	28207	146	356	4"	29	SD	28207	R104,A07,A08,A09,A10,B12	1-2301-G3-F46
058 *	CB	B/A	28127	28208	182	255	4"	153,174	SD	28208	RB58,A46	1-2301-G3-F30
059 *	CB	B/A	28128	28209	043	250	4"	83,175	SD	28209	RB45,A39	1-2301-G3-F29
060 *	CB	B	28123	28210	045	252	4"	52	SD	28210	RB60	1-2301-G3-F24
061	CB	B	28122	28211	183	253	2"	60	SD	28211	RB74	1-2301-G3-F25
062 *	CB	B	28119	28212	188	262	4"	61	SD	28212	RB78	1-2301-G3-F26
063 *	CB	B	28114	28213	200	264	6"	65,66,67	SD	28213	RB63,B73,B77	1-2301-G3-F27
064 *	CB	B	28116	28214	061	390	6"	73,72,80	SD	28214	RB60,B70,B80,B81	1-2301-G3-F27
065 *	CB,FHB	C	28129	28215	044	244	6"	42	SD	28215	CB RC01,C02,C03,C09,FHB RC06	1-2301-G3-F28
066 *	CB	A	28132	28216	041	245	6"	95	SD	28216	RA44	1-2301-G3-F31
067 *	CB	A	28101	28217	192	295	6"	95	SD	28217	RA44	1-2301-G3-F30
068 *	CB	A	28102	28218	191	294	6"	85,101	SD	28218	RA58	1-2301-G3-F31
069 *	CB	A	28103	28219	190	293	6"	84	SD	28219	RA70	1-2301-G3-F30
070 *	CB	A	28112	28220	189	299	6"	86,93,102	SD	28220	RA51,A52,A53,A55	1-2301-G3-F32
071 *	CB	A	28108	28221	193	305	6"	89	SD	28221	RA59,A62,A65	1-2301-G3-F33
072 *	CB	A/B	28109	28222	184	303	6"	151,173	SD	28222	RA47,B59	1-2301-G3-F30
075	CB	2	28097	28225	199	375	4"	133,182,127	SD	28225	R239,245,251	1-2301-G3-F35
076 *	CB	2	28096	28226	039	246	4"	177	SD	28226	R242	1-2301-G3-F37
077 *	CB	2	28095	28227	098	324	4"	176	SD	28227	R241,230,231,232,233, 237,247	1-2301-G3-F37
078 *	CB	2	28094	28228	194	326	6"	120	SD	28228	R225	1-2301-G3-F35
079	CB	2	28092	28229	099	329	4"	107	SD	28229	R228,178	1-2301-G3-F36
080	CB	2	28091	28230	100	331	4"	108	SD	28230	R227	1-2301-G3-F36
081 *	CB	2	28090	28231	195	332	6"	120	SD	28231	R225	1-2301-G3-F35
083	AB	2		28256	164	414	4"	55	SD	28256	R202	1-2301-G3-F23
085	AB	2		28257	127	389	4"	54	SD	28257	R203	1-2301-G3-F23
086 *	CB	3	28079	28235	177	344	4"	181	SD	28235	R217	1-2301-G3-F38
087 *	CB	3	28076	28236	178	346	4"	179	SD	28236	R312,313,314,316,319,322	1-2301-G3-F40
088	AB	2		28258	158	391	4"	23	SD	28258	R205	1-2301-G3-F23
089 *	CB	3	28072	28237	180	350	4"	178	SD	28237	R323,320,321,325	1-2301-G3-F39
090	CB	3	28071	28238	181	351	4"	180	SD	28238	R324	1-2301-G3-F38
091	AB	1	28245	28259	265	182	4"	44,45,48	SD	28259	R110	1-2301-G3-F20
092	AB	1	28246	28260	266	185	4"	49	SD	28260	R105	1-2301-G3-F20
093	AB	1	28247	28261	267	086	4"	43,46	SD	28261	R116,122,136	1-2301-G3-F21
094	AB	1	28248	28262	268	232	4"	52	SD	28262	R104	1-2301-G3-F21
095 *	AB	A/1	28263	28265	056	415	4"	39,45	SD	28265	RA109,108	1-2301-G3-F22
096 *	CB	A/B	28264	28266	038	247	4"	97,152	SD	28266	RB46,A40	1-2301-G3-F29
NOTE 5 A201 *	RSB	192'	18027	18038	012	436	6"	307	SD	18038	RA03,RA04,RA05,RA06	1-2301-G3-F60
A202 *	RSB	220'	18028	18033	020	437	6"	308,312,313	SD	18033	R152,R154,R155,R156, R166,R153,RA17,RA20	1-2301-G3-F61
A203 *	RSB	246'	18029	18034	032	438	6"	317,321, 322,323	SD	18034	R202,R203,R204,R205, R206,R217,RM03	1-2301-G3-F62
A204	RSB	192'	18030	18035	350	445	2 1/2"	305	EXP,HD	18035	RA09	1-2301-G3-F60
A205	RSB	192'	18031	18037	351	446	2 1/2"	306	EXP,HD	18037	RA14	1-2301-G3-F60
206	ARB	228'	—	—	C-14	453	4"	330	SD	16794	R101	—

PRE-ACTION SPRINKLER SYSTEM DATA



TYPICAL PRE-ACTION SPRINKLER SYSTEM
SYSTEM 1-2301-S4-062 SHOWN.
FOR ADDITIONAL DETAILS OF PRE-ACTION SYSTEM AND OTHER SYSTEMS DETAILS (MANUAL & WET PIPE) REFER TO V.P. AX44AX03-05352 AND AX44AX03-05353.

LEGEND:

1. BUILDINGS
   - AB- AUXILIARY BUILDING
   - CB- CONTROL BUILDING
   - RSB- RAD WASTE HANDLING BUILDING
   - DGB- DIESEL GENERATOR BUILDING
   - EB- ENCLOSURE BUILDING
   - CNT- CONTAINMENT BUILDING
   - AFH- AUXILIARY FEEDWATER PUMP HOUSE
   - RTB- RADWASTE TRANSFER BUILDING
   - RSB- RADWASTE SOLIDIFICATION BUILDING
2. FIRE ZONE NUMBERS - PER FIRE HAZARDS ANALYSIS
3. DETECTORS
   - TH- THERMAL ELECTRIC LINE-TYPE DETECTOR, INSTALLED IN CABLE TRAYS
   - HD- FIXED TEMPERATURE/ RATE-OF-RISE HEAT DETECTOR
   - SD- IONIZATION SMOKE DETECTOR
   - IR- INFRA-RED DETECTOR
   - PE- PHOTO-ELECTRIC SMOKE DETECTOR
   - EXP- EXPLOSIVE GAS LEVEL DETECTORS
4. ROOM NUMBERS - PER ARCHITECTURAL DRAWINGS, UNIT 1 ROOM NUMBERS SHOWN
5. AN 'A' PRECEDING THE SYSTEM NO. MEANS THE SYSTEM IS COMMON TO UNITS 1&2 AND THE COMPLETE SYSTEM NO IS A-2301-S4-XXX
6. LSIP- LOCAL FIRE SUPPRESSION INDICATING PANEL
   LZIP- LOCAL FIRE ZONE INDICATING PANEL, REFER TO SPEC. X3AQ01 FOR DETAILS.
   GFPC- FIRE PROTECTION CONSOLE, REFER TO SPEC. X3AQ09 FOR DETAILS.
7. ASTERISK (*) INDICATES SERVICE AIR SUPPLY IS REQUIRED FOR SUPERVISION OF PRE-ACTION SYSTEMS IN ACCORDANCE WITH NFPA 13. CONNECTION DETAILS ARE INDICATED ON VENDOR DRAWINGS AX44AX03-05352 AND AX44AX03-05353.

FIRE HOSE STATION DATA

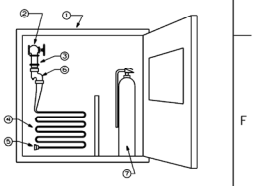
CABINET TAG NUMBER	LENGTH	AREA LOC.	LINE NO. SIZE	VALVE ASSY. TAG NO.
EQUIPMENT BLDG FHS				
1-2301-R4-022	100'	14E1	263-2 1/2"	A01
1-2301-R4-023	100'	14H1	264-2 1/2"	A02
1-2301-R4-024	100'	14G1	265-2 1/2"	A03
AUXILIARY BLDG LEVEL B FHS				
1-2301-R4-112	100'	13HB	189-2 1/2"	A05
1-2301-R4-113	100'	13HB	164-2 1/2"	A06
1-2301-R4-114	100'	13GB	188-2 1/2"	A07
1-2301-R4-115	100'	13HB	191-2 1/2"	A08
1-2301-R4-116	100'	13GB	186-2 1/2"	A09
1-2301-R4-117	100'	13CB	187-2 1/2"	A10
1-2301-R4-118	100'	13BB	194-2 1/2"	A11
1-2301-R4-120	100'	13EB	193-2 1/2"	A12
1-2301-R4-121	100'	13FB	190-2 1/2"	A13
A-2301-R4-121	100'	23BB	192-2 1/2"	A14
1-2301-R4-170	100'	13HB	165-2 1/2"	A15
1-2301-R4-171	100'	13FB	163-2 1/2"	A16
AUXILIARY BLDG LEVEL C FHS				
1-2301-R4-105	100'	13HC	156-2 1/2"	A17
1-2301-R4-106	100'	13HC	157-2 1/2"	A18
1-2301-R4-107	100'	13CC	158-2 1/2"	A19
1-2301-R4-108	100'	13GC	159-2 1/2"	A20
1-2301-R4-109	100'	13HC	160-2 1/2"	A21
1-2301-R4-110	100'	13EC	161-2 1/2"	A22
A-2301-R4-111	100'	23BC	162-2 1/2"	A23
1-2301-R4-167	100'	13GC	141-2 1/2"	A24
1-2301-R4-168	100'	13FC	142-2 1/2"	A25
1-2301-R4-169	100'	13BC	155-2 1/2"	A26
A-2301-R4-021	100'	13D3	106-2 1/2"	A27
AUXILIARY BLDG LEVEL D FHS				
1-2301-R4-163	100'	13FD	123-2 1/2"	A28
1-2301-R4-164	100'	13GD	124-2 1/2"	A29
1-2301-R4-165	100'	13BD	125-2 1/2"	A30
1-2301-R4-098	100'	13HD	098-2 1/2"	A31
1-2301-R4-099	100'	13GD	099-2 1/2"	A32
1-2301-R4-100	100'	13HD	126-2 1/2"	A33
1-2301-R4-101	100'	13FD	127-2 1/2"	A34
1-2301-R4-102	100'	13CD	128-2 1/2"	A35
1-2301-R4-103	100'	13ED	129-2 1/2"	A36
A-2301-R4-104	100'	23BD	262-2 1/2"	A37
AUXILIARY BLDG LEVEL 2 FHS				
1-2301-R4-156	100'	13F2	223-2 1/2"	A39
1-2301-R4-134	100'	13C2	224-2 1/2"	A40
1-2301-R4-135	100'	13B2	225-2 1/2"	A41
1-2301-R4-136	100'	13E2	226-2 1/2"	A42
1-2301-R4-137	100'	13H2	227-2 1/2"	A43
1-2301-R4-138	100'	13H2	228-2 1/2"	A44
1-2301-R4-139	100'	13B2	229-2 1/2"	A45
1-2301-R4-157	100'	13F2	230-2 1/2"	A46
AUXILIARY BLDG LEVEL 1 FHS				
A-2301-R4-160	100'	13B1	233-2 1/2"	A47
1-2301-R4-159	100'	13G1	211-2 1/2"	A48
1-2301-R4-158	100'	13B1	212-2 1/2"	A49
1-2301-R4-128	100'	13F1	213-2 1/2"	A50
1-2301-R4-129	100'	13E1	214-2 1/2"	A51
1-2301-R4-130	100'	13C1	215-2 1/2"	A52
1-2301-R4-131	100'	13H1	216-2 1/2"	A53
1-2301-R4-132	100'	13H1	217-2 1/2"	A54
1-2301-R4-133	100'	13G1	218-2 1/2"	A55
AUXILIARY BLDG LEVEL A FHS				
A-2301-R4-162	100'	23BA	203-2 1/2"	A56
1-2301-R4-122	100'	13FA	204-2 1/2"	A57
1-2301-R4-123	100'	13GA	205-2 1/2"	A58
1-2301-R4-124	100'	13GA	206-2 1/2"	A59
1-2301-R4-125	100'	13GA	207-2 1/2"	A60
1-2301-R4-126	100'	13HA	208-2 1/2"	A61
1-2301-R4-161	100'	13FA	209-2 1/2"	A62
1-2301-R4-127	100'	13CA	201-2 1/2"	A63
TUNNELS T15A & T15B FHS				
1-2301-R4-140	100'	7E	084-2 1/2"	A64
1-2301-R4-141	100'	7E	181-2 1/2"	A65
1-2301-R4-143	100'	7E	183-2 1/2"	A66
1-2301-R4-144	100'	7E	184-2 1/2"	A67
1-2301-R4-146	100'	7F	176-2 1/2"	A68
1-2301-R4-147	100'	7F	177-2 1/2"	A69
1-2301-R4-148	100'	7F	178-2 1/2"	A70
1-2301-R4-149	100'	7F	179-2 1/2"	A71
1-2301-R4-150	100'	7F	180-2 1/2"	A72
1-2301-R4-151	100'	7F	199-2 1/2"	A73
E-W TUNNEL FHS CONTROL BLDG.				
1-2301-R4-045	100'	12JA	315-2 1/2"	A74
1-2301-R4-046	100'	12JA	316-2 1/2"	A75
1-2301-R4-093	100'	12JA	314-2 1/2"	A76
1-2301-R4-094	100'	12JA	136-2 1/2"	A77
1-2301-R4-095	100'	12JA	313-2 1/2"	A78
1-2301-R4-096	100'	12JA	139-2 1/2"	A79
1-2301-R4-097	100'	12JA	312-2 1/2"	A80
1-2301-R4-166	100'	12JA	149-2 1/2"	A81
CONTROL BLDG LEVEL 1 FHS				
1-2301-R4-073	100'	12B1	322-2 1/2"	A82
1-2301-R4-074	100'	12B1	319-2 1/2"	A83
A-2301-R4-075	100'	22D1	114-2 1/2"	A84
1-2301-R4-078	100'	12A1	318-2 1/2"	A87
1-2301-R4-079	100'	12B1	317-2 1/2"	A88
RADWASTE TRANSFER BLDG EL 220' FHS				
A-2301-R4-020	100'	3J	410-2 1/2"	A91
A-2301-R4-186	100'	3H	411-2 1/2"	C09
RADWASTE TRANSFER BLDG EL 247' FHS				
A-2301-R4-018	100'	3J	408-2 1/2"	A89
A-2301-R4-019	100'	3K	409-2 1/2"	A90
RADWASTE SOLIDIFICATION BLDG EL 192' FHS				
A-2301-R4-187	100'	3P	426-2 1/2"	B88
A-2301-R4-188	100'	3M	427-2 1/2"	C10
A-2301-R4-189	100'	3N	428-2 1/2"	C11
RADWASTE SOLIDIFICATION BLDG EL 220' FHS				
A-2301-R4-190	100'	3P	429-2 1/2"	C12
A-2301-R4-191	100'	3M	430-2 1/2"	C13
A-2301-R4-192	100'	3N	431-2 1/2"	B90
A-2301-R4-193	100'	HPB	432-2 1/2"	B91
RADWASTE SOLIDIFICATION BLDG EL 246' FHS				
A-2301-R4-194	100'	3P	433-2 1/2"	B92
A-2301-R4-195	100'	3M	434-2 1/2"	B93
A-2301-R4-196	100'	3N	435-2 1/2"	B94
RADWASTE SOLIDIFICATION BLDG EL 247' FHS				
A-2301-R4-199	100'	3P	447-2 1/2"	B97

CABINET TAG NUMBER	LENGTH	AREA LOC.	LINE NO. SIZE	VALVE ASSY. TAG NO.
CONTROL BLDG LEVEL 2 FHS				
A-2301-R4-080	100'	22D2	333-2 1/2"	A92
1-2301-R4-081	100'	12B2	323-2 1/2"	A93
1-2301-R4-083	100'	12A2	328-2 1/2"	A95
1-2301-R4-084	100'	12A2	327-2 1/2"	A96
1-2301-R4-085	100'	12B2	325-2 1/2"	A97
CONTROL BLDG LEVEL 3 FHS				
A-2301-R4-086	100'	22B3	336-2 1/2"	A98
1-2301-R4-088	100'	12B3	352-2 1/2"	B01
1-2301-R4-089	100'	12B3	346-2 1/2"	B02
1-2301-R4-090	100'	12A3	345-2 1/2"	B03
1-2301-R4-091	100'	12A3	349-2 1/2"	B05
CONTROL BLDG LEVEL 4 FHS				
1-2301-R4-092	100'	12B4	049-2 1/2"	B04
CONTROL BLDG LEVEL A FHS				
1-2301-R4-056	100'	12BA	291-2 1/2"	B06
1-2301-R4-057	100'	12AA	296-2 1/2"	B07
1-2301-R4-058	100'	12DA	297-2 1/2"	B08
1-2301-R4-059	100'	12CA	298-2 1/2"	B09
1-2301-R4-060	100'	12DA	301-2 1/2"	B10
1-2301-R4-061	100'	12FA	302-2 1/2"	B11
1-2301-R4-062	100'	12FA	304-2 1/2"	B12
1-2301-R4-063	100'	12FA	306-2 1/2"	B13
1-2301-R4-064	100'	12EA	307-2 1/2"	B14
1-2301-R4-065	100'	12AA	308-2 1/2"	B15
1-2301-R4-066	100'	12FC	142-2 1/2"	B16
1-2301-R4-067	100'	12AA	310-2 1/2"	B17
1-2301-R4-068	100'	12BA	281-2 1/2"	B18
CONTROL BLDG LEVEL B FHS				
1-2301-R4-047	100'	12CB	407-2 1/2"	B19
1-2301-R4-048	100'	12GB	283-2 1/2"	B20
1-2301-R4-049	100'	12FB	284-2 1/2"	B21
1-2301-R4-050	100'	12FB	285-2 1/2"	B22
1-2301-R4-051	100'	12FB	286-2 1/2"	B23
1-2301-R4-052	100'	12FB	287-2 1/2"	B24
1-2301-R4-053	100'	12DB	289-2 1/2"	B25
1-2301-R4-054	100'	12BB	249-2 1/2"	B26
1-2301-R4-055	100'	12DB	251-2 1/2"	B27
1-2301-R4-069	100'	12AB	054-2 1/2"	B28
1-2301-R4-070	100'	12FB	288-2 1/2"	B29
1-2301-R4-071	100'	12FB	248-2 1/2"	B30
1-2301-R4-072	100'	12BB	281-2 1/2"	B31
DIESEL BLDG FHS				
A-2301-R4-172	100'	7D	276-2 1/2"	B32
1-2301-R4-173	100'	7D	269-2 1/2"	B33
AUX FEEDWATER PUMP HOUSE FHS				
A-2301-R4-152	100'	7C	116-2 1/2"	B34
1-2301-R4-153	100'	7C	119-2 1/2"	B35
1-2301-R4-154	100'	7C	115-2 1/2"	B36
CONTAINMENT FHS				
A-2301-R4-001	100'	14DC	237-2 1/2"	B37
1-2301-R4-002	100'	14DC	238-2 1/2"	B38
1-2301-R4-003	100'	14AA	239-2 1/2"	B39
1-2301-R4-004	100'	14A1	240-2 1/2"	B40
1-2301-R4-005	100'	14D2	241-2 1/2"	B41
1-2301-R4-006	100'	14D3	388-2 1/2"	B42
1-2301-R4-007	100'	14BC	393-2 1/2"	B43
1-2301-R4-008	100'	14BA	392-2 1/2"	B44
1-2301-R4-009	100'	14B1	393-2 1/2"	B45
1-2301-R4-010	100'	14B2	394-2 1/2"	B46
1-2301-R4-011	100'	14CC	396-2 1/2"	B47
1-2301-R4-012	100'	14CA	382-2 1/2"	B48
1-2301-R4-013	100'	14C1	383-2 1/2"	B49
1-2301-R4-014	100'	14C2	384-2 1/2"	B50
1-2301-R4-015	100'	14C3	385-2 1/2"	B51
1-2301-R4-016	100'	14BC	386-2 1/2"	B52
1-2301-R4-017	100'	14AC	387-2 1/2"	B53
1-2301-R4-201	100'	14AA	448-2 1/2"	B86
1-2301-R4-202	100'	14BB	449-2 1/2"	B87
1-2301-R4-203	100'	14CA	450-2 1/2"	B88
1-2301-R4-204	100'	14DA	451-2 1/2"	B89
TUNNELS T14A & T14B FHS				
1-2301-R4-174	100'	7C	271-2 1/2"	B54
1-2301-R4-175	100'	7C	273-2 1/2"	B55
1-2301-R4-176	100'	7C	275-2 1/2"	B56
1-2301-R4-177	100'	7C	279-2 1/2"	B57
1-2301-R4-178	100'	7C	280-2 1/2"	B58
1-2301-R4-179	100'	7C	374-2 1/2"	B59
1-2301-R4-180	100'	7C	150-2 1/2"	B60
1-2301-R4-181	100'	7C	151-2 1/2"	B61
1-2301-R4-182	100'	7C	152-2 1/2"	B62
1-2301-R4-183	100'	7C	153-2 1/2"	B63
1-2301-R4-184	100'	7C	154-2 1/2"	B64
1-2301-R4-185	100'	7C	210-2 1/2"	B65
FUEL HANDLING BLDG FHS				
1-2301-R4-034	100'	13DC	062-2 1/2"	B66
1-2301-R4-035	100'	13AB	061-2 1/2"	B67
1-2301-R4-036	100'	13AB	354-2 1/2"	B68
1-2301-R4-037	100'	13DA	355-2 1/2"	B69
1-2301-R4-038	100'	13AA	121-2 1/2"	B70
1-2301-R4-039	100'	13AA	357-2 1/2"	B71
1-2301-R4-040	100'	13AB	361-2 1/2"	B72
1-2301-R4-041	100'	13DB	360-2 1/2"	B73
1-2301-R4-042	100'	13A1	359-2 1/2"	B74
1-2301-R4-043	100'	13A1	074-2 1/2"	B75
N-S-ELECTRICAL TUNNEL FHS CONTROL BLDG.				
1-2301-R4-026	100'	13AC	261-2 1/2"	B77
1-2301-R4-027	100'	13AC	278-2 1/2"	B78
1-2301-R4-028	100'	13DC	260 411-2 1/2"	C09
1-2301-R4-029	100'	12BC	258-2 1/2"	B80
1-2301-R4-030	100'	12AC	041-2 1/2"	B81
1-2301-R4-031	100'	12HC	256-2 1/2"	B82
1-2301-R4-032	100'	12AC	259-2 1/2"	B83
1-2301-R4-033	100'	12AC	257-2 1/2"	B84
TECHNICAL SUPPORT CENTER FHS				
A-2301-R4-197	100'	12D1	418-2 1/2"	B95
A-2301-R4-198	100'	12D1	420-2 1/2"	B96
PLANT SECURITY ENTRANCE BUILDING FHS				
A-2301-R4-680	100'	17L1	454-2 1/2"	C16
A-2301-R4-681	100'	17L1	642-2 1/2"	C17

SEE NOTE 1

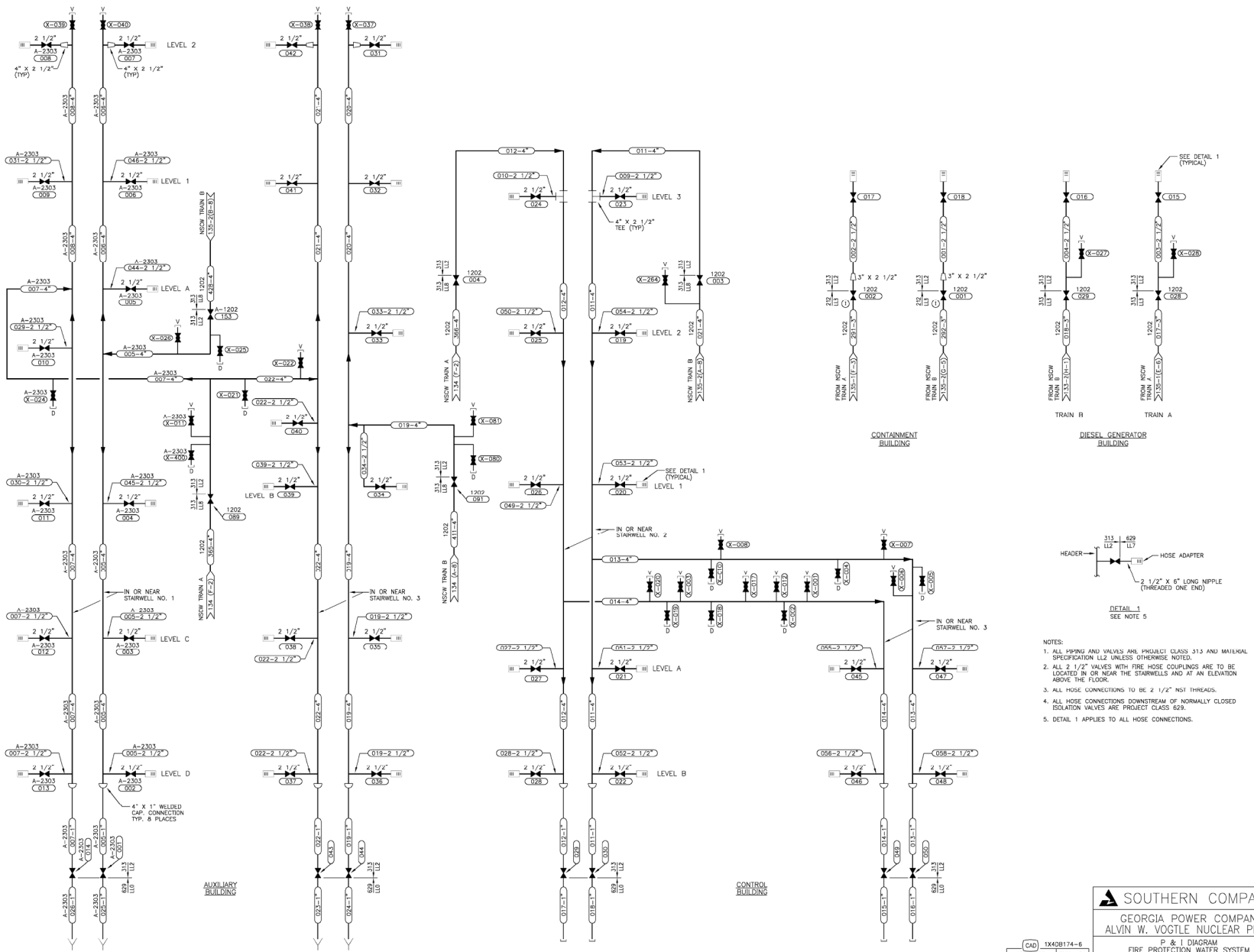
FIRE HOSE STATION DETAIL

SCALE: NONE



SCHEDULE OF MATERIALS

1. LARSEN HOSE CABINET (SURFACE MOUNTED) -HCS-3238-SM. LARSEN FIRE HOSE CABINET WITH STYLE H DUO PANEL DOOR AND DSA GLASS (OR PLEXIGLASS)WITH BRACKETS TO HOLD HOSE IN FLAT FOLD CONFIGURATION WITH 12 SPACE FOR FIRE EXTINGUISHER.
2. 1 1/2" ANGLE VALVE-ELKHART MODEL U-25.
3. DELETED
4. 1 1/2" X 100'-0" ANGUS REDSKIN FIRE HOSE WITH BARLITE ROCKER LUG STYLE COUPLINGS.
5. 1 1/2" ADJUSTABLE FOG NOZZLE-ELKHART MODEL "LE" SELECT-O-STREAM.
6. 1 1/2" DRAIN VENT VALVE-POHATAN MODEL



DETAIL 1
SEE NOTE 5

NOTES:

1. ALL PIPING AND VALVES ARE PROJECT CLASS 313 AND MATERIAL SPECIFICATION LL2 UNLESS OTHERWISE NOTED.
2. ALL 2 1/2" VALVES WITH FIRE HOSE COUPLINGS ARE TO BE LOCATED IN OR NEAR THE STAIRWELLS AND AT AN ELEVATION ABOVE THE FLOOR.
3. ALL HOSE CONNECTIONS TO BE 2 1/2" NST THREADS.
4. ALL HOSE CONNECTIONS DOWNSTREAM OF NORMALLY CLOSED ISOLATION VALVES ARE PROJECT CLASS 629.
5. DETAIL 1 APPLIES TO ALL HOSE CONNECTIONS.

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NO.		DATE	DR	CHK	APPV	DTL
15.0	REDRAWN PER ABN-52274, VER. 1.0	2-24-06	CYN	LCF	SHS	X

CAD 1X4DB174-6
ACAD2002 CYN- 01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
FIRE PROTECTION WATER SYSTEM
SYSTEM NO. 2303

SCALE: NONE

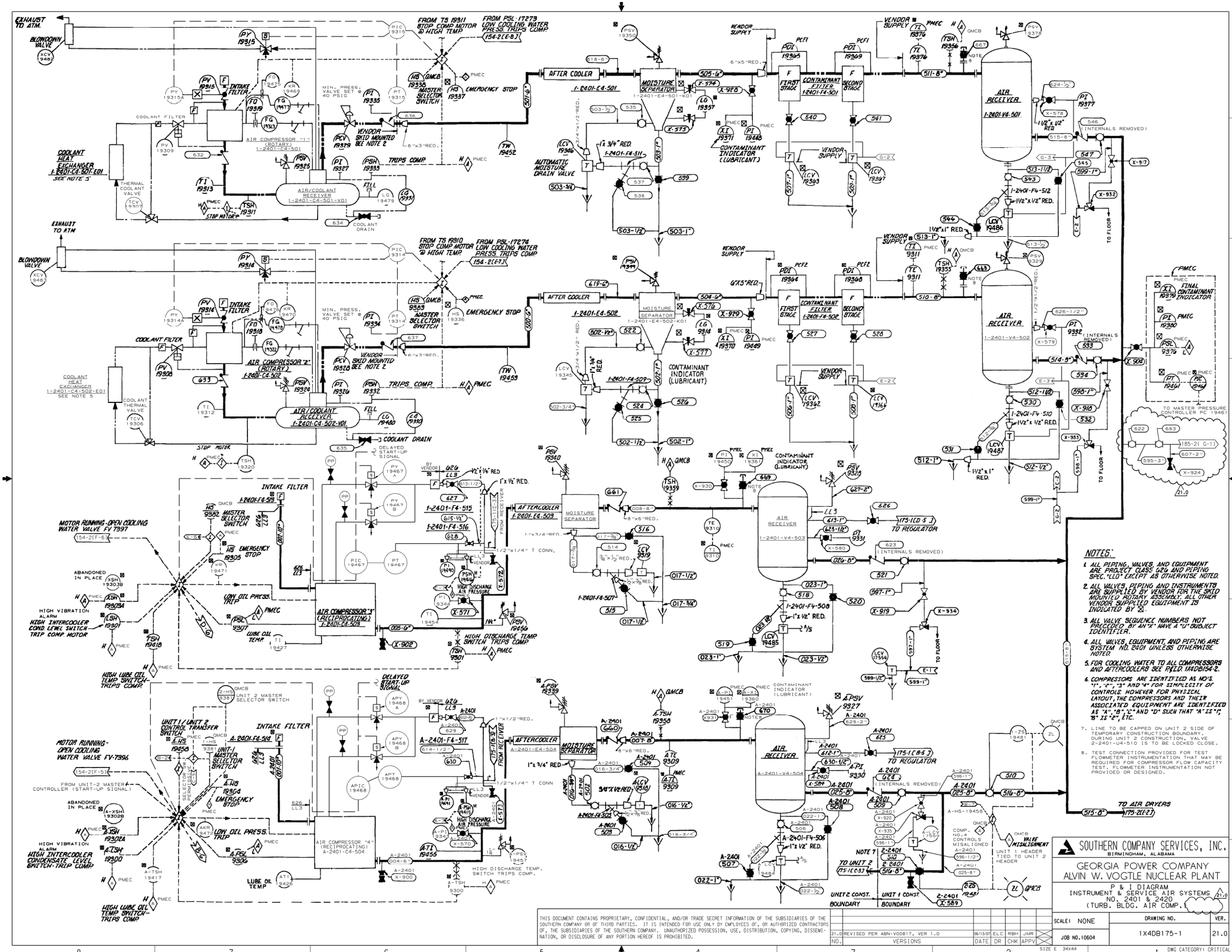
DRAWING NO. 1X4DB174-6

VER. 15.0

JOB NO.10604

SIZE E 34X44

1 DRAWING CATEGORY: CRITICAL


EXHAUST TO ATM.
BLOWDOWN VALVE
XCV 9482
FROM TS 19311 STOP COMP MOTOR @ HIGH TEMP.
FROM PSL-17273 LOW COOLING WATER PRESS. TRIPS COMP.
154-2(E-8)
INTAKE FILTER
COOLANT FILTER
AIR COMPRESSOR "1" (ROTARY) 1-2401-C4-501
MIN. PRESS. VALVE SET @ 40 PSIG
MASTER SELECTOR SWITCH
EMERGENCY STOP
VENDOR SKID MOUNTED SEE NOTE 2
TRIPS COMP
COOLANT HEAT EXCHANGER 1-2401-C4-501-E01 SEE NOTE 5
THERMAL COOLANT VALVE
AIR/COOLANT RECEIVER 1-2401-C4-501-V01
COOLANT DRAIN
STOP MOTOR
AFTER COOLER 1-2401-E4-501
MOISTURE SEPARATOR 1-2401-E4-501-K01
CONTAMINANT FILTER 1-2401-F4-501
FIRST STAGE
SECOND STAGE
CONTAMINANT INDICATOR (LUBRICANT)
AUTOMATIC MOISTURE DRAIN VALVE
AIR RECEIVER 1-2401-V4-501
VENDOR SUPPLY
TO FLOOR
EXHAUST TO ATM
BLOWDOWN VALVE
XCV 9483
FROM TS 19310 STOP COMP MOTOR @ HIGH TEMP.
FROM PSL-17274 LOW COOLING WATER PRESS. TRIPS COMP.
154-2(F-7)
AIR COMPRESSOR "2" (ROTARY) 1-2401-C4-502
COOLANT HEAT EXCHANGER 1-2401-C4-502-E01 SEE NOTE 5
AIR/COOLANT RECEIVER 1-2401-C4-502-V01
AFTER COOLER 1-2401-E4-502
MOISTURE SEPARATOR 1-2401-E4-502-K01
CONTAMINANT FILTER 1-2401-F4-502
AIR RECEIVER 1-2401-V4-502
FINAL CONTAMINANT INDICATOR
TO MASTER PRESSURE CONTROLLER PC 19461
MOTOR RUNNING-OPEN COOLING WATER VALVE FV 7397
HIGH VIBRATION ALARM
HIGH INTERCOOLER COND. LEVEL SWITCH TRIP COMP. MOTOR
ABANDONED IN PLACE
LOW OIL PRESS. TRIP
LUBE OIL TEMP
AIR COMPRESSOR "3" (RECIPROCATING) 1-2401-E4-503
HIGH DISCHARGE TEMP. SWITCH TRIPS COMP.
HIGH LUBE OIL TEMP. SWITCH-TRIPS COMP.
AFTERCOOLER 1-2401-E4-503
MOISTURE SEPARATOR
AIR RECEIVER 1-2401-V4-503
TO REGULATOR
UNIT 1 / UNIT 2 CONTROL TRANSFER SWITCH
UNIT-1 MASTER SELECTOR SWITCH
EMERGENCY STOP
FROM UNIT-2 MASTER CONTROLLER (START-UP SIGNAL)
HIGH INTERCOOLER CONDENSATE LEVEL SWITCH-TRIP COMP.
AIR COMPRESSOR "4" (RECIPROCATING) A-2401-C4-504
AFTERCOOLER A-2401-E4-504
MOISTURE SEPARATOR
AIR RECEIVER A-2401-V4-504
TO AIR DRYERS
UNIT 1 HEADER TIED TO UNIT 2 HEADER
VALVE MISALIGNMENT
UNIT 2 CONST. BOUNDARY
UNIT 1 CONST. BOUNDARY
NOTES:
1. ALL PIPING, VALVES, AND EQUIPMENT ARE PROJECT CLASS G2G AND PIPING SPEC. "LLO" EXCEPT AS OTHERWISE NOTED.
2. ALL VALVES, PIPING AND INSTRUMENTS ARE SUPPLIED BY VENDOR FOR THE SKID MOUNTED ROTARY SCREW/AIR, ALL OTHER VENDOR SUPPLIED EQUIPMENT IS INDICATED BY ☒.
3. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" HAVE A "U" SUBJECT IDENTIFIER.
4. ALL VALVES, EQUIPMENT, AND PIPING ARE SYSTEM NO. 2401 UNLESS OTHERWISE NOTED.
5. FOR COOLING WATER TO ALL COMPRESSORS AND AFTERCOOLERS SEE FIELD. 1X4DB154-2.
6. COMPRESSORS ARE IDENTIFIED AS NO'S. "1", "2", "3" AND "4" FOR SIMPLICITY OF CONTROLS. HOWEVER FOR PHYSICAL LAYOUT, THE COMPRESSORS AND THEIR ASSOCIATED EQUIPMENT ARE IDENTIFIED AS "A", "B", "C" AND "D" SUCH THAT "A" IS "1", "B" IS "2", ETC.
7. LINE TO BE CAPPED ON UNIT 2 SIDE OF TEMPORARY CONSTRUCTION BOUNDARY. DURING UNIT 2 CONSTRUCTION, VALVE 2-2401-U4-510 IS TO BE LOCKED CLOSE.
8. TEST CONNECTION PROVIDED FOR TEST FLOWMETER INSTRUMENTATION THAT MAY BE REQUIRED FOR COMPRESSOR FLOW CAPACITY TEST. FLOWMETER INSTRUMENTATION NOT PROVIDED OR DESIGNED.
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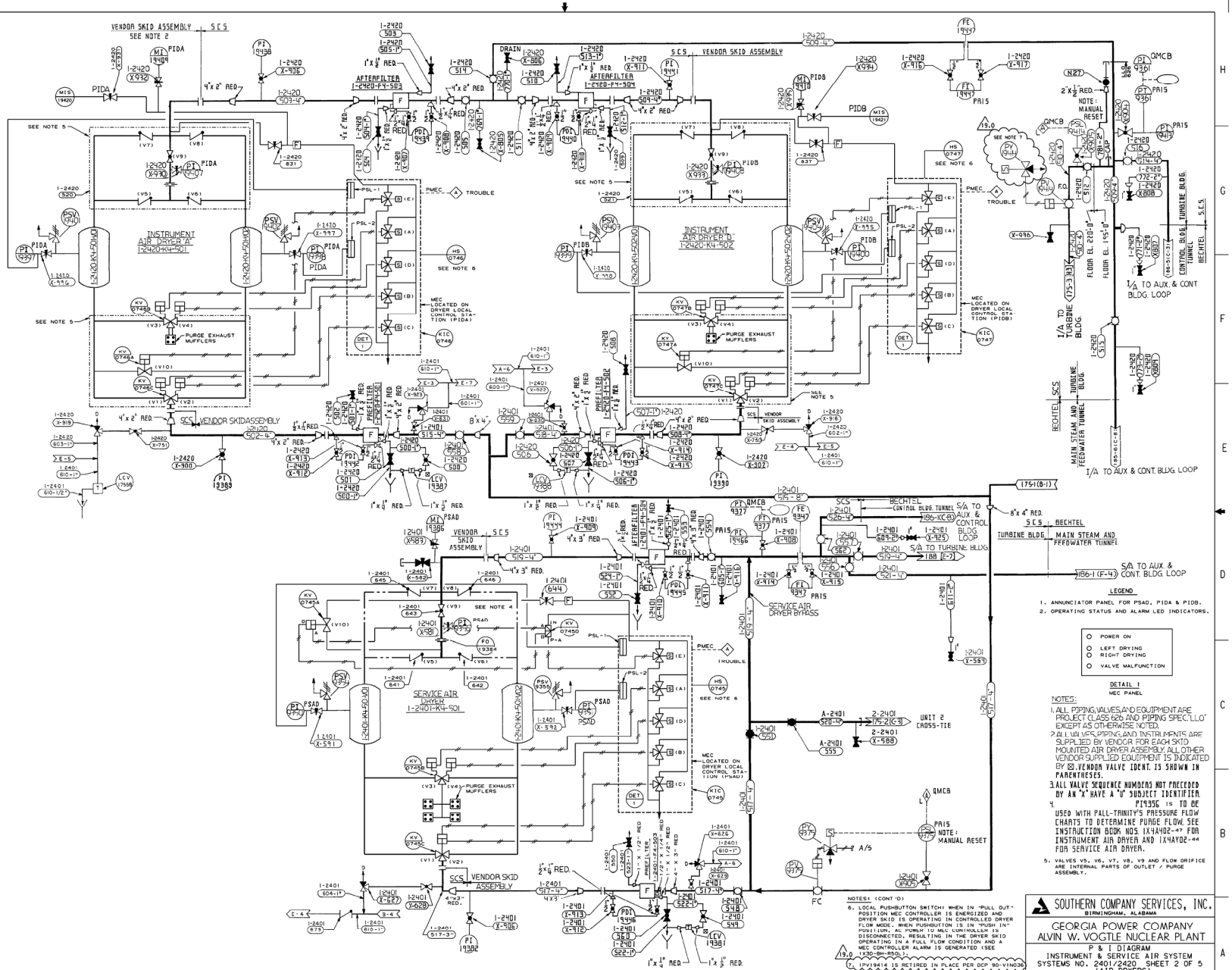
NO. VERSIONS DATE DR CHK APPV
21.0 REVISED PER ABN-V00817, VER 1.0 06/13/07 ELC RBH JMR
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
INSTRUMENT & SERVICE AIR SYSTEMS NO. 2401 & 2420 (TURB. BLDG. AIR COMP.)
21.0
SCALE: NONE
DRAWING NO. 1X4DB175-1
VER. 21.0
JOB NO. 10604
SIZE E 34X44
DWG CATEGORY: CRITICAL


INSTRUMENT AIR DRYER 'A' 1-2420-K4-501
INSTRUMENT AIR DRYER 'B' 1-2420-K4-502
SERVICE AIR DRYER 1-2401-K4-501
VENDOR SKID ASSEMBLY
SEE NOTE 2
PURGE EXHAUST MUFFLERS
MEC LOCATED ON DRYER LOCAL CONTROL STATION (PIDA)
MEC LOCATED ON DRYER LOCAL CONTROL STATION (PIDB)
MEC LOCATED ON DRYER LOCAL CONTROL STATION (PSAD)
SEE NOTE 5
SEE NOTE 6
SEE NOTE 4
PMEC
TROUBLE
AFTERFILTER 1-2420-FX-503
AFTERFILTER 1-2420-FX-504
PREFILTER 1-2420-FX-501
PREFILTER 1-2420-FX-502
PREFILTER 1-2401-FX-501
PREFILTER 1-2401-FX-502
PREFILTER 1-2401-FX-503
AFTERFILTER 1-2401-FX-504
SCS VENDOR SKID ASSEMBLY
SCS BECHTEL CONTROL BLDG. TUNNEL
SCS BECHTEL TURBINE BLDG. MAIN STEAM AND FEEDWATER TUNNEL
I/A TO AUX. & CONT BLDG. LOOP
I/A TO TURBINE BLDG
I/A TO AUX & CONT. BLDG. LOOP
S/A TO AUX & CONT. BLDG. LOOP
S/A TO TURBINE BLDG
SERVICE AIR DRYER BYPASS
UNIT 2 CROSS-TIE
QMCB
PA15 NOTE: MANUAL RESET
FC
A/S
LEGEND
1. ANNUNCIATOR PANEL FOR PSAD, PIDA & PIDB.
2. OPERATING STATUS AND ALARM LED INDICATORS.
POWER ON
LEFT DRYING
RIGHT DRYING
VALVE MALFUNCTION
DETAIL 1
MEC PANEL
NOTES:
1. ALL PIPING, VALVES, AND EQUIPMENT ARE PROJECT CLASS N2B AND PIPING SPEC "L1L0" EXCEPT AS OTHERWISE NOTED.
2. ALL VALVES, PIPING, AND INSTRUMENTS ARE SUPPLIED BY VENDOR FOR EACH SKID MOUNTED AIR DRYER ASSEMBLY. ALL OTHER VENDOR SUPPLIED EQUIPMENT IS INDICATED BY ⊠. VENDOR VALVE IDENT. IS SHOWN IN PARENTHESES.
3. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" HAVE A "V" SUBJECT IDENTIFIER.
4. P19356 IS TO BE USED WITH PALL-TRINITY'S PRESSURE FLOW CHARTS TO DETERMINE PURGE FLOW. SEE INSTRUCTION BOOK NOS. IX4A402-47 FOR INSTRUMENT AIR DRYER AND IX4AY02-44 FOR SERVICE AIR DRYER.
5. VALVES V5, V6, V7, V8, V9 AND FLOW ORIFICE ARE INTERNAL PARTS OF OUTLET / PURGE ASSEMBLY.
NOTES: (CONT'D)
6. LOCAL PUSHBUTTON SWITCH: WHEN IN "PULL OUT" POSITION MEC CONTROLLER IS ENERGIZED AND DRYER SKID IS OPERATING IN CONTROLLED DRYER FLOW MODE. WHEN PUSHBUTTON IS IN "PUSH IN" POSITION, AC POWER TO MEC CONTROLLER IS DISCONNECTED, RESULTING IN THE DRYER SKID OPERATING IN A FULL FLOW CONDITION AND A MEC CONTROLLER ALARM IS GENERATED (SEE 19.0 1X30-BH-R5QL).
7. 1PY19414 IS RETIRED IN PLACE PER DCP 90-V1N036
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
INSTRUMENT & SERVICE AIR SYSTEM
SYSTEMS NO. 2401/2420 SHEET 2 OF 5
(AIR DRYERS)
SCALE: NONE
DRAWING NO. 1X4DB175-2
VER. 19.0
JOB NO. 10604
SIZE E 34X44
1 DRAWING CATEGORY: CRITICAL
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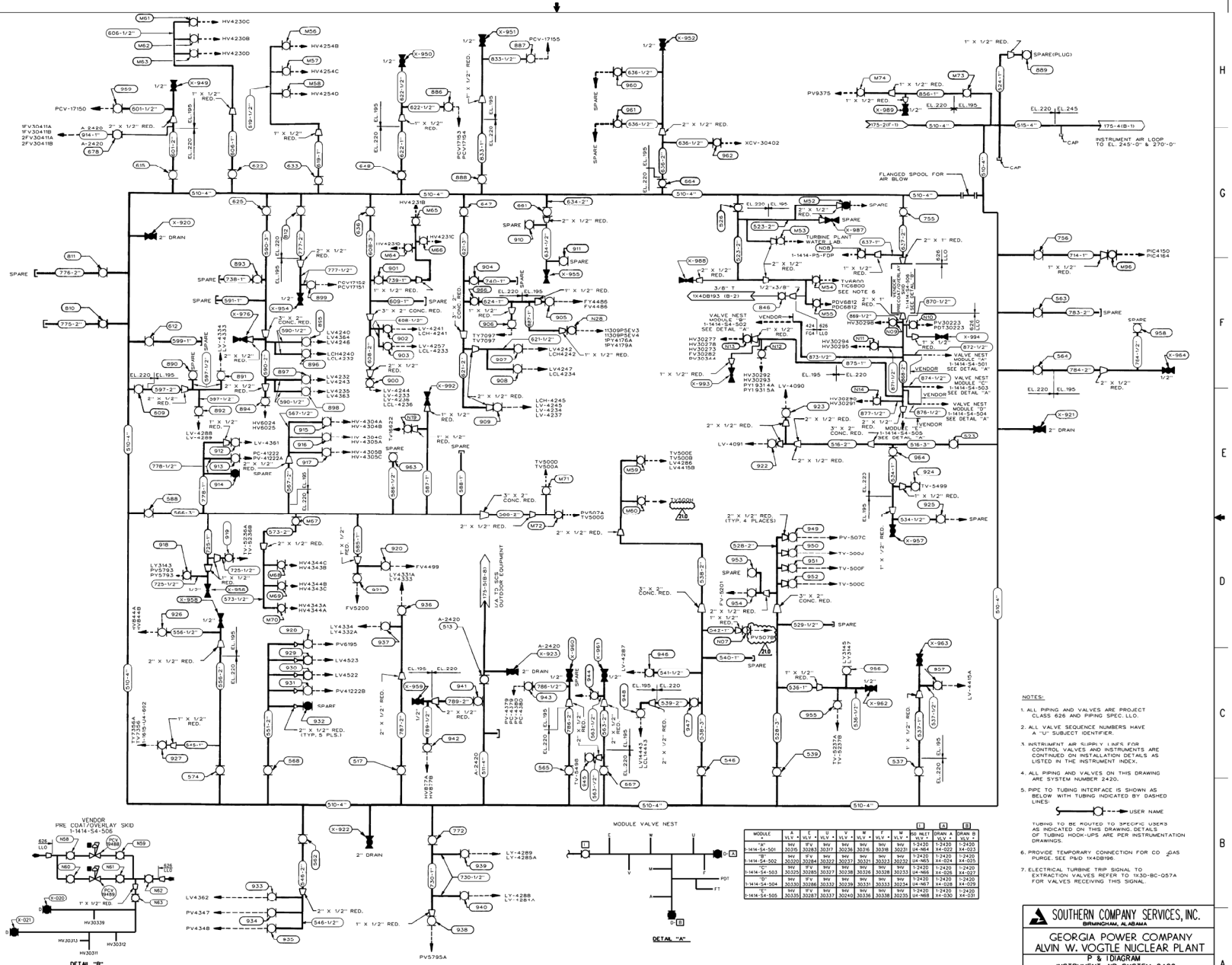
19.0 REVISED PER ABN-V03619, VERSION 1 06-11-14 PLB SO JCG
VERSIONS DATE DR CHK APPV



MODULE	A VLV *	C VLV *	U VLV *	V VLV *	W VLV *	F VLV *	M VLV *	RD INLET VLV *	[1] DRAIN A VLV *	[2] DRAIN B VLV *
"A" 1-1414-S4-501	SHV 30315	SFV 30283	SHV 30317	SHV 30236	SHV 30316	SHV 30318	SHV 30231	1-2420 U4-984	1-2420 X4-022	1-2420 X4-023
"B" 1-1414-S4-502	SHV 30320	SFV 30284	SHV 30322	SHV 30237	SHV 30321	SHV 30323	SHV 30232	1-2420 U4-985	1-2420 X4-024	1-2420 X4-025
"C" 1-1414-S4-503	SHV 30325	SFV 30285	SHV 30327	SHV 30238	SHV 30326	SHV 30328	SHV 30233	1-2420 U4-986	1-2420 X4-026	1-2420 X4-027
"D" 1-1414-S4-504	SHV 30330	SFV 30286	SHV 30332	SHV 30239	SHV 30331	SHV 30333	SHV 30234	1-2420 U4-987	1-2420 X4-028	1-2420 X4-029
"E" 1-1414-S4-505	SHV 30335	SFV 30287	SHV 30337	SHV 30240	SHV 30336	SHV 30338	SHV 30235	1-2420 U4-988	1-2420 X4-030	1-2420 X4-031

NOTES:

1. ALL PIPING AND VALVES ARE PROJECT CLASS 626 AND PIPING SPEC. LLO.
2. ALL VALVE SEQUENCE NUMBERS HAVE A "U" SUBJECT IDENTIFIER.
3. INSTRUMENT AIR SUPPLY LINES FOR CONTROL VALVES AND INSTRUMENTS ARE CONTINUED ON INSTALLATION DETAILS AS LISTED IN THE INSTRUMENT INDEX.
4. ALL PIPING AND VALVES ON THIS DRAWING ARE SYSTEM NUMBER 2420.
5. PIPE TO TUBING INTERFACE IS SHOWN AS BELOW WITH TUBING INDICATED BY DASHED LINES:

   ——— USER NAME

   TUBING TO BE ROUTED TO SPECIFIC USERS AS INDICATED ON THIS DRAWING. DETAILS OF TUBING HOOK-UPS ARE PER INSTRUMENTATION DRAWINGS.
6. PROVIDE TEMPORARY CONNECTION FOR CO GAS PURGE. SEE P&ID 1X4DB196.
7. ELECTRICAL TURBINE TRIP SIGNAL TO EXTRACTION VALVES REFER TO 1X30-BC-057A FOR VALVES RECEIVING THIS SIGNAL.

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NO.	VERSIONS	DATE	DR	CHK	APP'V
	(SEE MICROFILM FOR PREVIOUS REVISION SIGNATURES)				
21.0	REVISED PER ABN 107105340U029, VER. 1.0.	04/16/08	KB	CFC	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

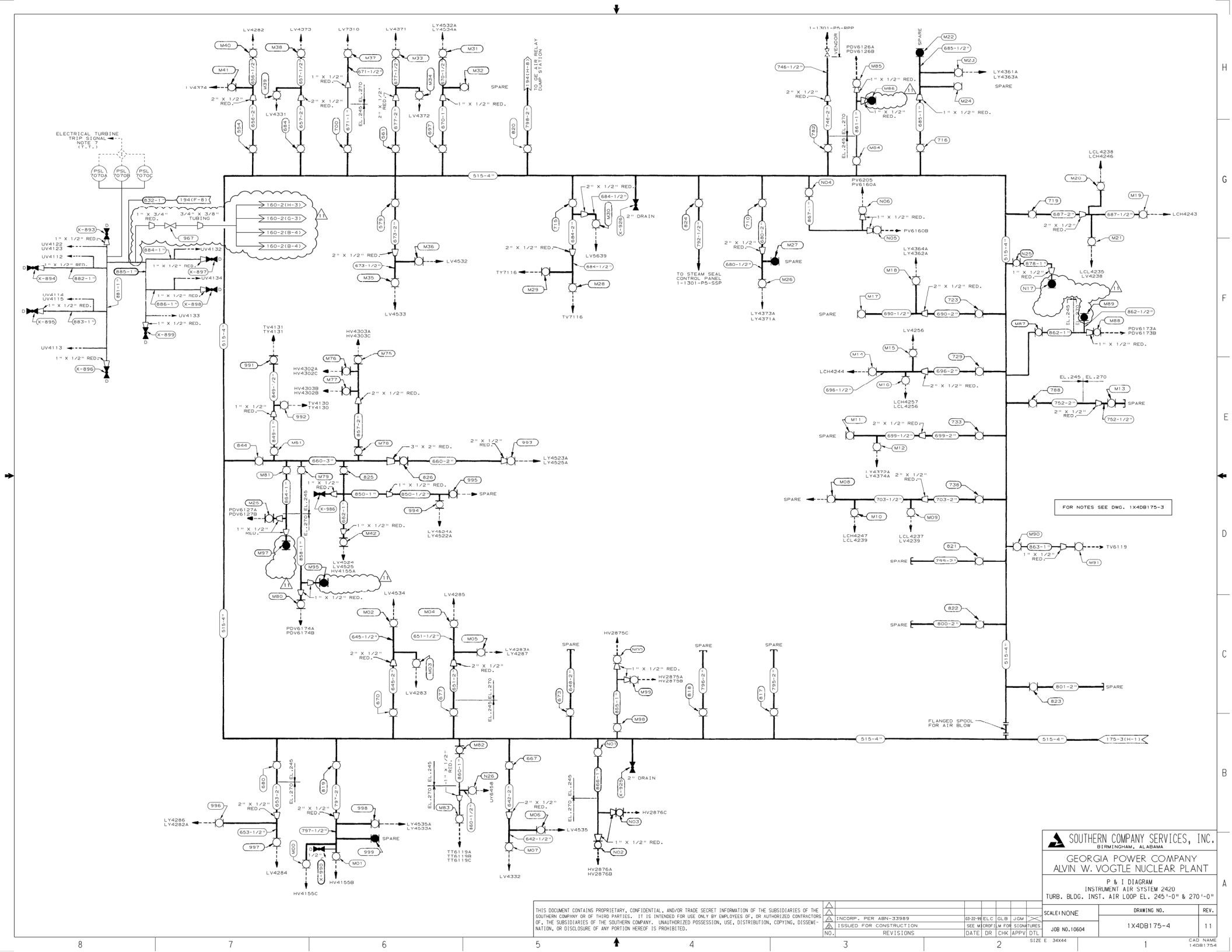
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
INSTRUMENT AIR SYSTEM 2420
TURB. BLDG. INST. AIR LOOP
EL. 195 & EL. 220

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	1X4DB175-3	21.0

SIZE E 34x44

DWG CATEGORY: CRITICAL

\Drawings\1X4DB175-3.DGN 4/15/2008 3:22:38 PM



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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP. PER ABN-33989	03-22-99	ELC	GLB	JGM	
2	ISSUED FOR CONSTRUCTION	SEE MICROFILM FOR SIGNATURES				

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

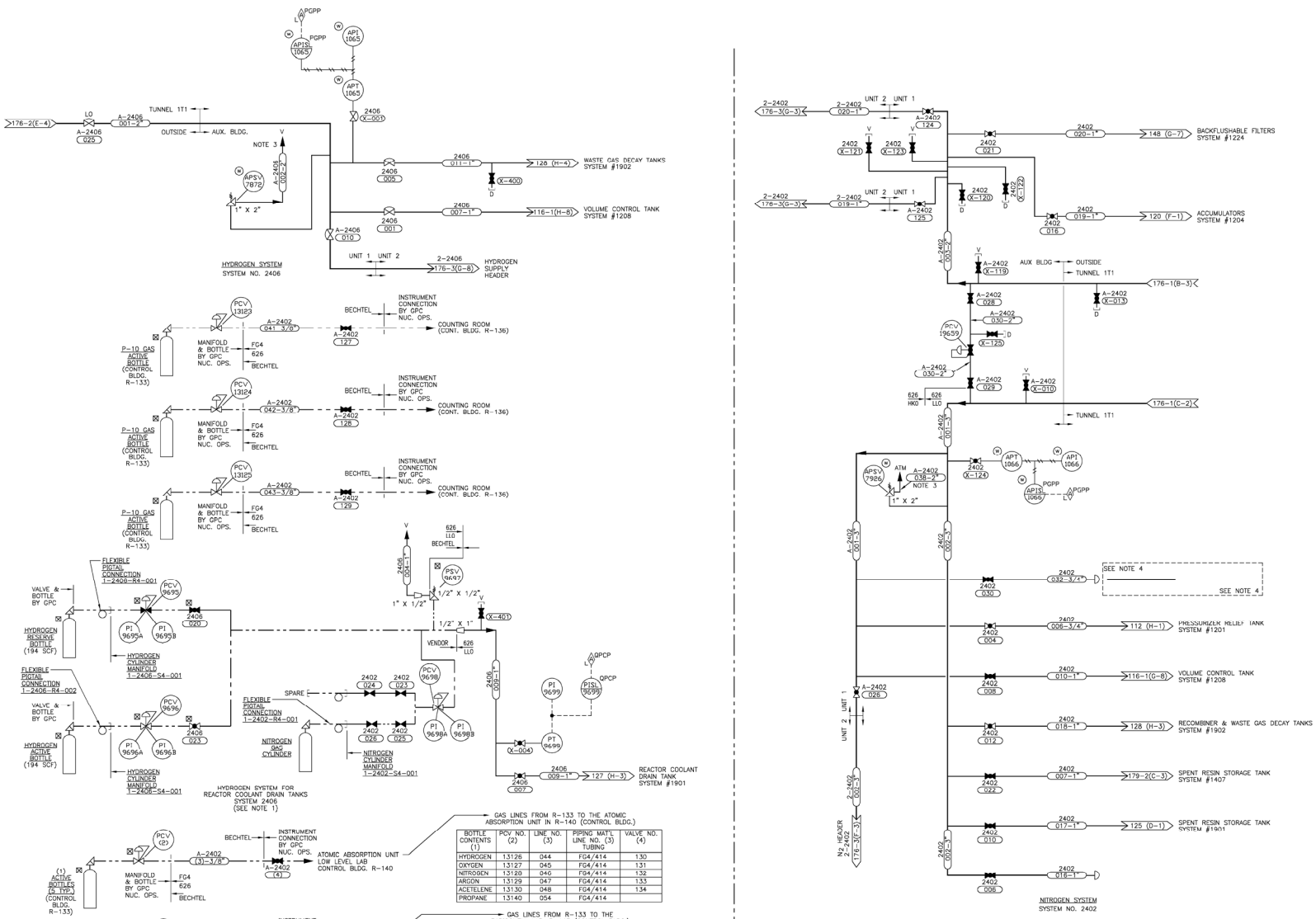
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
INSTRUMENT AIR SYSTEM 2420
TURB. BLDG. INST. AIR LOOP EL. 245'-0" & 270'-0"

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DB175-4	11

SIZE E 34X44

CAD NAME 1ADB1754



BOTTLE CONTENTS (1)	PCV NO. (2)	LINE NO. (3)	PIPING MAT'L LINE NO. (3) TUBING	VALVE NO. (4)
HYDROGEN	1.3126	044	FG4/414	130
OXYGEN	1.3127	045	FG4/414	131
NITROGEN	1.3128	046	FG4/414	132
ARGON	1.3129	047	FG4/414	133
ACETELENE	1.3130	048	FG4/414	134
PROPANE	1.3140	054	FG4/414	

GAS LINES FROM R-133 TO THE ATOMIC ABSORPTION UNIT IN R-140 (CONTROL BLDG.)

BOTTLE CONTENTS (1)	PCV NO. (2)	LINE NO. (3)	PIPING MAT'L LINE NO. (3) TUBING	VALVE NO. (4)
HYDROGEN	1.3131	049	FG4/414	135
OXYGEN	1.3132	050	FG4/414	136
NITROGEN	1.3133	051	FG4/414	137
ARGON	1.3134	052	FG4/414	138
ACETELENE	1.3135	053	FG4/414	139
PROPANE	1.3139	055	FG4/414	

GAS LINES FROM R-133 TO THE RADIOCHEM. LAB. R-143 (CONTROL BLDG.)

NOTES:
1. LOCATE REACTOR COOLANT DRAIN TANK HYDROGEN SUPPLY BOTTLES (2) EAST OF THE MAIN STEAM VALVE ROOM AT GRADE (EL. 220'-0").
2. DELETED.
3. VENT OUTSIDE BLDG.
4. "RETIRED IN PLACE".

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
AUXILIARY GAS SYSTEMS
SYSTEMS 2402 & 2406

CAD 1X4DB176-3
ACAD2004 CYN-01

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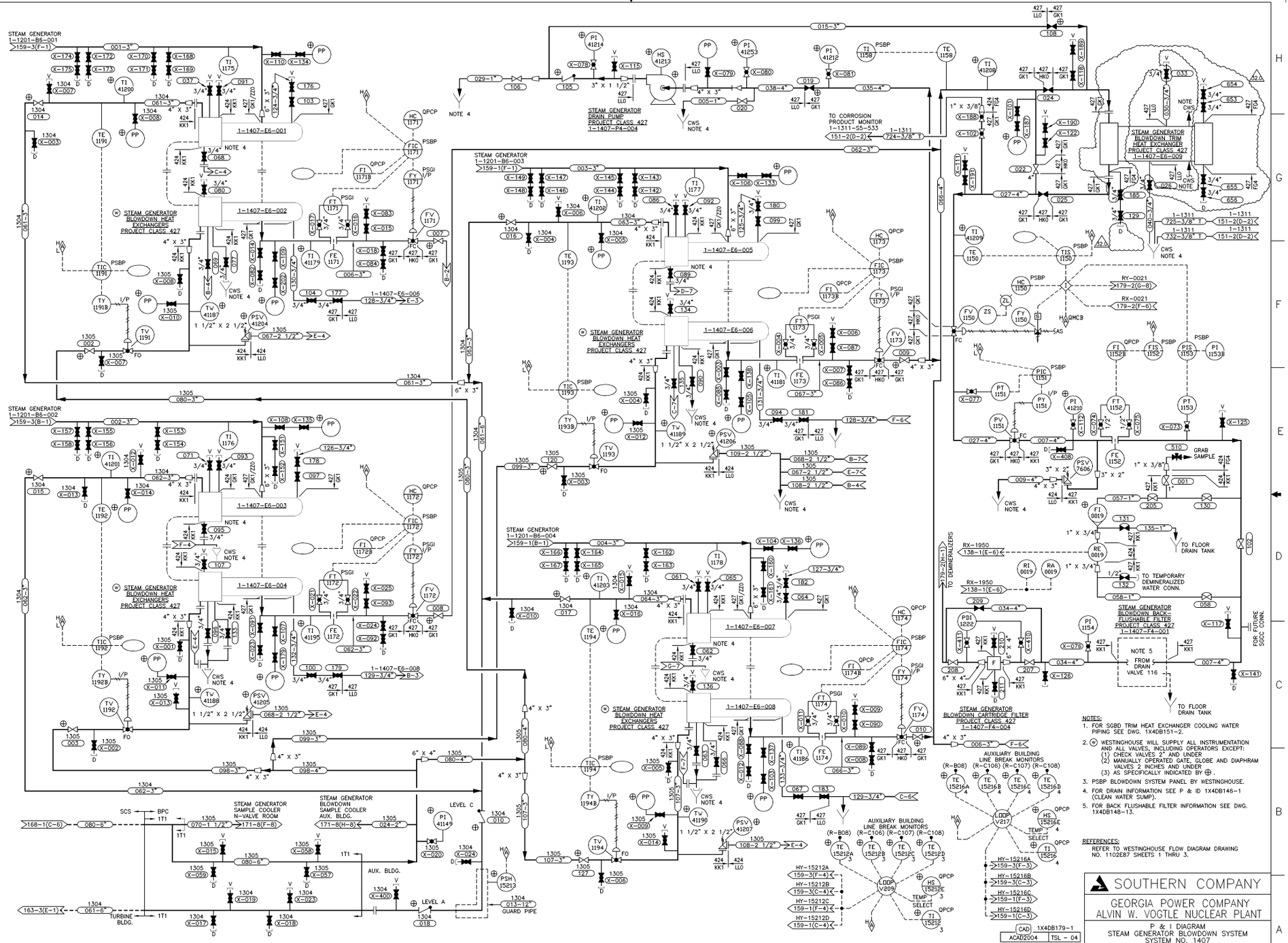
NO.	VERSIONS	DATE	DN	CHK	APPV	DTL
24.0	REDRAWN PER ABN-V70118, VER. 1.0	9/01/06	CYN	LCF	LKB	X

SIZE E 34X44

SCALE: NONE
DRAWING NO. 1X4DB176-3
VER. 24.0
JOB NO.10604
DRAWING CATEGORY: CRITICAL



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NO.	VERSION	DATE	DR	CHK	APPV	DTL
32.0	REVISED PER ABN 1071724201M002, VER. 3.0	11/03/09	TSL	MWD	JMR	X

SIZE E 34X44

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
STEAM GENERATOR BLOWDOWN SYSTEM
SYSTEM NO. 1407

CAD 1X4DB179-1
ACAD2004 TSL - 04

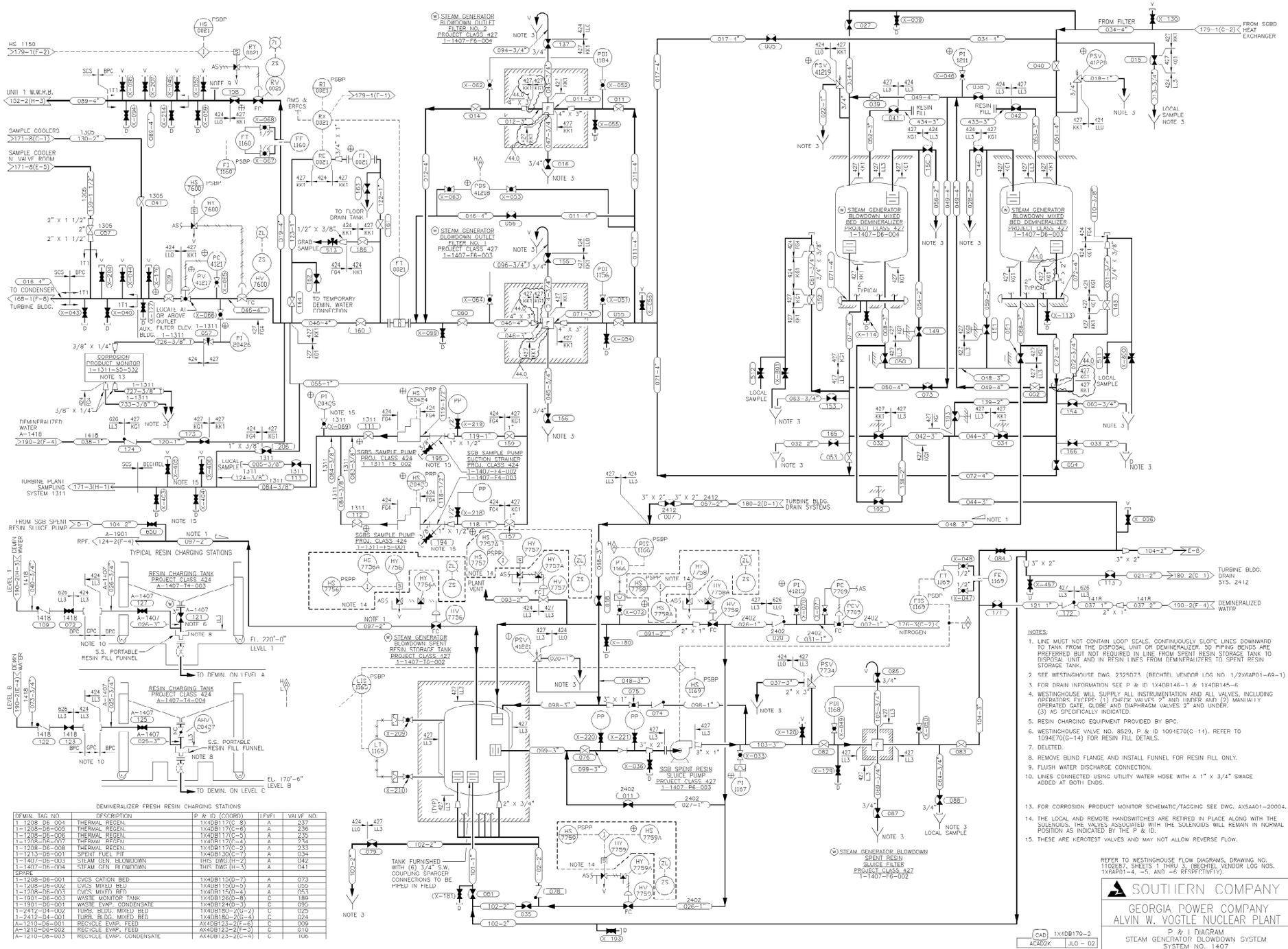
SCALE: NONE

DRAWING NO. 1X4DB179-1

VER. 32.0

JOB NO.10604

DRAWING CATEGORY: CRITICAL



DEMINERALIZER FRESH RESIN CHARGING STATIONS

DEMIN. TAG NO.	DESCRIPTION	P & ID (COORD)	LEVEL	VALVE NO.
1-1208-D6-004	THERMAL REGEN.	1X4DB117(C-8)	A	237
1-1208-D6-005	THERMAL REGEN.	1X4DB117(C-6)	A	236
1-1208-D6-006	THERMAL REGEN.	1X4DB117(C-5)	A	235
1-1208-D6-007	THERMAL REGEN.	1X4DB117(C-4)	A	234
1-1208-D6-008	THERMAL REGEN.	1X4DB117(C-2)	A	233
1-1213-D6-001	SPENT FUEL PIT	1X4DB130(C-7)	A	034
1-1407-D6-003	STEAM GEN. BLOWDOWN	THIS DWG.(H-2)	A	042
1-1407-D6-004	STEAM GEN. BLOWDOWN	THIS DWG.(H-3)	A	041
SPARE				
1-1208-D6-001	CVCS CATION BED	1X4DB115(D-7)	A	073
1-1208-D6-002	CVCS MIXED BED	1X4DB115(D-5)	A	020
1-1208-D6-003	CVCS MIXED BED	1X4DB115(D-4)	A	053
1-1901-D6-003	WASTE MONITOR TANK	1X4DB126(D-8)	C	189
1-1901-D6-001	WASTE EVAP. CONDENSATE	1X4DB124(D-2)	C	090
1-2412-D4-002	TURB. BLDG. MIXED BED	1X4DB150-2(G-2)	C	025
1-2412-D4-001	TURB. BLDG. MIXED BED	1X4DB180-2(G-4)	C	024
A-1210-D6-001	RECYCLE EVAP. FEED	AX4DB123-2(F-6)	C	009
A-1210-D6-002	RECYCLE EVAP. FEED	AX4DB123-2(F-3)	C	010
A-1210-D6-003	RECYCLE EVAP. CONDENSATE	AX4DB123-2(C-4)	C	106

NOTES:

1. LINE MUST NOT CONTAIN LOOP SEALS. CONTINUOUSLY SLOPE LINES DOWNWARD TO TANK FROM THE DISPOSAL UNIT OR DEMINERALIZER. NO PIPING BENDS ARE PREFERRED BUT NOT REQUIRED IN LINE FROM SPENT RESIN STORAGE TANK TO DISPOSAL UNIT AND IN RESIN LINES FROM DEMINERALIZERS TO SPENT RESIN STORAGE TANK.
2. SEE WESTINGHOUSE DWG. 232507J. (BECHTEL VENDOR LOG NO. 1/2XAP01-69-1)
3. FOR DRAIN INFORMATION SEE P & ID 1X4DB146-1 & 1X4DB145-6.
4. WESTINGHOUSE WILL SUPPLY ALL INSTRUMENTATION AND ALL VALVES, INCLUDING OPERATORS EXCEPT: (1) CHECK VALVES 2" AND UNDER AND (2) MANUALLY OPERATED GATE, GLOBE AND DIAPHRAGM VALVES 2" AND UNDER. (3) AS SPECIFICALLY INDICATED.
5. RESIN CHARGING EQUIPMENT PROVIDED BY BPC.
6. WESTINGHOUSE VALVE NO. 8520, P & ID 1094E70(C-11). REFER TO 1094E70(G-14) FOR RESIN FILL DETAILS.
7. DELETED.
8. REMOVE BLIND FLANGE AND INSTALL FUNNEL FOR RESIN FILL ONLY.
9. FLUSH WATER DISCHARGE CONNECTION.
10. LINES CONNECTED USING UTILITY WATER HOSE WITH A 1" X 3/4" SWAGE ADDED AT BOTH ENDS.
13. FOR CORROSION PRODUCT MONITOR SCHEMATIC/TAGGING SEE DWG. AX5AA01-20004.
14. THE LOCAL AND REMOTE HANDSWITCHES ARE RETIRED IN PLACE ALONG WITH THE SOLENOIDS. THE VALVES ASSOCIATED WITH THE SOLENOIDS WILL REMAIN IN NORMAL POSITION AS INDICATED BY THE P & ID.
15. THESE ARE HEROTEST VALVES AND MAY NOT ALLOW REVERSE FLOW.

REFER TO WESTINGHOUSE FLOW DIAGRAMS, DRAWING NO. 1102E87, SHEETS 1 THRU 3, (BECHTEL VENDOR LOG NOS. 1XRAP01-4, -5, AND -6 RESPECTIVELY).

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
STEAM GENERATOR BLOWDOWN SYSTEM
SYSTEM NO. 1407

CAD 1X4DB179-2
ACAD2K | JLO - 02

SCALE: NONE | DRAWING NO. 1X4DB179-2 | VER. 44.0 | JOB NO.10604

NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
44.0	REVISED PER ABN-V01137, VER. 1.0	4-11-08	JLO	JWM	JMR	X

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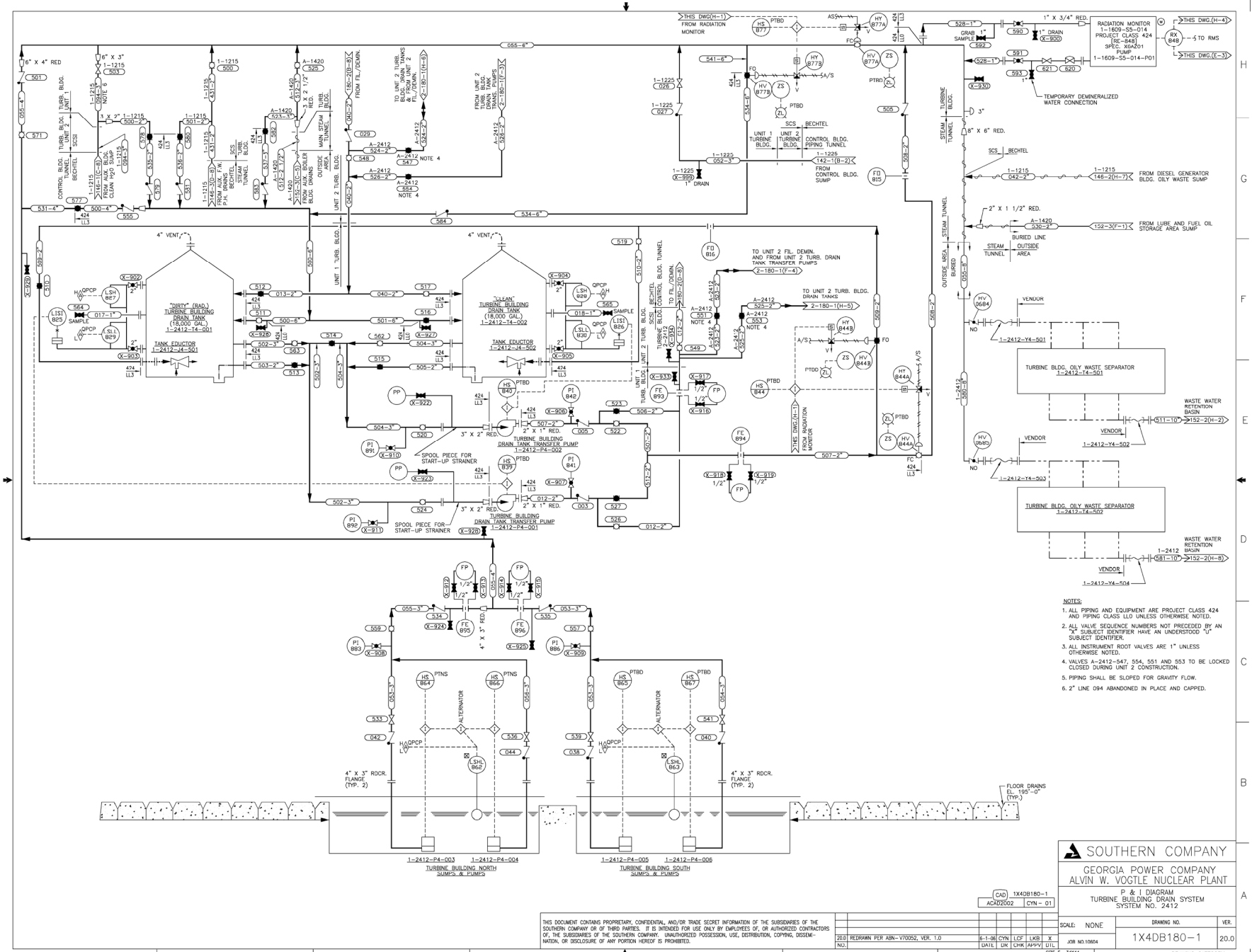
SIZE E 34X44 | DRAWING CATEGORY: CRITICAL



NOTES:

1. ALL PIPING AND EQUIPMENT ARE PROJECT CLASS 424 AND PIPING CLASS LLO UNLESS OTHERWISE NOTED.
2. ALL VALVE SEQUENCE NUMBERS NOT PRECEDED BY AN "X" SUBJECT IDENTIFIER HAVE AN UNDERSTOOD "U" SUBJECT IDENTIFIER.
3. ALL INSTRUMENT ROOT VALVES ARE 1" UNLESS OTHERWISE NOTED.
4. VALVES A-2412-547, 554, 551 AND 553 TO BE LOCKED CLOSED DURING UNIT 2 CONSTRUCTION.
5. PIPING SHALL BE SLOPED FOR GRAVITY FLOW.
6. 2" LINE 094 ABANDONED IN PLACE AND CAPPED.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE BUILDING DRAIN SYSTEM
SYSTEM NO. 2412

CAD 1X4DB180-1
ACAD2002 CYN- 01

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NO.		DATE	DR	CHK	APPV	DTL
20.0	REDRAWN PER ABN-V70052, VER. 1.0	6-1-06	CYN	LCF	LKB	X

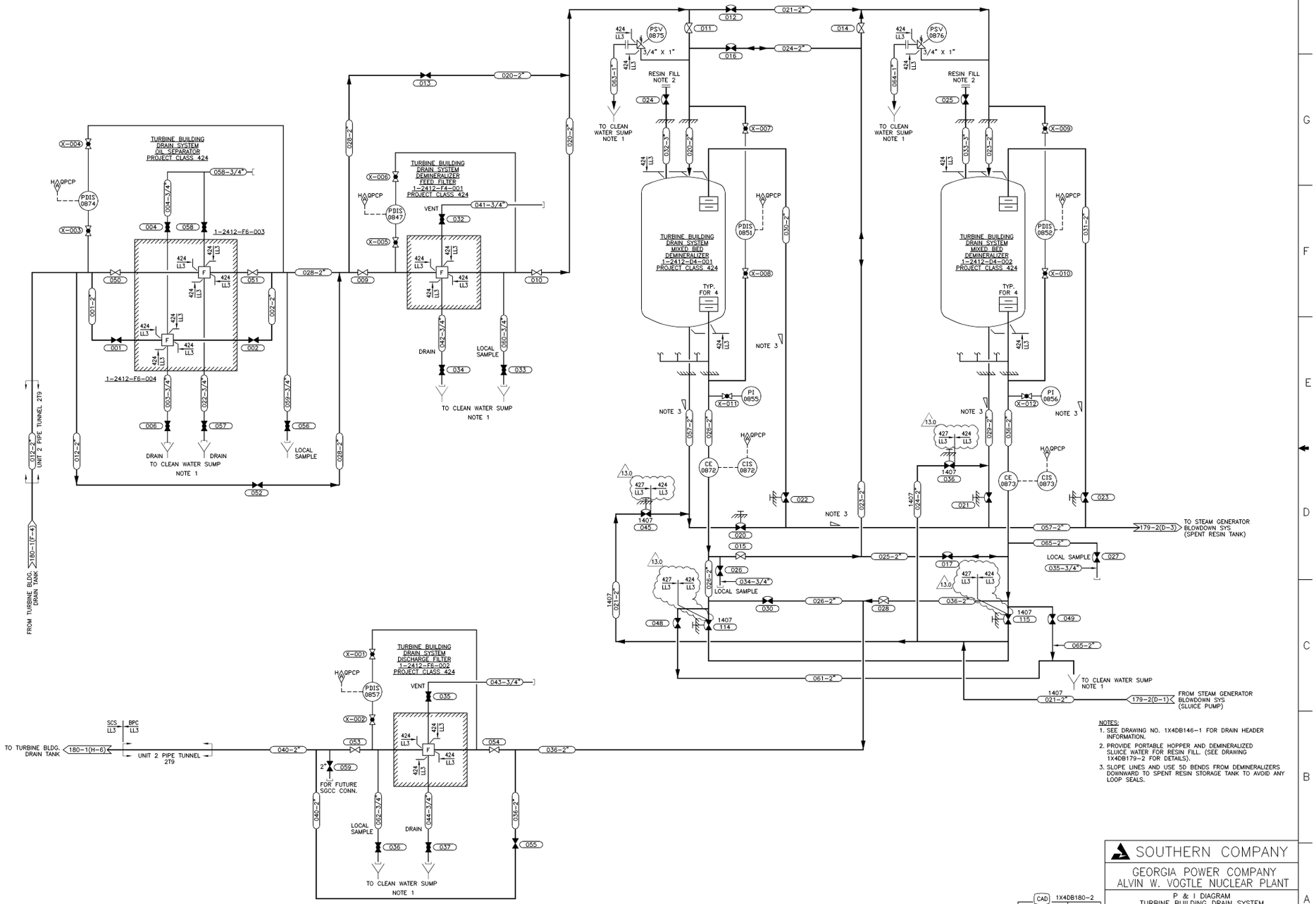
SIZE E 34X44

SCALE: NONE

DRAWING NO. 1X4DB180-1 VER. 20.0

JOB NO.10604

1 DRAWING CATEGORY: CRITICAL



NOTES:
1. SEE DRAWING NO. 1X4DB146-1 FOR DRAIN HEADER INFORMATION.
2. PROVIDE PORTABLE HOPPER AND DEMINERALIZED SLUICE WATER FOR RESIN FILL. (SEE DRAWING 1X4DB179-2 FOR DETAILS).
3. SLOPE LINES AND USE 5D BENDS FROM DEMINERALIZERS DOWNWARD TO SPENT RESIN STORAGE TANK TO AVOID ANY LOOP SEALS.

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE BUILDING DRAIN SYSTEM
SYSTEM NO. 2412

CAD 1X4DB180-2
ACAD2002 DCP- 01

SCALE: NONE
DRAWING NO. 1X4DB180-2
VER. 13.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL

NO.		DATE	DR	CHK	APPV	DTL
13.0	REVISED PER ABN-V00309, VER. 1.0	09-12-06	DCP	HAD	ASK	X

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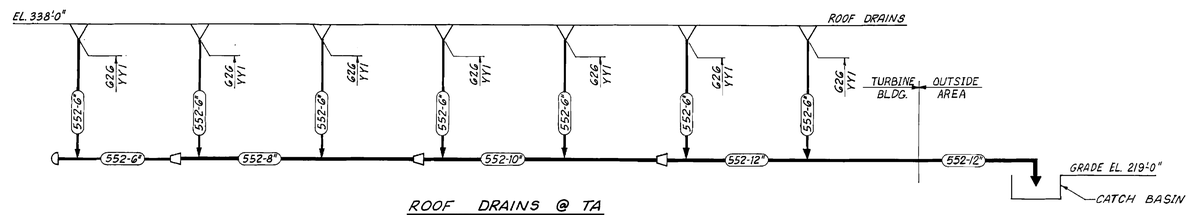
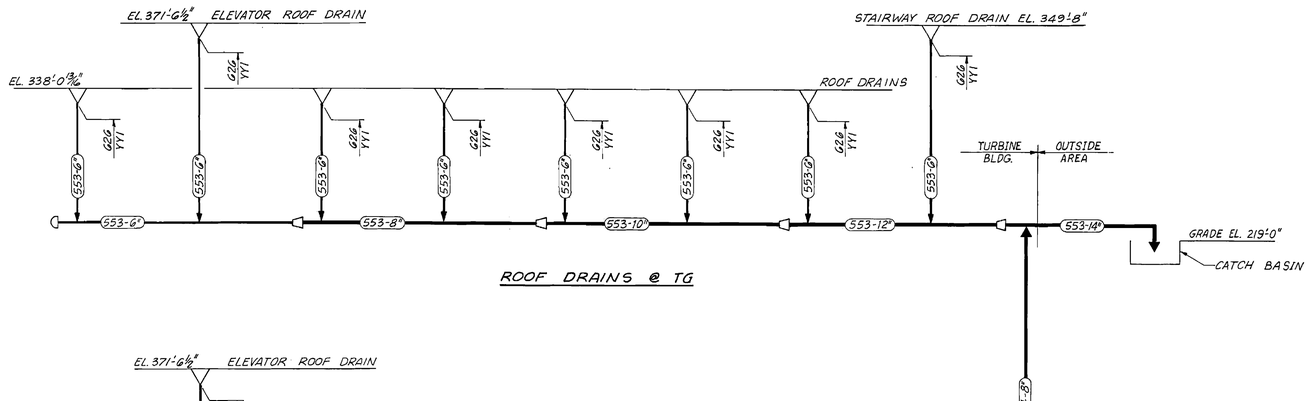
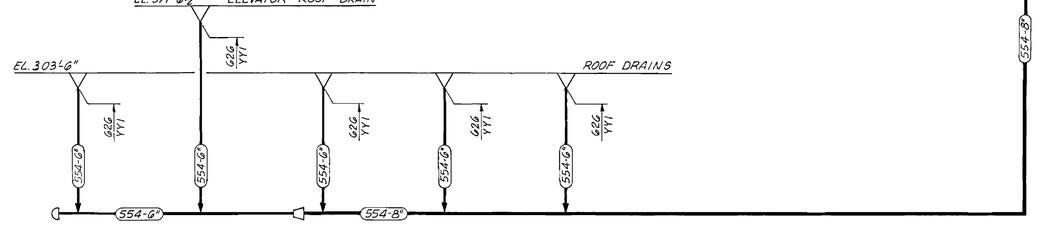


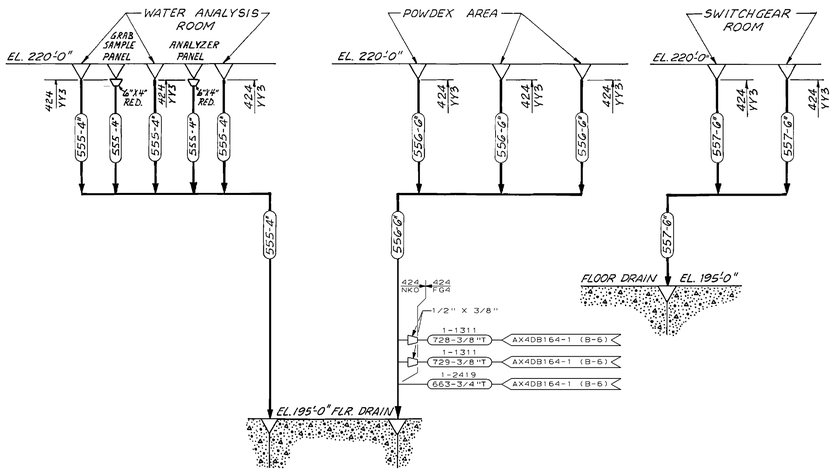
ROOF DRAINS @ TA

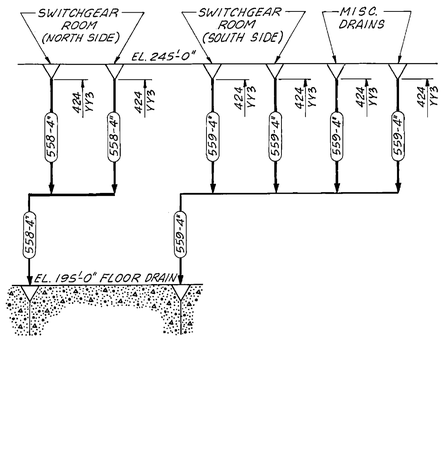


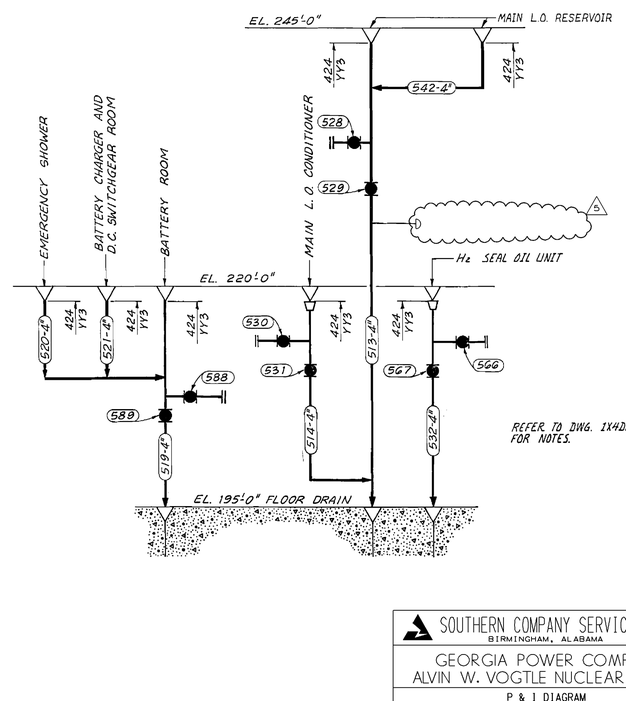
ROOF DRAINS @ TG



ROOF DRAINS @ TH







REFER TO DWG. 1X4DB180-1 FOR NOTES.

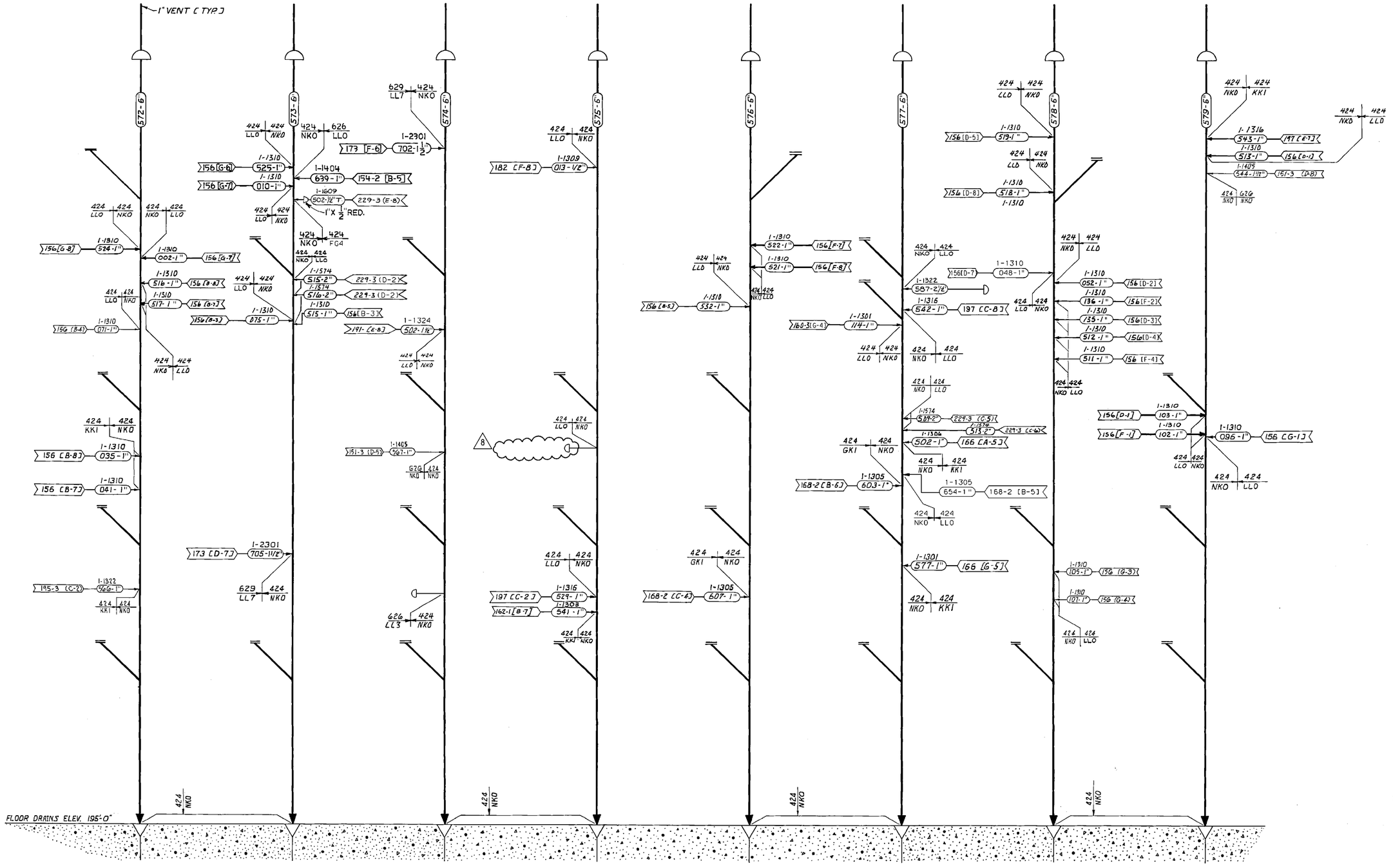
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE BUILDING DRAIN SYSTEM
(SUSPENDED & ROOF DRAINS)
SYSTEM NO. 2412 SHEET 3 OF 4

NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
2	INCORP. PER ABN-39557	12-23-03	KMH	CMM	DEW	

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB180-3
REV. 5
SIZE E 34X44



REFER TO DWG 1X4DB180-1 FOR NOTES.

SEE DWG 1X4DB180-4 REV. 5

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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
4	INCORP. PER ABN 39558	12-26-03	KMH	CMM	DEW	X

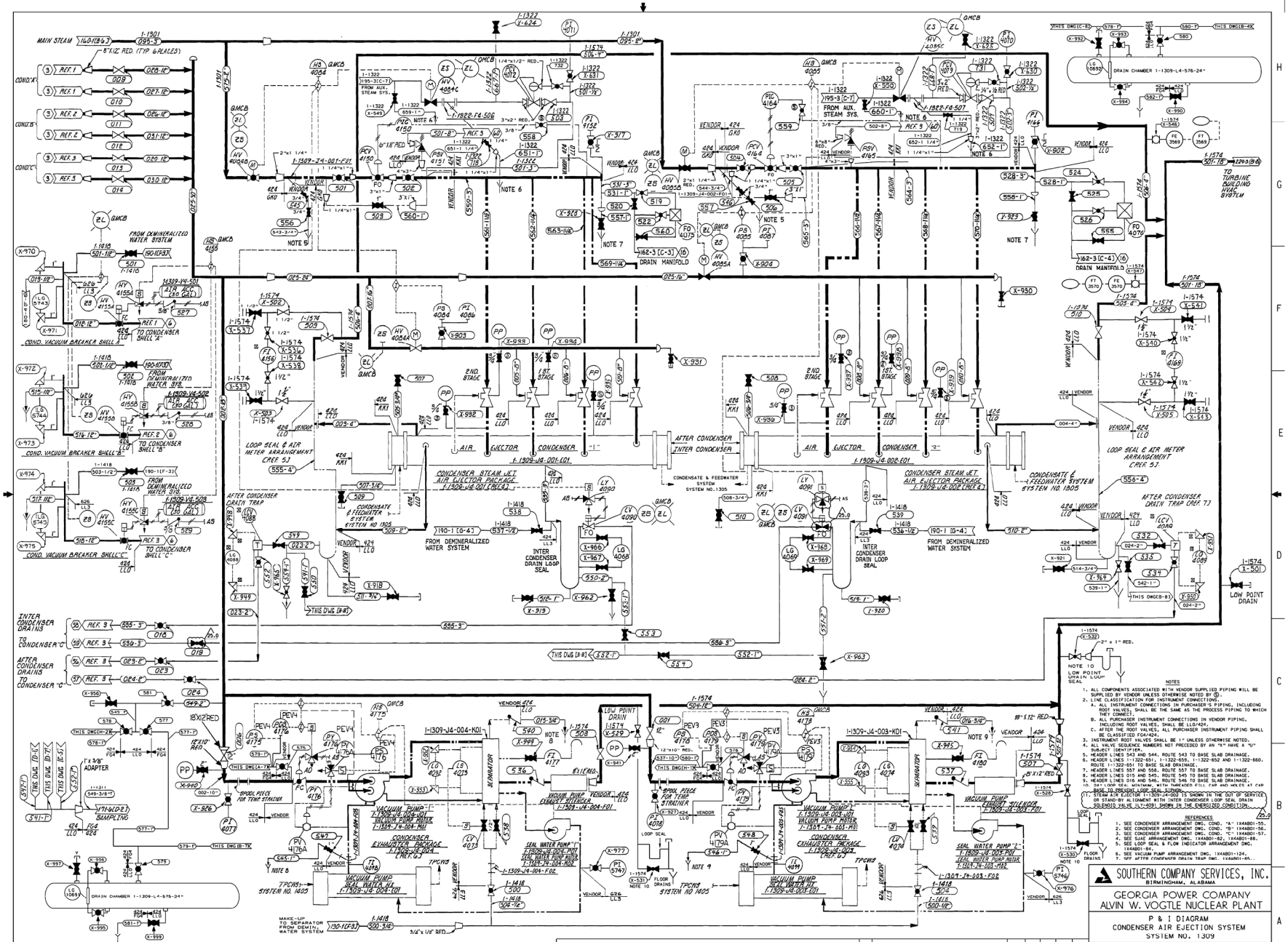
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE BUILDING DRAIN SYSTEM
SYSTEM NO. 2412 SHEET 4 OF 4

SCALE: NONE	DRAWING NO.	REV.
JOB NO.10604	1X4DB180-4	8

SIZE E 34X44 CAD NAME 1X4DB180-4



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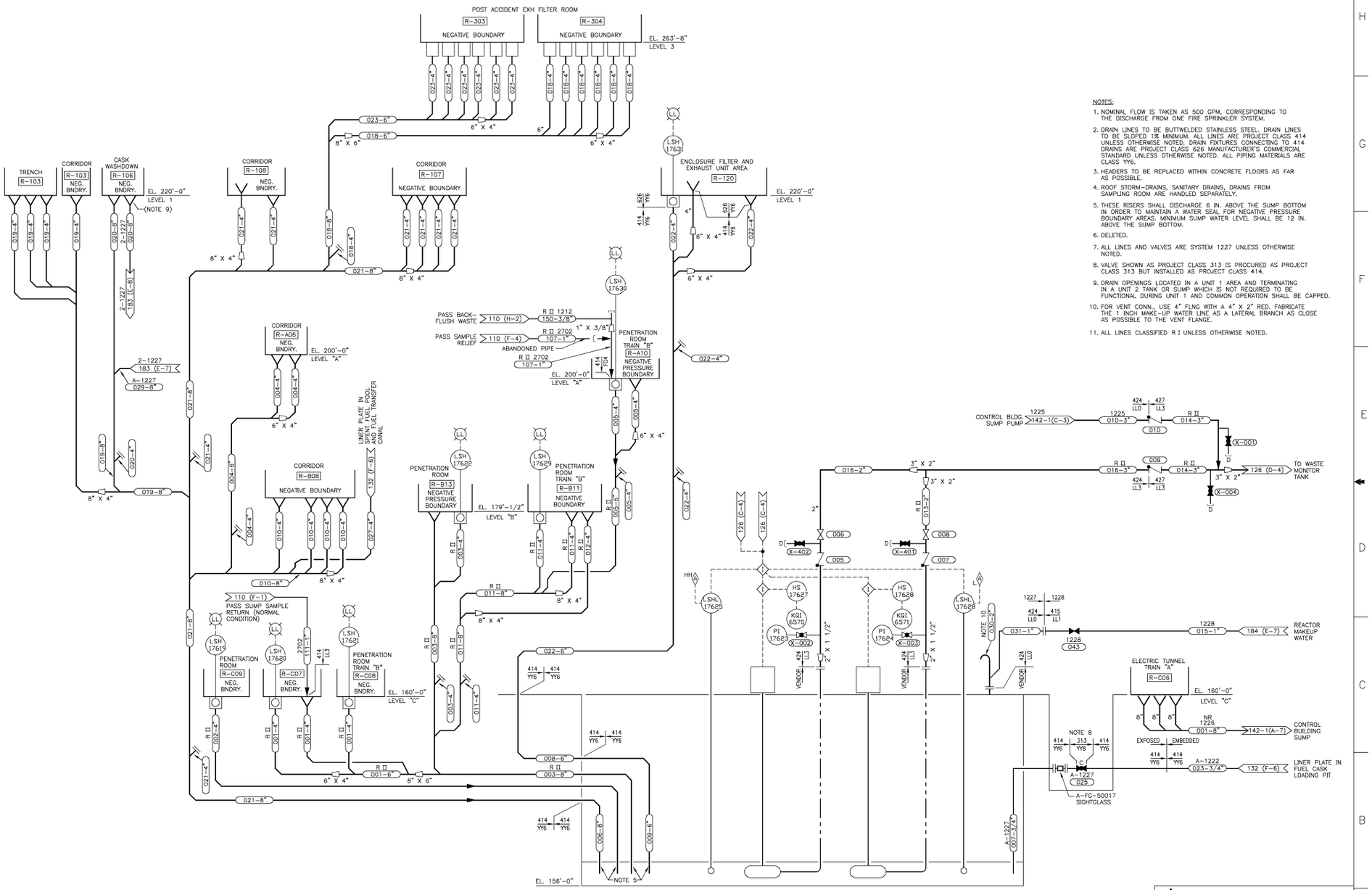
NO.	VERSIONS	DATE	DR	CHK	APP'V
25.0	REVISED PER ABN-V03258, VER.1.0	5/31/13	BgT	SO	RKF



NOTES:

1. NOMINAL FLOW IS TAKEN AS 500 GPM, CORRESPONDING TO THE DISCHARGE FROM ONE FIRE SPRINKLER SYSTEM.
2. DRAIN LINES TO BE BUTTWELDED STAINLESS STEEL. DRAIN LINES TO BE SLOPED 1% MINIMUM. ALL LINES ARE PROJECT CLASS 414 UNLESS OTHERWISE NOTED. DRAIN FIXTURES CONNECTING TO 414 DRAINS ARE PROJECT CLASS 626 MANUFACTURER'S COMMERCIAL STANDARD UNLESS OTHERWISE NOTED. ALL PIPING MATERIALS ARE CLASS YY6.
3. HEADERS TO BE REPLACED WITHIN CONCRETE FLOORS AS FAR AS POSSIBLE.
4. ROOF STORM-DRAINS, SANITARY DRAINS, DRAINS FROM SAMPLING ROOM ARE HANDLED SEPARATELY.
5. THESE RISERS SHALL DISCHARGE 6 IN. ABOVE THE SUMP BOTTOM IN ORDER TO MAINTAIN A WATER SEAL FOR NEGATIVE PRESSURE BOUNDARY AREAS. MINIMUM SUMP WATER LEVEL SHALL BE 12 IN. ABOVE THE SUMP BOTTOM.
6. DELETED.
7. ALL LINES AND VALVES ARE SYSTEM 1227 UNLESS OTHERWISE NOTED.
8. VALVE SHOWN AS PROJECT CLASS 313 IS PROCURED AS PROJECT CLASS 313 BUT INSTALLED AS PROJECT CLASS 414.
9. DRAIN OPENINGS LOCATED IN A UNIT 1 AREA AND TERMINATING IN A UNIT 2 TANK OR SUMP WHICH IS NOT REQUIRED TO BE FUNCTIONAL DURING UNIT 1 AND COMMON OPERATION SHALL BE CAPPED.
10. FOR VENT CONN., USE 4" FLNG WITH A 4" X 2" RED. FABRICATE THE 1 INCH MAKE-UP WATER LINE AS A LATERAL BRANCH AS CLOSE AS POSSIBLE TO THE VENT FLANGE.
11. ALL LINES CLASSIFIED R I UNLESS OTHERWISE NOTED.

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NO.		DATE	DR	CHK	APPV	DTL
15.0	REDRAWN PER ABN-V70013, VER. 1.0.	4-3-06	CYN	LCF	WPE	X

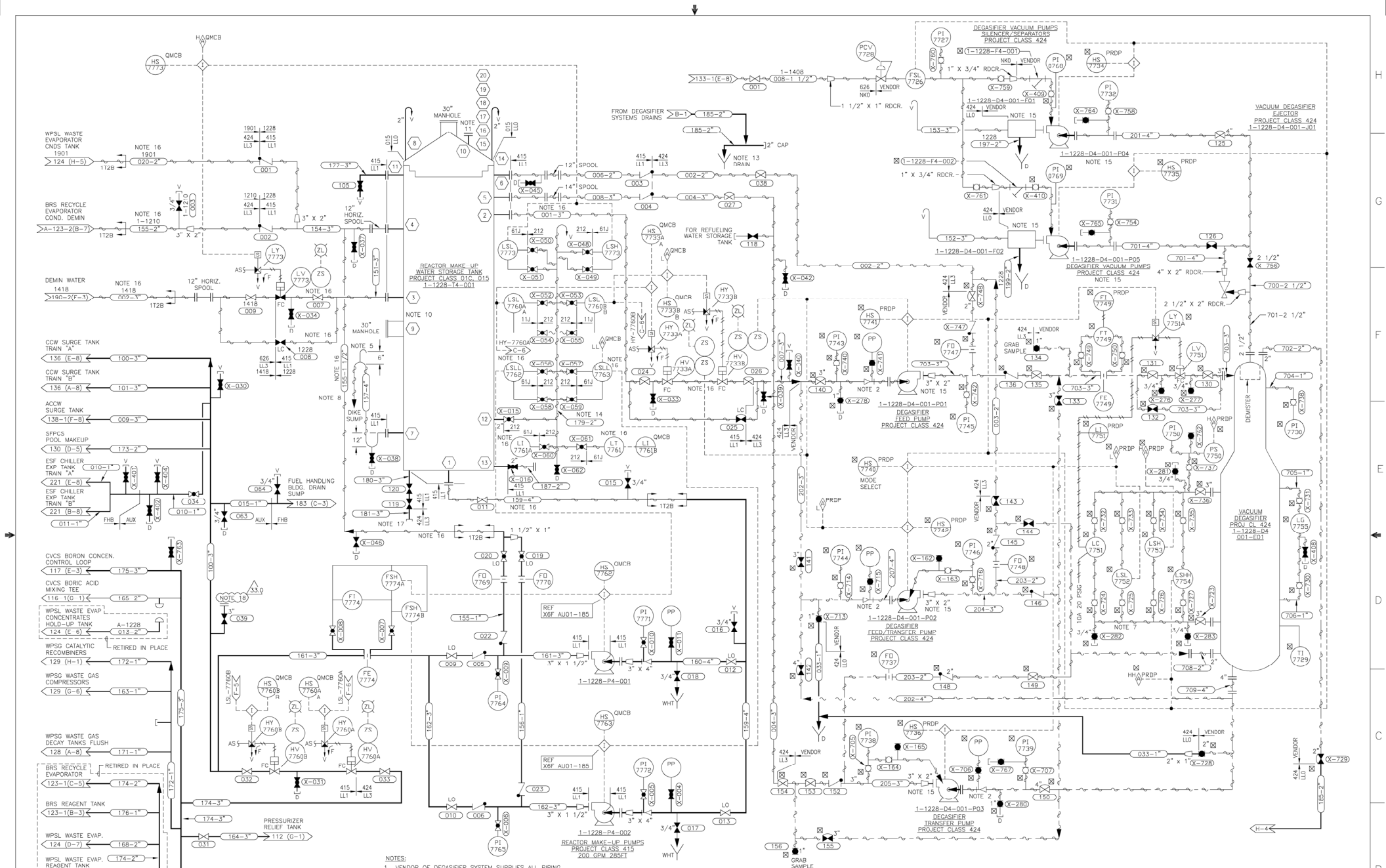
CAD 1X4DB183
ACADZ002 CYN-01

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
FUEL HANDLING BUILDING &
ELECTRICAL CHASE TUNNEL DRAINS
SYSTEM NO. 1226 & 1227

SCALE: NONE
DRAWING NO. 1X4DB183
VER. 15.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:

1. VENDOR OF DEGASIFIER SYSTEM SUPPLIES ALL PIPING, INSTRUMENTS AND CONTROLS WITHIN HIS PACKAGE PER SPEC X4AM10. STATUS LIGHTS ARE ON PRDP PANEL FOR PUMPS ON/OFF, AND FOR AUTOMATIC VALVES OPEN/CLOSED.
2. VENDOR SUPPLIES PIPE SPOOL PER SPEC X4AM10 FOR PURCHASER'S STRAINER.
3. DELETED.
4. ALL PIPING, VALVES AND COMPONENTS ARE RADIATION ZONE R 1.
5. THE SYPHON BREAK PIPE END TO EXTEND TO DIAPHRAM TO TANK SEAL ELEVATION.
6. DELETED.
7. HEAT TRACING ON DEGASIFIER SYSTEM INSTRUMENTS BY PURCHASER.
8. ALL SAFETY RELATED PROCESS, INSTRUMENT, AND SAMPLING LINES THAT ARE PHYSICALLY ROUTED IN AN AREA EXPOSED TO EXTREMELY COLD WEATHER SHALL BE PROVIDED WITH ADEQUATE HEAT TRACING.
9. COMPONENTS OF THE 415 LINES ARE PURCHASED PER 313 REQUIREMENTS.
10. ⟨ ⟩ INDICATES TANK NOZZLE CONNECTION NUMBER.
11. SIX EQUALLY SPACED 2" DIAPHRAM PLACEMENT NO771F, 150# RF FLANGE WITH BLIND FLANGE LOCATED 15 FEET FROM CENTER.
12. DELETED.
13. FOR DRAIN INFORMATION, SEE P & ID 1X4DB146-3.
14. ANALYSIS FOR LINE 1228-179-2" NEED NOT INCLUDE TORNADO WIND OR MISSILE LOADING.
15. DEGASIFIER PUMPS 1-1228-D4-001-P01, 1-1228-D4-001-P02, 1-1228-D4-001-P03, 1-1228-D4-001-P04 AND 1-1228-D4-001-P05 AND SEPARATORS 1-1228-D4-001-F01 AND 1-1228-D4-001-F02 SHALL BE HEAT TRACED TO PREVENT FREEZING.
16. HEAT TRACE ON THE FOLLOWING LINES, VALVES & INSTRUMENTATION IS RETIRED IN PLACE:
    1-1210-155-2"
    1-1228-001-3" (X-033)
    1-1228-151-3" (PARTIAL) (X-034)
    1-1228-155-1 1/2" (X-046)
    1-1228-159-4"
    1-1228-179-2" (PARTIAL) (LSLL 7762, 63)
    LI7761A
    LT7761, (X-056 THROUGH 062)
    1-1418-002-3"
    1-1901-020-2"
17. CONNECTION USED TO SUPPORT FLEX COPING STRATEGY PER NRC ORDER EA-12-049. CONNECTION LISTED BELOW IS ADEQUATELY ROBUST TO MEET FLEX REQUIREMENTS:

VALVE	REF. CALC
1-1228-U4-119	1K5122818001

18. CONNECTION USED TO SUPPORT FLEX COPING STRATEGY PER NRC ORDER EA-12-049. PER CALC AX4DT114: 33.0

CAD 1X4DB184
ACAD 2012 | SMD- 04

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
33.0	REVISED PER ABN SNC764664M001, VER. 2.0	02/23/16	SMD	MWD	WLN	X

SOUTHERN COMPANY

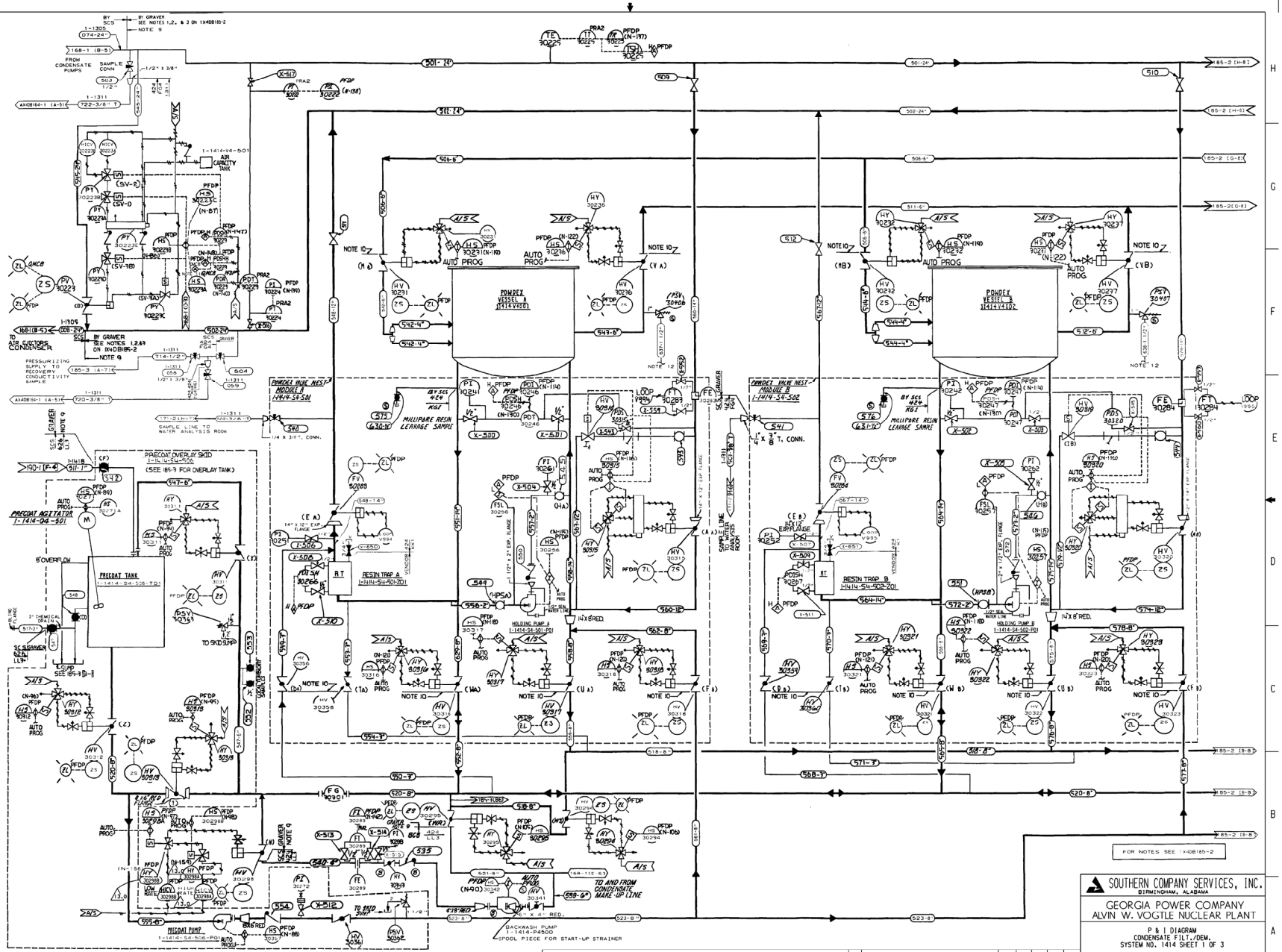
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
REACTOR MAKE-UP WATER STORAGE TANK AND DEGASIFIER SYSTEM NO. 1228

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB184	33.0

SIZE E 34X44

1 DRAWING CATEGORY: CRITICAL



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONDENSATE FILT./DEM.
SYSTEM NO. 1414 SHEET 1 OF 3

SCALE:	NONE	DRAWING NO.	VER.
JOB NO.		1X4DB185-1	13.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL

NO.	VERSIONS	DATE	DR	CHK	APPV
13.0	REVISED PER ABN SNC762689M001, VER. 1.0	04/09/18	TDA	MWD	WLN

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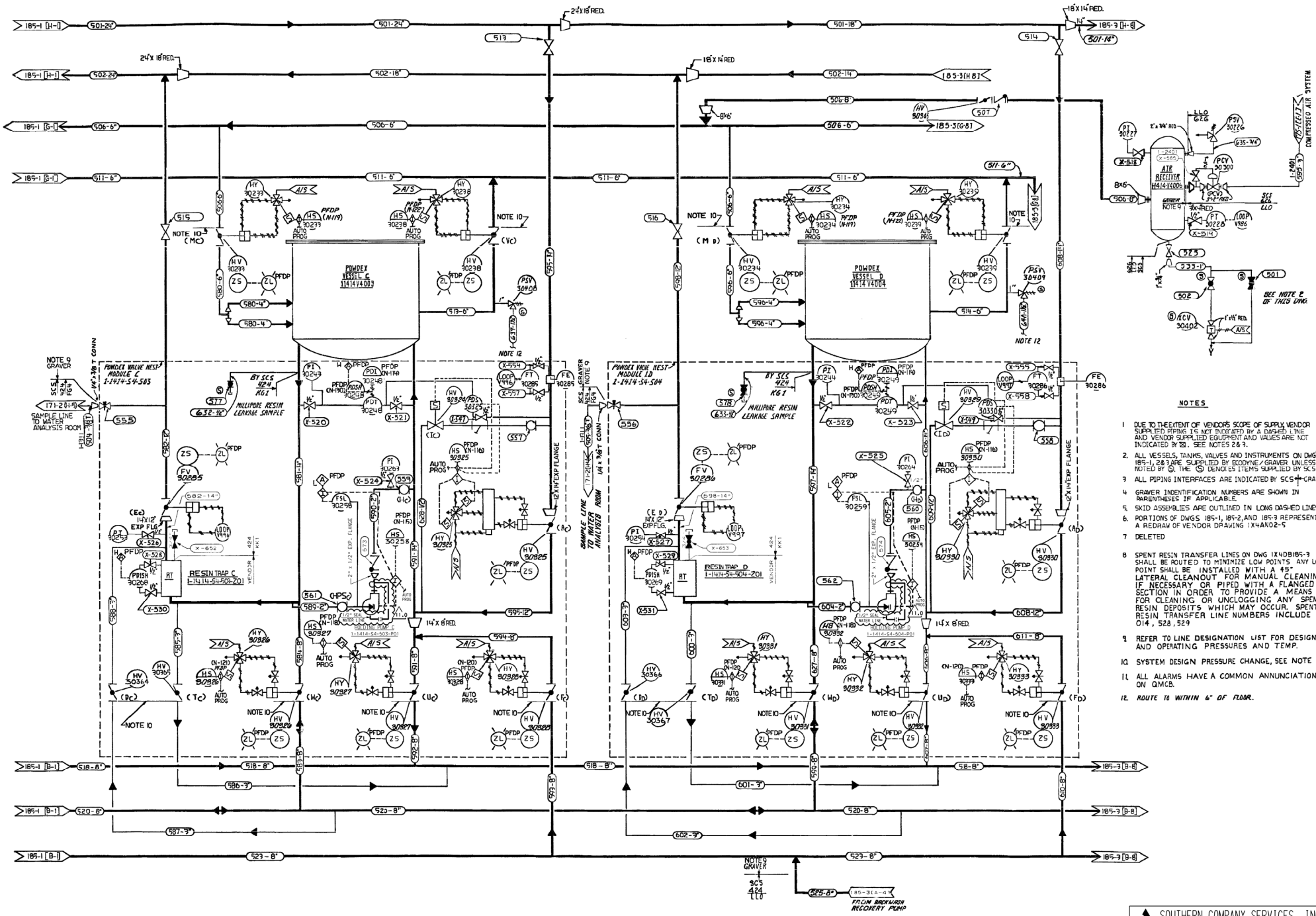

NOTES

1. DUE TO THE EXTENT OF VENDOR'S SCOPE OF SUPPLY, VENDOR SUPPLIED PIPING IS NOT INDICATED BY A DASHED LINE AND VENDOR SUPPLIED EQUIPMENT AND VALVES ARE NOT INDICATED BY ⊠. SEE NOTES 2 & 3.
2. ALL VESSELS, TANKS, VALVES AND INSTRUMENTS ON DWG'S. 185-1, 2 & 3 ARE SUPPLIED BY ECODYNE/GRAVER UNLESS NOTED BY Ⓢ. THE Ⓢ DENOTES ITEMS SUPPLIED BY SCS.
3. ALL PIPING INTERFACES ARE INDICATED BY SCS ⊣ GRAVER
4. GRAVER INDENTIFICATION NUMBERS ARE SHOWN IN PARENTHESES IF APPLICABLE.
5. SKID ASSEMBLIES ARE OUTLINED IN LONG DASHED LINES.
6. PORTIONS OF DWGS 185-1, 185-2, AND 185-3 REPRESENT A REDRAW OF VENDOR DRAWING 1X4A02-S
7. DELETED
8. SPENT RESIN TRANSFER LINES ON DWG 1X4DB185-3 SHALL BE ROUTED TO MINIMIZE LOW POINTS. ANY LOW POINT SHALL BE INSTALLED WITH A 45° LATERAL CLEANOUT FOR MANUAL CLEANING IF NECESSARY OR PIPED WITH A FLANGED SECTION IN ORDER TO PROVIDE A MEANS FOR CLEANING OR UNCLOGGING ANY SPENT RESIN DEPOSITS WHICH MAY OCCUR. SPENT RESIN TRANSFER LINE NUMBERS INCLUDE 014, 528, 529
9. REFER TO LINE DESIGNATION LIST FOR DESIGN AND OPERATING PRESSURES AND TEMP.
10. SYSTEM DESIGN PRESSURE CHANGE, SEE NOTE 9.
11. ALL ALARMS HAVE A COMMON ANNUNCIATION ON QMCB.
12. ROUTE TO WITHIN 6" OF FLOOR.

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NO.	VERSIONS	DATE	DR	CHK	APPV
11.0	REVISED PER ABN-V00063, VER. 1.0	04-11-06	KMH	MWD	ASK

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

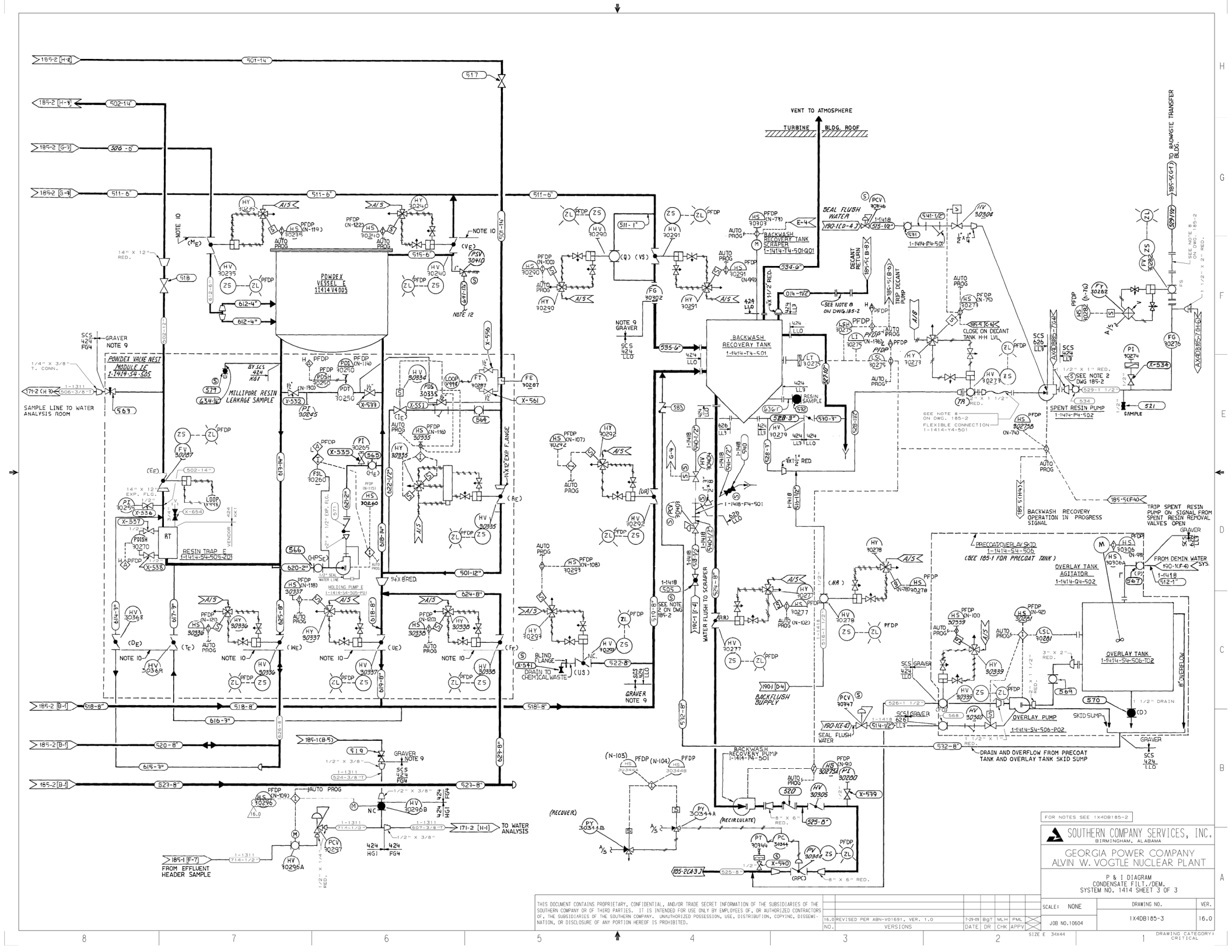
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONDENSATE FILT./DEM.
SYSTEM NO. 1414 SHEET 2 OF 3

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB185-2	11.0

SIZE E 34X44

DRAWING CATEGORY: CRITICAL


VENT TO ATMOSPHERE
TURBINE BLDG. ROOF
POWDEX VESSEL E 1414V4005
POWDEX VALVE NEST MODULE I.C. 1-1414-54-505
RESIN TRAP E 1-1414-54-505-Z01
HOLDING PUMP E 1-1414-54-505-P01
BACKWASH RECOVERY TANK 1-1414-T4-501
BACKWASH RECOVERY PUMP 1-1414-P4-501
SPENT RESIN PUMP 1-1414-P4-502
PRECOAT/OVERLAY SKID 1-1414-54-506
OVERLAY TANK AGITATOR 1-1414-Q4-502
OVERLAY TANK 1-1414-54-506-T02
OVERLAY PUMP 1-1414-54-506-P02
SAMPLE LINE TO WATER ANALYSIS ROOM
MILLIPORE RESIN LEAKAGE SAMPLE
FROM EFFLUENT HEADER SAMPLE
TO WATER ANALYSIS
TO RADWASTE TRANSFER BLDG.
SEAL FLUSH WATER
BACKFLUSH SUPPLY
WATER FLUSH TO SCRAPER
DRAIN TO CHEMICAL WASTE
BLIND FLANGE
FROM DEMIN WATER SYS.
SAMPLE
BACKWASH RECOVERY OPERATION IN PROGRESS SIGNAL
TRIP SPENT RESIN PUMP ON SIGNAL FROM SPENT RESIN REMOVAL VALVES OPEN
DRAIN AND OVERFLOW FROM PRECOAT TANK AND OVERLAY TANK SKID SUMP
SKID SUMP
8" OVERFLOW
1 1/2" DRAIN
GRAVER NOTE 9
NOTE 10
NOTE 12
SEE NOTE 8 ON DWG. 185-2
FLEXIBLE CONNECTION 1-1414-Y4-501
CLOSE ON DECANT TANK H-H LVL
TRIP DECANT PUMP
DECANT RETURN
AUTO PROG
PFDP
HOLDING PUMP E 1-1414-54-505-P01
(RECOVER)
(RECIRCULATE)
(RPC)
(SEE 185-1 FOR PRECOAT TANK)
FOR NOTES SEE 1X4DB185-2
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT
P & I DIAGRAM
CONDENSATE FILT./DEM.
SYSTEM NO. 1414 SHEET 3 OF 3
SCALE: NONE
DRAWING NO. 1X4DB185-3
VER. 16.0
JOB NO. 10604
SIZE E 34x44
DRAWING CATEGORY: CRITICAL
16.0 REVISED PER ABN-V01691, VER. 1.0 7-29-09 BGT MLH PML
NO. VERSIONS DATE DR CHK APPV
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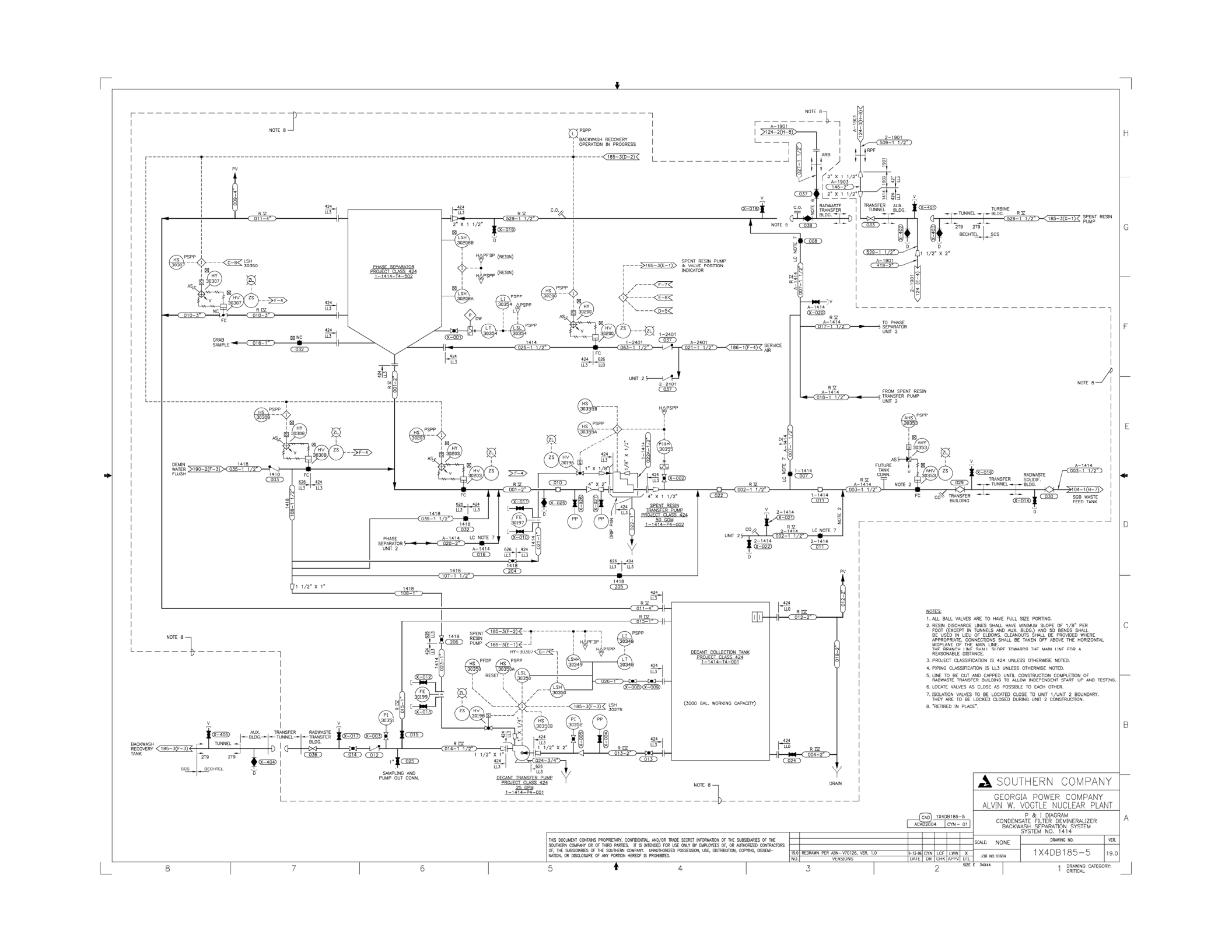

NOTES:

1. ALL BALL VALVES ARE TO HAVE FULL SIZE PORTING.
2. RESIN DISCHARGE LINES SHALL HAVE MINIMUM SLOPE OF 1/8" PER FOOT (EXCEPT IN TUNNELS AND AUX. BLDG.) AND 5D BENDS SHALL BE USED IN LIEU OF ELBOWS. CLEANOUTS SHALL BE PROVIDED WHERE APPROPRIATE. CONNECTIONS SHALL BE TAKEN OFF ABOVE THE HORIZONTAL MIDPLANE OF THE MAIN LINE. THE BRANCH LINE SHALL SLOPE TOWARDS THE MAIN LINE FOR A REASONABLE DISTANCE.
3. PROJECT CLASSIFICATION IS 424 UNLESS OTHERWISE NOTED.
4. PIPING CLASSIFICATION IS LL3 UNLESS OTHERWISE NOTED.
5. LINE TO BE CUT AND CAPPED UNTIL CONSTRUCTION COMPLETION OF RADWASTE TRANSFER BUILDING TO ALLOW INDEPENDENT START UP AND TESTING.
6. LOCATE VALVES AS CLOSE AS POSSIBLE TO EACH OTHER.
7. ISOLATION VALVES TO BE LOCATED CLOSE TO UNIT 1/UNIT 2 BOUNDARY. THEY ARE TO BE LOCKED CLOSED DURING UNIT 2 CONSTRUCTION.
8. "RETIRED IN PLACE".

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
19.0	REDRAWN PER ABN-V70126, VER. 1.0	9-13-06	CYN	LCF	LWW	X

CAD 1X4DB185-5 | ACAD2004 | CYN-01

SOUTHERN COMPANY

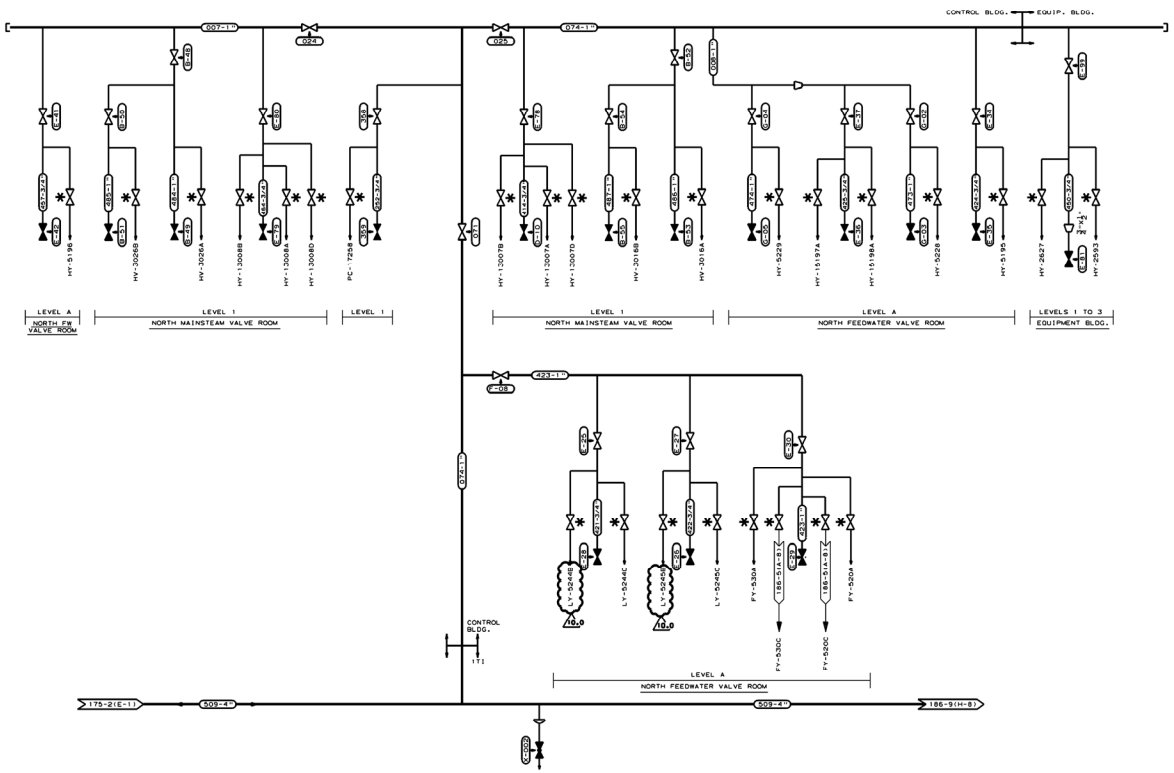
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
CONDENSATE FILTER DEMINERALIZER
BACKWASH SEPARATION SYSTEM
SYSTEM NO. 1414

SCALE: NONE | DRAWING NO. 1X4DB185-5 | VER. 19.0 | JOB NO. 10604

SIZE E 34X44

DRAWING CATEGORY: CRITICAL



NOTE:
1. * IDENTIFY EACH VALVE BY THE AIR USER'S TAG NO. SUFFIXED WITH "-EAI". EXAMPLE: FY-520A-EAI.

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NO.	VERSIONS	DATE	DR	CHK	APPV
10.0	REVISED PER ABN-V02020, VER. 1.0.	10/17/10	DRW	SRG	PML

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
INSTRUMENT AIR SYSTEM
CONTROL BUILDING LEVELS A & 1
SYSTEM 2420

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	1X4DB185-6	10.0

SIZE E 34x44

DRAWING CATEGORY: CRITICAL

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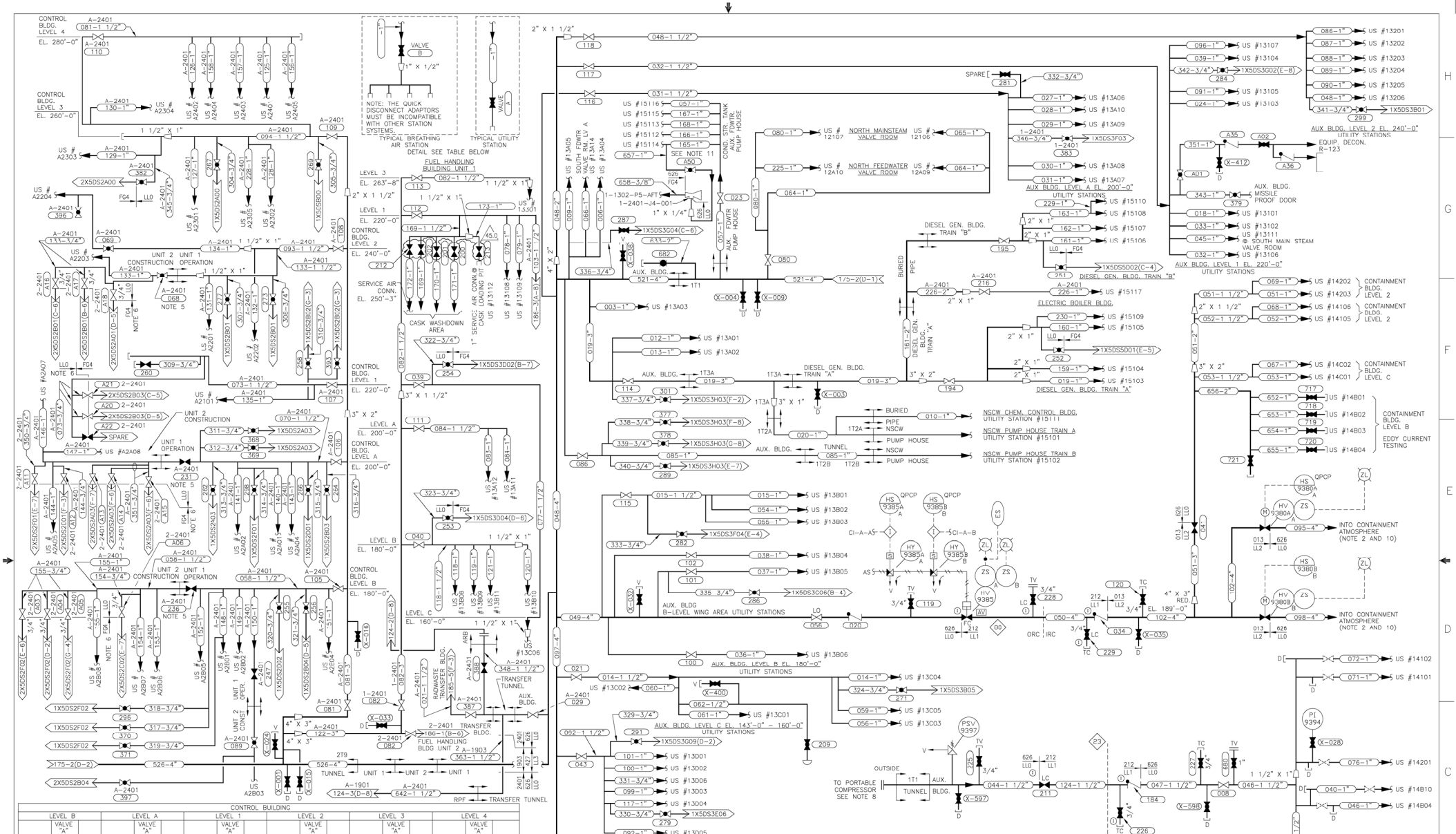

CONTROL BUILDING																					
LEVEL B			LEVEL A			LEVEL 1			LEVEL 2			LEVEL 3			LEVEL 4					
STATION	VALVE "A" A-2401	AREA	STATION	VALVE "A" A-2401	AREA	STATION	VALVE "A" A-2401	AREA	STATION	VALVE "A" A-2401	AREA	STATION	VALVE "A" A-2401	AREA	STATION	VALVE "A" A-2401	AREA			
A2B01	087	12FB	A2A01	075	12FA	A2101	070	12D1	A2201	066	12A2	A2301	062	12A3	A2401	060	12B4			
A2B02	088	12DB	A2A02	076	12FA	12106	050	12E1	A2202	067	12B2	A2302	063	12B3	A2402	061	22B4			
A2B03	395	22BB				12107	104	12E1	A2203	A19	22A2	A2303	064	22A3	A2403	127	12B4			
A2B04	090	12BB	A2A04	078	12AA				A2204	394	22B2	A2304	165	22B3	A2404	128	22B4			
A2B05	091	12DB	A2A05	079	22DA							A2305	219	12B3	A2405	126	12B4			
A2B06	092	22DB																			
A2B07	094	22DB	A2A07	084	22FA																
A2B08	095	22FB	A2A08	085	22FA																
			12A09	048	12EA																
			12A10	103	12EA																

FUEL BUILDING			AUXILIARY BUILDING																		
			LEVEL 2			LEVEL 1			LEVEL A			LEVEL B			LEVEL C			LEVEL D		
STATION	VALVE "A"	AREA	STATION	VALVE "A"	AREA	STATION	VALVE "A"	AREA	STATION	VALVE "A"	AREA	STATION	VALVE "A"	AREA	STATION	VALVE "A"	AREA	STATION	VALVE "A"	AREA
13C06	132	13DC				13101	145	13H1	13A01	151	13HA	13B01	157	13CB	13C01	162	13CC	13D01	167	13HD
13B08	133	13DB				13102	146	13H1	13A02	152	13HA	13B02	158	13EB	13C02	163	13HC	13D02	168	13HD
13B09	134	13AB				13103	147	13F1	13A03	153	13GA	13B03	159	13FB	13C03	164	13BC	13D03	169	13FD
13B10	135	13AB				13104	148	13E1	13A04	154	13GA	13B04	160	13HB	13C04	165	13CC	13D04	170	13BD
13B11	136	13AB				13105	149	13E1	13A05	155	13GA	13B05	161	13GB	13C05	166	13CC	13D05	171	13CD
13A11	137	13DA				13106	150	13G1	13A06	156	13FA	13B06	033	13HB				13D06	294	13HD
13A12	138	13AA				13107	026	13B1	13A07	027	13FA									
13108	139	13A1				13111	038	13F1	13A08	028	13BA									
13109	140	13A1							13A09	031	13BA									
13301	141	13A3							13A10	032	13CA									
13112	213	13AA							13A14	054	13AA									

MISCELLANEOUS				CONTAINMENT			BREATHING AIR STATIONS TABLE				
STATION	NO.	VALVE "A"	AREA	STATION	VALVE "A"	AREA & LEV	STATION	LOCATION	VALVE "B"	AREA & LEV
NSCW CHEM CNTL BLDG.	15111	172	17F1	14C01	178	14AC	14B01	R. C. PUMP #1	035	14AB
NSCW PMP HOUSE TR "A"	15101	173	17E1	14C02	179	14CC	14B02	R. C. PUMP #2	010	14BB
NSCW PMP HOUSE TR "B"	15102	174	17F1	14105	180	14B2	14B03	R. C. PUMP #3	011	14CB
				14106	181	14C2	14B04	R. C. PUMP #4	012	14DB
AUX FW PMP HSE TR "A"	15112	197	17C1	14202	182	14A2	14201	PRESSURIZER (TOP)	013	14D1
D. G. BLDG. TRAIN "A"	15103	191	17D1	14203	183	14D2	14101	REFUELING POOL	014	14D2
D. G. BLDG. TRAIN "A"	15104	189	17D1	14B01	717	14DB	14102	REFUELING POOL	015	14C1
D. G. BLDG. TRAIN "A"	15105	190	17D1	14B02	718	14DB	14103	REACTOR HD LAYDOWN	016	14A1
D. G. BLDG. TRAIN "A"	15109	192	17D1	14B03	719	14DB	14104	REACTOR HD LAYDOWN	017	14B1
D. G. BLDG. TRAIN "B"	15106	96	17D1	14B04	720	14DB	14B10	PRESSURE RELIEF TK	018	14DB
D. G. BLDG. TRAIN "B"	15107	97	17D1							
D. G. BLDG. TRAIN "B"	15108	98	17D1							
D. G. BLDG. TRAIN "B"	15110	99	17D1							
AUX. BOILER BLDG.	15117	217	17E1							
AUX FW PMP HSE TR "B"	15113	198	17C1							
AUX FW PMP HSE TR "C"	15114	196	17C1							
COND. STR. TK. EL. 220'	15115	199	17C1							
COND. STR. TK. EL. 238'	15116	220	17C1							

UTILITY STATION TABLE

NOTES:

1. ALL PIPING, VALVES AND EQUIPMENT ARE PROJECT CLASS 626 AND PIPING SPEC. "LLO" EXCEPT AS OTHERWISE NOTED.
2. AIR SUPPLY FOR CONTAINMENT POST LOCA PURGE PRESSURIZATION. REFER TO P & ID 1X4DB213 FOR POST LOCA PURGE FILTER EXHAUST.
3. FOR TUNNEL NUMBER IDENTIFICATION REFER TO DRAWING 1X4DB397.
4. HOLD WELDING OF CAP FOR START-UP AIR BLOW.
5. ISOLATION VALVES TO BE LOCATED CLOSE TO U1/U2 BOUNDARY. THEY ARE TO BE LOCKED CLOSED DURING UNIT 2 CONSTRUCTION.
6. CLASS BREAK HAS BEEN SHOWN OFF THE ENDS OF THE VALVES TO ALLOW CONSTRUCTION TO INSTALL UNIT 2 SYSTEM COMPONENTS TO THE VALVES WITHOUT HAVING TO INTERFACE WITH UNIT 1 NUCLEAR OPERATIONS.
7. A PORTABLE EMERGENCY BREATHING AIR BACK-UP SUPPLY MUST BE FURNISHED BY GPC HEALTH PHYSICS.
8. PORTABLE COMPRESSOR AND AIR PURIFIER (GRADE D AIR REQUIRED) TO BE SUPPLIED BY GPC.
9. SEE TYPICAL BREATHING AIR STATION DETAIL (H-6).
10. THREADED END OF PIPE TO BE USED FOR MOV TESTING OR TO PROVIDE SERVICE AIR OUTLETS IN CONTAINMENT DURING MODES 5, 6 OR DEFUELED ONLY.
11. VALVE 1-2401-U4-A50 MAY BE LEFT IN A THROTTLED POSITION.

CONTAINMENT BREATHING AIR STATIONS SEE NOTES 7 & 9

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
SERVICE AIR SYSTEM
SYSTEM NO. 2401

CAD 1X4DB186-1
ACAD07 JLO - 05

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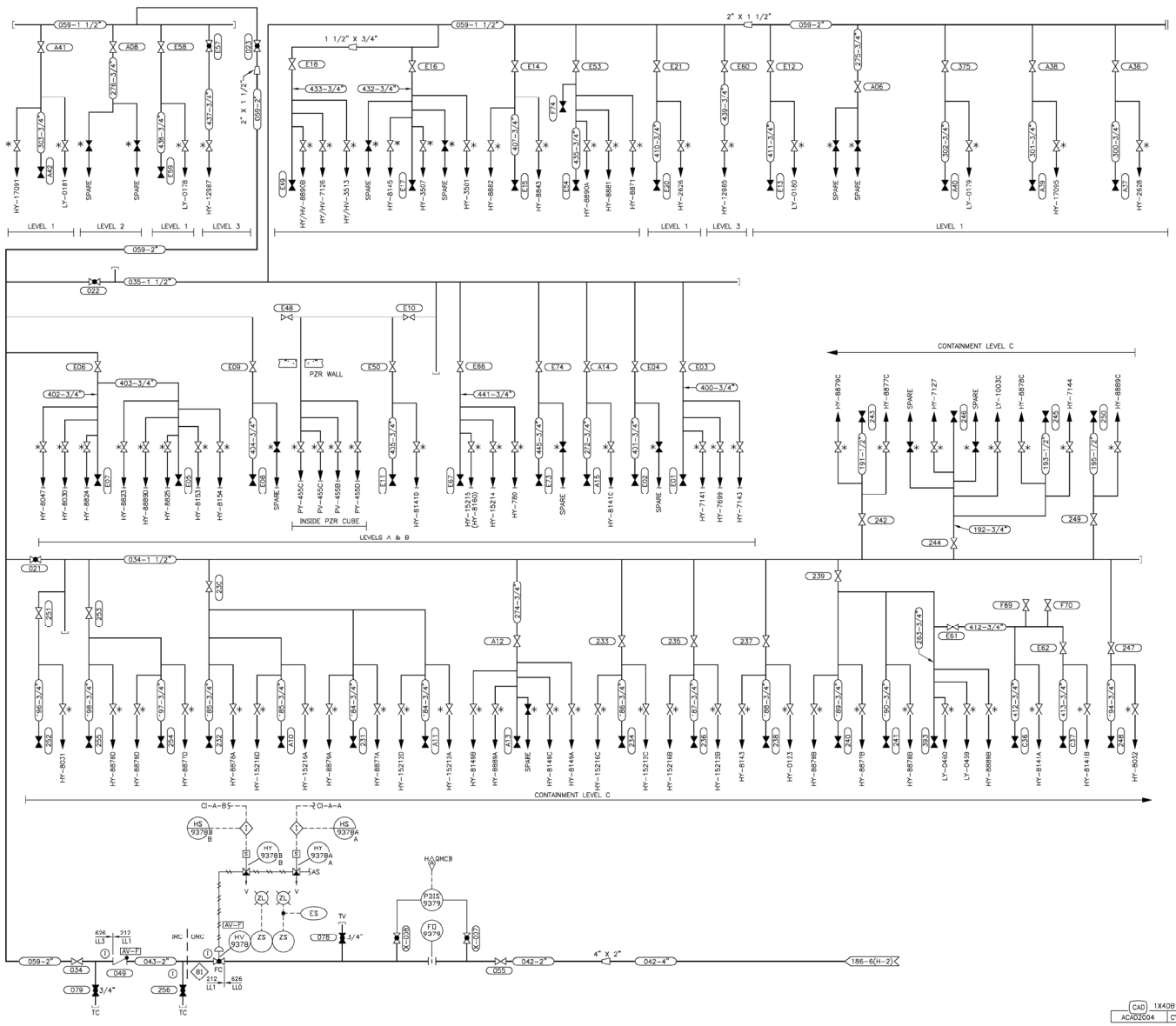
NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
45.0	REVISED PER ABN C100867701M002, VER. 2.0	7/9/13	JLO	ELC	JMR	X

SCALE: NONE
JOB NO.10604
DRAWING NO. 1X4DB186-1
VER. 45.0
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTES:
1. * IDENTIFY EACH VALVE BY THE AIR USER'S TAG NO. SUFFIXED WITH EAL EXAMPLE: LY-0178-EAL

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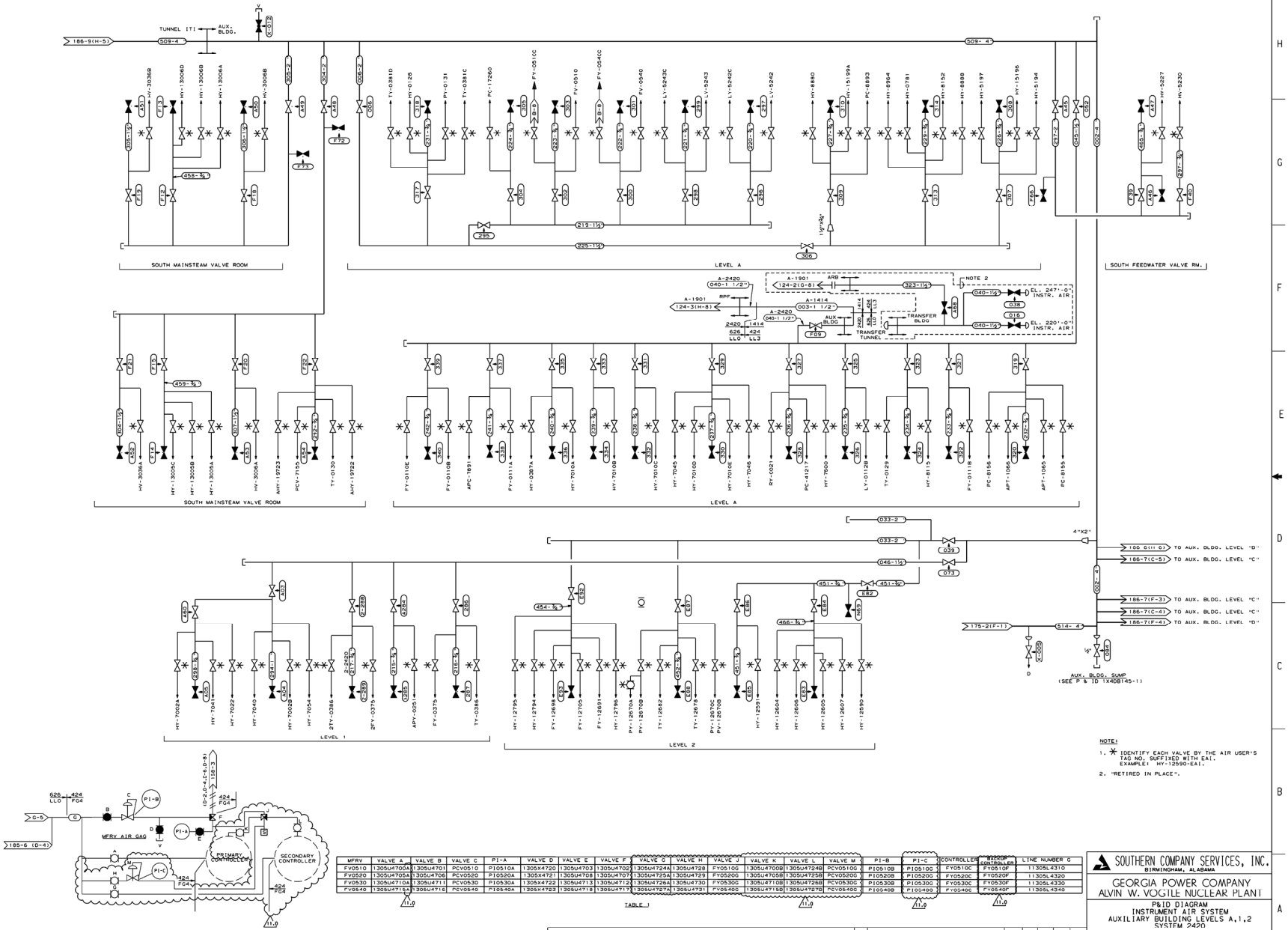
NO.	VERSIONS	DATE	DR	CHK	AP'VV	UTL
14.0	REDRAWN PER ABN-V70154, VER. 1.0	11-30-06	CYN	LCF	WJJ	X

CAD 1X4DB186-4
ACAD2004 CYN-01

SOUTHERN COMPANY
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
INSTRUMENT AIR SYSTEM
CONTAINMENT BUILDING
SYSTEM 2420

SCALE: NONE
SIZE E 34X44
DRAWING NO. 1X4DB186-4
VER. 14.0
JOB NO. 10604
DRAWING CATEGORY: CRITICAL



MFRV	VALVE A	VALVE B	VALVE C	PI-A	VALVE D	VALVE E	VALVE F	VALVE G	VALVE H	VALVE J	VALVE K	VALVE L	VALVE M	PI-B	PI-C	CONTROLLER	BACKUP CONTROLLER	LINE NUMBER G
FV0510	1305U4700A	1305U4701	PCV0510	P10510A	1305X4720	1305U4703	1305U4702	1305U4724A	1305U4728	FY0510G	1305U4700B	1305U4724B	PCV0510G	P10510B	P10510G	FY0510C	FY0510F	11305L4310
FV0520	1305U4705A	1305U4706	PCV0520	P10520A	1305X4721	1305U4708	1305U4707	1305U4725A	1305U4729	FY0520G	1305U4705B	1305U4725B	PCV0520G	P10520B	P10520G	FY0520C	FY0520F	11305L4320
FV0530	1305U4710A	1305U4711	PCV0530	P10530A	1305X4722	1305U4713	1305U4712	1305U4726A	1305U4730	FY0530G	1305U4710B	1305U4726B	PCV0530G	P10530B	P10530G	FY0530C	FY0530F	11305L4330
FV0540	1305U4715A	1305U4716	PCV0540	P10540A	1305X4723	1305U4718	1305U4717	1305U4727A	1305U4731	FY0540G	1305U4715B	1305U4727B	PCV0540G	P10540B	P10540G	FY0540C	FY0540F	11305L4340

TABLE I

NOTE:

1. * IDENTIFY EACH VALVE BY THE AIR USER'S TAG NO. SUFFIXED WITH EAI. EXAMPLE: HY-12590-EAI.
2. "RETIRED IN PLACE".

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NO.	VERSIONS	DATE	DR	CHK	APPV
11.0	REVISED PER ABN 1039002501J105, VER. 2.0	10/01/12	JLG	ELC	JWR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

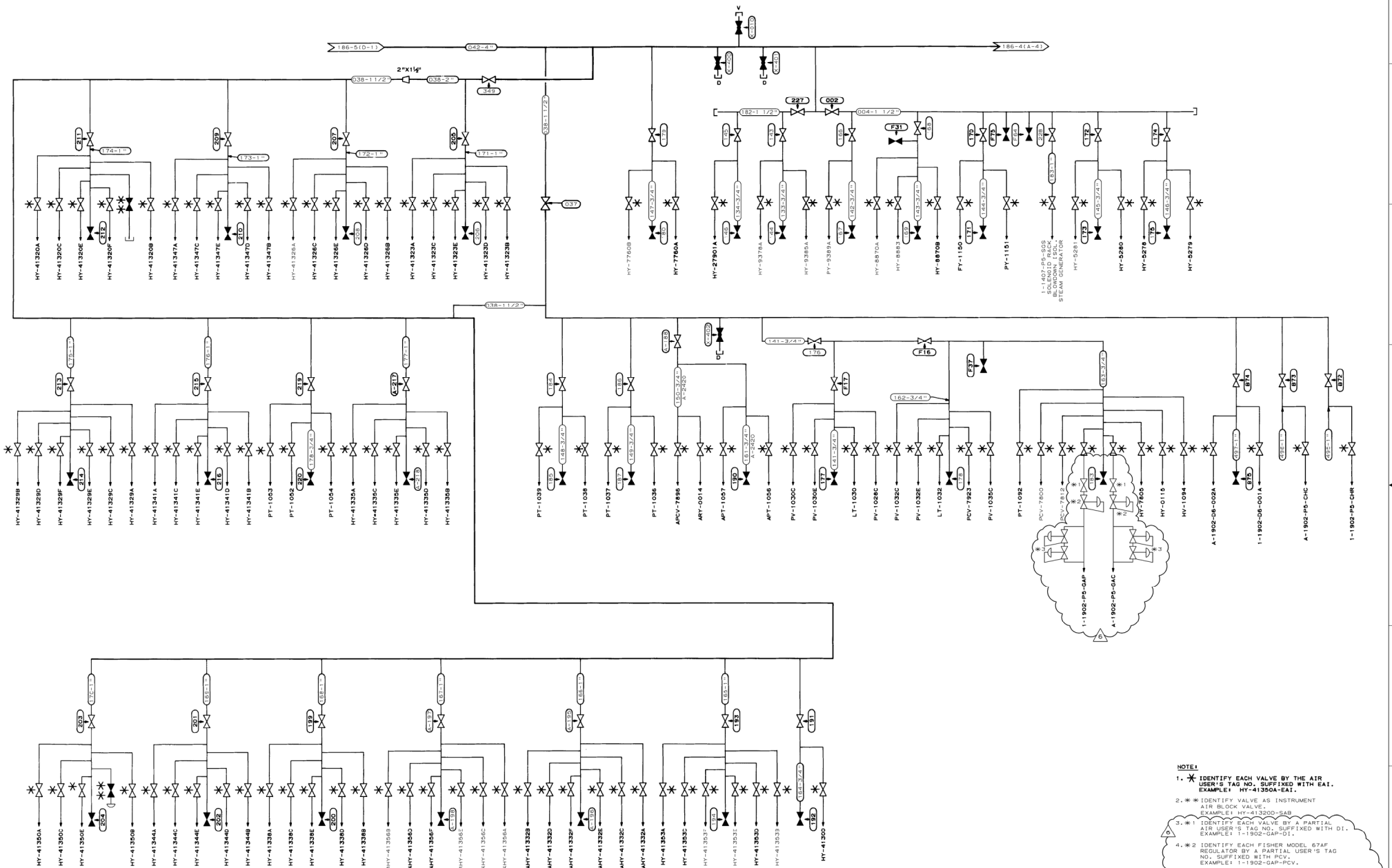
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P&ID DIAGRAM
INSTRUMENT AIR SYSTEM
AUXILIARY BUILDING LEVELS A,1,2
SYSTEM 2420

SCALE: NONE
JOB NO.10604
SIZE E 34X44

DRAWING NO. 1X4DB186-5
VER. 11.0

DRAWING CATEGORY: 1 CRITICAL



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NO.	REVISIONS	DATE	DR	CHK	APPV	DTL
3	INCORP. PER ABN-36791	11/09/01	PM	GLB	MWD	

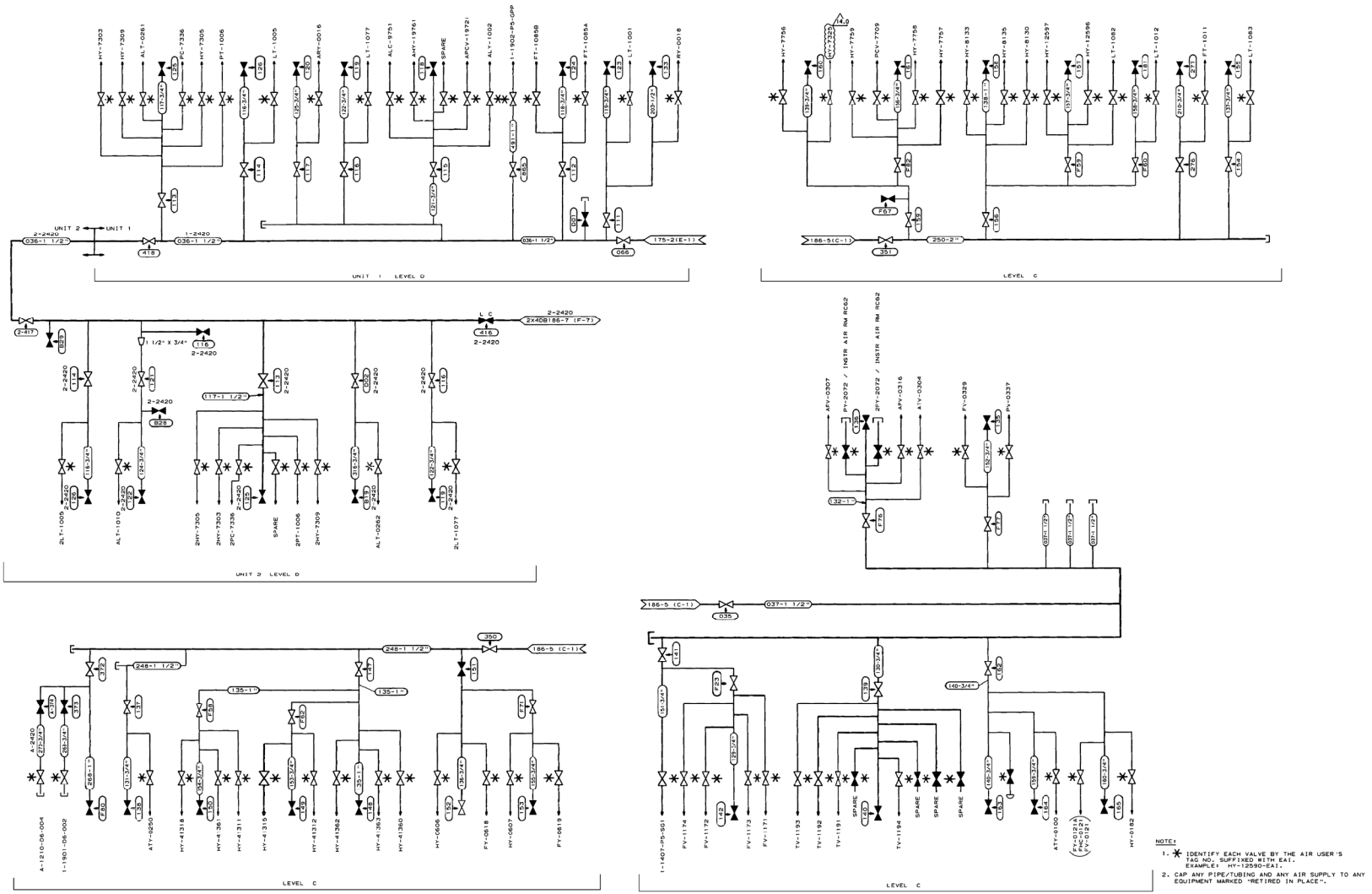
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
INSTRUMENT AIR SYSTEM
AUXILIARY BUILDING LEVEL B
SYSTEM 2420

SCALE: NONE	DRAWING NO.	REV.
JOB NO. 10604	1X4DB186-6	6

SIZE E 34X44



SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
INSTRUMENT AIR SYSTEM
AUXILIARY BUILDING LEVELS C & D
SYSTEM 2420

SCALE: NONE

DRAWING NO. 1X4DB186-7 VER. 14.0

JOB NO. 10604

SIZE E 34x44

DRAWING CATEGORY: CRITICAL

NO.	VERSIONS	DATE	DR	CHK	APPV
14.0	REVISED PER ABN-VO3618, VER.1.0	3-6-14	BGT	LCF	RKF

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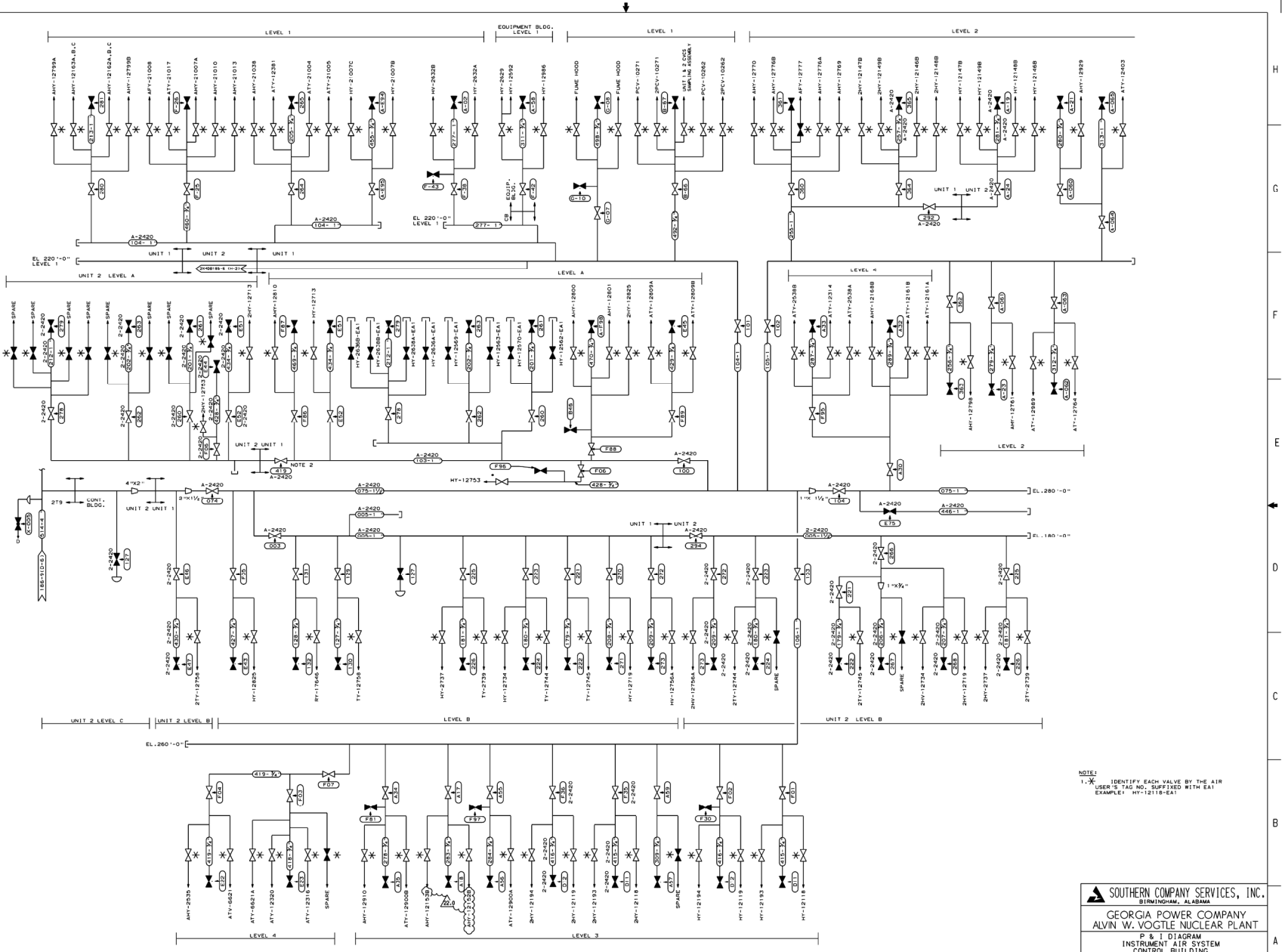

NOTE:
1. IDENTIFY EACH VALVE BY THE AIR USER'S TAG NO. SUFFIXED WITH EA1
EXAMPLE: HY-12118-EA1

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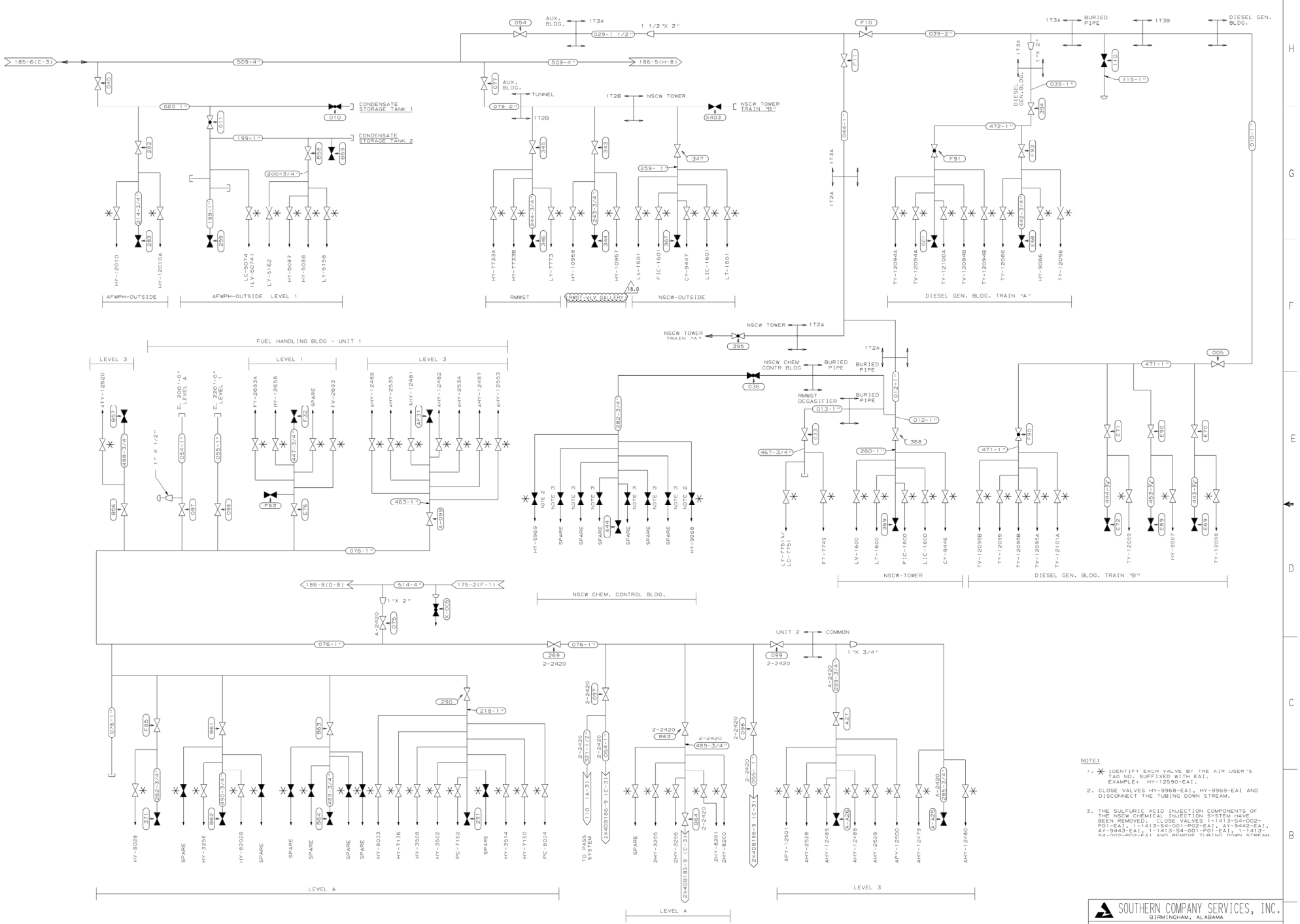
NO.	VERSIONS	DATE	DR	CHK	APPV
22.0	REVISED PER ABN-V03283, VER. 1.0	5/3/13	JJB	CJK	RKF

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
INSTRUMENT AIR SYSTEM
CONTROL BUILDING
SYSTEM 2420

SCALE: N.T.S.
DRAWING NO. 1X4DB186-8
VER. 22.0
JOB NO.10604
SIZE E 34X44
DRAWING CATEGORY: CRITICAL



NOTE:

1. X IDENTIFY EACH VALVE BY THE AIR USER'S TAG NO. SUFFIXED WITH EAI. EXAMPLE: HY-12590-EAI.
2. CLOSE VALVES HY-9968-EAI, HY-9969-EAI AND DISCONNECT THE TUBING DOWN STREAM.
3. THE SULFURIC ACID INJECTION COMPONENTS OF THE NSCW CHEMICAL INJECTION SYSTEM HAVE BEEN REMOVED. CLOSE VALVES 1-1413-S4-002-P01-EAI, 1-1413-S4-001-P02-EAI, AY-9442-EAI, AY-9443-EAI, 1-1413-S4-001-P01-EAI, 1-1413-S4-002-P02-EAI AND REMOVE TUBING DOWN STREAM.

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
INSTRUMENT AIR SYSTEM
AUX FEED WTR PUMP HSE & FUEL HNDLG BLDG
SYSTEM 2420

SCALE: NONE | DRAWING NO. 1X4DB186-9 | VER. 18.0 | JOB NO. 10604

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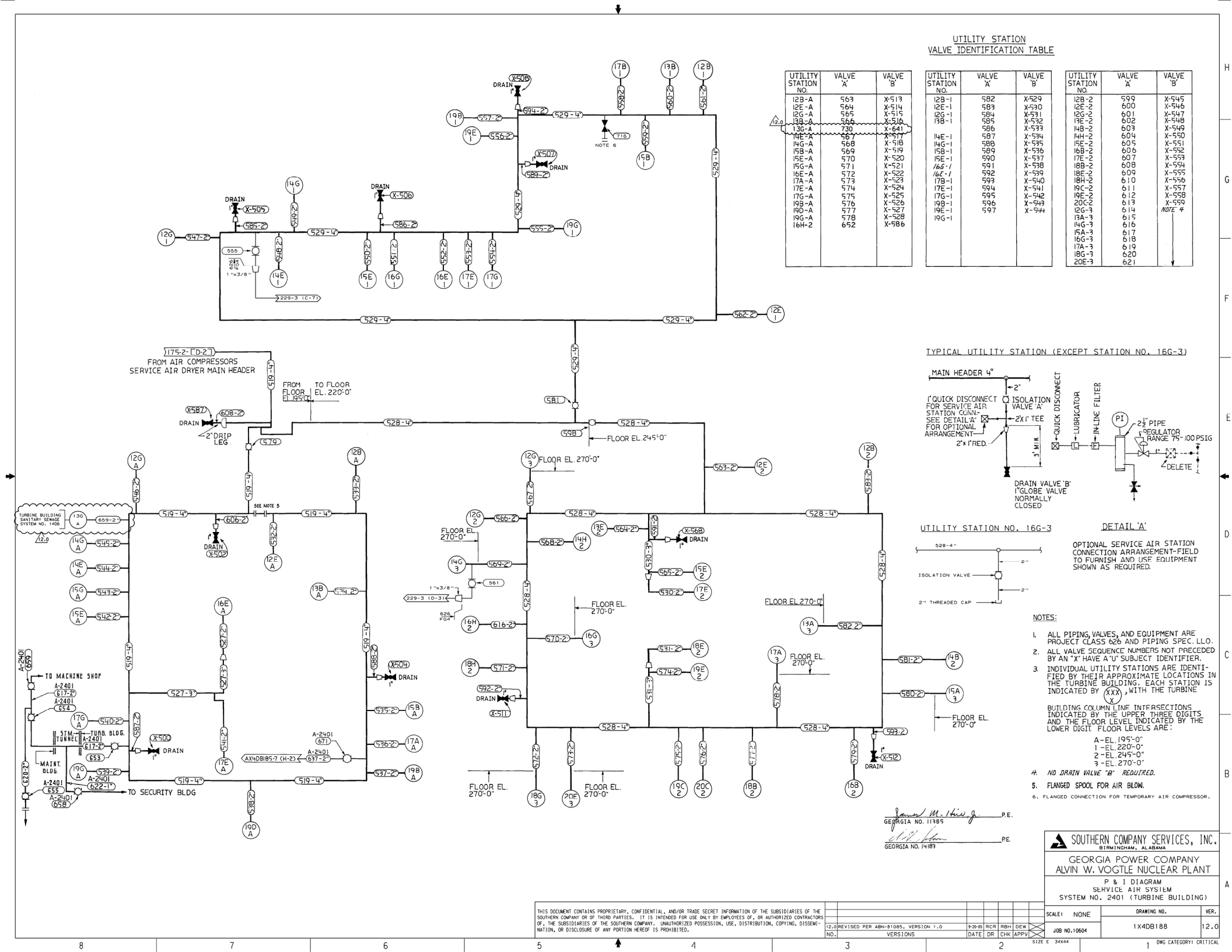
NO.	VERSIONS	DATE	DR	CHK	APPV
18.0	REVISED PER ABN-V01567, VER.1.0	4-8-09	BgT	LRP	PML

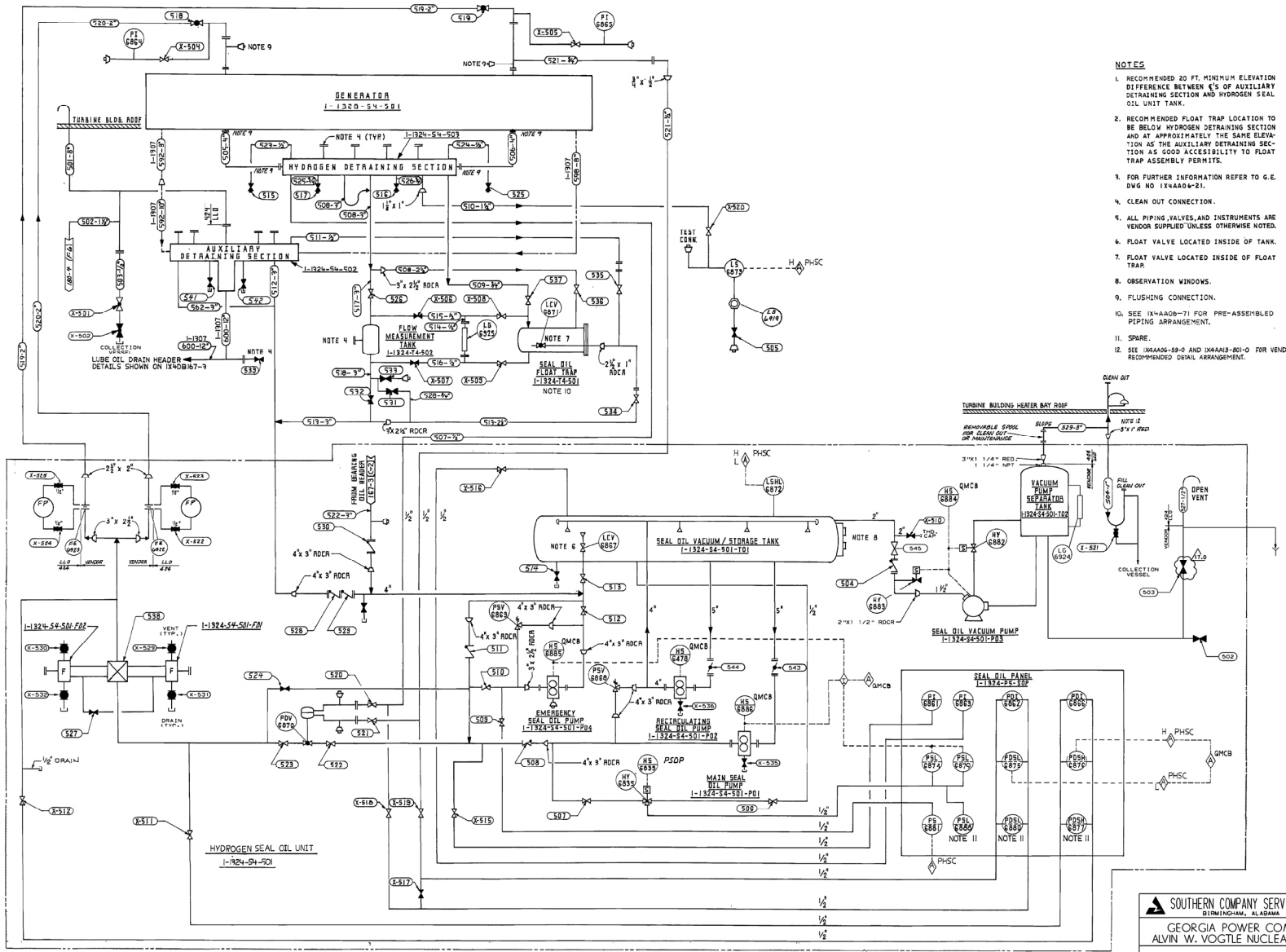
SIZE E 34x44 | DWG CATEGORY: CRITICAL



UTILITY STATION NO.	VALVE 'A'	VALVE 'B'
12B-A	563	X-513
12E-A	564	X-514
12G-A	565	X-515
13B-A	566	X-516
13G-A	730	X-641
14E-A	567	X-517
14G-A	568	X-518
15B-A	569	X-519
15E-A	570	X-520
15G-A	571	X-521
16E-A	572	X-522
17A-A	573	X-523
17E-A	574	X-524
17G-A	575	X-525
19B-A	576	X-526
19D-A	577	X-527
19G-A	578	X-528
16H-2	652	X-586

UTILITY STATION NO.	VALVE 'A'	VALVE 'B'
12B-1	582	X-529
12E-1	583	X-530
12G-1	584	X-531
13B-1	585	X-532
	586	X-533
14E-1	587	X-534
14G-1	588	X-535
15B-1	589	X-536
15E-1	590	X-537
16G-1	591	X-538
17B-1	592	X-539
17E-1	593	X-540
17G-1	594	X-541
19B-1	595	X-542
19E-1	596	X-543
19G-1	597	X-544

UTILITY STATION NO.	VALVE 'A'	VALVE 'B'
12B-2	599	X-545
12E-2	600	X-546
12G-2	601	X-547
13E-2	602	X-548
14B-2	603	X-549
14H-2	604	X-550
15E-2	605	X-551
16B-2	606	X-552
17E-2	607	X-553
18B-2	608	X-554
18E-2	609	X-555
18H-2	610	X-556
19C-2	611	X-557
19E-2	612	X-558
20C-2	613	X-559
12G-3	614	NOTE 4
13A-3	615	
14G-3	616	
15A-3	617	
16G-3	618	
17A-3	619	
18G-3	620	
20E-3	621	



NOTES

1. RECOMMENDED 20 FT. MINIMUM ELEVATION DIFFERENCE BETWEEN ℄'S OF AUXILIARY DETRAINING SECTION AND HYDROGEN SEAL OIL UNIT TANK.
2. RECOMMENDED FLOAT TRAP LOCATION TO BE BELOW HYDROGEN DETRAINING SECTION AND AT APPROXIMATELY THE SAME ELEVATION AS THE AUXILIARY DETRAINING SECTION AS GOOD ACCESSIBILITY TO FLOAT TRAP ASSEMBLY PERMITS.
3. FOR FURTHER INFORMATION REFER TO G.E. DWG NO 1X4AA06-21.
4. CLEAN OUT CONNECTION.
5. ALL PIPING, VALVES, AND INSTRUMENTS ARE VENDOR SUPPLIED UNLESS OTHERWISE NOTED.
6. FLOAT VALVE LOCATED INSIDE OF TANK.
7. FLOAT VALVE LOCATED INSIDE OF FLOAT TRAP.
8. OBSERVATION WINDOWS.
9. FLUSHING CONNECTION.
10. SEE 1X4AA06-71 FOR PRE-ASSEMBLED PIPING ARRANGEMENT.
11. SPARE.
12. SEE 1X4AA06-59-0 AND 1X4AA13-801-0 FOR VENDOR RECOMMENDED DETAIL ARRANGEMENT.

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NO.	REVISION	DATE	DR	CHK	APPV
17.0	REVISED PER ABN-V02795, VER. 1.0	11-17-11	DCP	ELC	JMR

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

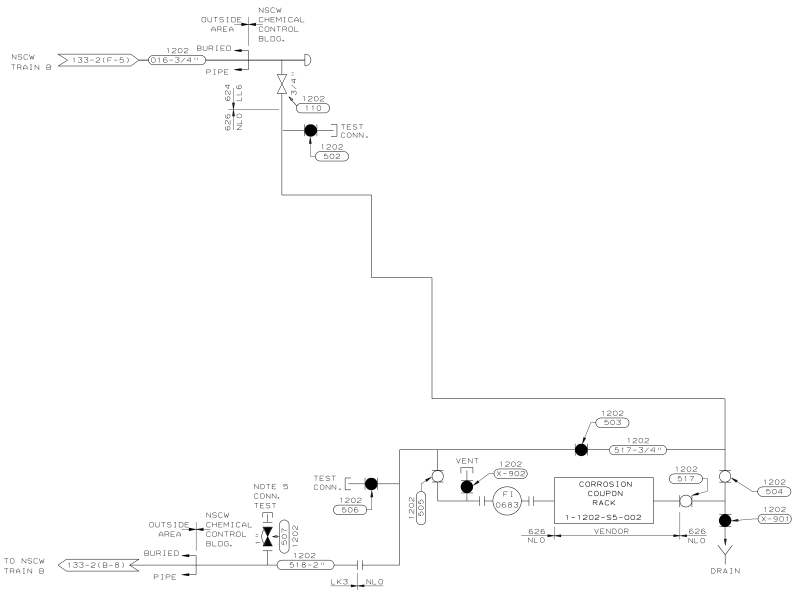
GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

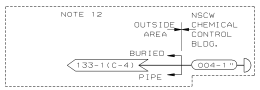
P&I DIAGRAM
TURBINE GENERATOR HYDROGEN SEAL OIL SYS
SYSTEM NO. 1324

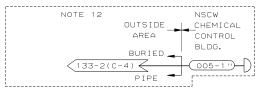
SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	1X4DB191	17.0

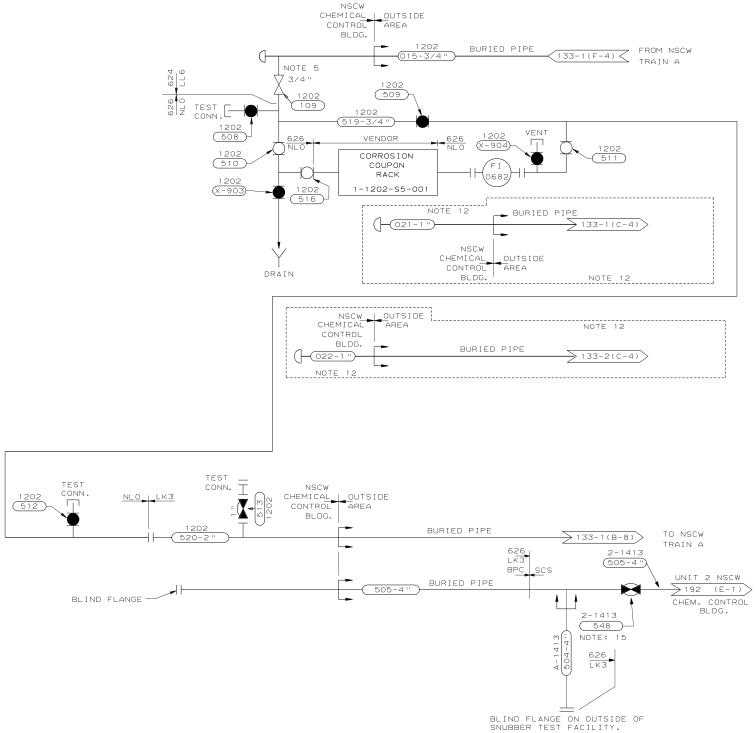
SIZE E 34x44

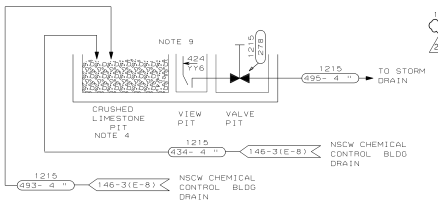
DRAWING CATEGORY: CRITICAL











NOTES:

1. DELETED
2. DELETED
3. DELETED
4. PIT AREA 450 SQ. FT.
5. ALL ISOLATION VALVES AND ALL TEST CONNECTION VALVES SHALL BE LOCATED AS CLOSE AS POSSIBLE TO THE BURIED PIPE.
6. DELETED
7. DELETED
8. DELETED
9. DRAIN PIPES PENETRATE 6" INTO CRUSHED LIMESTONE.
10. DELETED
11. DELETED
12. THE ACID AND DISPERSANT INJECTION COMPONENTS OF THE NSCW CHEMICAL INJECTION SYSTEM HAVE BEEN REMOVED. THIS PORTION OF PIPING IS "RETIRED IN PLACE".
13. LEASED/VENDOR FURNISHED EQUIPMENT
14. PROVIDE SEISMICALLY QUALIFIED SUPPORTS FOR ALL PIPING INSIDE NSCW TOWER SPLASH RING, INCLUDING INJECTION POINT AT EDGE OF BASIN.
15. LOCK CLOSE VALVE 2-1413-548.

21.0

THIS DRAWING WAS RENUMBERED FROM 1X4DB192 (SHEET 1)

WORK THIS DRAWING WITH: 1X4DB192-2

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21.0	REVISED PER ABN-V00621, VER. 2.0.	5-21-07	TJS	MWD	WLN

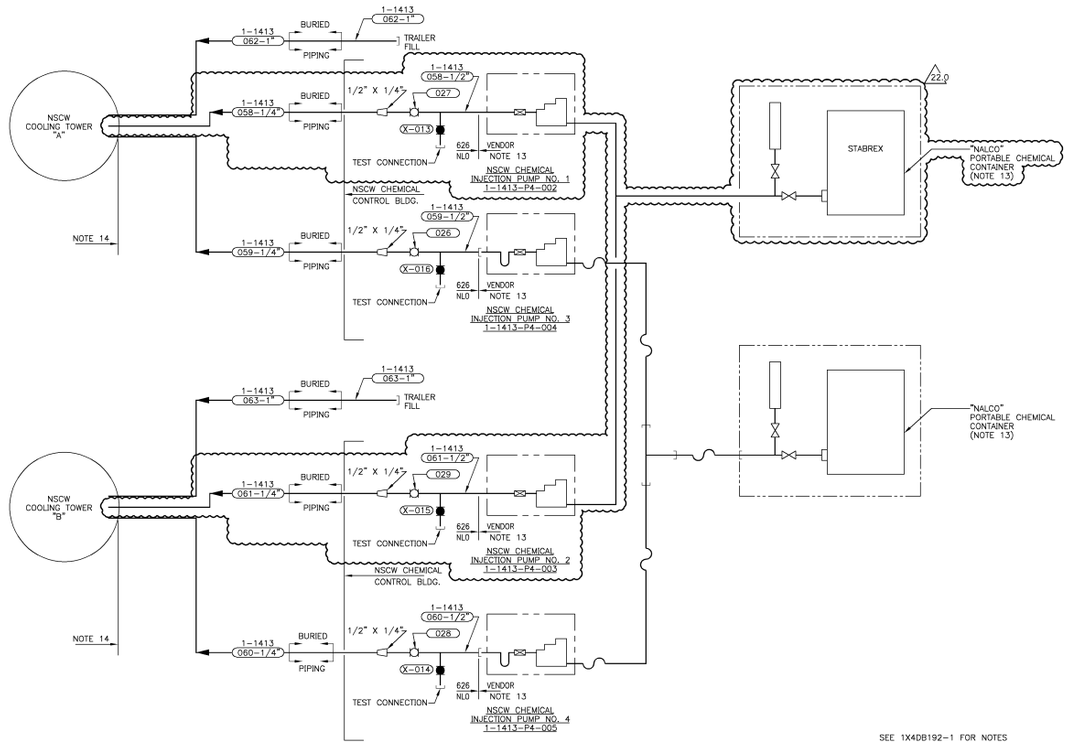
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
NUCLEAR SERVICE COOLING WATER
CHEM INJECTION SYSTEM &
CHLORINE ANALYSIS SYSTEM 1413

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB192-1	21.0

SIZE E 34X44



SEE 1X4DB192-1 FOR NOTES

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WORK THIS DRAWING WITH: 1X4DB192-1

CAD	1X4DB192-2	
Acad2004	TJS - 02	

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
22.0	REVISED PER ABN-V00621, VER. 2.0.	5-21-07	TJS	MWD	WLN	X

SIZE E 34X44

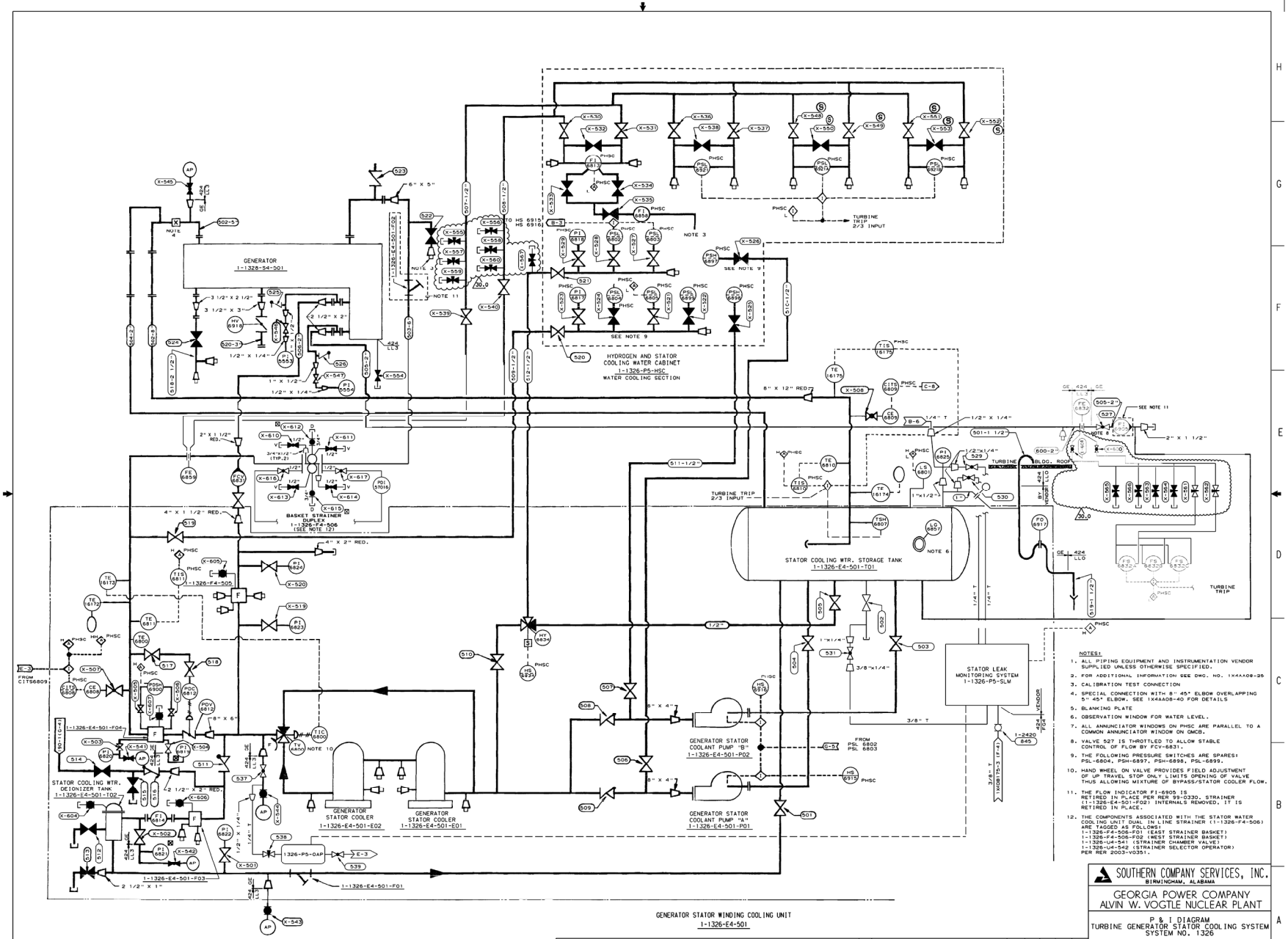
SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
NUCLEAR SERVICE COOLING WATER
CHEM. INJECTION SYSTEM &
CHLORINE ANALYSIS SYSTEM 1413

SCALE: NONE	DRAWING NO.	VER.
JOB NO.10604	1X4DB192-2	22.0

DRAWING CATEGORY: CRITICAL



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NO.	VERSIONS	DATE	DR	CHK	APPV
30.0	REVISED PER ABN SNC494736J482, VER. 1.0	10/26/15	SMO	SO	WLN

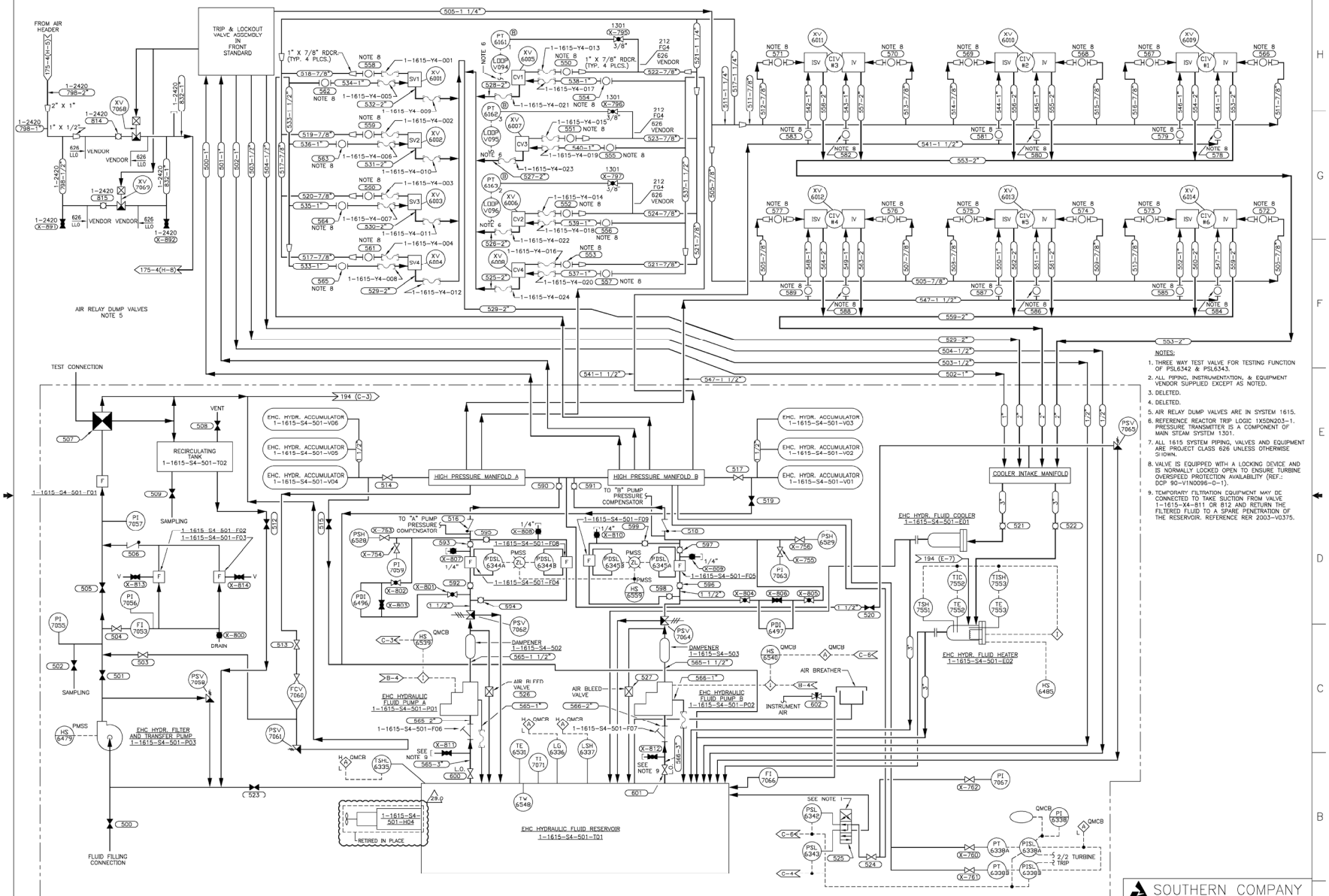
SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
TURBINE GENERATOR STATOR COOLING SYSTEM
SYSTEM NO. 1326

SCALE: NONE	DRAWING NO.	VER.
JOB NO. 10604	1X4DB193	30.0

SIZE E 34x44 — DWG CATEGORY: CRITICAL



NOTES:

1. THREE WAY TEST VALVE FOR TESTING FUNCTION OF PSL6342 & PSL6343.
2. ALL PIPING, INSTRUMENTATION, & EQUIPMENT VENDOR SUPPLIED EXCEPT AS NOTED.
3. DELETED.
4. DELETED.
5. AIR RELAY DUMP VALVES ARE IN SYSTEM 1615.
6. REFERENCE REACTOR TRIP LOGIC 1XSDN203-1. PRESSURE TRANSMITTER IS A COMPONENT OF MAIN STEAM SYSTEM 1301.
7. ALL 1615 SYSTEM PIPING, VALVES AND EQUIPMENT ARE PROJECT CLASS 626 UNLESS OTHERWISE SHOWN.
8. VALVE IS EQUIPPED WITH A LOCKING DEVICE AND IS NORMALLY LOCKED OPEN TO ENSURE TURBINE OVERSPEED PROTECTION AVAILABILITY (REF.: DCP 90-V1N0096-0-1).
9. TEMPORARY FILTRATION EQUIPMENT MAY BE CONNECTED TO TAKE SUCTION FROM VALVE 1-1615-X4-811 OR 812 AND RETURN THE FILTERED FLUID TO A SPARE PENETRATION OF THE RESERVOIR. REFERENCE RER 2003-V0375.

EHC HYDR. POWER UNIT
1-1615-S4-501

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
ELECTRO-HYDRAULIC CONTROL (EMC) SYSTEM
SYSTEM NO. 1615

CAD 1X4DB194
Acad2004 | TJS - 02

SCALE: NONE

DRAWING NO. 1X4DB194 VER. 29.0

JOB NO.10604

SIZE E 34X44

1 DRAWING CATEGORY: CRITICAL

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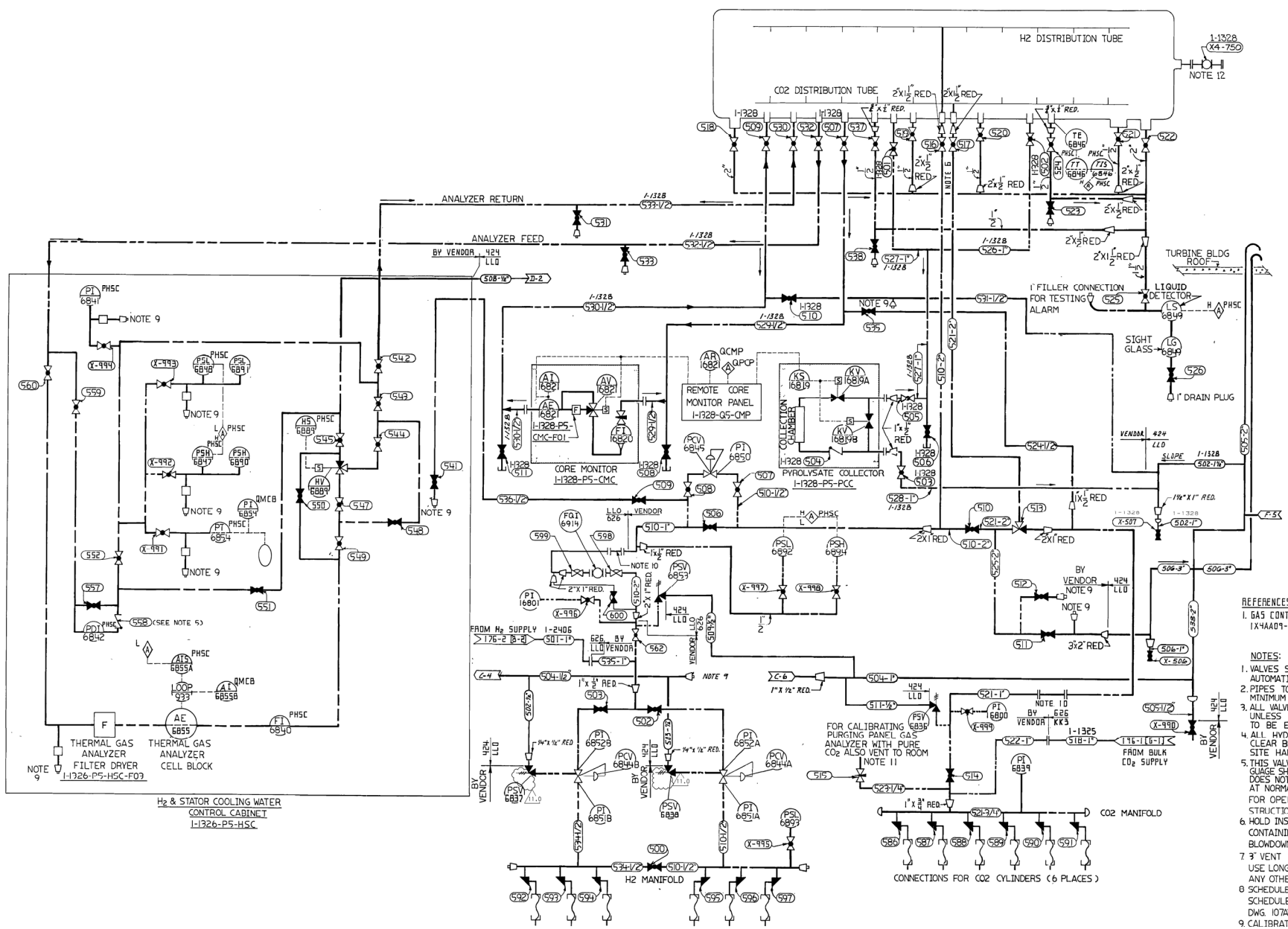
NO.	VERSIONS	DATE	DR	CHK	APPV	UTL
29.0	REVISED PER ABN- V00852, VER. 1.0.	8-07-07	TJS	MWD	WLN	X



REFERENCES

1. GAS CONTROL PIPING DIAGRAM [BY VENDOR] 1X4AA09-2.

NOTES:

1. VALVES SHOWN IN NORMAL POSITION FOR AUTOMATIC OPERATION IN HYDROGEN.
2. PIPES TO PITCH DOWN IN DIRECTION OF ARROWS MINIMUM 1/4" PER FOOT.
3. ALL VALVES AND PIPE FITTINGS WELDED STEEL UNLESS NOTED. ALL DRAIN PLUGS AND VALVES TO BE EASILY ACCESSIBLE LOCATIONS.
4. ALL HYDROGEN GAS PIPING TO BE BLOWN CLEAR BEFORE AND DURING INSTALLATION SITE HANGING.
5. THIS VALVE PREVENTS LOSS OF LIQUID FROM THE GUAGE SHOULD THE GUAGE LIMITS BE EXCEEDED & DOES NOT RESTRICT FLOW IN EITHER DIRECTION AT NORMAL GUAGE DIFFERENTIAL PRESSURE. FOR OPERATING INSTRUCTIONS REFER TO INSTRUCTION BOOK.
6. HOLD INSTALLATION OF LAST SPOOL OF 2" LINES CONTAINING VALVES 516 & 517 UNTIL AFTER BLOWDOWN OF PIPING.
7. 3" VENT TO BE SHIELDED OUTSIDE OF BUILDING USE LONG R. ELBOWS AND DO NOT TIE IN WITH ANY OTHER PIPING.
8. SCHEDULE 80 PIPE-1/2" THRU 2" LARGER SIZE SCHEDULE 40, PIPING INSTRUCTIONS PER GE. DWG. 107A3231 (SCS NO. 1X4AA06-4)
9. CALIBRATION OR TEST CONNECTION.
10. REMOVE SPOOL PIECE AND ASSEMBLE BLANKING FLANGES TO GAS SUPPLY SIDE WHEN GENERATOR IS DE-GASSED.
11. CONNECTION FOR LOCAL AIR SUPPLY FOR PURGING GENERATOR OF CO_2 FOR MAINTENANCE. DO NOT MAKE CONNECTION PERMANENT. DISCONNECT AFTER EACH USE. USE ONLY INSTRUMENT QUALITY AIR.
12. FLUX PROBE VALVE LOCATED ON NORTH SIDE OF THE GENERATOR.

SOUTHERN COMPANY SERVICES, INC.
BIRMINGHAM, ALABAMA

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
GENERATOR GAS PURGING SYSTEM
SYS. NO. 1325

SCALE: NONE | DRAWING NO. 1X4DB196 | VER. 11.0 | JOB NO. 10604 | SIZE E 34x44

NO.	VERSIONS	DATE	DR	CHK	APPV
11.0	REVISED PER ABN-V01096, VER.1.0	3-14-08	BgT	WMW	PML

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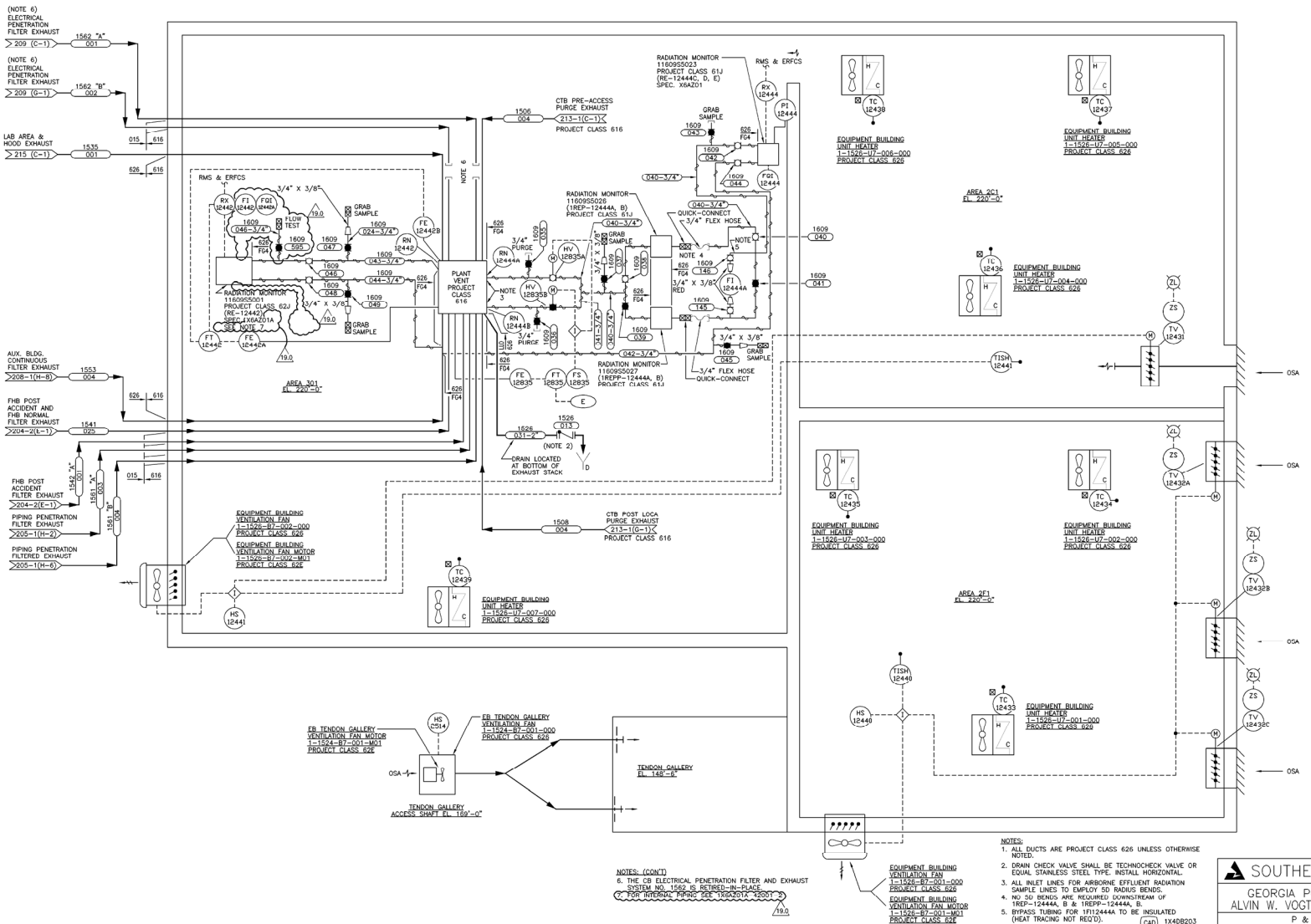

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NO.	VERSIONS	DATE	DR	CHK	APPV	DTL
19.0	REVISED PER ABN 108013180\J002, VER. 2.0	09-26-12	ELC	TSL	JMR	X

SOUTHERN COMPANY

GEORGIA POWER COMPANY
ALVIN W. VOGTLE NUCLEAR PLANT

P & I DIAGRAM
EQUIPMENT BUILDING HVAC SYSTEM
SYSTEM NO. 1526 & 1524

CAD 1X4DB203
ACAD2004 | ELC – 02

SCALE: NONE
SIZE E 34X44
JOB NO.10604

DRAWING NO. 1X4DB203
VER. 19.0

DRAWING CATEGORY: CRITICAL